

750+ **BLOCKBUSTER** **PROBLEMS in** **CHEMISTRY** **for JEE Main**

Corporate
Office

DISHA PUBLICATION

45, 2nd Floor, Maharishi Dayanand Marg,
Corner Market, Malviya Nagar, New Delhi - 110017
Tel : 49842349 / 49842350

No part of this publication may be reproduced in any form without prior permission of the publisher. The author and the publisher do not take any legal responsibility for any errors or misrepresentations that might have crept in. We have tried and made our best efforts to provide accurate up-to-date information in this book.

All Right Reserved

© Copyright
Disha

Typeset by Disha DTP Team



www.dishapublication.com

Books &
ebooks for
School &
Competitive
Exams



www.mylearninggraph.com

Etests
for
Competitive
Exams



Write to us at feedback_disha@aiets.co.in

Contents

S.no.	Chapter Name	Page no.	
		Questions	Solutions
1.	Some Basic Concepts of Chemistry	1 – 3	103-106
2.	Structure of Atom	4 – 7	107-110
3.	Classification of Elements and Periodicity in Properties	8 – 10	111-112
4.	Chemical Bonding and Molecular Structure	11 – 14	113-117
5.	States of Matter	15 – 17	118-121
6.	Thermodynamics	18 – 21	122-125
7.	Equilibrium	22 – 25	126-130
8.	Redox Reactions	26 – 27	131-132
9.	Hydrogen	28 – 29	133-134
10.	The s-Block Elements	30 – 32	135-136
11.	The p-Block Elements (Group-13 and 14)	33 – 35	137-138
12.	Organic Chemistry — Some Basic Principles & Techniques	36 – 40	139-143
13.	Hydrocarbons	41 – 45	144-147
14.	Environmental Chemistry	46 – 47	148
15.	The Solid State	48 – 50	149-152
16.	Solutions	51 – 54	153-158

17.	Electrochemistry	55 – 58	159-164
18.	Chemical Kinetics	59 – 62	165-170
19.	Surface Chemistry	63 – 66	171-74
20.	General Principles and Processes of Isolation of Elements	67 – 68	175
21.	The p-Block Elements (Group 15, 16, 17 & 18)	69 – 70	176-177
22.	The d-and f-Block Elements	71 – 72	178-179
23.	Co-ordination Compounds	73 – 75	180-182
24.	Haloalkanes and Haloarenes	76 – 78	183-185
25.	Alcohols, Phenols and Ethers	79 – 82	186-188
26.	Aldehydes, Ketones and Carboxylic Acids	83 – 86	189-192
27.	Amines	87 – 90	193-195
28.	Biomolecules	91 – 94	196-198
29.	Polymers	95 – 97	199-200
30.	Chemistry in Everyday Life	98 – 99	201
31.	Analytical Chemistry	100 – 102	202-204

SOME BASIC CONCEPTS OF CHEMISTRY

1

MCQs with One Correct Answer

- Which of the following statements is correct about the reaction given below ?
$$4\text{Fe(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{Fe}_2\text{O}_3\text{(g)}$$
 - Total mass of iron and oxygen in reactants = total mass of iron and oxygen in product therefore, it follows law of conservation of mass.
 - Total mass of reactants = total mass of product; therefore, law of multiple proportions is followed.
 - Amount of Fe_2O_3 can be increased by reducing the amount of any one of the reactants (iron or oxygen).
 - Amount of Fe_2O_3 produced will decrease if the amount of any one of the reactants (iron or oxygen) is taken in excess.
- Which of the following statements indicates that law of multiple proportion is being followed.
 - Sample of carbon dioxide taken from any source will always have carbon and oxygen in the ratio 1 : 2.
 - Carbon forms two oxides namely CO_2 and CO , where masses of oxygen which combine with fixed mass of carbon are in the simple ratio 2 : 1.
 - When magnesium burns in oxygen, the amount of magnesium taken for the reaction is equal to the amount of magnesium in magnesium oxide formed.
 - At constant temperature and pressure, 200 mL of hydrogen will combine with 100 mL oxygen to produce 200 mL of water vapour.
- The number of water molecules in 250 mL of water is closest to [Given, density of water is 1.0 g mL^{-1} ; Avogadro's number = 6.023×10^{23}]
 - 83.6×10^{23}
 - 13.9×10^{23}
 - 1.5×10^{23}
 - 33.6×10^{23}
- A metal oxide has the formula Z_2O_3 . It can be reduced by hydrogen to give free metal and water. 0.1596 g of the metal oxide requires 6 mg of hydrogen for complete reduction. The atomic weight of the metal is
 - 27.9
 - 159.6
 - 79.8
 - 55.8
- 4 g of a hydrated crystal of formula $\text{A} \cdot x\text{H}_2\text{O}$ has 0.8 g of water. If the molar mass of the anhydrous crystal (A) is 144 g mol^{-1} , The value of x is
 - 4
 - 1
 - 2
 - 3
- A solution of 20.2 g of 1,2-dibromopropane in MeOH upon heating with excess Zn produces 3.58 g of an unsaturated compound X. The yield (%) of X is closest to [Atomic weight of Br is 80]
 - 18
 - 85
 - 89
 - 30

7. 5.0 g of a certain element X forms 10.0 g of its oxide having the formula X_4O_6 . The atomic mass of X is
 (a) 12.0 amu (b) 24.0 amu
 (c) 30.0 amu (d) 32.0 amu
8. The maximum number of molecules are present in
 (a) 15 L of H_2 gas at STP
 (b) 5 L of N_2 gas at STP
 (c) 0.5 g of H_2 gas
 (d) 10 g of O_2 gas
9. Number of moles of MnO_4^- required to oxidize one mole of ferrous oxalate completely in acidic medium will be
 (a) 0.6 moles (b) 0.4 moles
 (c) 7.5 moles (d) 0.2 moles
10. A metallic chloride contain 47.22% metal. Calculate the equivalent weight of metal.
 (a) 39.68 (b) 31.76 (c) 36.35 (d) 33.46
11. The number of moles of water present in a spherical water droplet of radius 1.0 cm is [Given: density of water in the droplet = 1.0 g cm^{-3}]
 (a) $\frac{\pi}{18}$ (b) $\frac{2\pi}{27}$ (c) 24π (d) $\frac{2\pi}{9}$
12. If potassium chlorate is 80% pure, then 48 g of oxygen would be produced from (atomic mass of K = 39)
 (a) 153.12 g of $KClO_3$ (b) 122.5 g of $KClO_3$
 (c) 245 g of $KClO_3$ (d) 98 g of $KClO_3$
13. 10 g $CaCO_3$ were dissolved in 250 mL of 100 M HCl or the solution was boiled. What volume of 2M KOH would be required to equivalence point after boiling? Assume no change in volume during boiling.
 (a) 50 mL (b) 25 mL (c) 75 mL (d) 60 mL
14. The density of 3M solution of sodium chloride is 1.252 g mL^{-1} . The molality of the solution will be :
 (molar mass, $NaCl = 58.5 \text{ g mol}^{-1}$)
 (a) 260m (b) 2.18m (c) 2.79m (d) 3.00m
15. 2 g of a mixture of CO and CO_2 on reaction with excess I_2O_5 produced 2.54 g of I_2 . What would be the mass % of CO_2 in the original mixture?
 (a) 60 (b) 30 (c) 70 (d) 35
16. 3.0 g of oxalic acid $[(CO_2H)_2 \cdot 2H_2O]$ is dissolved in a solvent to prepare a 250 mL solution. The density of the solution is 1.9 g/mL . The molality and normality of the solution, respectively, are closest to
 (a) 0.10 and 0.38
 (b) 0.10 and 0.19
 (c) 0.05 and 0.19
 (d) 0.05 and 0.09
17. How many moles of P_4O_6 and P_4O_{10} will be produced by the combustion of 12.4 g of phosphorous (atomic mass 31) in 12.8 g of oxygen, leaving no P_4 or O_2 ?
 (a) 0.1 and 0.3 mol
 (b) 0.15 mol and 0.25 mol
 (c) 0.05 mol each
 (d) 0.1 mol each
18. 5 g sample contain only Na_2CO_3 and Na_2SO_4 . This sample is dissolved and the volume made up to 250 mL. 25 mL of this solution neutralizes 20 mL of 0.1 M H_2SO_4 . Calculate the % of Na_2SO_4 in the sample.
 (a) 42.4 (b) 57.6 (c) 36.2 (d) 0.576
19. 1 mole of equimolar mixture of ferric oxalate and ferrous oxalate requires x mole of $KMnO_4$ in acidic medium for complete oxidation. x is:
 (a) 0.5 mole (b) 0.9 mole
 (c) 1.2 mole (d) 4.5 mole
20. Two solutions of a substance (non electrolyte) are mixed in the following manner. 480 mL of 1.5 M first solution + 520 mL of 1.2 M second solution. What is the molarity of the final mixture?
 (a) 2.70 M (b) 1.344 M
 (c) 1.50 M (d) 1.20 M

Numeric Value Answer

21. A mixture of $HCOOH$ and $H_2C_2O_4$ is heated with concentrated H_2SO_4 . The gas produced is collected and on treating with KOH solution, the volume of gas decreases by one-sixth. Calculate the molar ratio of the two acids ($HCOOH : H_2C_2O_4$) in the original mixture.
22. One gram of a metallic chloride was found to contain 0.835 g of chlorine. Its vapour density is 85.5. If its molecular formula is M_xCl_y , then what is value of $(x + y)$?

23. 0.7875 g of crystalline barium hydroxide is dissolved in water. For the neutralization of this solution 20 mL of $N/4$ HNO_3 is required. How many moles of water of crystallization are present in one mole of this base? (Given : Atomic mass Ba = 137, O = 16, N = 14, H = 1)
24. A mixture contains 1.0 mole each of NaOH , Na_2CO_3 and NaHCO_3 . When half of mixture is titrated with HCl , it required x mole of HCl in presence of phenolphthalein. In another experiment, half of mixture required y mole of same HCl in presence of methyl orange. Find the value of $(x + y)$.
25. A 0.276 g impure sample of copper ore is dissolved and Cu^{2+} is titrated with KI solution. I_2 liberated required 40 mL of 0.1M $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration. What is the % of impurities in the ore?
26. Density of a sulphuric acid solution is 1.225g/mL. and it is 40% H_2SO_4 by weight. Determine molarity of this solution.
27. A 16.24 mL sample of vinegar of density 1.06 g/mL required 48.24 mL of 0.36 N solution of a standard alkali. Determine percentage weight of acetic acid in vinegar.
28. How much volume of sulphur dioxide at STP will be obtained by completely burning 10 g of pure sulphur?
29. How much volume of 4.0 M HNO_3 is required to prepare 60 mL of 0.2 M HNO_3 from a stock solution of 4.0 M HNO_3 ?
30. The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is _____.

ANSWER KEY

1	(a)	4	(d)	7	(b)	10	(b)	13	(b)	16	(c)	19	(b)	22	(5)	25	(8)	28	(7)
2	(b)	5	(c)	8	(a)	11	(b)	14	(c)	17	(c)	20	(b)	23	(8)	26	(5)	29	(3)
3	(a)	6	(b)	9	(b)	12	(a)	15	(b)	18	(b)	21	(4)	24	(3)	27	(6)	30	(18)

STRUCTURE OF ATOM

2

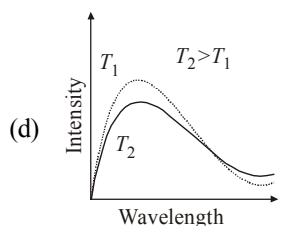
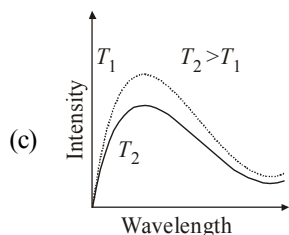
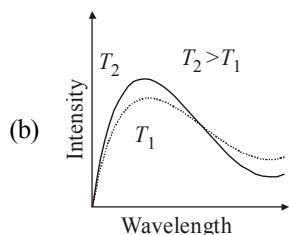
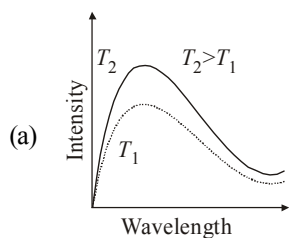
MCQs with One Correct Answer

- The potential energy of electron present in ground state of Li^{2+} ion is represented by :
 - $\frac{+3e^2}{4\pi\epsilon_0 r}$
 - $\frac{-3e}{4\pi\epsilon_0 r}$
 - $\frac{-3e^2}{4\pi\epsilon_0 r^2}$
 - $\frac{-3e^2}{4\pi\epsilon_0 r}$
- Let m_p be the mass of a proton, m_n that of a neutron, M_1 that of a $^{20}_{10}\text{Ne}$ nucleus and M_2 that of a $^{40}_{20}\text{Ca}$ nucleus. Then
 - $M_2 = 2M_1$
 - $M_1 < 10(m_p + m_n)$
 - $M_2 > 2M_1$
 - $M_1 = M_2$
- Which of the following pairs of nucleides are isodiaphers ?
 - $^{13}_6\text{C}$ and $^{16}_8\text{O}$
 - ^1_1H and ^2_1H
 - ^3_1H and ^4_2He
 - $^{55}_{25}\text{Mn}$ and $^{65}_{30}\text{Zn}$
- Based on equation $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$, certain conclusions are written. Which of them is not correct ?
 - Larger the value of n , the larger is the orbit radius.
 - Equation can be used to calculate the change in energy when the electron changes orbit.
 - For $n = 1$, the electron has a more negative energy than it does for $n = 6$ which mean that the electron is more loosely bound in the smallest allowed orbit.
 - The negative sign in equation simply means that the energy or electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus.
- An electron in the ground state of hydrogen was excited to a higher energy level using monochromatic radiations of wave length (λ) 975 Å. The longest wave length that appears in the resulting spectrum is due to transition from:
 - $n_4 \longrightarrow n_1$
 - $n_4 \longrightarrow n_3$
 - $n_5 \longrightarrow n_4$
 - $n_5 \longrightarrow n_1$

6. If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:

(a) $\frac{1}{2} \frac{e^2}{r}$ (b) $-\frac{e^2}{r}$
 (c) $\frac{me^2}{r}$ (d) $-\frac{1}{2} \frac{e^2}{r}$

7. The correct representation of wavelength intensity relationship of an ideal black body radiation at two different temperatures T_1 and T_2 is



8. Among the following, the incorrect statement is
 (a) No two electrons in an atom can have the same set of four quantum numbers

(b) The maximum number of electrons in the shell with principal quantum number, n is equal to $n^2 + 2$

(c) Electrons in an orbital must, have opposite spin

(d) In the ground state, atomic orbitals are filled in the order of their increasing energies

9. What transition in He^+ ion shall have the same wave number as the first line in Balmer series of H atom?

(a) $7 \rightarrow 5$ (b) $6 \rightarrow 4$

(c) $5 \rightarrow 3$ (d) $4 \rightarrow 2$

10. The velocity of an electron in excited state of H-atom is 1.093×10^6 m/s. What is the circumference of this orbit?

(a) 3.32×10^{-10} m (b) 6.64×10^{-10} m

(c) 13.30×10^{-10} m (d) 13.28×10^{-8} m

11. The angular momentum of an electron in a Bohr's orbit of He^+ is 3.1652×10^{-34} kg-m²/sec. What is the wave number in terms of Rydberg constant (R) of the spectral line emitted when an electron falls from this level to the first excited state. [Use $h = 6.626 \times 10^{-34}$ J · s]

(a) $3R$ (b) $\frac{5R}{9}$

(c) $\frac{3R}{4}$ (d) $\frac{8R}{9}$

12. A electron in a hydrogen atom in its ground state absorbs 1.5 times as much energy as the minimum required for it to escape from the atom. What is the velocity of the emitted electron?

(a) 1.54×10^6 m/s

(b) 1.54×10^8 m/s

(c) 1.54×10^3 m/s

(d) 1.54×10^4 m/s

13. For a multi-electron atom, the highest energy level among the following is

(a) $n = 5, l = 0, m = 0, s = +\frac{1}{2}$

(b) $n = 4, l = 2, m = 0, s = +\frac{1}{2}$

(c) $n = 4, l = 1, m = 0, s = +\frac{1}{2}$

(d) $n = 5, l = 1, m = 0, s = +\frac{1}{2}$

14. For Balmer series in the spectrum of atomic hydrogen, the wave number of each line is given

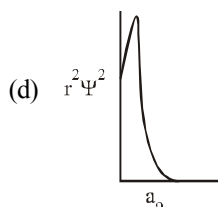
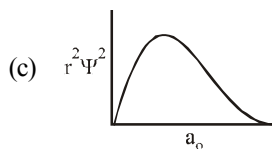
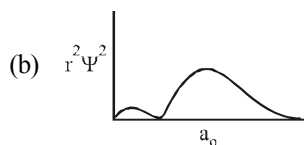
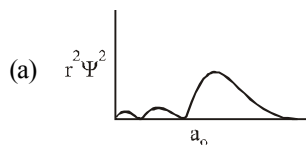
$$\bar{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ where } R_H \text{ is a constant}$$

and n_1 and n_2 are integers. Which of the following statement(s) is (are) correct?

- I. As wavelength decreases, the lines in the series converge.
 - II. The integer n_1 is equal to 2.
 - III. The ionization energy of hydrogen can be calculated from the wave number of these lines.
 - IV. The line of longest wavelength corresponds to $n_2 = 3$.
- (a) I, II and III
 - (b) II, III and IV only
 - (c) I, II and IV
 - (d) II and IV only
15. Calculate the minimum and maximum number of electrons which may have magnetic quantum number, $m = +1$ and spin quantum number,

$$s = -\frac{1}{2} \text{ in chromium (Cr):}$$

- (a) 0, 1
 - (b) 1, 2
 - (c) 4, 6
 - (d) 2, 3
16. The correct order of energy of $2s$ -orbitals in H, Li, Na and K, is
- (a) $K < Na < Li < H$
 - (b) $Na < Li < K < H$
 - (c) $Na < K < H < Li$
 - (d) $H < Na < Li < K$
17. Which of the following radial distribution graphs correspond to $\ell = 2$ for the H atom ?



18. The electrons, identified by quantum numbers n and ℓ ,
- (I) $n = 3, \ell = 2$
 - (II) $n = 5, \ell = 0$
 - (III) $n = 4, \ell = 1$
 - (IV) $n = 4, \ell = 2$
- (V) $n = 4, \ell = 0$ can be placed in order of increasing energy, as
- (a) $I < V < III < IV < II$
 - (b) $I < V < III, II, IV$
 - (c) $V < I < III < II < IV$
 - (d) $V < I < II < III < IV$
18. The energies E_1 and E_2 of two radiations are 25 eV and 50 eV, respectively. The relation between their wavelengths i.e., λ_1 and λ_2 will be :
- (a) $\lambda_1 = \lambda_2$
 - (b) $\lambda_1 = 2\lambda_2$
 - (c) $\lambda_1 = 4\lambda_2$
 - (d) $\lambda_1 = \frac{1}{2}\lambda_2$
19. Among the following, the correct statement about cathode ray discharge tube is
- (a) the electrical discharge can only be observed at high pressure and at low voltages.
 - (b) in the absence of external electrical or magnetic field, cathode rays travel in straight lines.
 - (c) the characteristics of cathode rays depend upon the material of electrodes.
 - (d) the characteristics of cathode rays depend upon the gas present in the cathode ray tube.
20. A 600 W mercury lamp emits monochromatic radiation of wavelength 331.3 nm. How many photons are emitted from the lamp per second?
- $$h = 6.62 \times 10^{-34} \text{ Js velocity of light} = 3 \times 10^8 \text{ ms}^{-1}$$
- (a) 1.0×10^{19}
 - (b) 1.0×10^{23}
 - (c) 1.0×10^{21}
 - (d) 2.0×10^{23}

Numeric Value Answer

21. An electron has a speed of $30,000 \text{ cm sec}^{-1}$ accurate upto 0.001%. What is the uncertainty in locating it's position.
22. If the energies of two radiations of wavelength 800 nm and 400 nm are E_1 and E_2 respectively. Then calculate the value of E_2/E_1 .
23. Determine the Bohr orbit of Li^{2+} ion in which electron is moving at speed equal to the speed of electron in the first Bohr orbit of H-atom.
24. A hydrogen atom with an electron in the first shell absorbs ultraviolet light with a wavelength of $1.03 \times 10^{-7} \text{ m}$. To what shell does the electron jump?
25. What is the sum of radial node(s) and nodal plane(s) for $4d_{z^2}$ -orbital?
26. An electron with de Broglie wavelength $4 \times 10^{-6} \text{ m}$ makes transition to a state with de Broglie wavelength $6 \times 10^{-6} \text{ m}$. The wavelength of photon generated is $\frac{mc}{h} \times x \times 10^{-11} \text{ m}$. What is the value of x ?
27. Calculate the ratio of wavelength of first line of Balmer series of H-atom to the wavelength of first line of Lyman series of 10 times ionized sodium atom.
28. To stop the flow of photoelectrons produced by electromagnetic radiation incident on a certain metal, a negative potential of 300 V is required. If the photoelectric threshold of metal is $1500 \text{ \AA}$, what is the frequency of the incident radiation (in terms of 10^{16} Hz)?
29. The energy of the electron in the second and the third Bohr's orbits of the hydrogen atom is $-5.42 \times 10^{-12} \text{ erg}$ and $-2.41 \times 10^{-12} \text{ erg}$ respectively. Calculate the wavelength of the emitted radiation when the electron drops from the third to the second orbit.
30. Naturally occurring boron consists of two isotopes whose atomic weights are 10.01 and 11.01. The atomic weight of natural boron is 10.81. Calculate the percentage of isotope with atomic weight 11.01 in natural boron.

ANSWER KEY

1	(d)	4	(c)	7	(a)	10	(c)	13	(d)	16	(a)	19	(b)	22	(2)	25	(1)	28	(7.45)
2	(a)	5	(b)	8	(b)	11	(b)	14	(c)	17	(c)	20	(c)	23	(3)	26	(576)	29	(6.604)
3	(d)	6	(d)	9	(b)	12	(a)	15	(d)	18	(b)	21	(2)	24	(3)	27	(653.4)	30	(80)

CLASSIFICATION OF ELEMENTS & PERIODICITY IN PROPERTIES

3

MCQs with One Correct Answer

- Which form coloured salts?
 - Non-metals
 - Metals
 - p-block elements
 - Transitional elements
- The ionic radii of Na^+ , F^- , O^{2-} , N^{3-} follow the order
 - $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{N}^{3-}$
 - $\text{N}^{3-} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$
 - $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+$
 - $\text{Na}^+ > \text{F}^- > \text{O}^{2-} > \text{N}^{3-}$
- The electronic configuration of four elements are given below. Which elements does not belong to the same family as others?
 - $[\text{Xe}]4f^{14}5d^{10}1s^2$
 - $[\text{Kr}]4d^{10}5s^2$
 - $[\text{Ne}]3s^23p^5$
 - $[\text{Ar}]3d^{10}4s^2$
- The elements with atomic numbers 9, 17, 35, 53 and 85 belong to
 - alkali metals
 - alkaline earth metals
 - halogens
 - noble gases
- Eka-aluminium and Eka-silicon are known as
 - gallium and germanium
 - aluminium and silicon
 - iron and sulphur
 - manganese and magnesium
- What is the atomic number of the next halogen if discovered?
 - 85
 - 117
 - 167
 - 104
- The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^3$. The atomic number and the group number of the element 'X' which is just below the above element in the periodic table are respectively
 - 23 and 5
 - 23 and 15
 - 33 and 15
 - 33 and 5
- Which one of the following ions has the highest value of ionic radius ?
 - O^{2-}
 - B^{3+}
 - Li^+
 - F^-

9. The first ionisation potential in electron volts of nitrogen and oxygen atoms are respectively given by
 (a) 14.6, 13.6 (b) 13.6, 14.6
 (c) 13.6, 13.6 (d) 14.6, 14.6
10. In which of the following process highest energy is absorbed?
 (a) $\text{Cu} \rightarrow \text{Cu}^+$ (b) $\text{Br} \rightarrow \text{Br}^-$
 (c) $\text{I} \rightarrow \text{I}^-$ (d) $\text{Li} \rightarrow \text{Li}^+$
11. Which ionisation potential (IP) in the following equations involves the greatest amount of energy?
 (a) $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$
 (b) $\text{K}^+ \rightarrow \text{K}^{2+} + \text{e}^-$
 (c) $\text{C}^{2+} \rightarrow \text{C}^{3+} + \text{e}^-$
 (d) $\text{Ca}^+ \rightarrow \text{Ca}^{2+} + \text{e}^-$
12. Which of the following order is wrong?
 (a) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ – Acidic
 (b) $\text{Li} < \text{Be} < \text{B} < \text{C}$ – First IP
 (c) $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$ – Basic
 (d) $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Cs}^+$ – Ionic radius
13. The incorrect statement among the following is
 (a) The first ionization potential of Al is less than the first ionization potential of Mg
 (b) The second ionization potential of Mg is greater than the second ionization potential of Na
 (c) The first ionization potential of Na is less than the first ionization potential of Mg
 (d) The third ionization potential of Mg is greater than the third ionization potential of Al.
14. The ease of loss of electron from Ne and Na^+ ion (both isoelectronic) are
 (a) 4562 kJ mol⁻¹, 2081 kJ mol⁻¹
 (b) 2081 kJ mol⁻¹, 4562 kJ mol⁻¹
 (c) 2081 kJ mol⁻¹ each
 (d) 4562 kJ mol⁻¹ each
15. The electron affinity for the inert gases is
 (a) zero (b) high
 (c) negative (d) positive
16. The correct order of acidic strength :
 (a) $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$
 (b) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
 (c) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 (d) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
17. The formation of the oxide ion $\text{O}^{2-}(\text{g})$ requires first an exothermic and then an endothermic step as shown below :

$$\text{O}(\text{g}) + \text{e}^- = \text{O}^-(\text{g}) \quad \Delta H^\circ = -142 \text{ kJmol}^{-1}$$

$$\text{O}^-(\text{g}) + \text{e}^- = \text{O}^{2-}(\text{g}) \quad \Delta H^\circ = 844 \text{ kJmol}^{-1}$$
 This is because
 (a) O^- ion will tend to resist the addition of another electron
 (b) Oxygen has high electron affinity
 (c) Oxygen is more electronegative
 (d) O^- ion has comparatively larger size than oxygen atom
18. Amphoteric-oxide combinations are in
 (a) $\text{ZnO}, \text{K}_2\text{O}, \text{SO}_3$
 (b) $\text{ZnO}, \text{P}_2\text{O}_5, \text{Cl}_2\text{O}_7$
 (c) $\text{SnO}_2, \text{Al}_2\text{O}_3, \text{ZnO}$
 (d) $\text{PbO}_2, \text{SnO}_2, \text{SO}_3$
19. Electronic configurations of four elements A, B, C and D are given below :
 (i) $1s^2 2s^2 2p^6$
 (ii) $1s^2 2s^2 2p^4$
 (iii) $1s^2 2s^2 2p^6 3s^1$
 (iv) $1s^2 2s^2 2p^5$
 Which of the following is the correct order of increasing tendency to gain electron ?
 (a) (i) < (iii) < (ii) < (iv)
 (b) (i) < (ii) < (iii) < (iv)
 (c) (iv) < (ii) < (iii) < (i)
 (d) (iv) < (i) < (ii) < (iii)

20. The basic character of MgO , BaO , Na_2O and FeO follow the order
- $\text{MgO} < \text{FeO} < \text{BaO} < \text{Na}_2\text{O}$
 - $\text{FeO} < \text{MgO} < \text{Na}_2\text{O} < \text{BaO}$
 - $\text{FeO} < \text{MgO} < \text{BaO} < \text{Na}_2\text{O}$
 - $\text{Na}_2\text{O} < \text{MgO} < \text{FeO} < \text{BaO}$
21. In the IUPAC nomenclature of elements, what is atomic number for the element 'Unh'?
22. The first ionisation potential of Cu, Ag and Au are 731, 745 and 890 KJ/mol (not in the same order). What is the numerical value of I.E. for Ag?
23. How many completely filled periods are there in the long form of the periodic table?
24. What is the number of elements in the third period of the periodic table except metalloids?
25. How many pairs are, in which first species has lower ionisation energy than second species:
- N and O
 - Br and K
 - Be and B
 - I and I^-
 - Li and Li^+
 - O and S
 - Ba and Sr

Numeric Value Answer

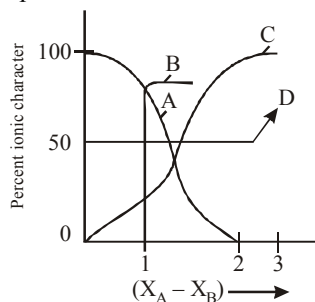
ANSWER KEY																	
1	(d)	4	(c)	7	(c)	10	(a)	13	(b)	16	(a)	19	(a)	22	(731)	25	(2)
2	(c)	5	(a)	8	(a)	11	(b)	14	(b)	17	(a)	20	(c)	23	(7)		
3	(c)	6	(b)	9	(a)	12	(b)	15	(a)	18	(c)	21	(106)	24	(7)		

CHEMICAL BONDING AND MOLECULAR STRUCTURE

4

MCQs with One Correct Answer

1. For AB bond if percent ionic character is plotted against electronegativity difference ($X_A - X_B$), the shape of the curve would look like



The correct curve is

- (a) (A) (b) (B)
(c) (C) (d) (D)
2. The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5 D) is larger than that of NF_3 (0.2D). This is because
- (a) in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions
- (b) in NH_3 as well as NF_3 the atomic dipole and bond dipole are in opposite directions
- (c) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction
- (d) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction
3. Among the following, the species with identical bond order are
- (a) CO and O_2^{2-}
(b) O_2^- and CO
(c) O_2^{2-} and B_2
(d) CO and N_2^+
4. Which of the following statements is/are true
- PH_5 and BiCl_5 do not exist
 - $p\pi - d\pi$ bond is present in SO_2
 - I_3^+ has bent geometry
 - SeF_4 and CH_4 have same shape
- (a) 1, 2, 3 (b) 1, 3
(c) 1, 3, 4 (d) 1, 2, 4

5. The number of sigma (σ) and pi (π) bonds present in 1,3,5,7 octatetraene respectively are
 (a) 14 and 3 (b) 17 and 4
 (c) 16 and 5 (d) 15 and 4
6. The hybridizations of N, C and O shown in the following compound
- $$\begin{array}{c} \text{R} \\ \diagdown \\ \text{N}=\text{C}=\text{O} \end{array}$$
- respectively, are
 (a) sp^2 , sp , sp^2 (b) sp^2 , sp^2 , sp^2
 (c) sp^2 , sp , sp (d) sp , sp , sp^2
7. The bond dissociation energy of B–F in BF_3 is 646 kJ mol^{-1} whereas that of C–F in CF_4 is 515 kJ mol^{-1} . The correct reason for higher B–F bond dissociation energy as compared to that of C–F is
 (a) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4 .
 (b) significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4 .
 (c) lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4 .
 (d) smaller size of B-atom as compared to that of C-atom.
8. The most polarizable ion among the following is
 (a) F^- (b) I^-
 (c) Na^+ (d) Cl^-
9. Minimum F–S–F bond angle present in :
 (a) SSF_2 (b) SF_6
 (c) SF_2 (d) F_3SSF
10. Which of the following statement is correct about I_3^+ and I_3^- molecular ions?
 (a) Number of lone pairs at central atoms are same in both molecular ions
 (b) Hybridization of central atoms in both ions are same
 (c) Both are polar species
 (d) Both are planar species
11. In which of the following species, d -orbitals having xz and yz two nodal planes involved in hybridization of central atom?
 (a) IO_2F_2^-
 (b) ClF_4^-
 (c) IF_7
 (d) None of these
12. The type of bonds present in sulphuric anhydride is,
 (a) 3σ and three $p\pi - d\pi$
 (b) 3σ , one $p\pi - p\pi$ and two $p\pi - d\pi$
 (c) 2σ and three $p\pi - d\pi$
 (d) 2σ and two $p\pi - d\pi$
13. The correct statement with regard to H_2^+ and H_2^- is
 (a) Both H_2^+ and H_2^- do not exist
 (b) H_2^- is more stable than H_2^+
 (c) H_2^+ is more stable than H_2^-
 (d) Both H_2^+ and H_2^- are equally stable
14. The correct order of bond energies in NO , NO^+ and NO^- is:
 (a) $\text{NO}^- > \text{NO} > \text{NO}^+$
 (b) $\text{NO} > \text{NO}^- > \text{NO}^+$
 (c) $\text{NO}^+ > \text{NO} > \text{NO}^-$
 (d) $\text{NO}^+ > \text{NO}^- > \text{NO}$
15. Which of the following represents the correct order of Cl–O bond lengths in ClO^- , ClO_2^- , ClO_3^- , ClO_4^- ?
 (a) $\text{ClO}_4^- = \text{ClO}_3^- = \text{ClO}_2^- = \text{ClO}^-$
 (b) $\text{ClO}^- < \text{ClO}_2^- < \text{ClO}_3^- < \text{ClO}_4^-$
 (c) $\text{ClO}_4^- < \text{ClO}_3^- < \text{ClO}_2^- < \text{ClO}^-$
 (d) $\text{ClO}_3^- < \text{ClO}_4^- < \text{ClO}_2^- < \text{ClO}^-$

16. Which one of the following molecule will have all equal X—F bond length? (where X = Central atom)

(a) SOCl_2F_2 (b) SeF_4
(c) PBr_2F_3 (d) IF_7

17. Select the incorrect statement about N_2F_4 and N_2H_4 :

- (I) In N_2F_4 , d -orbitals are contracted by electronegative fluorine atoms, but d -orbital contraction is not possible by H-atom in N_2H_4
(II) The N-N bond energy in N_2F_4 is more than N-N bond energy in N_2H_4
(III) The N-N bond length in N_2F_4 is more than that of in N_2H_4
(IV) The N-N bond length in N_2F_4 is less than that of in N_2H_4
(a) I, II and III
(b) I and III
(c) II and IV
(d) II and III

18. The correct order of 'S—O' bond length is

- (a) $\text{SO}_3^{2-} > \text{SO}_4^{2-} > \text{SO}_3 > \text{SO}_2$
(b) $\text{SO}_3^{2-} > \text{SO}_4^{2-} > \text{SO}_2 > \text{SO}_3$
(c) $\text{SO}_4^{2-} > \text{SO}_3^{2-} > \text{SO}_2 > \text{SO}_3$
(d) $\text{SO}_4^{2-} > \text{SO}_3^{2-} > \text{SO}_3 > \text{SO}_2$

19. Among the following transformations, the hybridization of the central atom remains unchanged in

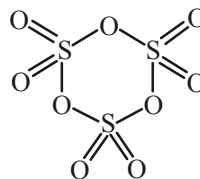
- (a) $\text{CO}_2 \rightarrow \text{HCOOH}$
(b) $\text{BF}_3 \rightarrow \text{BF}_4^-$
(c) $\text{NH}_3 \rightarrow \text{NH}_4^+$
(d) $\text{PCl}_3 \rightarrow \text{PCl}_5$

20. In which species, X—O bond order is 1.5 and contains $p\pi - d\pi$ bond(s).

- (a) IO_2F_2^- (b) HCOO^-
(c) SO_3^{2-} (d) XeO_2F_2

Numeric Value Answer

21. Consider the following molecule



Calculate the value of $p \div q$, here p and q are total number of $d\pi-p\pi$ bonds and total number of sp^3 hybridised atoms respectively in given molecule.

22. Calculate the value of " $x + y - z$ " here x , y and z are total number of non-bonded electron pair(s), π bond(s) and sigma (σ) bonds in hydrogen phosphite ion respectively.
23. Total number of species which used all three p -orbitals in hybridisation of central atoms and should be non-polar also are

XeO_2F_2 , SnCl_2 , IF_5 , I_3^+ , XeO_4 , SO_2 , XeF_7^+ , SeF_4

24. Consider the following orbitals $3s$, $2p_x$, $4d_{xy}$, $4d_z^2$, $3d_{x^2-y^2}$, $3p_y$, $4s$, $4p_z$ and find total number of orbital(s) having even number of nodal plane.
25. For the following molecules:

PCl_5 , BrF_3 , ICl_2^- , XeF_5^- , NO_3^- , XeO_2F_2 , PCl_4^+ , CH_3^+

Calculate the value of $\frac{a+b}{c}$

a = Number of species having sp^3 d -hybridisation

b = Number of species which are planar

c = Number of species which are non-planar

26. Find total number of orbital which can overlap colaterally, (if inter nuclear axis is z) s , p_x , p_y , p_z , d_{xy} , d_{yz} , d_{xz} , d_{z^2} , $d_{x^2-y^2}$

27. In O_2^- , O_2 and O_2^{2-} molecular species, the sum of the total number of antibonding electrons is _____

28. Dipole moment of  X is 1.6 D. What is the dipole moment of following compound:

29. What is the % of *p*-character with central atom in SF_6 molecule?

30. Consider the following values for an ionic compound NaCl.

$$\Delta H_f(\text{NaCl}) = -200 \text{ KJ/mol}$$

$$\Delta H_{\text{sub}}(\text{Na(s)}) = 650 \text{ KJ/mol}$$

$$\Delta H_{\text{diss}}(\text{Cl}_2(\text{g})) = 400 \text{ KJ/mol}$$

$$\text{I.E.}_1(\text{Na(g)}) = 500 \text{ KJ/mol}$$

$$\text{Electron gain enthalpy}(\text{Cl(g)}) = -350 \text{ KJ/mol}$$

Using Born Haber Cycle, find the value of lattice energy (*U*) in KJ/mol.

ANSWER KEY

1	(c)	4	(a)	7	(b)	10	(d)	13	(c)	16	(a)	19	(c)	22	(3)	25	(3)	28	(1.6)
2	(a)	5	(b)	8	(b)	11	(c)	14	(c)	17	(b)	20	(a)	23	(2)	26	(6)	29	(50)
3	(c)	6	(a)	9	(d)	12	(b)	15	(c)	18	(b)	21	(1)	24	(5)	27	(21)	30	(1200)

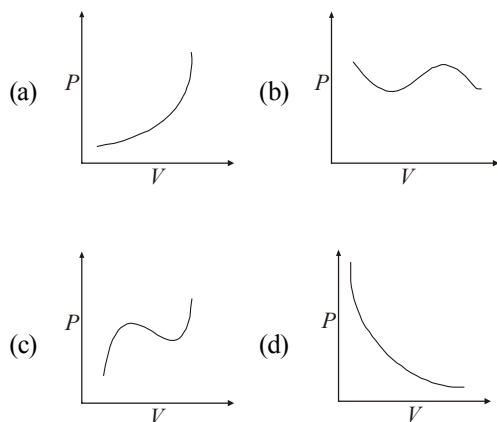
STATES OF MATTER



MCQs with One Correct Answer

- 600 c.c. of a gas at a pressure of 750 mm of Hg is compressed to 500 c.c. Taking the temperature to remain constant, the increase in pressure, is
 - 150 mm of Hg
 - 250 mm of Hg
 - 350 mm of Hg
 - 450 mm of Hg
- 16 g of oxygen and 3 g of hydrogen are mixed and kept at 760 mm of Hg pressure and 0°C . The total volume occupied by the mixture will be nearly
 - 22.4 litres
 - 33.6 litres
 - 448 litres
 - 44800 mL
- Equal weights of methane and hydrogen are mixed in an empty container at 25°C . The fraction of the total pressure exerted by hydrogen is :
 - $\frac{1}{2}$
 - $\frac{8}{9}$
 - $\frac{1}{9}$
 - $\frac{16}{17}$
- X mL of H_2 gas effuses through a hole in a container in 5 seconds. The time taken for the effusion of the same volume of the gas specified below under identical conditions is :
 - 10 seconds : He
 - 20 seconds : O_2
 - 25 seconds : CO
 - 55 seconds : CO_2
- A container contains 1 mole of a gas at 1 atm pressure and 27°C , while its volume is 24.6 litres. If its pressure is 10 atm and temperature 327°C , then new volume is
 - 2.56 litres
 - 3.15 litres
 - 4.92 litres
 - 5.44 litres
- 14 g of N_2 and 36 g of ozone are at the same pressure and temperature. Their volumes will be related as
 - $2V_{\text{N}_2} = 3V_{\text{O}_3}$
 - $3V_{\text{N}_2} = 2V_{\text{O}_3}$
 - $3V_{\text{N}_2} = 4V_{\text{O}_3}$
 - $4V_{\text{N}_2} = 3V_{\text{O}_3}$

7. The pressure (P) and volume (V) isotherm of a van der Waals' gas, at the temperature at which it undergoes gas to liquid transition, is correctly represented by



8. One mole of gas A and three moles of a gas B are placed in flask of volume 100 litres at 27°C . Calculate the total partial pressure of the gases in the mixture.
 (a) 1.0 atm. (b) 0.9 atm
 (c) 0.985 atm (d) 10.850 atm.
9. Oxygen is present in 1 litre flask at a pressure of 7.6×10^{-10} mm of Hg. Calculate the number of oxygen molecules in the flask at 0°C .
 (a) 2.8×10^8 (b) 2.7×10^{10}
 (c) 0.27×10^{10} (d) 3.0×10^9
10. A spherical balloon of 21 cm diameter is to be filled up with hydrogen at N.T.P. from a cylinder containing the gas at 20 atmospheres at 27°C . If the cylinder can hold 2.82 litres of water, calculate the number of balloons that can be filled up.
 (a) 9 (b) 8
 (c) 12 (d) 10
11. A given volume of ozonised oxygen (containing 60% oxygen by volume) required 220 sec to effuse while an equal volume of oxygen took 200 sec only under identical conditions. If density of O_2 is 1.6 g/L then find density of O_3 .
 (a) 1.936 g/L (b) 2.16 g/L
 (c) 3.28 g/L (d) 2.24 g/L

12. A container of volume 2.24 L can withstand a maximum pressure of 2 atm at 298 K before exploding. The maximum amount of nitrogen (in g) that can be safely put in this container at this temperature is closest to

(a) 2.8 (b) 5.6
 (c) 1.4 (d) 4.2

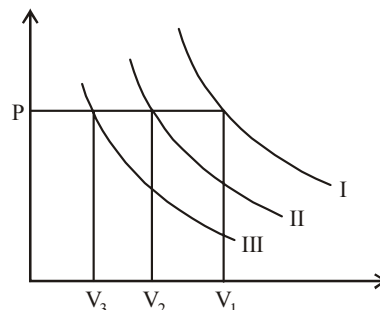
13. At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen (N_2) at 4 bar. The molar mass of gaseous molecule is :

(a) 28 g mol^{-1} (b) 56 g mol^{-1}
 (c) 112 g mol^{-1} (d) 224 g mol^{-1}

14. Calculate the total pressure in a 10.0 L cylinder which contains 0.4 g helium, 1.6 g oxygen and 1.4 g nitrogen at 27°C .

(a) 0.492 atm (b) 49.2 atm
 (c) 4.52 atm (d) 0.0492 atm

15. I, II, and III are three isotherms, respectively, at T_1 , T_2 and T_3 . Temperature will be in order



(a) $T_1 = T_2 = T_3$ (b) $T_1 < T_2 < T_3$
 (c) $T_1 > T_2 > T_3$ (d) $T_1 > T_2 = T_3$

16. The root mean square speeds at STP for the gases H_2 , N_2 , O_2 and HBr are in the order :

(a) $\text{H}_2 < \text{N}_2 < \text{O}_2 < \text{HBr}$
 (b) $\text{HBr} < \text{O}_2 < \text{N}_2 < \text{H}_2$
 (c) $\text{H}_2 < \text{N}_2 = \text{O}_2 < \text{HBr}$
 (d) $\text{HBr} < \text{O}_2 < \text{H}_2 < \text{N}_2$

17. Two flasks A and B of equal volumes maintained at temperatures 300K and 600K contain equal mass of H_2 and CH_4 respectively. The ratio of total translational kinetic energy of gas in flask A to that in flask B is

(a) unity (b) 2
 (c) 4 (d) 0.25

18. The root mean square speed of hydrogen is $\sqrt{5}$ times than that of nitrogen. If T is the temperature of the gas, then :
- (a) $T_{H_2} = T_{N_2}$ (b) $T_{H_2} > T_{N_2}$
 (c) $T_{H_2} < T_{N_2}$ (d) $T_{H_2} = \sqrt{7}T_{N_2}$
19. For one mole of a van der Waals gas, the compressibility factor $Z \left(= \frac{PV}{RT} \right)$ at a fixed volume will certainly decrease if [Given: "a", "b" are standard parameters for van der Waals gas]
- (a) "b" increases and "a" decreases at constant temperature
 (b) "b" decreases and "a" increases at constant temperature
 (c) temperature increases at constant "a" and "b" values
 (d) "b" increases at constant "a" and temperature
20. A mixture of Ne and Ar kept in a closed vessel at 250 K has a total K.E. = 3 kJ. The total mass of Ne and Ar is 30 g. Find mass % of Ne in gaseous mixture at 250 K.
- (a) 61.63 (b) 38.37
 (c) 50% (d) 28.3
21. A gas diffuses 1/5 times as fast as hydrogen. If its molecular weight is 10y. What will be the value of y?
22. The rate of diffusion of methane at a given temperature is twice that of a gas X. If the molecular weight of X is 8y then what will be the value of y?
23. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is
24. One mole of a gas changed from its initial state (15L, 2 atm) to final state (4L, 10 atm) reversibly. If this change can be represented by a straight line in P – V curve maximum temperature (approximate), the gas attained is $x \times 10^2$ K. Then find the value of x.
25. A flask containing air at 107°C and 722 mm of Hg is cooled to 100 K and 760 mm of Hg. If density in the initial condition 1 g/cm³, then what is the final density (g/cm³)?
26. If an ideal gas at 100 K is heated to 109 K in a rigid container, the pressure increases by X%. What is the value of X?
27. A flask has 10 gas particles, out of which four particles are moving at 7 ms⁻¹ and the remaining are moving at the same speed of 'X' ms⁻¹. If the r.m.s. of the gas is 5 ms⁻¹, what is the value of X?
28. At 300 K, the average kinetic energy of a deuterium molecule is n times than that of a hydrogen molecule. Find the value of n .
29. The root mean square speed of 4g of helium gas is 450 m/s. What is the value of total kinetic energy (J) of this sample?
30. If the ratio of critical temperature to critical pressure for a real gas is mb/R, then what is the integer value of m?

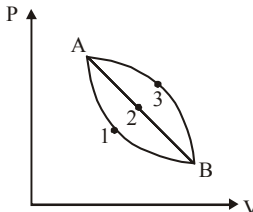
Numeric Value Answer

ANSWER KEY																			
1	(a)	4	(b)	7	(b)	10	(d)	13	(c)	16	(b)	19	(b)	22	(8)	25	(4)	28	(1)
2	(d)	5	(c)	8	(c)	11	(d)	14	(a)	17	(c)	20	(d)	23	(4)	26	(9)	29	(405)
3	(b)	6	(b)	9	(b)	12	(d)	15	(c)	18	(c)	21	(5)	24	(7)	27	(3)	30	(8)

THERMODYNAMICS

6

MCQs with One Correct Answer

- The internal energy change when a system goes from state A to B is 40 kJ/mol. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy ?
 (a) > 40 kJ (b) < 40 kJ
 (c) Zero (d) 40 kJ
- A gas absorbs 200 J heat and undergoes simultaneous expansion against a constant external pressure of 10^5 Pa. The volume changes from 4L to 5L. The change in internal energy is
 (a) 300 J (b) 100 J
 (c) -300 J (d) -100 J
- When 110 g of manganese (At. mass = 55) dissolves in dilute HNO_3 at 27°C under atmospheric pressure, the work done in the process is :
 (a) 2494.2 J (b) -2494.2 J
 (c) -4988.4 J (d) 4988.4 J
- Workdone for the conversion of 0.5 mole of water at 100°C to steam at 1 atm pressure is (heat of vaporisation of water at 100°C is 40670 J mol^{-1})
 (a) -1.54 kJ (b) 1.54 kJ
 (c) 1.25 kJ (d) -1.35 kJ
- The amount of work done by 2 mole of an ideal gas at 298 K in reversible isothermal expansion from 10 litre to 20 litre will be
 (a) 3.4 kJ (b) -3.4 kJ
 (c) 4.0 kJ (d) -4.0 kJ
- A gas expands reversibly from state A to state B by three paths 1, 2 and 3 as shown in the figure. If w_1 , w_2 and w_3 respectively be the work done by the gas along three paths then

 (a) $w_1 > w_2 > w_3$ (b) $w_1 < w_2 < w_3$
 (c) $w_1 = w_2 = w_3$ (d) $w_2 < w_3 < w_1$
- 10 mole of an ideal gas expand isothermally and reversibly from a pressure of 10 atm to 1 atm at 300 K. What is the largest mass which can be lifted through a height of 100 meter?
 (a) 31842 kg (b) 58.55 kg
 (c) 342.58 kg (d) 6051 kg

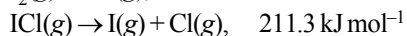
8. Given that $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 : \Delta H^\circ = -x \text{ kJ}$



the enthalpy of formation of carbon monoxide will be

- (a) $\frac{2x-y}{2}$ (b) $\frac{y-2x}{2}$
 (c) $2x-y$ (d) $y-2x$

9. The enthalpy changes for the following processes are listed below :

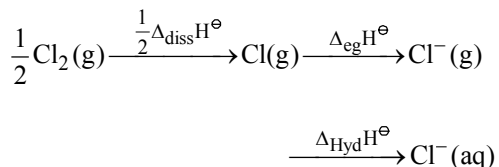


Given that the standard states for iodine and chlorine are $\text{I}_2(\text{s})$ and $\text{Cl}_2(\text{g})$, the standard enthalpy of formation for $\text{ICl}(\text{g})$ is :

- (a) $+16.8 \text{ kJ mol}^{-1}$
 (b) $+244.8 \text{ kJ mol}^{-1}$
 (c) $-14.6 \text{ kJ mol}^{-1}$
 (d) $-16.8 \text{ kJ mol}^{-1}$

10. The quantity of heat (in J) required to raise the temperature of 1.0 kg of ethanol from 293.45 K to the boiling point and then change to liquid to vapour at that temperature is closest to [Given, boiling point of ethanol 351.45 K. Specific heat capacity of liquid ethanol $2.44 \text{ J g}^{-1} \text{ K}^{-1}$. Latent heat of vaporisation of ethanol 855 J g^{-1}]
- (a) 142×10^2 (b) 9.97×10^2
 (c) 142×10^5 (d) 9.97×10^5

11. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:



using the data,

$$\Delta_{\text{diss}}H^\ominus_{\text{Cl}_2} = 240 \text{ kJ mol}^{-1},$$

$$\Delta_{\text{eg}}H^\ominus_{\text{Cl}} = -349 \text{ kJ mol}^{-1},$$

$$\Delta_{\text{hyd}}H^\ominus_{\text{Cl}^\ominus} = -381 \text{ kJ mol}^{-1}, \text{ will be}$$

- (a) $+152 \text{ kJ mol}^{-1}$ (b) -850 kJ mol^{-1}
 (c) -610 kJ mol^{-1} (d) $+120 \text{ kJ mol}^{-1}$

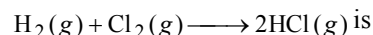
12. The specific heat of a monoatomic gas at constant pressure is $248.2 \text{ J kg}^{-1} \text{ K}^{-1}$ and at constant volume it is $149.0 \text{ J kg}^{-1} \text{ K}^{-1}$. The mean molar mass of the gas is

- (a) 83.8 g mol^{-1} (b) 84.0 g mol^{-1}
 (c) 82 g mol^{-1} (d) 80 g mol^{-1}

13. The values of standard enthalpy of formation of $\text{SF}_6(\text{g})$, $\text{S}(\text{g})$ and $\text{F}(\text{g})$ are : -1100 , 275 and 80 kJ mol^{-1} respectively. The average S-F bond energy in SF_6 will be

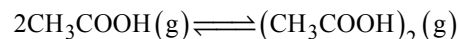
- (a) 309 kJ (b) 315 kJ
 (c) 320 kJ (d) 300 kJ

14. Given the following entropy values (in $\text{J K}^{-1} \text{ mol}^{-1}$) at 298 K and 1 atm : $\text{H}_2(\text{g})$: 130.6, $\text{Cl}_2(\text{g})$: 223.0, $\text{HCl}(\text{g})$: 186.7. The entropy change (in $\text{J K}^{-1} \text{ mol}^{-1}$) for the reaction



- (a) $+540.3$ (b) $+727.0$
 (c) -166.9 (d) $+19.8$

15. One mole of CH_3COOH undergoes dimerization in vapour phase at 127°C as:



if dimer formation is due to two H-bonds involved in dimer, each of 33 kJ strength and the degree of dimerisation of acetic acid 98.2%. The change in standard entropy during dimerization is

- (a) $-104 \text{ JK}^{-1} \text{ mol}^{-1}$ (b) $-120 \text{ JK}^{-1} \text{ mol}^{-1}$
 (c) $135 \text{ JK}^{-1} \text{ mol}^{-1}$ (d) $108 \text{ JK}^{-1} \text{ mol}^{-1}$

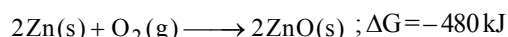
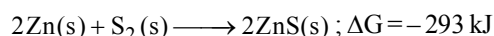
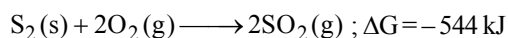
16. For a spontaneous process

- (a) enthalpy change of the system must be negative.
 (b) entropy change of the system must be positive.
 (c) entropy change of the surrounding must be positive.
 (d) entropy change of the system plus surrounding must be positive.

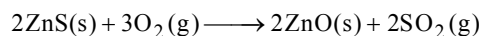
17. At the sublimation temperature, for the process $\text{CO}_{2(s)} \rightleftharpoons \text{CO}_{2(g)}$

(a) $\Delta H, \Delta S$ and ΔG are all positive
 (b) $\Delta H > 0, \Delta S > 0$ and $\Delta G < 0$
 (c) $\Delta H < 0, \Delta S > 0$ and $\Delta G < 0$
 (d) $\Delta H > 0, \Delta S > 0$ and $\Delta G = 0$

18. The factor of ΔG values is important in metallurgy. The ΔG values for the following reactions at 800°C are given as :



Then ΔG for the reaction :



will be :

(a) -357 kJ (b) -731 kJ
 (c) -773 kJ (d) -229 kJ

19. Given that the molar combustion enthalpy of benzene, cyclohexane, and hydrogen are $x, y,$ and z , respectively, the molar enthalpy of hydrogenation of benzene to cyclohexane is

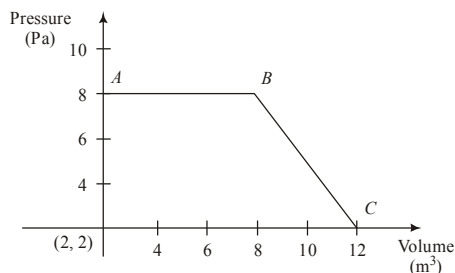
(a) $x - y + z$ (b) $x - y + 3z$
 (c) $y - x + z$ (d) $y - x + 3z$

20. For vaporization of water at 1 atmospheric pressure, the values of ΔH and ΔS are $40.63 \text{ kJ mol}^{-1}$ and $108.8 \text{ JK}^{-1} \text{ mol}^{-1}$, respectively. The temperature when Gibbs energy change (ΔG) for this transformation will be zero, is:

(a) 293.4 K (b) 273.4 K
 (c) 393.4 K (d) 373.4 K

Numeric Value Answer

21. For a dimerization reaction, $2\text{A}(\text{g}) \rightarrow \text{A}_2(\text{g})$, at 298 K , $\Delta U^\ominus = -20 \text{ kJ mol}^{-1}$, $\Delta S^\ominus = -30 \text{ JK}^{-1} \text{ mol}^{-1}$, then the $\Delta G^\ominus$ will be _____ J.
22. The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____



23. At constant volume, 4 mol of an ideal gas when heated from 300 K to 500 K changes its internal energy by 5000 J . The molar heat capacity at constant volume is _____.

24. A perfect gas undergoes a reversible adiabatic expansion from $(300 \text{ K}, 200 \text{ atm})$ to $(90 \text{ K}, 10 \text{ atm})$. The atomicity of gas will be _____.

25. A mixture of ferric oxide, Fe_2O_3 , and aluminium, Al , is used in solid fuel rockets. The fuel value per cc of the mixture is _____.

Heats of formation and densities are as follows :

$H_f(\text{Al}_2\text{O}_3) = 399 \text{ kcal/mol}$;

$H_f(\text{Fe}_2\text{O}_3) = 199 \text{ kcal/mol}$;

Density of $\text{Fe}_2\text{O}_3 = 5.2 \text{ g/cc}$;

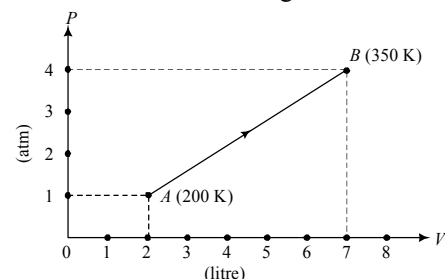
Density of $\text{Al} = 2.7 \text{ g/cc}$.

26. $x \text{ g}$ sample of NH_4NO_3 is decomposed in a Bomb calorimeter. The temperature of calorimeter increase by 4°C . The heat capacity of the system is $125 \text{ kJ/}^\circ\text{C}$. The value of x is _____.

(Given molar heat of decomposition of NH_4NO_3 at constant volume is 400 kJ mol^{-1})

27. A heat engine is operating between 500 K to 400 K . If the engine absorbs 100 J heat, the impossible amount of heat rejected by the engine is 20 J , then the value of T is _____.

28. One mole of a real gas undergoes a change of state A to B with a change in internal energy

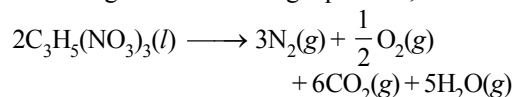


$\Delta E = 50 \text{ L - atm}$. What is the value of change in enthalpy in L - atm ?

29. A 20g piece of iron (specific heat capacity = 0.5 J/g °C) at 200°C is immersed in 100g water of 27°C.

What will be the temperature (°C) of this system at thermal equilibrium?

30. Nitroglycerine (MW = 227.1) detonated according to the following equations,



The standard molar enthalpies of formation,

ΔH_f° for all the compounds are given below.

$$\Delta H_f^\circ[\text{C}_3\text{H}_5(\text{NO}_3)_3] = -364 \text{ kJ/mol}$$

$$\Delta H_f^\circ[\text{CO}_2(g)] = -393.5 \text{ kJ/mol}$$

$$\Delta H_f^\circ[\text{H}_2\text{O}(g)] = -241.8 \text{ kJ/mol}$$

$$\Delta H_f^\circ[\text{N}_2(g)] = 0 \text{ kJ/mol}$$

$$\Delta H_f^\circ[\text{O}_2(g)] = 0 \text{ kJ/mol}$$

The enthalpy change when 10 g of nitroglycerine is detonated as

ANSWER KEY

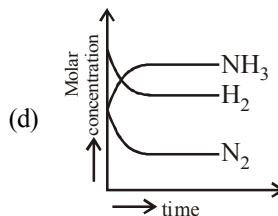
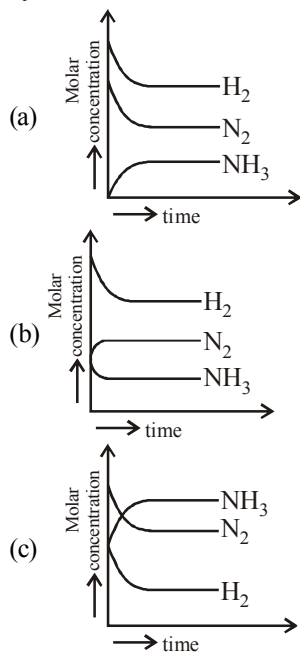
1	(c)	4	(a)	7	(b)	10	(d)	13	(a)	16	(d)	19	(b)	22	(48.00)	25	(4)	28	(76)
2	(b)	5	(b)	8	(b)	11	(c)	14	(d)	17	(d)	20	(d)	23	(6.25)	26	(1)	29	(31)
3	(c)	6	(b)	9	(a)	12	(a)	15	(a)	18	(b)	21	(-13538)	24	(1)	27	(4)	30	(62.5)

EQUILIBRIUM

7

MCQs with One Correct Answer

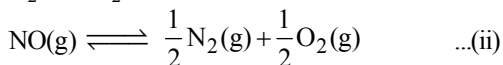
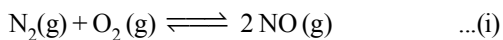
1. For the synthesis of ammonia by the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ in the Haber process, the attainment of equilibrium is correctly predicted by the curve



2. Two moles of PCl_5 were heated in a closed vessel of 2L. At equilibrium 40% of PCl_5 is dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is

(a) 0.53 (b) 0.267
(c) 2.63 (d) 5.3

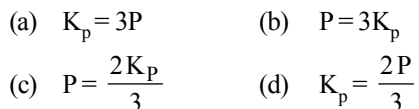
3. K_1 and K_2 are equilibrium constant for reactions (i) and (ii)



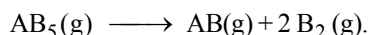
Then,

(a) $K_1 = \left(\frac{1}{K_2}\right)^2$ (b) $K_1 = K_2^2$
(c) $K_1 = \frac{1}{K_2}$ (d) $K_1 = (K_2)^0$

4. PCl_5 is dissociating 50% at 250°C at a total pressure of P atm. If equilibrium constant is K_p , then which of the following relation is numerically correct –



5. 1.0 mole of $\text{AB}_5(\text{g})$ is placed in a closed container under one atmosphere and at 300K . It is heated to 600K , when 20% by mass of it dissociates as

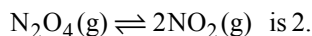


The resultant pressure is

- (a) 1.2 atm (b) 2.4 atm
 (c) 1.4 atm (d) 2.8 atm
6. For the reaction $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightarrow 2\text{CO}(\text{g})$, $K_p = 63$ atm at 1000K . If at equilibrium : $P_{\text{CO}} = 10 P_{\text{CO}_2}$, then the total pressure of the gases at equilibrium is



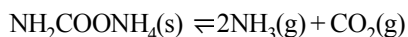
7. The value of K_p for the equilibrium reaction



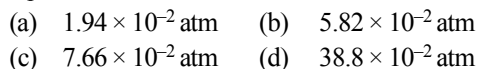
The percentage dissociation of $\text{N}_2\text{O}_4(\text{g})$ at a pressure of 0.5 atm is



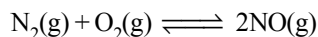
8. For the decomposition of the compound, represented as



the $K_p = 2.9 \times 10^{-5} \text{ atm}^3$. If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be :



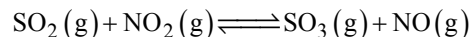
9. Air containing 79% of nitrogen and 21% of oxygen by volume is heated at 2200K and 1 atm until equilibrium is established according to the reaction



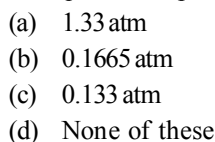
If the K_p of the reaction is 1.1×10^{-3} , calculate the amount of nitric oxide produced in terms of volume percent.



10. At a certain temperature and 2 atm pressure equilibrium constant (K_p) is 25 for the reaction



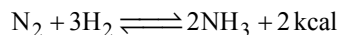
Initially if we take 2 moles of each of the four gases and 2 moles of inert gas, what would be the equilibrium partial pressure of NO_2 ?



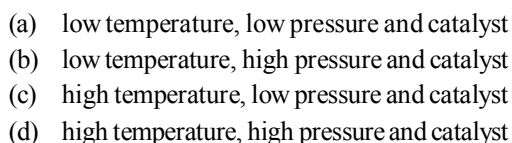
11. The equilibrium constant for a reaction, $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ is 4×10^{-4} at 2000K . In the presence of catalyst, the equilibrium is attained 10 times faster. The equilibrium constant in presence of catalyst at 2000K is :



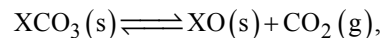
12. For the manufacture of ammonia by the reaction



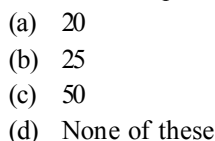
the favourable conditions are



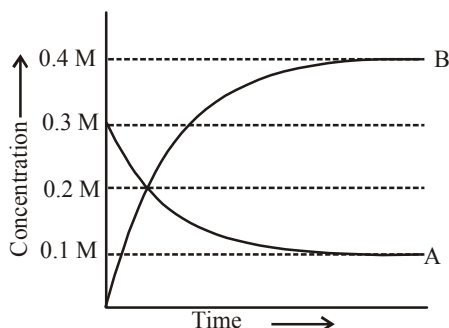
13. For the reaction



$K_p = 1.642 \text{ atm}$ at 727°C . If 4 moles of $\text{XCO}_3(\text{s})$ was put into a 50 litre container and heated to 727°C . What mole percent of the XCO_3 remains unreacted at equilibrium?



14. The figure shows the change in concentration of species A and B as a function of time. The equilibrium constant K_c for the reaction



$A(g) \rightleftharpoons 2B(g)$ is :

- (a) $K_c > 1$ (b) $K < 1$
 (c) $K = 1$ (d) data insufficient
15. The degree of dissociation of acetic acid in a 0.1 M solution is 1.32×10^{-2} . Find out the dissociation constant of the acid.
 (a) 1.50×10^{-4} (b) 1.80×10^{-16}
 (c) 1.76×10^{-5} (d) 1.2×10^{-3}
16. At 25°C , the solubility product of Hg_2Cl_2 in water is $3.2 \times 10^{-17} \text{ mol}^3 \text{ dm}^{-9}$. What is the solubility of Hg_2Cl_2 in water at 25°C ?
 (a) $1.2 \times 10^{-12} \text{ M}$ (b) $3.0 \times 10^{-6} \text{ M}$
 (c) $2 \times 10^{-6} \text{ M}$ (d) $1.2 \times 10^{-16} \text{ M}$
17. Solid AgNO_3 is slowly added to a solution containing each of 0.01 M NaCl and 0.001 M NaBr . What will be the concentration of Cl^- ions in solution when AgBr will just start to precipitate? $K_{sp}(\text{AgBr}) = 3.6 \times 10^{-13}$, $K_{sp}(\text{AgCl}) = 1.8 \times 10^{-10}$.
 (a) 1.8×10^{-7} (b) 3.6×10^{-10}
 (c) 0.01 (d) 2×10^{-4}
18. A buffered solution is prepared by mixing equal volumes of
 (a) 0.2 M NH_4OH and 0.1 M HCl
 (b) 0.2 M NH_4OH and 0.2 M HCl
 (c) 0.2 M NaOH and 0.1 M CH_3COOH
 (d) 0.1 M NH_4OH and 0.2 M HCl

19. Among the following, the correct statement is:
 (a) pH decreases when solid ammonium chloride is added to a dilute aqueous solution of NH_3
 (b) pH decreases when solid sodium acetate is added to a dilute aqueous solution of acetic acid
 (c) pH decreases when solid NaCl is added to a dilute aqueous solution of NaOH
 (d) pH decreases when solid sodium oxalate is added to a dilute aqueous solution of oxalic acid
20. Solubility products of CuI and Ag_2CrO_4 have almost the same value ($\sim 4 \times 10^{-12}$). The ratio of solubilities of the two salts ($\text{CuI}:\text{Ag}_2\text{CrO}_4$) is closest to
 (a) 0.01 (b) 0.02
 (c) 0.03 (d) 0.10

Numeric Value Answer

21. In the reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$, the amount of each PCl_5 , PCl_3 and Cl_2 is 2 mole at equilibrium and total pressure is 3 atm. What will be the value of K_p ?
22. For the reaction : $\text{SnO}_2(\text{s}) + 2\text{H}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g}) + \text{Sn}(\text{s})$
 The value of $2 \times K_p$ at 900 K where the equilibrium steam hydrogen mixture was 45% H_2 by volume is
23. For a reaction $\text{X} + \text{Y} \rightleftharpoons 2\text{Z}$, 1.0 mol of X, 1.5 mol of Y and 0.5 mol of Z were taken in a 1 L vessel and allowed to react. At equilibrium, the concentration of Z was 1.0 mol L^{-1} . The equilibrium constant of the reaction is $\frac{x}{15}$. The value of x is _____.
24. A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature. The partial pressure of CO_2 over the solution approaches a value of 30 bar when 44g of CO_2 is dissolved in 1 kg of water at room temperature. The approximate pH of the soft drink is _____ $\times 10^{-1}$.
 (First dissociation constant of $\text{H}_2\text{CO}_3 = 4.0 \times 10^{-7}$; $\log 2 = 0.3$; density of the soft drink = 1 g mL^{-1})

25. If the solubility product of AB_2 is $3.20 \times 10^{-11} \text{ M}^3$, then the solubility of AB_2 in pure water is $\text{_____} \times 10^{-4} \text{ mol L}^{-1}$.
[Assuming that neither kind of ion reacts with water]
26. Calculate the pH at the equivalence point when a solution of 0.01 M CH_3COOH is titrated with a solution of 0.01 M NaOH. $\text{p}K_a$ of CH_3COOH is 4.74.
27. A buffer solution is prepared by mixing 0.1 M ammonia and 1.0 M ammonium chloride. At 298 K, the $\text{p}K_b$ of NH_4OH is 5.0. The pH of the buffer is
28. Calculate the pOH of a solution at 25°C that contains $1 \times 10^{-10} \text{ M}$ of hydronium ions, i.e. H_3O^+ .
29. Calculate pH of a resultant solution of 25 mL of 0.1 M HCl, 50 mL of 0.02 M HNO_3 and 25 mL of 0.1 M NaOH.
30. 0.1 M NaOH is titrated with 0.1 M HA till the end point; K_a for HA is 5.6×10^{-6} and degree of hydrolysis is less compared to 1. Calculate pH of the resulting solution at the end point.

ANSWER KEY

1	(a)	4	(b)	7	(d)	10	(c)	13	(d)	16	(c)	19	(a)	22	(3)	25	(2.0)	28	(4)
2	(b)	5	(d)	8	(b)	11	(c)	14	(a)	17	(c)	20	(b)	23	(16)	26	(8.22)	29	(2)
3	(a)	6	(b)	9	(c)	12	(b)	15	(c)	18	(a)	21	(1)	24	(3)	27	(8)	30	(8.98)

REDOX REACTIONS



MCQs with One Correct Answer

- Which reaction involves neither oxidation nor reduction?
 - $\text{CrO}_4^{2-} \longrightarrow \text{Cr}_2\text{O}_7^{2-}$
 - $\text{Cr} \longrightarrow \text{CrCl}_3$
 - $\text{Na} \longrightarrow \text{Na}^+$
 - $2\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S}_4\text{O}_6^{2-}$
- In the reaction
$$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{I}_2$$
Which element is reduced
 - I
 - O
 - H
 - Cr
- Which one of the following reaction involves oxidation-reduction ?
 - $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$
 - $\text{NaBr} + \text{HCl} \rightarrow \text{NaCl} + \text{HBr}$
 - $\text{HBr} + \text{AgNO}_3 \rightarrow \text{AgBr} + \text{HNO}_3$
 - $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
- Oxidation numbers of chlorine atoms in CaOCl_2 are
 - 0, 0
 - 1, -1
 - 1, +1
 - None of these
- In the reaction
$$3\text{Br}_2 + 6\text{CO}_3^{2-} + 3\text{H}_2\text{O} \rightarrow 5\text{Br}^- + \text{BrO}_3^- + 6\text{HCO}_3^-$$
 - bromine is oxidised and carbonate is reduced.
 - bromine is reduced and water is oxidised
 - bromine is neither reduced nor oxidised
 - bromine is both reduced and oxidised
- The oxidation state of the most electronegative element in the products of the reaction between BaO_2 and H_2SO_4 are
 - 0 and -1
 - 1 and -2
 - 2 and 0
 - 2 and +1
- Which of the following substances acts as an oxidising as well as a reducing agent?
 - Na_2O
 - SnCl_2
 - Na_2O_2
 - NaNO_2

8. Which of the following reactions depicts the oxidising property of SO_2 ?
- $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$
 - $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
 - $\text{Cl}_2 + \text{SO}_2 \rightarrow \text{SO}_2\text{Cl}_2$
 - $2\text{MnO}_4^- + 5\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 5\text{SO}_4^{2-} + 2\text{Mn}^{2+} + 4\text{H}^+$
9. The most powerful oxidizing agent from the following is
- H_3BO_3
 - HPO_3
 - H_3PO_4
 - H_2SO_4
10. Which one of the following reactions involves disproportionation?
- $2\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow \text{CuSO}_4 + 2\text{H}_2\text{O} + \text{SO}_2$
 - $\text{As}_2\text{O}_3 + 3\text{H}_2\text{S} \rightarrow \text{As}_2\text{S}_3 + 3\text{H}_2\text{O}$
 - $2\text{KOH} + \text{Cl}_2 \rightarrow \text{KCl} + \text{KOC} + \text{H}_2\text{O}$
 - $\text{Ca}_3\text{P}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Ca}(\text{OH})_2 + 2\text{PH}_3$
11. In the standardization of $\text{Na}_2\text{S}_2\text{O}_3$ using $\text{K}_2\text{Cr}_2\text{O}_7$ by iodometry, the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ is
- Molecular weight/2
 - Molecular weight/6
 - Molecular weight/3
 - Same as molecular weight
12. Which of the following statements are correct concerning redox properties?
- A metal M for which E° for the half life reaction $\text{M}^{n+} + ne^- \rightleftharpoons \text{M}$ is very negative will be a good reducing agent.
 - The oxidizing power of the halogens decreases from chlorine to iodine.
 - The reducing power of hydrogen halides increases from hydrogen chloride to hydrogen iodide
- (i), (ii) and (iii)
 - (i) and (ii)
 - (i) only
 - (ii) and (iii)
13. $E^\ominus$ Values of some redox couples are given below. On the basis of these values choose the correct option.
- $E^\ominus$ values : $\text{Br}_2/\text{Br}^- = +1.90$; $\text{Ag}^+/\text{Ag(s)} = +0.80$
 $\text{Cu}^{2+}/\text{Cu(s)} = +0.34$; $\text{I}_2(\text{s})/\text{I}^- = 0.54$
- Cu will reduce Br^-
 - Cu will reduce Ag
 - Cu will reduce I^-
 - Cu will reduce Br_2
14. Standard reduction electrode potentials of three metals A, B & C are respectively +0.5 V, -3.0 V & -1.2 V. The reducing powers of these metals are
- $A > B > C$
 - $C > B > A$
 - $A > C > B$
 - $B > C > A$
15. The formal oxidation numbers of Cr and Cl in the ions $\text{Cr}_2\text{O}_7^{2-}$ and ClO_3^- , respectively are
- +6 and +7
 - +7 and +5
 - +6 and +5
 - +8 and +7

Numeric Value Answer

16. In the chemical reaction,
- $$\text{K}_2\text{Cr}_2\text{O}_7 + \text{X H}_2\text{SO}_4 + \text{Y SO}_2 \rightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{Z H}_2\text{O}$$
- X, Y and Z is
17. Atomic number of an element is 22. The highest O.S. exhibited by it in its compounds is
18. The number of electrons required to balance the following equation
- $$\text{NO}_3^- + 4\text{H}^+ + e^- \longrightarrow 2\text{H}_2\text{O} + \text{NO}$$
- is -
19. The O.N. of nitrogen in NO_3^- is
20. The oxidation state of iodine in H_4IO_6^- is

ANSWER KEY

1	(a)	3	(a)	5	(d)	7	(d)	9	(d)	11	(b)	13	(d)	15	(c)	17	(4)	19	(5)
2	(d)	4	(c)	6	(b)	8	(b)	10	(c)	12	(a)	14	(d)	16	(5)	18	(3)	20	(7)

HYDROGEN

9

MCQs with One Correct Answer

- When same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide solution the ratio of volumes of hydrogen evolved is
 - 1 : 1
 - 1 : 2
 - 2 : 1
 - 9 : 4
- 2 g of aluminium is treated separately with excess of dilute H_2SO_4 and excess of NaOH . The ratio of the volumes of hydrogen evolved is
 - 2 : 3
 - 1 : 1
 - 2 : 1
 - 1 : 2
- Consider the following statements :
 - Atomic hydrogen is obtained by passing hydrogen through an electric arc.
 - Hydrogen gas will not reduce heated aluminium oxide.
 - Finely divided palladium adsorbs large volume of hydrogen gas
 - Pure nascent hydrogen is best obtained by reacting Na with $\text{C}_2\text{H}_5\text{OH}$Which of the above statements is/are correct ?
 - only 1
 - only 2
 - 1, 2 and 3
 - 2, 3 and 4
- Which hydride is an ionic hydride?
 - H_2S
 - $\text{TiH}_{1.73}$
 - NH_3
 - NaH
- The reagent commonly used to determine hardness of water titrimetrically is
 - oxalic acid
 - sodium thiosulphate
 - sodium citrate
 - disodium salt of EDTA
- Pure water can be obtained from sea water by
 - centrifugation
 - plasmolysis
 - reverse osmosis
 - sedimentation
- Chemical A is used for water softening to remove temporary hardness. A reacts with sodium carbonate to generate caustic soda. When CO_2 is bubbled through a solution of A, it turns cloudy. What is the chemical formula of A.
 - CaCO_3
 - CaO
 - Ca(OH)_2
 - $\text{Ca(HCO}_3)_2$

8. Which of the following statements do not define the characteristic property of water "Water is a universal solvent"
- It can dissolve maximum number of compounds
 - It has very low dielectric constant
 - It has high liquid range
 - None of these
9. The amount of H_2O_2 present in 1 litre of 1.5 N H_2O_2 solution, is
- 25.5 g
 - 3.0 g
 - 8.0 g
 - 2.5 g
10. The volume strength of 1.5 N H_2O_2 solution is
- 8.4
 - 8.0
 - 4.8
 - 3.0
11. In lab H_2O_2 is prepared by
- cold $\text{H}_2\text{SO}_4 + \text{BaO}_2$
 - $\text{HCl} + \text{BaO}_2$
 - conc. $\text{H}_2\text{SO}_4 + \text{Na}_2\text{O}_2$
 - $\text{H}_2 + \text{O}_2$
12. The strength in volumes of a solution containing 30.36 g/L of H_2O_2 is
- 10 V
 - 5 V
 - 20 V
 - None of these
13. 20 Volume H_2O_2 solution has a strength of about
- 30%
 - 6%
 - 3%
 - 10%
14. H_2O_2 is a
- weak acid
 - weak base
 - neutral
 - None of these
15. $\text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{O}_2 + 2\text{e}^-$; $E^\circ = -0.68 \text{ V}$. This equation represents which of the following behaviour of H_2O_2 ?
- Reducing
 - Oxidising
 - Acidic
 - Catalytic
16. Heavy water reacts respectively with CO_2 , SO_3 , P_2O_5 and N_2O_5 to give the compounds
- D_2CO_3 , D_2SO_4 , D_3PO_2 , DNO_2
 - D_2CO_3 , D_2SO_4 , D_3PO_4 , DNO_2
 - D_2CO_3 , D_2SO_3 , D_3PO_4 , DNO_2
 - D_2CO_3 , D_2SO_4 , D_3PO_4 , DNO_3
17. A, 6 volume sample of H_2O_2
- will contain 6% V/V of H_2O_2
 - will contain 6% W/V of H_2O_2
 - would give 6 volumes of oxygen per unit volume of H_2O_2 sample at STP
 - would give 6 volumes of oxygen per unit weight of H_2O_2 sample at STP
18. In water-gas shift reaction, hydrogen gas is produced from the reaction of steam with
- methane
 - coke
 - carbon monoxide
 - carbon dioxide
19. Treatment with lime can remove hardness of water caused by
- CaCl_2
 - CaSO_4
 - $\text{Ca}(\text{HCO}_3)_2$
 - CaCO_3
20. Permanent hardness of water can be removed by
- heating
 - treating with sodium acetate ($\text{CH}_3\text{CO}_2\text{Na}$)
 - treating with $\text{Ca}(\text{HCO}_3)_2$
 - treatment with sodium hexametaphosphate ($\text{Na}_6\text{P}_6\text{O}_{18}$)

Numeric Value Answer

21. At room temperature, the ratio of ortho to para hydrogen is $n : 1$. what is the value of n ?
22. If 1 L of hard water (contains CaSO_4) requires 0.286 g of washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) for softening process. What is the hardness of water in ppm?
23. 3 Samples of H_2O_2 are labelled as 10 vol., 15 vol. and 20 vol. 100 mL. of each sample is mixed. What is the volume strength of resulting solution?
24. Ordinary water contains traces of heavy water. This is about 1 part in $n \times 1000$ parts. What is the integer value of n ?
25. What is the amount of energy released (in KJ) on combustion of 1 g of H_2 gas?

ANSWER KEY

1	(a)	4	(d)	7	(c)	10	(a)	13	(b)	16	(d)	19	(c)	22	(100)	25	(143)
2	(b)	5	(d)	8	(b)	11	(a)	14	(a)	17	(c)	20	(d)	23	(15)		
3	(c)	6	(c)	9	(a)	12	(a)	15	(a)	18	(c)	21	(3)	24	(6)		

THE S-BLOCK ELEMENT



MCQs with One Correct Answer

- Which of the following is not correct ?
 - $2\text{Li}_2\text{O} \xrightarrow[673\text{K}]{\text{heat}} \text{Li}_2\text{O}_2 + 2\text{Li}$
 - $2\text{K}_2\text{O} \xrightarrow[673\text{K}]{\text{heat}} \text{K}_2\text{O}_2 + 2\text{K}$
 - $2\text{Na}_2\text{O} \xrightarrow[673\text{K}]{\text{heat}} \text{Na}_2\text{O}_2 + 2\text{Na}$
 - $2\text{Rb}_2\text{O} \xrightarrow[673\text{K}]{\text{heat}} \text{Rb}_2\text{O}_2 + 2\text{Rb}$
- An inorganic compound which on heating first melts, then solidifies and liberates O_2 gas, the inorganic compound is
 - Al_2O_3
 - KMnO_4
 - MnO_2
 - KClO_3
- The element which on burning in air gives peroxide is
 - lithium
 - sodium
 - rubidium
 - caesium
- The stability of the following alkali metal chlorides follows the order
 - $\text{LiCl} > \text{KCl} > \text{NaCl} > \text{CsCl}$
 - $\text{CsCl} > \text{KCl} > \text{NaCl} > \text{LiCl}$
 - $\text{NaCl} > \text{KCl} > \text{LiCl} > \text{CsCl}$
 - $\text{KCl} > \text{CsCl} > \text{NaCl} > \text{LiCl}$
- Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because
 - the hydration energy of sodium sulphate is more than its lattice energy
 - the lattice energy of sodium sulphate is equal to its hydration energy
 - the hydration energy of sodium sulphate is less than its lattice energy
 - None of these
- Sodium thiosulphate, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ is used in photography to
 - reduce the silver bromide grains to metallic silver
 - convert the metallic silver to silver salt
 - remove undecomposed AgBr as soluble silver thiosulphate complex.
 - remove reduced silver

7. Li has the maximum value of ionisation potential among alkali metals i.e. lithium has the minimum tendency to ionise to give Li^+ ion. Thus, in aq. solution lithium is
- Strongest reducing agent
 - Poorest reducing agent
 - Strongest oxidising agent
 - Poorest oxidising agent
8. Which of the following statements about Na_2O_2 is not correct?
- It is diamagnetic in nature
 - It is derivative of H_2O_2
 - Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium.
 - It is the super oxide of sodium
9. Causticisation is a process in which
- soap is prepared by treating vegetable oils with caustic soda
 - paper is formed
 - petroleum is treated with caustic soda before fractional distillation
 - caustic soda is manufactured by treating sodium carbonate with lime.
10. $\text{KO}_2 + \text{CO}_2 + \text{H}_2\text{O} \xrightarrow{\text{more CO}_2} [\text{X}] + [\text{Y}]$
Products [X] and [Y] are respectively :
- $\text{K}_2\text{CO}_3, \text{O}_2$
 - $\text{KHCO}_3, \text{O}_2$
 - $\text{KOH}, \text{K}_2\text{CO}_3$
 - $\text{KHCO}_3, \text{H}_2\text{O}$
11. Which property of sodium is being used in street lights ?
- It shows photoelectric effect
 - It has low melting point
 - Sodium atom emits photons in the yellow region of visible spectrum, due to electrically stimulated transitions
 - Sodium vapours show golden colour
12. Which of the following compounds are paramagnetic in nature?
- KO_2
 - K_2O_2
 - Na_2O_2
 - Rb_2O_2
13. Which one of the following salts does not impart colour to the flame ?
- KI
 - LiCl
 - CaCl_2
 - MgCl_2
14. Lithopone is
- $\text{BaO} + \text{ZnSO}_4$
 - $\text{BaS} + \text{ZnSO}_4$
 - $\text{ZnS} + \text{BaSO}_4$
 - $\text{ZnO} + \text{BaSO}_4$
15. Philosopher's wool on heating with BaO at 1100°C produces:
- $\text{Ba} + \text{ZnCl}_2$
 - BaCdO_2
 - BaZnO_2
 - $\text{BaO}_2 + \text{Zn}$
16. Metallic magnesium is prepared by
- reduction of MgO by coke
 - electrolysis of aqueous solution of $\text{Mg}(\text{NO}_3)_2$
 - displacement of Mg by iron from MgSO_4 solution
 - electrolysis of molten MgCl_2
17. Which of the following sulphates have the highest solubility in water?
- MgSO_4
 - BaSO_4
 - CaSO_4
 - BeSO_4
18. Which of the following metal carbonate is decomposed on heating?
- MgCO_3
 - Na_2CO_3
 - K_2CO_3
 - Rb_2CO_3
19. Which of the following has correct increasing basic strength?
- $\text{MgO} < \text{BeO} < \text{CaO} < \text{BaO}$
 - $\text{BeO} < \text{MgO} < \text{CaO} < \text{BaO}$
 - $\text{BaO} < \text{CaO} < \text{MgO} < \text{BeO}$
 - $\text{CaO} < \text{BaO} < \text{BeO} < \text{MgO}$
20. Thermal stability of alkaline earth metal carbonates decreases in order
- $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$
 - $\text{BaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$
 - $\text{CaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{BaCO}_3$
 - None of these

21. Chemical A is used for water softening to remove temporary hardness. A reacts with Na_2CO_3 to generate caustic soda. When CO_2 is bubbled through A, it turns cloudy. What is the chemical formula of A
 (a) CaCO_3 (b) CaO
 (c) Ca(OH)_2 (d) $\text{Ca(HCO}_3)_2$
22. Which of the following is used as a scavenger in metallurgy?
 (a) Be (b) Mg
 (c) Ca (d) Sr
23. Hesse's process is a method for the manufacture of
 (a) NaOH (b) HNO_3
 (c) H_2SO_4 (d) Bleaching powder
24. Equimolar solutions of the following compounds are prepared separately in water. Which will have the lowest pH value?
 (a) BeCl_2 (b) SrCl_2
 (c) CaCl_2 (d) MgCl_2
25. 100 mL of tap water containing $\text{Ca(HCO}_3)_2$ was titrated with N/50 HCl with methyl orange as indicator. If 30 mL of HCl were required, calculate the temporary hardness as parts of CaCO_3 per 10^6 parts of water.
 (a) 150 ppm (b) 300 ppm
 (c) 450 ppm (d) 600 ppm

ANSWER KEY

1	(a)	4	(d)	7	(a)	10	(b)	13	(d)	16	(d)	19	(b)	22	(c)	25	(b)
2	(d)	5	(a)	8	(d)	11	(c)	14	(c)	17	(d)	20	(a)	23	(d)		
3	(b)	6	(c)	9	(d)	12	(a)	15	(c)	18	(a)	21	(c)	24	(a)		

THE p-BLOCK ELEMENTS (GROUPS 13 AND 14)



MCQs with One Correct Answer

- H_3BO_3 on heating up to 373 K yields
 - boric anhydride
 - orthoboric acid
 - metaboric acid
 - tetraboric acid
- Aluminium chloride exists as dimer, (Al_2Cl_6) in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives
 - $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$
 - $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
 - $\text{Al}^{3+} + 3\text{Cl}^-$
 - $\text{Al}_2\text{O}_3 + 6\text{HCl}$
- In reaction $\text{BF}_3 + 3\text{LiBH}_4 \rightarrow 3\text{LiF} + \text{X}$; X is
 - B_4H_{10}
 - B_2H_6
 - BH_3
 - B_3H_8
- The role of fluor spar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is
 - as a catalyst
 - to make the fused mixture very conducting
 - to increase the temperature of the melt.
 - to decrease the rate of oxidation of carbon at the anode.
- The dissolution of $\text{Al}(\text{OH})_3$ by a solution of NaOH results in the formation of
 - $[\text{Al}(\text{H}_2\text{O})_4(\text{OH})]^{2+}$
 - $[\text{Al}(\text{H}_2\text{O})_2(\text{OH})_4]^-$
 - $[\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3]$
 - $[\text{Al}(\text{H}_2\text{O})_6(\text{OH})_3]$
- $\text{BX}_3 + \text{NH}_3 \xrightarrow{\text{R.T.}} \text{BX}_3 \cdot \text{NH}_3 + \text{Heat of adduct formation } (\Delta H)$
The numerical value of ΔH is found to be maximum for :
 - BF_3
 - BCl_3
 - BBR_3
 - BI_3

7. When aluminium is heated in atmosphere of nitrogen it forms :
 (a) AlN (b) Al_2N
 (c) Al_3N (d) Al_2N_3
8. Diborane is prepared on large scale by
 (a) $2\text{BF}_3(\text{g}) + 6\text{LiH}(\text{s}) \xrightarrow{450\text{ K}} \text{B}_2\text{H}_6(\text{g}) + 6\text{LiF}(\text{s})$
 (b) $2\text{BCl}_3(\text{g}) + 6\text{LiH}(\text{s}) \xrightarrow{450\text{ K}} \text{B}_2\text{H}_6(\text{g}) + 6\text{LiCl}(\text{s})$
 (c) $2\text{BF}_3(\text{g}) + 6\text{NaH} \xrightarrow{450\text{ K}} \text{B}_2\text{H}_6 + 6\text{NaF}$
 (d) $2\text{BCl}_3 + 6\text{NaH} \xrightarrow{450\text{ K}} \text{B}_2\text{H}_6 + 6\text{NaCl}$
9. Silicon dioxide is formed by the reaction of
 (a) $\text{SiCl}_4 + \text{H}_2\text{O}$
 (b) $\text{SiO}_2 + \text{HF}$
 (c) $\text{SiO}_2 + \text{NaOH}$
 (d) $\text{SiCl}_4 + \text{NaOH}$
10. SiF_4 gets hydrolysed giving.....
 (a) SiO_2 (b) $\text{Si}(\text{OH})_2\text{F}_2$
 (c) H_2SiF_6 (d) $\text{Si}(\text{OH})_4$
11. Soldiers of Napoleon army while at alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey powder. This transformation is related to
 (a) a change in the partial pressure of oxygen in the air
 (b) a change in the crystalline structure of tin
 (c) an interaction with nitrogen of the air at very low temperatures
 (d) an interaction with water vapours contained in the humid air
12. The tendency of X in BX_3 ($X = \text{F}, \text{Cl}, \text{OMe}, \text{NMe}$) to form a π -bond with boron follows the order
 (a) $\text{BCl}_3 < \text{BF}_3 < \text{B}(\text{OMe})_3 < \text{B}(\text{NMe})_3$
 (b) $\text{BF}_3 < \text{BCl}_3 < \text{B}(\text{OMe})_3 < \text{B}(\text{NMe})_3$
 (c) $\text{BCl}_3 < \text{B}(\text{NMe})_3 < \text{B}(\text{OMe})_3 < \text{BF}_3$
 (d) $\text{BCl}_3 < \text{BF}_3 < \text{B}(\text{NMe})_3 < \text{B}(\text{OMe})_3$
13. PbO_2 is obtained from
 (a) the reaction of PbO with HCl
 (b) thermal decomposition of $\text{Pb}(\text{NO}_3)_2$ at 200°C
 (c) the reaction of Pb_3O_4 with HNO_3
 (d) the reaction of Pb with air at room temperature
14. An allotrope of carbon which exhibits only two types of C – C bond distance of 143.5 pm and 138.3 pm, is
 (a) charcoal (b) graphite
 (c) diamond (d) fullerene
15. A solid element (symbol Y) conducts electricity and forms two chlorides YCl_n (colourless volatile liquid) and YCl_{n-2} (a colourless solid). To which one of the following groups of the periodic table does Y belong?
 (a) 13 (b) 14
 (c) 15 (d) 16
16. The reducing power of divalent species decreases in the order
 (a) $\text{Ge} > \text{Sn} > \text{Pb}$
 (b) $\text{Sn} > \text{Ge} > \text{Pb}$
 (c) $\text{Pb} > \text{Sn} > \text{Ge}$
 (d) None of these
17. The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4 , on hydrolysis gives a white gelatinous precipitate. The precipitate is
 (a) hydrofluorosilicic acid
 (b) silica gel
 (c) silicic acid
 (d) calciumfluorosilicate
18. On heating $\text{Pb}(\text{NO}_3)_2$, the products formed are:
 (a) $\text{PbO}, \text{N}_2, \text{O}_2$ (b) $\text{Pb}(\text{NO}_2)_2, \text{O}_2$
 (c) $\text{PbO}, \text{NO}_2, \text{O}_2$ (d) $\text{Pb}, \text{N}_2, \text{O}_2$
19. The product of the following reaction are:

$$\text{SiO}_2 + \text{C} \xrightarrow{\Delta} \text{Products}$$
 (a) SiC and CO_2 (b) SiO and CO
 (c) SiC and CO (d) Si and CO

20. What is not correct about carbon monoxide ?
 (a) Carbon in CO is sp hybridised and is linear molecule
 (b) CO is a ligand because it has a lone pair of electrons on the carbon atom
 (c) CO is the reducing agent in metallurgy of iron
 (d) CO has the structure C = O
21. In aluminates, the coordination number of Al is
22. How many oxides of the following are non-amphoteric in nature?
 CO_2 , SiO_2 , SnO_2 and CaO
23. How many orbitals of boron are involved in hybridisation in B_2H_6 ?
24. In a molecule of C_{60} Buckminsterfullerene, what is the sum of six-membered and five-membered rings?
25. What is the sum for the molecular masses of constituents of producer gas?

Numeric Value Answer

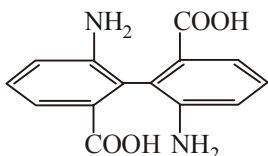
ANSWER KEY																	
1	(c)	4	(b)	7	(a)	10	(d)	13	(c)	16	(a)	19	(c)	22	(3)	25	(56)
2	(b)	5	(b)	8	(a)	11	(b)	14	(d)	17	(d)	20	(d)	23	(4)		
3	(b)	6	(d)	9	(a)	12	(a)	15	(b)	18	(c)	21	(6)	24	(32)		

ORGANIC CHEMISTRY- SOME BASIC PRINCIPLES AND TECHNIQUES

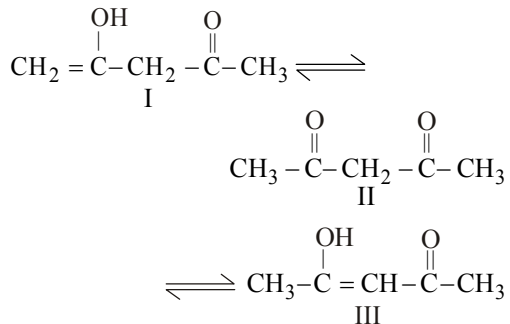
12

MCQs with One Correct Answer

1. The IUPAC name of

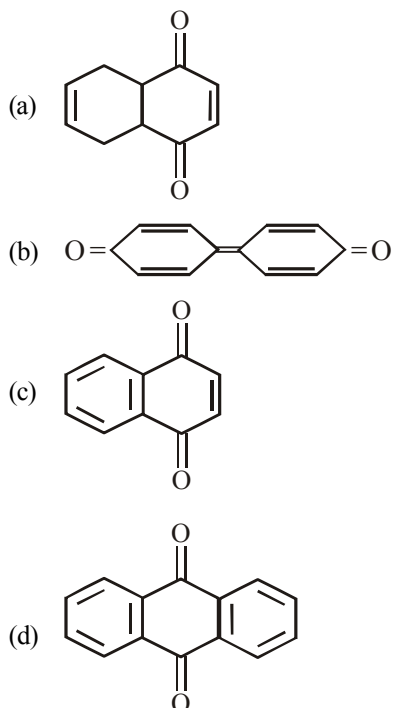


- (a) *o, o'* - diamino - *o, o'* - dicarboxy biphenyl
(b) 6, 6' - diamino diphenyl - 2, 2' - dicarboxylic acid
(c) diamino dicarboxy biphenyl
(d) none of these
2. The order of stability of the following tautomeric compounds is :

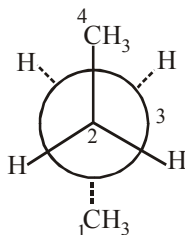


- (a) III > II > I (b) II > I > III
(c) II > III > I (d) I > II > III

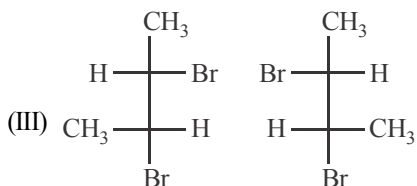
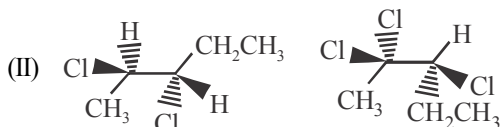
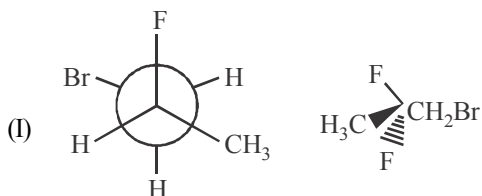
3. Which of the following will show tautomerism ?



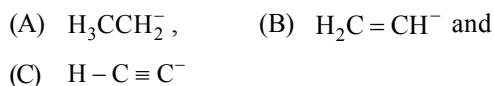
4. In the given conformation, if C_2 is rotated about $C_2 - C_3$ bond anticlockwise by an angle of 120° then the conformation obtained is



- (a) fully eclipsed conformation
(b) partially eclipsed conformation
(c) gauche conformation
(d) staggered conformation
5. Indicate whether each of the following pairs are identical or not?

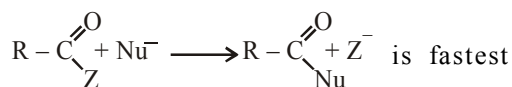


- (a) I-enantiomers; II-diastereomers; III-enantiomers
(b) I-identical; II-enantiomers; III-enantiomers
(c) I-enantiomers; II-diastereomers; III-identical
(d) I-enantiomers; II-identical; III-identical
6. Base strength of:



is in the order of:

- (a) $(B) > (A) > (C)$ (b) $(C) > (B) > (A)$
(c) $(A) > (C) > (B)$ (d) $(A) > (B) > (C)$
7. Chlorine in vinyl chloride is less reactive because
(a) sp^2 - hybridised carbon has more acidic character than sp^3 - hybridised carbon
(b) C - Cl bond develops partial double bond character
(c) of resonance
(d) All are correct
8. Most stable carbocation is
(a) $p\text{-NO}_2\text{-C}_6\text{H}_4\text{-CH}_2^+$
(b) $\text{C}_6\text{H}_5\text{CH}_2^+$
(c) $p\text{-Cl-C}_6\text{H}_4\text{-CH}_2^+$
(d) $p\text{-CH}_3\text{O-C}_6\text{H}_4\text{-CH}_2^+$
9. Rate of the reaction



when Z is

- (a) OC_2H_5 (b) NH_2
(c) Cl (d) OCOCH_3
10. The replacement of chlorine of chlorobenzene to give phenol requires drastic conditions, but the chlorine of 2,4-dinitrochlorobenzene is readily replaced since,
(a) nitro groups make the aromatic ring electron rich at ortho/para positions
(b) nitro groups withdraw electrons from the meta position of the aromatic ring
(c) nitro groups donate electrons at meta position
(d) nitro groups withdraw electrons from ortho/para positions of the aromatic ring

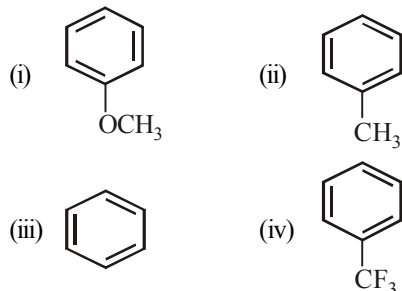
11. $\text{CH}_3\text{Br} + \text{Nu}^- \longrightarrow \text{CH}_3 - \text{Nu} + \text{Br}^-$

The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is



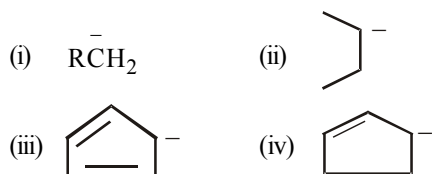
- (a) $A > B > C > D$
(b) $B > D > C > A$
(c) $D > C > A > B$
(d) $D > C > B > A$

12. In the below mentioned compounds the decreasing order of reactivity towards electrophilic substitution is



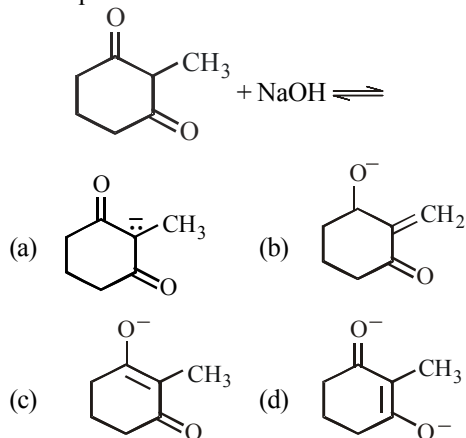
- (a) (iv) > (i) > (ii) > (iii)
 (b) (ii) > (iii) > (i) > (iv)
 (c) (iii) > (i) > (iv) > (ii)
 (d) (i) > (ii) > (iii) > (iv)

13. The correct order of stability of the following carbanions is

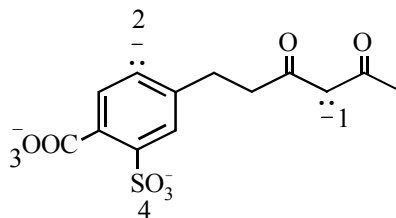


- (a) III > IV > I > II (b) I > II > III > IV
 (c) I > III > II > IV (d) IV > III > II > I

14. Which of the following is not a resonance form of the enolate ion formed in the following acid base equilibrium?

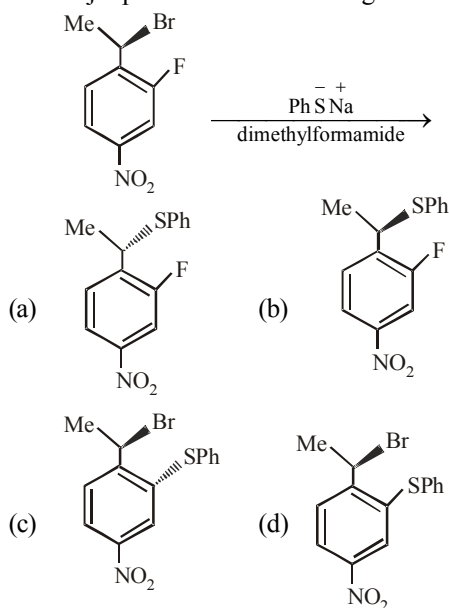


15. The structure drawn below has four nucleophilic sites, arrange them in order of decreasing nucleophilicity.

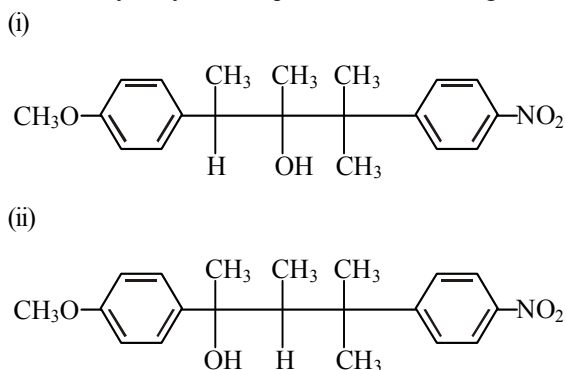
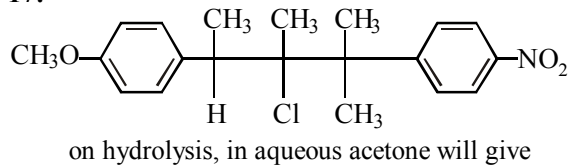


- (a) $3 > 4 > 1 > 2$ (b) $4 > 3 > 2 > 1$
 (c) $4 > 3 > 1 > 2$ (d) $3 > 4 > 2 > 1$

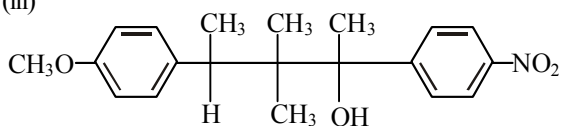
16. The major product of the following reaction is –



- 17.

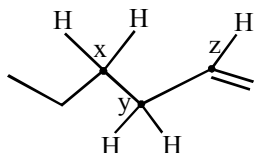


(iii)



- (a) Mixture of (i) and (ii)
 (b) Mixture of (i) and (iii)
 (c) Only (iii)
 (d) Only (i)

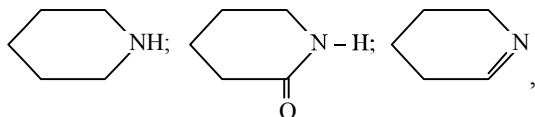
18.



Arrange the (C - H) bonds x, y and z in decreasing order of their bond dissociation energies in homolysis.

- (a) $y > x > z$ (b) $z > x > y$
 (c) $z > y > x$ (d) $y > z > x$

19.

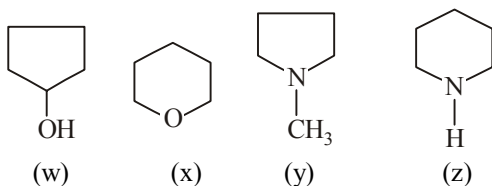


(x) (y) (z)

The correct order of decreasing basic strengths of x, y and z is:

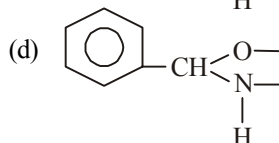
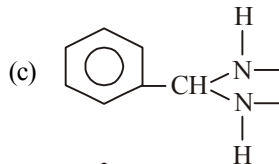
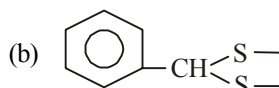
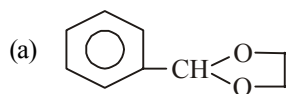
- (a) $x > y > z$ (b) $x > z > y$
 (c) $y > x > z$ (d) $y > z > x$

20. Arrange the following (w, x, y, z) in decreasing order of their boiling points:

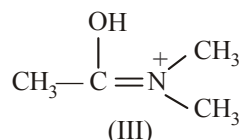
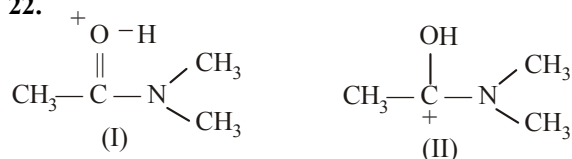


- (a) $w > x > z > y$ (b) $w > x > y > z$
 (c) $w > z > y > x$ (d) $w > z > x > y$

21. Which of the following compounds has most acidic hydrogen?



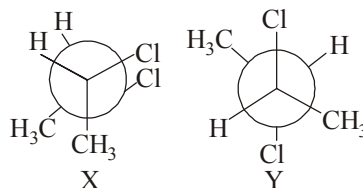
22.



The correct stability order of the given canonical structure is :

- (a) $I > II > III$ (b) $III > I > II$
 (c) $I > III > II$ (d) $II > III > I$

23. X and Y



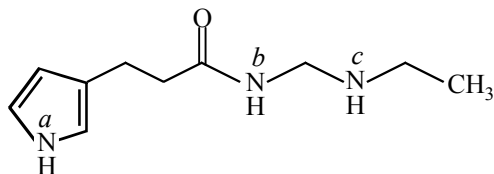
are

- (a) enantiomers
 (b) diastereomers
 (c) constitutional isomers
 (d) conformers

24. The higher stabilities of *tert*-butyl cation over *iso*-propyl cation and *trans*-2-butene over propene, respectively, are due to orbital interactions involving.

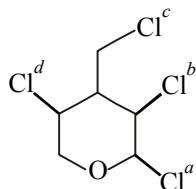
- (a) $\sigma \rightarrow \pi$ and $\sigma \rightarrow \pi^*$
 (b) $\sigma \rightarrow$ vacant p and $\sigma \rightarrow \pi$
 (c) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$
 (d) $\sigma \rightarrow$ vacant p and $\sigma \rightarrow \pi^*$

25. The most acidic proton and the strongest, nucleophilic nitrogen in the following compound



respectively, are

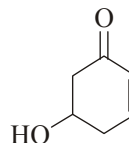
- (a) $N^a - H$; N^b (b) $N^b - H$; N^c
 (c) $N^a - H$; N^c (d) $N^c - H$; N^a
26. The chlorine atom of the following compound



that reacts most readily with $AgNO_3$ to give a precipitate is

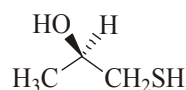
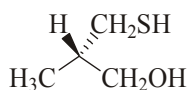
- (a) Cl^a (b) Cl^b
 (c) Cl^c (d) Cl^d
27. The lower stability of ethyl anion compared to methyl anion and the higher stability of ethyl radical compared to methyl radical, respectively, are due to
- (a) +I effect of the methyl group in ethyl anion
 $\sigma \rightarrow p$ -orbital conjugation in ethyl radical
 (b) -I effect of the methyl group in ethyl anion and $\sigma \rightarrow \sigma^*$ p-conjugation in ethyl radical
 (c) +I effect of the methyl group in both cases
 (d) +I effect of the methyl group in ethyl anion and $\sigma \rightarrow \sigma^*$ conjugation in ethyl radical.

28. IUPAC name of the following compound



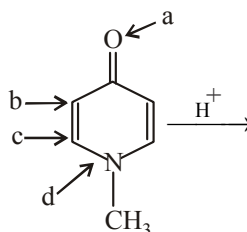
is

- (a) 1-hydroxycyclohex-4-en-3-one
 (b) 1-hydroxycyclohex-3-en-5-one
 (c) 3-hydroxycyclohex-5-en-1-one
 (d) 5-hydroxycyclohex-2-en-1-one
29. The absolute configurations of the following compounds



respectively, are

- (a) *R* and *R* (b) *S* and *S*
 (c) *R* and *S* (d) *S* and *R*
- 30.



Identify the site, where attack of H^+ is most favourable.

- (a) a (b) b
 (c) c (d) d

ANSWER KEY

1	(b)	4	(c)	7	(c)	10	(d)	13	(a)	16	(a)	19	(b)	22	(b)	25	(b)	28	(d)
2	(a)	5	(c)	8	(d)	11	(c)	14	(b)	17	(a)	20	(d)	23	(d)	26	(a)	29	(d)
3	(a)	6	(d)	9	(c)	12	(d)	15	(c)	18	(b)	21	(b)	24	(d)	27	(a)	30	(a)

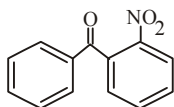
HYDROCARBONS

13

MCQs with One Correct Answer

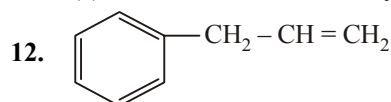
- On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be
 (a) pentane (b) isopentane
 (c) neopentane (d) propane
- cis*-2-Butene has higher boiling point than the *trans*-analog. It is because of
 (a) hydrogen bonding in the *cis*-isomer
 (b) higher molecular mass of the *cis*-isomer
 (c) van der Waal forces in the *cis*-isomer
 (d) polarity in the *cis*-isomer.
- 2-Butene can be obtained by the electrolysis of an aqueous solution of
 (a) 1, 2-Dimethylmaleic acid
 (b) 2, 2-Dimethylbutandioic acid
 (c) 3, 3-Dimethylbutandioic acid
 (d) 2, 3-Dimethylbutandioic acid
- The decreasing order of reactivity of hydrogen bromide on the following alkenes is
 $\text{CH}_2=\text{CH}_2$, $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$, $\text{CH}_2=\text{CHCl}$,
 I II III
 $(\text{CH}_3)_2\text{C}=\text{CH}_2$
 IV
- Which of the following reagents convert propene to 1-propanol?
 (a) H_2O , H_2SO_4
 (b) Aqueous KOH
 (c) MgSO_4 , $\text{NaBH}_4/\text{H}_2\text{O}$
 (d) B_2H_6 , H_2O_2 , OH^-
- When hydrochloric acid gas is treated with propene in presence of benzoyl peroxide, it gives
 (a) 2-Chloropropane
 (b) Allyl chloride
 (c) *n*-Propyl chloride.
 (d) No reaction
- 1, 3-Butadiene when treated with Br_2 gives
 (a) 1, 4-dibromo-2-butene
 (b) 1, 3-dibromo-2-butene
 (c) 3, 4-dibromo-1-butene
 (d) 2, 3-dibromo-2-butene
- Addition of Br_2 to *trans*-2-butene would give a product which is
 (a) chiral (b) meso
 (c) racemic (d) optically active

9. Reaction of hydrogen bromide with propene in the absence of peroxide is a/an
- free radical addition
 - nucleophilic addition
 - electrophilic substitution
 - electrophilic addition
10. One mole of propanone and one mole of formaldehyde are the products of ozonolysis of one mole of an alkene. The alkene may be
- 2-methylpropene
 - 2,2-dimethyl-1-butene
 - propene
 - 2-butene
11. The compound shown below



can be readily prepared by Friedel-Craft's reaction between

- benzene and 2-nitrobenzoyl chloride
- benzyl chloride and nitrobenzene
- nitrobenzene and benzoyl chloride
- benzene and 2-nitrobenzyl chloride



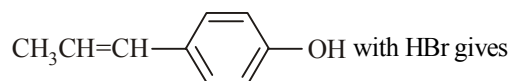
on mercuration-demercuration produces the major product:

-
-
-
-

13. In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition of alkenes because:

- One of the steps is endothermic in HCl and HI
- Both HCl and HI are strong acids
- HCl is oxidizing and the HI is reducing
- All the steps are exothermic in HCl and HI

14. The reaction of

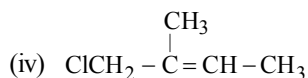
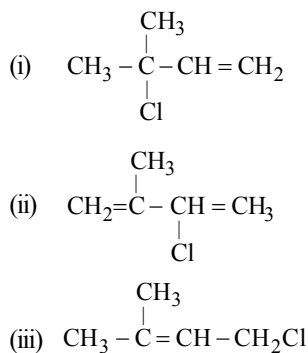


- $\text{CH}_3\text{CHBrCH}_2-$ OH
- $\text{CH}_3\text{CH}_2\text{CHBr}-$ OH
- $\text{CH}_3\text{CHBrCH}_2-$ Br
- $\text{CH}_3\text{CH}_2\text{CHBr}-$ Br

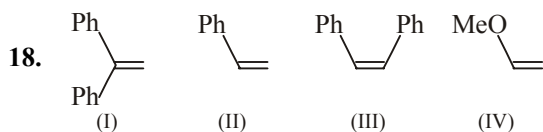
15. Among the following sets, the most stable ionic species are

- and
- and
- and
- and

16. Isopropene, $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}=\text{CH}_2$ is treated with HCl, the products formed will be a mixture of

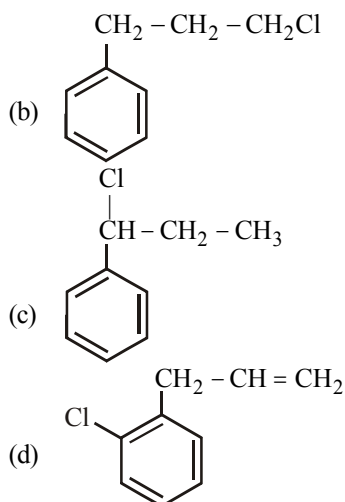
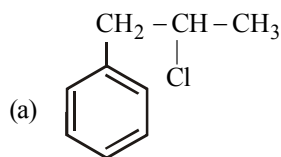
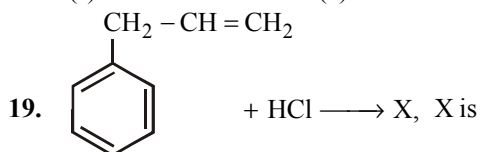


- (a) (i) and (iii) (b) (ii) and (iii)
 (c) (ii) and (iv) (d) All the four.
17. 1, 3-Butadiene is treated with hot $\text{KMnO}_4/\text{NaIO}_4$, the product(s) formed is(are)
- (a) $2\text{HCHO} + \text{OHC}-\text{CHO}$
 (b) $2\text{HCOOH} + \text{HOOC}-\text{COOH}$
 (c) $2\text{CO}_2 + \text{HOOC}-\text{COOH}$
 (d) 4CO_2

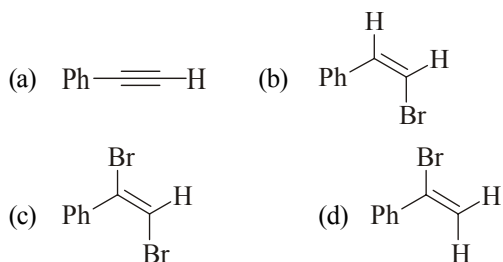
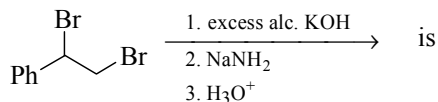


Order of rate of electrophilic addition reaction with HBr will be

- (a) $\text{IV} > \text{I} > \text{III} > \text{II}$ (b) $\text{I} > \text{II} > \text{III} > \text{IV}$
 (c) $\text{I} > \text{III} > \text{II} > \text{IV}$ (d) $\text{IV} > \text{I} > \text{II} > \text{III}$



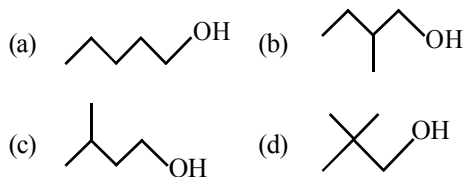
20. The major product of the following reaction



21. The addition of HI in the presence of peroxide catalyst does not follow anti-Markovnikov's rule because

- (a) HI is a strong reducing agent.
 (b) H-I bond is too strong to be broken homolytically.
 (c) I atom combines with H atom to give back HI.
 (d) Iodine atom is not reactive enough to add across a double bond.

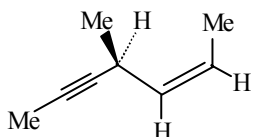
22. Which of the following alcohols cannot be prepared from hydration of an alkene?



23. In which of the following reaction, Markovnikov's rule is violated?

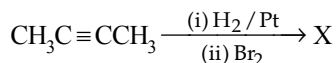
- (a) $\text{CH}_3\text{—O—CH=CH}_2 \xrightarrow[\text{CCl}_4]{\text{HBr}}$
 (b) $\text{CH}_3\text{—NH—CH=CH}_2 \xrightarrow[\text{CCl}_4]{\text{HBr}}$
 (c) $\text{CH}_3\text{—S—CH=CH}_2 \xrightarrow[\text{CCl}_4]{\text{HBr}}$
 (d) $\text{O}_2\text{N—CH=CH}_2 \xrightarrow[\text{CCl}_4]{\text{HBr}}$

24.



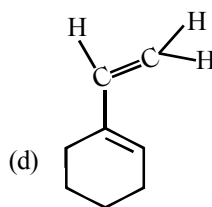
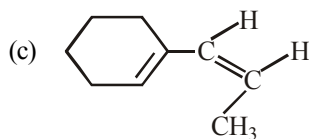
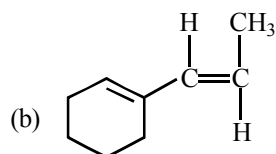
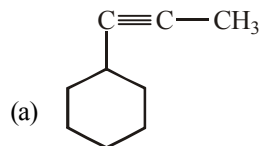
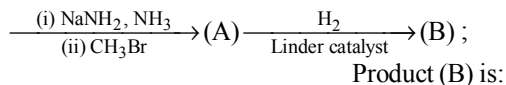
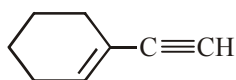
Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives

- (a) an optically active compound
 (b) an optically inactive compound
 (c) a racemic mixture
 (d) a diastereomeric mixture
25. The compound X in the following reaction is

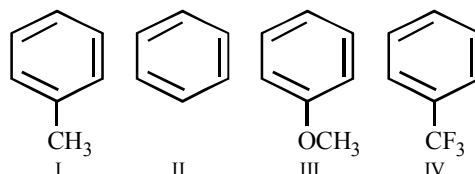


- (a) *d*-2, 3-Dibromobutane
 (b) *l*-2, 3-Dibromobutane
 (c) *dl*-2, 3-Dibromobutane
 (d) *meso*-2, 3 Dibromobutane.

26.

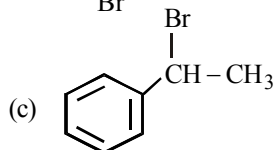
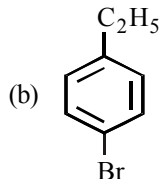
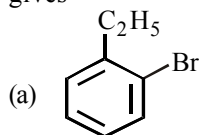


27. Among the following compounds, the decreasing order of reactivity towards electrophilic substitution is



- (a) III > I > II > IV
 (b) IV > I > II > III
 (c) I > II > III > IV
 (d) II > I > III > IV

28. Bromination of ethyl benzene in presence of light gives

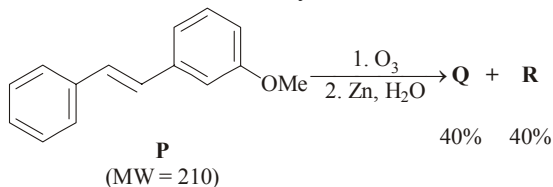


- (d) Both (a) and (b)

29. Which of the following would not give 2-phenylbutane as the major product in a Friedel-Crafts alkylation reaction ?

- (a) 1-butene + HF
 (b) 2-butanol + H_2SO_4
 (c) Butanoyl chloride + AlCl_3 then Zn, HCl
 (d) Butyl chloride + AlCl_3

30. In the following reaction, **P** gives two products **Q** and **R**, each in 40% yield



If the reaction is carried out with 420 mg of **P**, the reaction yields 108.8 mg of **Q**. The amount of **R** produced in the reaction is closest to

- (a) 97.6 mg (b) 108.8 mg
(c) 84.8 mg (d) 121.6 mg

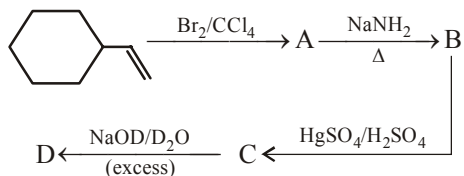
Numeric Value Answer

31. On conversion into the Grignard reagent followed by treatment with water, how many alkyl bromides would yield isopentane?
32. Number of hydrocarbons formed when C_2H_5Br and $CH_3-CH_2-CH_2-Br$ are treated with Na in the presence of dry ether is _____.

33. How many number of moles of H_2 is used for the complete hydrogenation of the given compound in the presence of a metal catalyst?



34. Consider the following reaction scheme



How many deuterium are present in the compound D?

35. Total no. of alkynes that on catalytic reduction gives 3-ethyl-4-methylheptane _____.

ANSWER KEY

1	(c)	5	(d)	9	(d)	13	(a)	17	(c)	21	(b)	25	(c)	29	(c)	33	(7)
2	(d)	6	(a)	10	(a)	14	(b)	18	(d)	22	(d)	26	(c)	30	(c)	34	(4)
3	(d)	7	(a)	11	(a)	15	(d)	19	(c)	23	(d)	27	(a)	31	(4)	35	(3)
4	(b)	8	(b)	12	(a)	16	(a)	20	(a)	24	(b)	28	(c)	32	(7)		

ENVIRONMENTAL CHEMISTRY



MCQs with One Correct Answer

- In Antarctica, ozone depletion is due to the formation of following compound
 - acrolein
 - peroxyacetyl nitrate
 - SO_2 and SO_3
 - chlorine nitrate
- Classical smog occurs in places of
 - excess SO_2
 - low temperature
 - high temperature
 - excess NH_3
- Which gas is responsible for 'Bhopal Gas Tragedy' in 1984?
 - CO
 - Methyl isocyanate
 - SO_2 and NO_2
 - Ethyl isocyanate
- CFC which is a main reason behind air pollution, is produced by
 - sewage pollutant
 - aerosols
 - industrial remains
 - all the these
- Which is a dangerous radiological pollutant?
 - C^{14}
 - S^{35}
 - Sr^{90}
 - P^{32}
- Which of the following is the coldest region of atmosphere?
 - Thermosphere
 - Mesosphere
 - Troposphere
 - Stratosphere
- Green house gases –
 - allow shorter wavelength to enter earth's atmosphere while doesn't allow longer wavelength to leave the earth's atmosphere.
 - allow longer wavelength to enter earth atmosphere while doesn't allow shorter wavelength to leave the surface
 - don't have wavelength specific behaviour.
 - show wavelength specific behaviour near the earth while far from earth these have wavelength independent behaviour.
- Formation of London smog takes place in
 - winter during day time
 - summer during day time
 - summer during morning time
 - winter during morning time

9. Which of the following statement is false ?
 (a) The average residence time of NO is one month
 (b) Limestone acts as a sink for SO_x
 (c) SO_x can be removed from fuel gases by passing through a solution of citrate ions
 (d) Ammonia acts as a sink for NO_x
10. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value
 (a) slightly lower than that of rain water without thunderstorm
 (b) slightly higher than that when the thunderstorm is not there
 (c) uninfluenced by occurrence of thunderstorm
 (d) which depends upon the amount of dust in air
11. Which causes death of fish in water bodies polluted by sewage?
 (a) Foul smell (b) Pathogens
 (c) Herbicides (d) Decrease in D.O.
12. Negative soil pollution is
 (a) reduction in soil productivity due to erosion and over use
 (b) reduction in soil productivity due to addition of pesticides and industrial wastes
 (c) converting fertile land into barren land by dumping ash, sludge and garbage
 (d) None of the above
13. The quantity of DDT in food chain
 (a) decreases (b) remains same
 (c) increases (d) changes
14. Minamata disease is a pollution related disease results from
 (a) oil spills into sea
 (b) accumulation of arsenic into atmosphere
 (c) release of industrial waste mercury into water bodies
 (d) release human organic waste into drinking water
15. Lichens do not like to grow in cities
 (a) because of absence of the right type of algae and fungi
 (b) because of lack of moisture
 (c) because of SO₂ pollution
 (d) because natural habitat is missing

ANSWER KEY

1	(a)	3	(b)	5	(c)	7	(a)	9	(a)	11	(d)	13	(c)	15	(c)
2	(b)	4	(b)	6	(b)	8	(d)	10	(a)	12	(a)	14	(c)		

THE SOLID STATE



MCQs with One Correct Answer

- The pure crystalline substance on being heated gradually first forms a turbid liquid at constant temperature and still at higher temperature turbidity completely disappears. This behaviour is a characteristic property of substance forming.
(a) Allotropic (b) Liquid
(c) Isomeric (d) Isomorphous
- Which of crystal systems contains the maximum number of Bravais lattices?
(a) Cubic
(b) Hexagonal
(c) Triclinic
(d) Orthorhombic
- How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00 g ?
[Atomic masses : Na = 23, Cl = 35.5]
(a) 5.14×10^{21} unit cells
(b) 1.28×10^{21} unit cells
(c) 1.71×10^{21} unit cells
(d) 2.57×10^{21} unit cells
- The second order Bragg diffraction of X-rays with $\lambda = 1.00 \text{ \AA}$ from a set of parallel planes in a metal occurs at an angle 60° . The distance between the scattering planes in the crystal is
(a) $0.575 \text{ \AA}$
(b) $1.00 \text{ \AA}$
(c) $2.00 \text{ \AA}$
(d) $1.15 \text{ \AA}$
- A compound is formed by elements A and B. The crystalline cubic structure has the A atoms at the corners of the cube and B atoms at the body centre. The simplest formula of the compound is
(a) AB (b) A_6B
(c) AB_6 (d) A_8B_4
- Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius)
(a) $\frac{12}{3}\pi r^3$ (b) $\frac{16}{3}\pi r^3$
(c) $\frac{20}{3}\pi r^3$ (d) $\frac{24}{3}\pi r^3$

7. If calcium crystallizes in bcc arrangement and the radius of Ca atom is 96 pm, then the volume of unit cell of Ca is
 - (a) $10.9 \times 10^{-36} \text{ m}^3$
 - (b) $10.9 \times 10^{-30} \text{ m}^3$
 - (c) $21.8 \times 10^{-30} \text{ m}^3$
 - (d) $21.8 \times 10^{-36} \text{ m}^3$
8. The number of unit cells in the Ca atom lies on the surface of a cubic crystal that is 1.0 cm in length is
 - (a) 9.17×10^{23}
 - (b) 9.17×10^{22}
 - (c) $2 \times 9.17 \times 10^{23}$
 - (d) $2 \times 9.17 \times 10^{22}$
9. The radii of Na^+ and Cl^- ions are 95 pm and 181 pm respectively. The edge length of NaCl unit cell is
 - (a) 276 pm
 - (b) 138 pm
 - (c) 552 pm
 - (d) 415 pm
10. An alloy of copper, silver and gold is found to have cubic lattice in which Cu atoms constitute *ccp*. If Ag atoms are located at the edge centres and Au atom is present at body centre, the alloy will have the formula
 - (a) CuAgAu
 - (b) $\text{Cu}_4\text{Ag}_4\text{Au}$
 - (c) $\text{Cu}_4\text{Ag}_3\text{Au}$
 - (d) $\text{Cu}_4\text{Ag}_6\text{Au}$
11. A solid is made of two elements X and Z. The atoms Z are in *ccp* arrangement while the atoms X occupy all the tetrahedral sites. What is the formula of the compound?
 - (a) XZ
 - (b) XZ_2
 - (c) X_2Z
 - (d) X_2Z_3
12. The pyknometric density of sodium chloride crystal is $2.165 \times 10^3 \text{ kg m}^{-3}$ while its X-ray density is $2.178 \times 10^3 \text{ kg m}^{-3}$. The fraction of unoccupied sites in sodium chloride crystal is
 - (a) 5.96×10^{-3}
 - (b) 5.96×10^4
 - (c) 5.96×10^{-2}
 - (d) 5.96×10^{-1}
13. The edge length of unit cell of a metal having molecular weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm.
 - (a) 217 pm
 - (b) 210 pm
 - (c) 220 pm
 - (d) 205 pm
14. The number of atoms in 100 g of an *fcc* crystal with density, $d = 10 \text{ g/cm}^3$ and cell edge equal to 100 pm, is equal to
 - (a) 1×10^{25}
 - (b) 2×10^{25}
 - (c) 3×10^{25}
 - (d) 4×10^{25}
15. The cubic unit cell of a metal (molar mass = 63.55 g mol^{-1}) has an edge length of 362 pm. Its density is 8.92 g cm^{-3} . The type of unit cell is
 - (a) primitive
 - (b) face centered
 - (c) body centered
 - (d) end centered
16. In A^+B^- ionic compound, radii of A^+ and B^- ions are 180 pm and 187 pm respectively. The crystal structure of this compound will be
 - (a) NaCl type
 - (b) CsCl type
 - (c) ZnS type
 - (d) similar to diamond
17. Which set of following characteristics for ZnS crystal is correct?
 - (a) Coordination number (4 : 4); *ccp*; Zn^{2+} ion in the alternate tetrahedral voids
 - (b) Coordination number (6 : 6); *hcp*; Zn^{2+} ion in all tetrahedral voids.
 - (c) Coordination number (6 : 4); *hcp*; Zn^{2+} ion in all octahedral voids
 - (d) Coordination number (4 : 4); *ccp*; Zn^{2+} ion in all tetrahedral voids.
18. In a normal spinel type structure, the oxide ions are arranged in *ccp* whereas 1/8 tetrahedral holes are occupied by Zn^{2+} ions and 50% of octahedral holes are occupied by Fe^{3+} ions. The formula of the compound is
 - (a) $\text{Zn}_2\text{Fe}_2\text{O}_4$
 - (b) ZnFe_2O_3
 - (c) ZnFe_2O_4
 - (d) ZnFe_2O_2

19. A molecule A_2B (Mwt. = 166.4) occupies triclinic lattice with $a = 5 \text{ \AA}$, $b = 8 \text{ \AA}$, and $c = 4 \text{ \AA}$. If the density of AB_2 is 5.2 g cm^{-3} , the number of molecules present in one unit cell is
 (a) 2 (b) 3
 (c) 4 (d) 5
20. A mineral having the formula AB_2 crystallizes in ccp lattice with A atoms occupying the lattice points. Pick out the correct statements of the following
 (a) 100% occupancy of tetrahedral voids, C.N. of B = 4
 (b) 100% occupancy of octahedral voids, C.N. of B = 4
 (c) 50% occupancy of tetrahedral voids, C.N. of A = 4
 (d) 100% occupancy of octahedral voids, C.N. of A = 4
21. Schottky defect in crystals is observed when
 (a) an ion leaves its normal site and occupies an interstitial site
 (b) unequal number of cations and anions are missing from the lattice
 (c) density of the crystal increases
 (d) equal number of cations and anions are missing from the lattice
22. To get a n -type semiconductor, the impurity to be added to silicon should have which of the following number of valence electrons
 (a) 1 (b) 2
 (c) 3 (d) 5
23. Non stoichiometric defects are formed by
 (a) s -block elements
 (b) p -block elements
 (c) either s -block elements or d -block elements
 (d) only d -block elements.
24. If NaCl is doped with 10^{-4} mol % of $SrCl_2$, the concentration of cation vacancies will be ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)
 (a) $6.02 \times 10^{16} \text{ mol}^{-1}$
 (b) $6.02 \times 10^{17} \text{ mol}^{-1}$
 (c) $6.02 \times 10^{14} \text{ mol}^{-1}$
 (d) $6.02 \times 10^{15} \text{ mol}^{-1}$
25. In a solid lattice the cation has left a lattice site and is located at an interstitial position, the lattice defect is
 (a) Interstitial defect (b) Vacancy defect
 (c) Frenkel defect (d) Schottky defect

Numeric Value Answer

26. Potassium has a bcc structure with nearest neighbour distance $4.52 \text{ \AA}$. Its atomic weight is 39. Its density (in kg m^{-3}) will be
27. KCl crystallises in the same type of lattice as does NaCl. Given that $r_{Na^+}/r_{Cl^-} = 0.55$ and $r_{K^+}/r_{Cl^-} = 0.74$. Calculate the ratio of the edge length of the unit cell for KCl to that of NaCl.
28. A metal has a fcc lattice. The edge length of the unit cell is 404 pm . The density of the metal is 2.72 g cm^{-3} . The molar mass of the metal is : (N_A Avogadro's constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)
29. The edge length of unit cell of a metal having molecular weight 75 g/mol is $5 \text{ \AA}$ which crystallizes in cubic lattice. If the density is 2 g/cc and the radius of metal atom is 4.33 xpm . ($N_A = 6 \times 10^{23}$). Find the value of x .
30. In face centred cubic (fcc) crystal lattice, edge length is 400 pm . The diameter of greatest sphere is 29.29 d pm which can be fit into the interstitial void without distortion of lattice. Find the value of d .

ANSWER KEY

1	(b)	4	(d)	7	(b)	10	(c)	13	(a)	16	(b)	19	(b)	22	(d)	25	(c)	28	(27)
2	(d)	5	(a)	8	(b)	11	(c)	14	(d)	17	(a)	20	(a)	23	(d)	26	(910)	29	(5)
3	(d)	6	(b)	9	(c)	12	(a)	15	(b)	18	(c)	21	(d)	24	(b)	27	(1.123)	30	(4)

SOLUTIONS

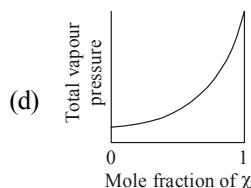
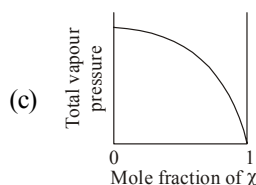
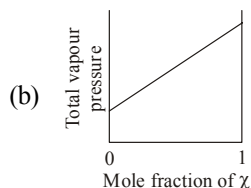
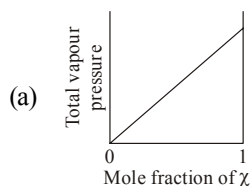
16

MCQs with One Correct Answer

1. If $\frac{N}{10}$ 50 mL H_2SO_4 , $\frac{N}{3}$ 30 mL HNO_3 , $\frac{N}{2}$ 10 mL HCl is mixed and solution is made to 1L. Then normality of resultant solution is
 - (a) $\frac{N}{20}$
 - (b) $\frac{N}{40}$
 - (c) $\frac{N}{50}$
 - (d) N
2. When a gas is bubbled through water at 298 K, a very dilute solution of the gas is obtained. Henry's law constant for the gas at 298 K is 100 kbar. If the gas exerts a partial pressure of 1 bar, the number of millimoles of the gas dissolved in one litre of water is
 - (a) 0.555
 - (b) 5.55
 - (c) 0.0555
 - (d) 55.5
3. Two liquids X and Y form an ideal solution. At 300 K, vapour pressure of the solution containing 1 mol of X and 3 mol of Y is 550 mm Hg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. Vapour pressure (in mm Hg) of X and Y in their pure states will be, respectively:
 - (a) 300 and 400
 - (b) 400 and 600
 - (c) 500 and 600
 - (d) 200 and 300
4. Equal masses of a solute are dissolved in equal amount of two solvents A and B , respective molecular masses being M_A and M_B . The relative lowering of vapour pressure of solution in solvent A is twice that of the solution in solvent B . If the solutions are dilute, M_A and M_B are related as
 - (a) $M_A = M_B$
 - (b) $2M_A = M_B$
 - (c) $M_A = 2M_B$
 - (d) $M_A = 4M_B$
5. When two liquids A and B are mixed they form an ideal solution. At certain temperature the V.P. of the solution, that contains 3 moles of A and 1 mole of B was found to be 550 mm of Hg. The vapour pressure of a solution containing 4 moles of A and 1 mole of B was found to be 560 mm of Hg. The vapour pressure of pure liquid A is
 - (a) 200mm
 - (b) 300mm
 - (c) 400mm
 - (d) 600mm

6. Two components A and B form an ideal solution. The mole fractions of A and B in ideal solution are X_A and X_B , while that of in vapour phase, these components have their mole fractions as Y_A and Y_B . Then, the slope and intercept of plot of $\frac{1}{Y_A}$ vs. $\frac{1}{X_A}$ will be:
- (a) $\frac{P_A^0}{P_B^0}, \frac{P_B^0 - P_A^0}{P_B^0}$ (b) $\frac{P_B^0}{P_A^0}, \frac{P_A^0 - P_B^0}{P_A^0}$
- (c) $\frac{P_B^0}{P_A^0}, \frac{P_B^0}{P_B^0 - P_A^0}$ (d) $P_A^0 - P_B^0, \frac{P_A^0}{P_B^0}$
7. The total vapour pressure of a 4 mole % solution of NH_3 in water at 293 K is 50.0 torr. The vapour pressure of pure water is 17.0 torr at this temperature. Applying Henry's and Raoult's laws, the total vapour pressure for a 5 mole % solution is
- (a) 58.25 torr (b) 33 torr
(c) 42.1 torr (d) 52.25 torr
8. Vapour pressure of benzene at 30°C is 121.8 mm Hg. When 15 g of a non-volatile solute is dissolved in 250 g of benzene its vapour pressure decreased to 120.2 mm Hg. The molecular weight of the solute is (Mo. wt. of solvent = 78)
- (a) 356.2 (b) 456.8
(c) 530.1 (d) 656.7
9. A solution of urea (mol. mass 56 g mol^{-1}) boils at 100.18°C at the atmospheric pressure. If K_f and K_b for water are 1.86 and $0.512 \text{ K kg mol}^{-1}$ respectively, the above solution will freeze at
- (a) 0.654°C (b) -0.654°C
(c) 6.54°C (d) -6.54°C
10. π_1, π_2, π_3 and π_4 atm are the osmotic pressures of 5% (mass/volume) solutions of urea, fructose, sucrose and KCl respectively at certain temperature. The correct order of their magnitudes is:
- (a) $\pi_1 > \pi_4 > \pi_2 > \pi_3$ (b) $\pi_1 > \pi_4 > \pi_2 > \pi_3$
(c) $\pi_4 > \pi_1 > \pi_2 > \pi_3$ (d) $\pi_4 > \pi_1 > \pi_3 > \pi_2$
11. 0.010M solution of an acid HA freezes at -0.0205°C . If K_f for water is $1.860 \text{ K kg mol}^{-1}$, the ionization constant of the conjugate base of the acid will be (assume $0.010 \text{ M} = 0.010 \text{ m}$)
- (a) 1.1×10^{-4} (b) 1.1×10^{-3}
(c) 9.0×10^{-11} (d) 9.0×10^{-12}
12. $\Delta T_f/K_f$ has the same value of 1 mol kg^{-1} for 8% AB_2 and 10% A_2B by mass of solvent, both AB_2 and A_2B being non-electrolytes. Atomic masses of A and B will be respectively
- (a) 20, 40 (b) 20, 50
(c) 40, 20 (d) 50, 40
13. The vapour pressure of a solvent decreases by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of the solute in the solution is 0.2. What should be the mole fraction of the solvent if the decrease in the vapour pressure is to be 20 mm of Hg?
- (a) 0.8 (b) 0.6
(c) 0.4 (d) 0.2
14. Which of the following pairs of solution are isotonic at the same temperature?
- (a) 0.1 M $\text{Ca}(\text{NO}_3)_2$ and 0.1 M Na_2SO_4
(b) 0.1 M NaCl and 0.1 M Na_2SO_4
(c) 0.1 M urea and 0.1 M MgCl_2
(d) 0.2 M urea and 0.1 M NaCl
15. 23.5 g of phenol was dissolved in 500 g of a solvent having depression constant K_f of $12.0 \text{ K kg mol}^{-1}$. If the depression in freezing point of solution was found to be 3.60 K, the percent dimerization of phenol in the solvent is
- (a) 89 (b) 85
(c) 80 (d) 92
16. A certain non-volatile electrolyte contains 40% carbon, 6.7% hydrogen and 53.3% oxygen. An aqueous solution containing 5% by mass of the solute boils at 100.15°C . The molecular formula of the compound is ($K_b = 0.51^\circ\text{C/m}$)
- (a) HCHO (b) CH_3OH
(c) $\text{C}_2\text{H}_5\text{OH}$ (d) $\text{C}_6\text{H}_{12}\text{O}_6$
17. A storage battery contains a solution of H_2SO_4 38% by weight. At this concentration, van't Hoff factor is 2.50. The temperature of which the battery contents freeze is ($K_f = 1.86 \text{ K kg mol}^{-1}$)
- (a) 225.45 (b) 235.85
(c) 249.92 (d) 243.92

18. An industrial waste water is found to contain 8.2% Na_3PO_4 and 12% MgSO_4 by mass in solution. If % ionisation of Na_3PO_4 and MgSO_4 are 50 and 60 respectively then its normal boiling point is [$K_b(\text{H}_2\text{O}) = 0.50 \text{ K kg mol}^{-1}$]:
- (a) 102.3°C (b) 103.35°C
 (c) 101.78°C (d) None of these
19. The plot of total vapour pressure as a function of mole fraction of the components of an ideal solution formed by mixing liquids X and Y is



20. The values of the Henry's law constant of Ar, CO_2 , CH_4 , and O_2 in water at 25°C are 40.30, 1.67, 0.41 and 34.86 kbar, respectively. The order of their solubility in water at the same temperature and pressure is
- (a) $\text{Ar} > \text{O}_2 > \text{CO}_2 > \text{CH}_4$
 (b) $\text{CH}_4 > \text{CO}_2 > \text{Ar} > \text{O}_2$
 (c) $\text{CH}_4 > \text{CO}_2 > \text{O}_2 > \text{Ar}$
 (d) $\text{Ar} > \text{CH}_4 > \text{O}_2 > \text{CO}_2$

Numeric Value Answer

21. Two liquids A and B are miscible in all proportions and form ideal solution. At 350 K the vapour pressure of pure A is 24.0 kPa and that of pure B is 12.0 kPa . A mixture of 60% (by mole) of A and 40% of B is distilled at this temperature. A small amount of distillate is collected and redistilled at the same temperature. The mole percent of A in the second distillate at initial stage is _____.
22. Two beaker A and B present in a closed vessel. Beaker A contains 152.4 g aqueous solution of urea, containing 12 g of urea. beaker B contains 196.2 g glucose solution, containing 18 g of glucose. Both solutions allowed to attain the equilibrium. The mass % of glucose in its solution at equilibrium is _____.
23. A solution containing 10 g per dm^3 of urea (molecular mass = 60 g mol^{-1}) is isotonic with a 5% solution of a non-volatile solute. The molecular mass of this non-volatile solute is _____.
24. The freezing point of $0.08 \text{ molal NaHSO}_4$ is -0.345°C . If assume 100% ionisation of NaHSO_4 , calculate the percentage of HSO_4^- ions that transfers a proton to water. (K_f for $\text{H}_2\text{O} = 1.86 \text{ K molality}^{-1}$).
25. 2 g of benzoic acid dissolved in 25 g of C_6H_6 shows a depression in freezing point equal to 1.62 K . Molal depression constant of C_6H_6 is $4.9 \text{ K mol}^{-1} \text{ kg}$. The percentage association of acid if it forms double molecule in solution is _____.
26. The amount of urea to be dissolved in 500 c.c. of water ($K = 1.86^\circ\text{C mol}^{-1}$) to produce a depression of 0.186°C in the freezing point is _____.
27. 0.400 g of an acid HA (mol. mass = 80) was dissolved in 100 g of water. The solution showed a depression of freezing point of 0.12 K . The dissociation constant (in multiple of 10^{-3}) of the acid at about 0°C , given $K_f(\text{water}) = 1.86 \text{ K Kg mol}^{-1}$ is (Assume molarity of solution $\approx$ molality)

28. An element X (Atomic mass = 25) exists as X_4 is benzene. 51g of saturated solution of X in benzene was added to 50.0 g of pure benzene. The resulting solution showed a depression of freezing point of 0.55 K. The solubility of X per 100 g of benzene is (K_f for benzene = 5.5 K kg mol⁻¹)
29. At 10°C, the osmotic pressure of urea solution is 500 mm. The solution is diluted and the temperature is raised to 25°C, when the osmotic pressure is found to be 105.3 mm. Determine extent of dilution.
30. A solution containing 28 g phosphorus in 315 g CS₂ (b. pt. 46.3°C) boils at 47.98°C. K_b for CS₂ is 2.34 K mol⁻¹ kg. _____ are the no. of atom of phosphorus present in its molecular formula.

ANSWER KEY

1	(c)	4	(c)	7	(a)	10	(c)	13	(b)	16	(d)	19	(b)	22	(14.49)	25	(99.2)	28	(2)
2	(a)	5	(d)	8	(a)	11	(c)	14	(a)	17	(d)	20	(c)	23	(300)	26	(3)	29	(5)
3	(b)	6	(b)	9	(b)	12	(c)	15	(c)	18	(c)	21	(0.857)	24	(31.9)	27	(6)	30	(4)

ELECTROCHEMISTRY

17

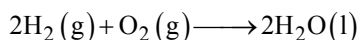
MCQs with One Correct Answer

- Small quantities of solutions of compounds TX, TY and TZ are put into separate test tubes containing X, Y and Z solution. TX does not react with any of these. TY reacts with both X and Z. TZ reacts with X. The decreasing order of state of oxidation of the anions X^- , Y^- , Z^- is
 (a) Y^-, Z^-, X^- (b) Z^-, X^-, Y^-
 (c) Y^-, X^-, Z^- (d) X^-, Z^-, Y^-
- Following cell has EMF 0.7995V.
 $Pt | H_2 (1 \text{ atm}) | HNO_3 (1M) || AgNO_3 (1M) | Ag$
 If we add enough KCl to the Ag cell so that the final Cl^- is 1M. Now the measured emf of the cell is 0.222V. The K_{sp} of AgCl would be –
 (a) $1 \times 10^{-9.8}$ (b) $1 \times 10^{-19.6}$
 (c) 2×10^{-10} (d) 2.64×10^{-14}
- $Sn^{4+} + 2e^- \longrightarrow Sn^{2+}$ $E^\circ = 0.13 \text{ V}$
 $Br_2 + 2e^- \longrightarrow 2Br^-$ $E^\circ = 1.08 \text{ V}$
 Calculate K_{eq} for the cell formed by two electrodes
 (a) 10^{41} (b) 10^{32}
 (c) 10^{-32} (d) 10^{-42}
- The standard reduction potential for Cu^{2+}/Cu is +0.34. The reduction potential at pH = 14 for the above couple. ($K_{sp} Cu(OH)_2 = 1 \times 10^{-19}$) is:
 (a) –0.22 V (b) +0.22 V
 (c) –0.44 V (d) +0.44 V
- The equilibrium constant for the disproportionation reaction
 $2Cu^+(aq) \longrightarrow Cu(s) + Cu^{2+}(aq)$
 at 25°C ($E^\circ Cu^+ / Cu = 0.52 \text{ V}$, $E^\circ Cu^{2+} / Cu = 0.16 \text{ V}$) is
 (a) 6×10^4 (b) 6×10^6
 (c) 1.2×10^6 (d) 1.2×10^{-6}
- The electrode potentials for
 $Cu^{2+}(aq) + e^- \longrightarrow Cu^+(aq)$
 and $Cu^+(aq) + e^- \longrightarrow Cu(s)$
 are +0.15 V and +0.50, respectively. The value of $E^\circ_{Cu^{2+}/Cu}$ will be :
 (a) 0.500 V (b) 0.325 V
 (c) 0.650 V (d) 0.150 V
- Calculate ΔG° for the reaction :
 $Cu^{2+}(aq) + Fe(s) \rightleftharpoons Fe^{2+}(aq) + Cu(s)$
 Given that : $E^\circ_{Cu^{2+}/Cu} = +0.34 \text{ V}$, $E^\circ_{Fe^{2+}/Fe} = -0.44 \text{ V}$
 (a) 180.55 kJ (b) 140.35 kJ
 (c) –130.15 kJ (d) –150.54 kJ

8. The e.m.f. of the cell $\text{Zn} | \text{Zn}^{2+} (0.01\text{M}) || \text{Fe}^{2+} (0.001\text{M}) | \text{Fe}$ at 298 K is 0.2905 then the value of equilibrium for the cell reaction is

$$\begin{array}{ll} \text{(a)} \quad e^{\frac{0.32}{0.0295}} & \text{(b)} \quad 10^{\frac{0.32}{0.0295}} \\ \text{(c)} \quad 10^{\frac{0.26}{0.0295}} & \text{(d)} \quad 10^{\frac{0.32}{0.0591}} \end{array}$$

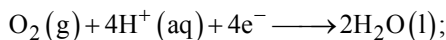
9. For a cell reaction



$$\Delta_r S_{298}^\circ = -0.32 \text{ kJ/K}.$$

What is the value of $\Delta_r S_{298}^\circ (\text{H}_2\text{O}, \text{l})$?

Given :



$$E^\circ = 1.23 \text{ V}$$

- (a) -285.07 kJ/mol (b) -570.14 kJ/mol
 (c) 285.07 kJ/mol (d) None of these
10. If 0.01 M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of 0.4 cm^{-1} , then its molar conductance in $\text{ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ is
 (a) 10^2 (b) 10^4
 (c) 10 (d) 10^3
11. When electric current is passed through acidified water, 112 mL of hydrogen gas at STP collected at the cathode in 965 seconds. The current passed in amperes is
 (a) 1.0 (b) 0.5
 (c) 0.1 (d) 2.0
12. $\Lambda_{\text{ClCH}_2\text{COONa}} = 224 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$,
 $\Lambda_{\text{NaCl}} = 38.2 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$,
 $\Lambda_{\text{HCl}} = 203 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$,
 What is the value of $\Lambda_{\text{ClCH}_2\text{COOH}}$
 (a) $288.5 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$
 (b) $289.5 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$
 (c) $388.8 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$
 (d) $59.5 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$
13. Equivalent conductance at infinite dilution, λ° of NH_4Cl , NaOH and NaCl are 128.0, 217.8 and $109.3 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$ respectively. The

equivalent conductance of $0.01 \text{ N NH}_4\text{OH}$ is $9.30 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$, then the degree of ionization of NH_4OH at this temperature would be

- (a) 0.04 (b) 0.1
 (c) 0.39 (d) 0.62

14. Dissociation constant of a weak acid (HA) in terms of Λ_m^∞ and Λ_m is:

$$\text{(a)} \quad K_a = \frac{c\Lambda_m^\infty}{(\Lambda_m - \Lambda^\infty)}$$

$$\text{(b)} \quad K_a = \frac{c\Lambda_m^2}{\Lambda_m^\infty (\Lambda_m^\infty - \Lambda_m)}$$

$$\text{(c)} \quad K_a = \frac{c(\Lambda_m^\infty)^2}{\Lambda_m^\infty (\Lambda_m^\infty - \Lambda_m)}$$

- (d) None of these

15. Given the ionic equivalent conductivities for the following ions:

$$\lambda_{\text{eq}}^\circ \text{K}^+ = 73.5 \text{ cm}^2 \text{ ohm}^{-1} \text{ eq}^{-1}$$

$$\lambda_{\text{eq}}^\circ \text{Al}^{3+} = 149 \text{ cm}^2 \text{ ohm}^{-1} \text{ eq}^{-1}$$

$$\lambda_{\text{eq}}^\circ \text{SO}_4^{2-} = 85.8 \text{ cm}^2 \text{ ohm}^{-1} \text{ eq}^{-1}$$

The $\Lambda_{\text{eq}}^\circ$ for potash alum ($\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$) is

- (a) 215.92 (b) 348.3
 (c) 368.2 (d) 108.52

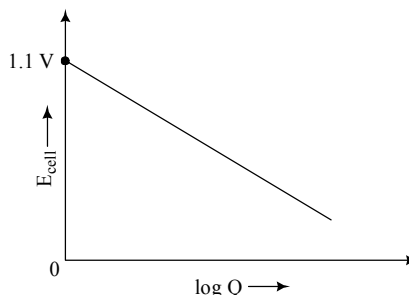
16. On passing a current of 1.0 ampere for 16 min and 5 sec through one litre solution of CuCl_2 , all copper of the solution was deposited at cathode. The strength of CuCl_2 solution was (Molar mass of $\text{Cu} = 63.5$; Faraday constant = 96500 Cmol^{-1})

- (a) 0.01 N (b) 0.01 M
 (c) 0.02 M (d) 0.2 N

17. Given the following molar conductivities at 25°C : HCl , $426 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$; NaCl , $126 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$; NaC (sodium crotonate), $83 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of crotonic acid, if the conductivity of a 0.001 M crotonic acid solution is $3.83 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$?

- (a) 10^{-5} (b) 1.11×10^{-5}
 (c) 1.11×10^{-4} (d) 0.01

18. A lead storage battery containing 5.0 L of (1N) H_2SO_4 solution is operated for 9.65×10^5 s with a steady current of 100 mA. Assuming volume of the solution remaining constant, normality of H_2SO_4 will
 (a) remain unchanged (b) increases by 0.20
 (c) increase by unity (d) decrease by 0.40
19. In an electrolysis experiment current was passed for 5 hours through two cells connected in series. The first cell contains a solution of gold and the second contains copper sulphate solution. 9.85 g of gold was deposited in the first cell. If the oxidation number of gold is +3, the amount of copper deposited on the cathode of the second cell and magnitude of the current in amperes is.
 (1 faraday = 96,500 coulombs)
 (a) 4.95 g, 0.8 A (b) 5.5 g, 0.9 A
 (c) 4.76 g, 0.8 A (d) 5.85 g, 0.5 A
20. For the electrochemical cell shown below
 $\text{Pt} | \text{H}_2 (p = 1 \text{ atm}) | \text{H}^+ (\text{aq.}, x \text{ M}) |$
 $| \text{Cu}^{2+} (\text{aq.}, 1.0 \text{ M}) | \text{Cu}(\text{s})$
 The potential is 0.49 V at 298 K. The pH of the solution is closest to:
 [Given, standard reduction potential, E° for Cu^{2+}/Cu is 0.34 V.
 Gas constant, R is $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
 Faraday constant, F is $9.65 \times 10^4 \text{ J V}^{-1} \text{ mol}^{-1}$]
 (a) 1.2 (b) 8.3
 (c) 2.5 (d) 8.2
21. What is the potential of an electrode (in V) which originally contained 0.1 M NO_3^- and 0.4 M H^+ and which has been treated by 80% of the cadmium necessary to reduce all the NO_3^- to $\text{NO}(\text{g})$ at 1 bar?
 Given : $\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \longrightarrow \text{NO} + 2\text{H}_2\text{O};$
 $E^\circ = 0.96 \text{ V}; \log 2 = 0.3$
22. On passing current through two cells, connected in series containing solution of AgNO_3 and CuSO_4 , 0.18 g of Ag is deposited. The amount of the Cu (in g) deposited is:
23. $E^\circ_{\text{cell}} = 0.74 \text{ V}$ for the cell,
 $\text{Cr} | \text{Cr}^{3+} (1 \text{ M}) || \text{H}^+ (1 \text{ M}) | \text{Pt}(\text{H}_2) (1 \text{ bar})$ and
 $E^\circ_{\text{cell}} = 0.80 \text{ V}$ for the cell (1 bar) $\text{Pt}(\text{H}_2) | \text{H}^+ (1 \text{ M}) || \text{Ag}^+ (1 \text{ M}) | \text{Ag}$
 What is the value of cell EMF for the cell
 $\text{Ag} | \text{Ag}^+ (0.1 \text{ M}) || \text{Cr}^{3+} (0.1 \text{ M}) | \text{Cr} ?$
24. For Daniell's cell, the reaction quotient is Q . A variation of E_{cell} with $\log Q$ has been plotted. At start, the molar concentration of CuSO_4 and ZnSO_4 are equal. When concentration of Zn^{2+} increases to 1.5 mol/L, the cell EMF becomes 1.07 V. What is the value of $[\text{Cu}^{2+}]$ at this point. Take $2.303 RT/F = 0.06$



Numeric Value Answer

21. What is the standard electrode potential for the reduction of HClO ?
 $\text{HClO}(\text{aq}) + \text{H}^+(\text{aq}) + 2\text{e}^- \longrightarrow$
 $\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 Given : $\text{Cr}^{2+}(\text{aq}) \longrightarrow \text{Cr}^{3+}(\text{aq}) + \text{e}^-$,
 $E^\circ = 0.41 \text{ V}$
 $\text{HClO}(\text{aq}) + \text{H}^+(\text{aq}) + 2\text{Cr}^{2+}(\text{aq}) \longrightarrow$
 $2\text{Cr}^{3+}(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}), E^\circ = 1.80$
26. For a hydrazine fuel cell, following data are given at 298 K:
 $E^\circ_{\text{N}_2\text{H}_4/\text{N}_2} = -0.9 \text{ V}$, $E^\circ_{\frac{1}{2}\text{O}_2/\text{H}_2\text{O}} = +1.23 \text{ V}$
 Assuming 60% efficiency, what is the maximum amount of energy (in KJ) that can be obtained?
27. During discharging of lead storage battery, the density of 38% H_2SO_4 (w/w) fell from 1.274 g/mL to 1.2 g/mL of 19.6 % (w/w) H_2SO_4 . What is the number of faradays that have been exchanged during the discharging process? (vol. of H_2SO_4 in lead storage battery = 2.36 L)

28. The photoelectric current from Na (work function, $w_0 = 2.3 \text{ eV}$) is stopped by the output voltage of the cell $\text{Pt(s)}|\text{H}_2(\text{g}, 1 \text{ bar})|\text{HCl(aq., pH=1)}|\text{AgCl(s)}|\text{Ag(s)}$. The pH of aq. HCl required to stop the photoelectric current from K ($w_0 = 2.25 \text{ eV}$), all other conditions remaining the same, is $\times 10^{-2}$ (to the nearest integer).
Given,
 $2.303 \frac{RT}{F} = 0.06 \text{ V}$; $E_{\text{AgCl}|\text{Ag}|\text{Cl}^-}^0 = 0.22 \text{ V}$
29. For an electrochemical cell $\text{Sn(s)}|\text{Sn}^{2+}(\text{aq}, 1 \text{ M})||\text{Pb}^{2+}(\text{aq}, 1 \text{ M})|\text{Pb(s)}$ the ratio $\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$ when this cell attains equilibrium is _____.
30. When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of *p*-amino-phenol (in g) produced is _____.

ANSWER KEY

1	(a)	4	(a)	7	(d)	10	(d)	13	(a)	16	(a)	19	(c)	22	(0.84)	25	(0.15)	28	(142)
2	(a)	5	(c)	8	(b)	11	(a)	14	(b)	17	(b)	20	(c)	23	(0.0529)	26	(76.43)	29	(2.15)
3	(b)	6	(b)	9	(a)	12	(c)	15	(a)	18	(d)	21	(1.39)	24	(1.5)	27	(6)	30	(9.81)

CHEMICAL KINETICS

18

MCQs with One Correct Answer

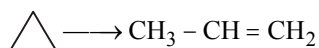
1. The rate of the reaction $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ can be written in three ways :

$$\frac{-d[\text{N}_2\text{O}_5]}{dt} = k [\text{N}_2\text{O}_5]; \quad \frac{d[\text{NO}_2]}{dt} = k' [\text{N}_2\text{O}_5]$$

$$\frac{d[\text{O}_2]}{dt} = k'' [\text{N}_2\text{O}_5]$$

The relationship between k and k' and between k and k'' are:

- (a) $k' = 2k$; $k' = k$ (b) $k' = 2k$; $k'' = k/2$
 (c) $k' = 2k$; $k'' = 2k$ (d) $k' = k$; $k'' = k$
2. A reaction proceeds by first order, 75% of this reaction was completed in 32 min. The time required for 50% completion is
- (a) 8 min (b) 16 min
 (c) 20 min (d) 24 min
3. Cyclopropane rearranges to form propene



This follows first order kinetics. The rate constant is $2.714 \times 10^{-3} \text{ sec}^{-1}$. The initial concentration of cyclopropane is 0.29 M. What will be the concentration of cyclopropane after 100 sec?

- (a) 0.035 M (b) 0.22 M
 (c) 0.145 M (d) 0.0018 M

4. The time taken for 90% of a first order reaction to complete is approximately

- (a) 1.1 times that of half-life
 (b) 2.2 times that of half-life
 (c) 3.3 times that of half-life
 (d) 4.4 times that of half-life

5. The initial rates of reaction

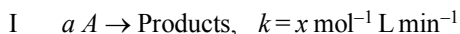
$3\text{A} + 2\text{B} + \text{C} \longrightarrow \text{Products}$, at different initial concentrations are given below:

Initial rate, Ms^{-1}	$[\text{A}]_0, \text{M}$	$[\text{B}]_0, \text{M}$	$[\text{C}]_0, \text{M}$
5.0×10^{-3}	0.010	0.005	0.010
5.0×10^{-3}	0.010	0.005	0.015
1.0×10^{-2}	0.010	0.010	0.010
1.25×10^{-3}	0.005	0.005	0.010

The order with respect to the reactants, A, B and C are respectively

- (a) 3, 2, 0 (b) 3, 2, 1
 (c) 2, 2, 0 (d) 2, 1, 0

6. Consider the two hypothetical reactions given below :



The half-lives of both the reactions are the same, equal to 1 hr when molar concentration of the reactant is 1.0 M in each case. If these reactions

are started at the same time taking 1M of the reactant in each case, the ratio $[A]/[B]$ after 3 hr will be :

- (a) 0.5 (b) 4
(c) 1 (d) 2

7. For a homogeneous gaseous reaction $A \longrightarrow 3B$, if pressure after time t was P_t and after completion of reaction, pressure was P_∞ then select correct relation

(a) $k = \frac{1}{t} \ln \left(\frac{P_\infty}{3(P_\infty - P_t)} \right)$

(b) $k = \frac{1}{t} \ln \left(\frac{2P_\infty}{(P_\infty - P_t)} \right)$

(c) $k = \frac{1}{t} \ln \left(\frac{3P_\infty}{2P_\infty - P_t} \right)$

(d) $k = \frac{1}{t} \ln \left(\frac{2P_\infty}{3(P_\infty - P_t)} \right)$

8. For a first order homogeneous gaseous reaction, $A \longrightarrow 2B + C$ then initial pressure was P_i while total pressure after time ' t ' was P_t . The right expression for the rate constants k in terms of P_i , P_t and t is :

(a) $k = \frac{2.303}{t} \log \left(\frac{2P_i}{3P_i - P_t} \right)$

(b) $k = \frac{2.303}{t} \log \left(\frac{2P_i}{2P_t - P_i} \right)$

(c) $k = \frac{2.303}{t} \log \left(\frac{P_i}{P_i - P_t} \right)$

(d) None of these

9. The reaction $A(g) \longrightarrow B(g) + 2C(g)$ is a first order reaction with rate constant $2.772 \times 10^{-3} \text{ s}^{-1}$. Starting with 0.1 mole of A in 2 litre vessel, find the concentration of A after 250 sec when the reaction is allowed to take place at constant pressure and at 300 K?

- (a) 0.0125 M (b) 0.025 M
(c) 0.05 M (d) None of these

10. $A(aq) \longrightarrow B(aq) + C(aq)$ is a first order reaction.

Time t ∞

mole of reagent n_1 n_2

Reaction progress is measured with the help of titration of reagent 'R'. If all A, B and C react with reagent and have 'n' factors [n factor; eq.

$\text{mass} = \frac{\text{mol. mass}}{n}$] in the ratio of 1 : 2 : 3 with the reagent, the k in terms of t , n_1 and n_2 is :

(a) $k = \frac{1}{t} \ln \left(\frac{n_2}{n_2 - n_1} \right)$

(b) $k = \frac{1}{t} \ln \left(\frac{2n_2}{n_2 - n_1} \right)$

(c) $k = \frac{1}{t} \ln \left(\frac{4n_2}{n_2 - n_1} \right)$

(d) $k = \frac{1}{t} \ln \left(\frac{4n_2}{5(n_2 - n_1)} \right)$

11. The gaseous decomposition reaction, $A(g) \rightarrow 2B(g) + C(g)$ is observed to first order over the excess of liquid water at 25°C . It is found that after 10 minutes the total pressure of system is 188 torr and after very long time it is 388 torr. The rate constant of the reaction (in hr^{-1}) is : [Given : vapour pressure of H_2O at 25° is 28 torr ($\ln 2 = 0.7$, $\ln 3 = 1.1$, $\ln 10 = 2.3$)]

- (a) 0.02 (b) 1.2
(c) 0.2 (d) 0.5

12. The decomposition of N_2O_5 according to the equation :



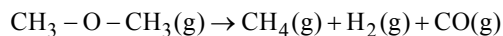
is a first order reaction. After 30 min. from the start of the decomposition in a closed vessel, the total pressure developed is found to be 284.5 mm of Hg and on complete decomposition, the total pressure is 584.5 mm of Hg. the rate constant of the reaction is:

- (a) 5.2 min^{-1} (b) $520 \times 10^{-3} \text{ min}^{-1}$
(c) $5.2 \times 10^{-3} \text{ min}^{-1}$ (d) $0.52 \times 10^{-3} \text{ min}^{-1}$

13. The activation energy for a simple chemical reaction $A \rightarrow B$ is E_a in forward direction. The activation energy for reverse reaction
- is always double of E_a
 - is negative of E_a
 - is always less than E_a
 - can be less than or more than E_a
14. The rate constant, the activation energy and the arrhenius parameter of a chemical reaction at 25°C are $3.0 \times 10^{-4}\text{s}^{-1}$, 104.4 kJ mol^{-1} and $6.0 \times 10^{14}\text{s}^{-1}$ respectively. The value of the rate constant as $T \rightarrow \infty$ is
- $2.0 \times 10^{18}\text{s}^{-1}$
 - $6.0 \times 10^{14}\text{s}^{-1}$
 - Infinity
 - $3.6 \times 10^{30}\text{s}^{-1}$
15. The activation energy for a reaction is 9.0 kcal/mol . The increase in the rate constant when its temperature is increased from 298K to 308K is
- 63%
 - 50%
 - 100%
 - 10%
16. A reaction rate constant is given by
- $$k = 1.2 \times 10^{14} e^{-25000/RT} \text{sec}^{-1}$$
- It means
- $\log k$ versus $\log T$ will give a straight line with a slope as -25000
 - $\log k$ versus T will give a straight line with slope as 25000
 - $\log k$ versus $\log 1/T$ will give a straight line with slope as -25000
 - $\log k$ versus $1/T$ will give a straight line
17. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 20°C to 35°C ? ($R = 8.314\text{ J mol}^{-1}\text{K}^{-1}$)
- 269 kJ mol^{-1}
 - 34.5 kJ mol^{-1}
 - 15.1 kJ mol^{-1}
 - 342 kJ mol^{-1}
18. The rate of decomposition for methyl nitrite and ethyl nitrite can be given in terms of rate constant (in sec^{-1}) k_1 and k_2 respectively. The energy of activations for the two reactions are $152.30\text{ kJ mol}^{-1}$ and 157.7 kJ mol^{-1} as well as frequency factors are 10^{13} and 10^{14} respectively for the decomposition of methyl and ethyl nitrite. The temperature at which rate constant same for the two reactions will be:
- 285 K
 - 290 K
 - 282 K
 - 275 K
19. A first order reaction $A \rightarrow B$ requires activation energy of 70 kJ mol^{-1} . When a 20% solution of A was kept at 25°C for 20 minute, 25% decomposition took place. What will be the per cent decomposition in the same time in a 30% solution maintained at 40°C ? Assume that activation energy remains constant in this range of temperature.
- 67.21
 - 65.21
 - 70.50
 - 64.25
20. The activity of a radio nuclide (X^{100}) is 6.023 curie at a time t . If the disintegration constant is $3.7 \times 10^4\text{ sec}^{-1}$, the mass of X after t seconds is—
- 10^{-3}g
 - 10^{-6}g
 - 10^{-14}g
 - 10^{-15}g

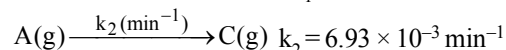
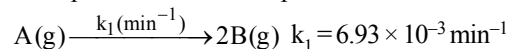
Numeric Value Answer

21. The gas phase decomposition of dimethyl ether follows first order kinetics.



The reaction is carried out in a constant volume container at 500°C and has a half life of 14.5 minutes. Initially, only dimethyl ether is present at a pressure of 0.40 atmosphere. What is the total pressure (in atm) of the system after 12 minutes? Assume ideal gas behaviour.

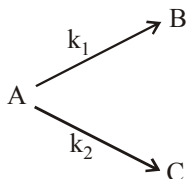
22. A compound A dissociates by two parallel first order paths at certain temperature



The reaction is started with 1 mole of pure 'A' in 1 litre closed container with initial pressure 2 atm. What is the pressure (in atm) developed in container after 50 minutes from start of experiment?

23. The time required for 10% completion of a first order reaction at 298 K is equal to that required for its 25% completion at 308 K . If the pre-exponential factor for the reaction is $3.56 \times 10^9\text{ s}^{-1}$, the rate constant (in kcal mol^{-1}) at 318 K is:

24. For first order parallel reactions k_1 and k_2 are 4 and 2 min^{-1} respectively at 300 K. If the activation energies for the formation of B and C are respectively 30,000 and 38,314 joule/mol respectively, the temperature (in K) at which B and C will be obtained in equimolar ratio is :



25. The rate constant of a reaction increases by 7% when its temperature is raised from 300 K to 310 K, while its equilibrium constant increases by 3%. The ratio of activation energy of the forward and reverse reactions is:
26. In the age of the earth is 4.5 billion years and the half-life of ^{40}K is 1.26 billion years, what percent of the earth's original amount of ^{40}K remains today ?
27. A reaction $A + B \rightarrow \text{Products}$ is first order w.r.t. each reactant, k being $5.052 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$. If start is made with initial concentration of A to be 0.1 M acid that of $B = 5.0 \text{ M}$, calculate the concentration of A (in mM) remaining after 100 s.

28. Carbon monoxide reacts with O_2 to form CO_2 : $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$. Informations about this reaction are given in the table below.

[CO] mol/L	[O ₂] mol/L	Rate of reaction (mol/L min)
0.02	0.02	4×10^{-5}
0.04	0.02	1.6×10^{-4}
0.02	0.04	8×10^{-5}

What is the value for the rate constant for the reaction in properly related unit ?

29. Two I order reactions having same reactant concentrations proceed at 25°C at the same rate. The temperature coefficient of the rate of the first reaction is 2 and that of second reaction is 3. Find the ratio of the rates of these reactions at 75°C .
30. For the reaction, $2\text{NO} + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2$, the value of $-\frac{dP}{dt}$ was found to be 1.5 Pa s^{-1} for a pressure of 359 Pa of NO and 0.25 Pa s^{-1} for a pressure of 152 Pa of NO. The pressure of H_2 being constant. If pressure of NO was kept constant, the value of $-\frac{dP}{dt}$ was found 1.60 Pa s^{-1} for a pressure of H_2 289 Pa and 0.79 Pa s^{-1} for a pressure of 147 Pa of H_2 . Calculate the order of reaction.

ANSWER KEY

1	(b)	4	(c)	7	(d)	10	(d)	13	(d)	16	(c)	19	(a)	22	(2.5)	25	(1.8)	28	(5)
2	(b)	5	(d)	8	(a)	11	(b)	14	(b)	17	(b)	20	(d)	23	(18.39)	26	(8.42)	29	(8)
3	(b)	6	(d)	9	(a)	12	(c)	15	(a)	18	(c)	21	(0.75)	24	(378.74)	27	(8)	30	(3)

SURFACE CHEMISTRY

19

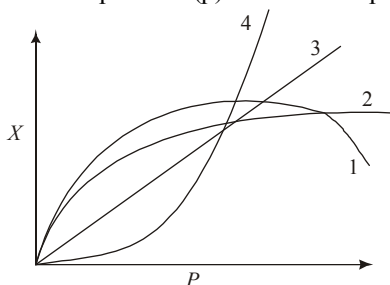
MCQs with One Correct Answer

- The adsorption of a gas on a solid surface varies with pressure of the gas in which of the following manner
 - fast $\rightarrow$ slow $\rightarrow$ independent of the pressure
 - slow $\rightarrow$ fast $\rightarrow$ independent of the pressure
 - independent of the pressure $\rightarrow$ fast $\rightarrow$ slow
 - independent of the pressure $\rightarrow$ slow $\rightarrow$ fast
- Consider the following statements about Langmuir isotherm.
 - The free gas and adsorbed gas are in dynamic equilibrium.
 - All adsorption sites are equivalent.
 - The initially adsorbed layer can act as a substrate for further adsorption
 - The ability of a molecule to get adsorbed at a given site is independent of the occupation of neighboring sites.The correct statements are
 - I, II, III and IV
 - I, II and IV
 - I, III and IV
 - I, II and III
- 1g of charcoal adsorbs 100 mL 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of the charcoal covered by each molecule of acetic acid. Surface area of charcoal $= 3.01 \times 10^2 \text{ m}^2/\text{g}$.
 - $0.5 \times 10^{-19} \text{ m}^2$
 - $4 \times 10^{-20} \text{ m}^2$
 - $0.5 \times 10^{-10} \text{ m}^2$
 - $5 \times 10^{-19} \text{ m}^2$
- 3.6 gram of oxygen is adsorbed on 1.2 g of metal powder. What volume of oxygen adsorbed per gram of the adsorbent at 1 atm and 273 K?
 - 0.19 L g^{-1}
 - 1 L g^{-1}
 - 2.1 L g^{-1}
 - None of these
- Which of the following is true in respect of chemical adsorption?
 - $\Delta H < 0, \Delta S > 0, \Delta G > 0$
 - $\Delta H < 0, \Delta S < 0, \Delta G < 0$
 - $\Delta H > 0, \Delta S > 0, \Delta G < 0$
 - $\Delta H > 0, \Delta S < 0, \Delta G > 0$
- In an experiment, 200 mL of 0.5 M oxalic acid is shaken with 10 g of activated charcoal and filtered. The concentration of the filtrate is reduced to 0.4 M. The amount of adsorption $\left(\frac{x}{m}\right)$ is
 - 0.9
 - 1.8
 - 0.18
 - 0.09

7. For a gas-solid adsorbent system, the adsorption isotherm $\frac{x}{m} = K p^{1/n}$ is applicable, where $n=5$ in the specified case. Point out the correct statement :

- $\frac{x}{m}$ Vs. p plot is a linear graph with slope equal to 5
- $\log \frac{x}{m}$ Vs. $\log p$ plot is a straight line with slope equal to 5
- $\log \frac{x}{m}$ Vs. $\log p$ is a straight line with slope equal to 0.2
- none of these

8. The curve that best describes the adsorption of a gas (X g) on 1.0 g of a solid substrate as a function of pressure (p) at a fixed temperature



is

- 1
 - 2
 - 3
 - 4
9. Which one of the following is an example of homogeneous catalysis ?
- Haber's process of synthesis of ammonia
 - Catalytic conversion of SO_2 to SO_3 in contact process
 - Catalytic hydrogenation of oils
 - Acid hydrolysis of methyl acetate
10. Which of the following statements about a catalyst is/are true ?
- A catalyst accelerates the reaction by bringing down the free energy of activation
 - A catalyst also takes part in the reaction mechanism

- A catalyst makes the reaction more feasible by making the ΔG° more negative
- A catalyst makes the equilibrium constant of the reaction more favourable for the forward reaction.

11. Flocculation value of BaCl_2 is much less than that of KCl for sol A and flocculation value of Na_2SO_4 is much less than that of NaBr for sol B. The correct statement among the following is

- Both the sols A and B are negatively charged.
- Sol A is positively charged and Sol B is negatively charged.
- Both the sols A and B are positively charged.
- Sol A is negatively charged and sol B is positively charged.

12. Freshly obtained SnO_2 is added to water containing a little HCl . The sol obtained would be due to preferential adsorption of ions

- positively charged, H^+
- negatively charged, SnO_3^{2-}
- positively charged, Sn^{4+}
- positively charged, Sn^{2+}

13. One desires to prepare a positively charged sol of silver iodide. This can be achieved by

- adding a little AgNO_3 solution to KI solution in slight excess
- adding a little KI solution to AgNO_3 solution in slight excess
- mixing equal volumes of equimolar solutions of AgNO_3 and KI
- None of these

14. 100 mL each of two sols of AgI , one obtained by adding AgNO_3 to slight excess of KI and another obtained by adding KI to slight excess of AgNO_3 , are mixed together. Then

- The two sols will stabilize each other
- The sol particles will acquire more electric charge
- The sols will coagulate each other mutually
- A true solution will be obtained

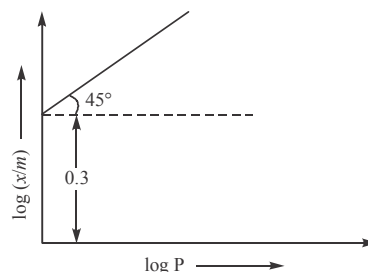
15. Under the influence of an electric field, the particles in a sol migrate towards cathode. The coagulation of the same sol is studied using NaCl, Na_2SO_4 and Na_3PO_4 solutions. Their coagulating values will be in the order
- $\text{NaCl} > \text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4$
 - $\text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4 > \text{NaCl}$
 - $\text{Na}_3\text{PO}_4 > \text{Na}_2\text{SO}_4 > \text{NaCl}$
 - $\text{Na}_2\text{SO}_4 > \text{NaCl} > \text{Na}_3\text{PO}_4$
16. To an emulsion of olive oil and water, stabilized by soap of alkali metals, a small amount of salts of bivalent or trivalent cations is added. Then
- stabilization of emulsion enhances
 - the emulsion breaks into the component liquids
 - reversal of phase to W/O type takes place
 - the emulsion turns into a true solution
17. The density of gold is 19 g/cm^3 . If $1.9 \times 10^{-4} \text{ g}$ of gold is dispersed in one litre of water to give a sol having spherical gold particles of radius 10 nm, then the number of gold particles per mm^3 of the sol will be :
- 1.9×10^{12}
 - 6.3×10^{14}
 - 6.3×10^{10}
 - 2.4×10^6
18. The volume of a colloidal particle, V_C as compared to the volume of a solute particle in a true solution V_S , could be
- $\frac{V_C}{V_S} \approx 10^3$
 - $\frac{V_C}{V_S} \approx 10^{-3}$
 - $\frac{V_C}{V_S} \approx 10^{23}$
 - $\frac{V_C}{V_S} \approx 1$
19. In an experiment, addition of 4.0 mL of 0.005M BaCl_2 to 16.0 mL of arsenious sulphide sol just causes the complete coagulation in 2 hrs. The flocculating value of the effective ion is
- Cl^- , 1.0
 - Cl^- , 2.0
 - Ba^{2+} , 1.0
 - Ba^{2+} , 0.5
20. The most appropriate method of making egg-albumin sol is :
- Break an egg carefully and transfer the transparent part of the content to 100 mL of 5% w/V saline solution and stir well.
 - Keep the egg in boiling water for 10 minutes. After removing the shell, transfer

the yellow part of the content to 100 mL of 5% w/V saline solution and homogenize with a mechanical shaker.

- Keep the egg in boiling water for 10 minutes. After removing the shell, transfer the white part of the content to 100 mL of 5% w/V saline solution and homogenize with a mechanical shaker
- Break an egg carefully and transfer only the yellow part of the content to 100 mL of 5% w/V saline solution and stir well.

Numeric Value Answer

21. In an adsorption experiment, a graph between $\log (x/m)$ versus $\log P$ was found to be linear with a slope of 45° . The intercept on the y axis was found to be 0.301. The amount of the gas adsorbed per gram of charcoal under a pressure of 3.0 atm is _____.
22. 10% sites of catalyst bed have adsorbed by H_2 . On heating H_2 gas is evolved from sites and collected at 0.03 atm and 300 K in a small vessel of 2.46 cm^3 .
No. of sites available is 5.4×10^{16} per cm^2 and surface area is 1000 cm^2 . Find out the no. of surface sites occupied per molecule of H_2 . (Given $N_A = 6 \times 10^{23}$)
23. Considering Freundlich adsorption isotherm in this plot, find the extent of adsorption at a pressure of 4 atm.



24. The density of gold is 19 g/cm^3 . $1.9 \times 10^{-3} \text{ g}$ of gold is dispersed in one litre of water to give a sol having spherical gold particles of radius 10.5 nm. The gold particles are occupying 50% of the available space. The number of gold particles per mm^3 is about 10^n . Find the value of n .

25. A cationic colloidal electrolyte forms micelle at 10^{-3} M concentration in water. Its 1 mm^3 solution contains 2×10^{13} micelle structures. What is the number of cations contained in one micelle? Take $N_A = 6 \times 10^{23}$.
26. The coagulation of 400 mL of a positive colloid took place when 1.46 g of HCl was added to it without changing the volume much. What is the flocculation value of HCl for the colloid?
27. The coagulation of 100 mL of a colloidal sol of gold is completely prevented by addition of 250 mg of a substance to it before adding 10 mL of 1% NaCl solution. What is the gold number of this substance?
28. A sample of 16 g charcoal was brought into contact with CH_4 gas contained in a vessel of 1 litre at 27°C . The pressure of gas was found to fall from 760 to 608 torr. The density of charcoal sample is 1.6 g/cm^3 . What is the volume of the CH_4 gas adsorbed per gram of the adsorbent at 608 torr and 27°C ?
29. One gram of activated carbon has a surface area of 1000 m^2 . Considering complete coverage as well as monomolecular adsorption, how much ammonia at 1 atm and 273 K would be adsorbed on the surface of $\frac{44}{7}$ g carbon if radius of ammonia molecules is 10^{-8} cm .
30. A detergent ($\text{C}_{12}\text{H}_{25}\text{SO}_4\text{Na}^+$) solution becomes a colloidal sol at a concentration of 10^{-3} M . On an average 10^{13} colloidal particles are present in 1 mm^3 . What is the average number of ions which are contained by one colloidal particle (micelle)?

ANSWER KEY

1	(a)	4	(c)	7	(c)	10	(a)	13	(b)	16	(c)	19	(c)	22	(3)	25	(30)	28	(16.25)
2	(b)	5	(b)	8	(b)	11	(b)	14	(c)	17	(d)	20	(a)	23	(8)	26	(100)	29	(7.46)
3	(d)	6	(c)	9	(d)	12	(c)	15	(a)	18	(a)	21	(6)	24	(7)	27	(25)	30	(60)

GENERAL PRINCIPLES AND PROCESSES OF ISOLATION OF ELEMENTS



MCQs with One Correct Answer

- Calcination is used in metallurgy for removal of
(a) moisture (b) water and CO_2
(c) CO_2 and H_2S (d) H_2O and H_2S
- Hydro-metallurgical process of extraction of metals is based on
(a) complex formation (b) hydrolysis
(c) dehydration (d) dehydrogenation
- The method of zone refining of metals is based on the principle of
(a) greater solubility of the impurities in the molten state than in the solid
(b) greater mobility of the pure metal than that of the impurity
(c) higher melting point of the impurities than that of the pure metal
(d) greater noble character of the solid metal than that of the impurities
- In froth flotation process many chemicals (frother, collector, activator, and depressant) are used. Which of the following is a frother:
(a) CuSO_4
(b) NaCN + alkali
(c) Pine oil
(d) Potassium xanthate.
- Thomas slag is
(a) calcium silicate
(b) calcium phosphate
(c) tricalcium phosphate and calcium silicate
(d) calcium ammonium phosphate
- Which of the following pairs of metals is purified by van Arkel method?
(a) Ga and In (b) Zr and Ti
(c) Ag and Au (d) Ni and Fe
- Calcination is the process in which
(a) ore is heated above its melting point to expel H_2O or CO_2 or SO_2
(b) ore is heated below its melting point to expel volatile impurities
(c) ore is heated above its melting point to remove S, As and Sb as SO_2 , As_2O_3 and Sb_2O_3 respectively
(d) ore is heated below its melting point to expel H_2O or CO_2
- A sulphide ore is generally roasted to the oxide before reduction, because
(a) the enthalpy of formation of CO_2 is more than that of CS_2
(b) a metal sulphide is generally more stable than the metal oxide
(c) no reducing agent is found suitable for reducing a sulphide ore
(d) a sulphide ore cannot be reduced at all

9. Nickel is purified by thermal decomposition of its
 (a) hydride (b) chloride
 (c) azide (d) carbonyl
10. The process of converting hydrated alumina into anhydrous alumina is called
 (a) roasting (b) smelting
 (c) dressing (d) calcination
11. The reduction of metal oxides is sometimes done by aluminium in the
 (a) Bayer reaction
 (b) Zilch reaction
 (c) Goldschmidt reaction
 (d) Silberschmidt reaction
12. Which of the following is chalcopyrite?
 (a) CuFeS_2 (b) FeS_2
 (c) $\text{KMgCl}_3 \cdot 6\text{H}_2\text{O}$ (d) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
13. Composition of azurite mineral is
 (a) $\text{CuCO}_3 \cdot \text{CuO}$
 (b) $\text{Cu}(\text{HCO}_3)_2 \cdot \text{Cu}(\text{OH})_2$
 (c) $2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 (d) $\text{CuCO}_3 \cdot 2\text{Cu}(\text{OH})_2$
14. Cassiterite is concentrated by
 (a) levigation
 (b) electromagnetic separation
 (c) floatation
 (d) liquefaction
15. One of the ores of lead is known as lead pyromorphite. What is approximate composition?
 (a) $\text{Pb}_3\text{O}_4 \cdot \text{PbCl}_2$ (b) $\text{PbSO}_4 \cdot \text{Pb}(\text{OH})_2$
 (c) $3\text{Pb}_3(\text{PO}_4)_2 \cdot \text{PbCl}_2$ (d) $\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$
16. Consider the following statements –
 (A) In the Aluminothermite process, aluminium acts as reducing agent.
 (B) The process of extraction of gold involves the formation of $[\text{Au}(\text{CN})_2]^-$ and $[\text{Zn}(\text{CN})_4]^{2-}$.
- (C) In the extractive metallurgy of zinc, partial fusion of ZnO with coke is called sintering and reduction of ore to the molten metal is called smelting.
- (D) Extractive metallurgy of silver from its ore argentite involves complex formation and displacement by more electropositive metal.
- Choose the correct options –
 (a) A and B (b) B and C
 (c) A, B and C (d) A, B, C and D
17. Carbon cannot be used in the reduction of Al_2O_3 because
 (a) the enthalpy of formation of CO_2 is more than that of Al_2O_3
 (b) pure carbon is not easily available
 (c) the enthalpy of formation of Al_2O_3 is very high
 (d) it is an expensive proposition
18. Which of the following process is generally used for the extraction of tin and lead from their chief ores ?
 (a) carbon reduction and self-reduction
 (b) self reduction and carbon-reduction
 (c) electrolysis and self-reduction
 (d) self-reduction and electrolysis
19. In order to refine “blister copper” it is melted in a furnace and is stirred with green logs of wood. The purpose is
 (a) to expel the dissolved gases in blister copper.
 (b) to bring the impurities to surface and oxidize them.
 (c) to increase the carbon content of copper.
 (d) to reduce the metallic oxide impurities with hydrocarbon gases liberated from the wood.
20. Muffle furnace is used in the metallurgy of
 (a) Zn (b) Sn
 (c) Pb (d) Cu

ANSWER KEY

1	(b)	3	(a)	5	(c)	7	(d)	9	(d)	11	(c)	13	(c)	15	(c)	17	(c)	19	(d)
2	(a)	4	(c)	6	(b)	8	(a)	10	(d)	12	(a)	14	(b)	16	(d)	18	(a)	20	(a)

THE P-BLOCK ELEMENTS (GROUP 15, 16, 17 AND 18)



MCQs with One Correct Answer

- $\text{NH}_4\text{ClO}_4 + \text{HNO}_3(\text{dil.}) \longrightarrow \text{HClO}_4 + [\text{X}]$
 $[\text{X}] \xrightarrow{\Delta} \text{Y}(\text{g})$
 $[\text{X}]$ and $[\text{Y}]$ are respectively –
 (a) NH_4NO_3 & N_2O (b) NH_4NO_2 & N_2
 (c) HNO_4 & O_2 (d) None of these
- In the reaction $4\text{HNO}_3 + \text{P}_4\text{O}_{10} \rightarrow 4\text{HPO}_3 + \text{X}$, the product X is
 (a) N_2O_5 (b) N_2O_3
 (c) NO_2 (d) H_2O
- Ammonia is generally manufactured for fertilizers by the reaction
 (a) $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCl}_2 + 2\text{H}_2\text{O} + 2\text{NH}_3$
 (b) By passing an electric discharge in a mixture of N_2 and H_2
 (c) By passing a mixture of N_2 and H_2 under high pressure and moderate temperature over a catalyst
 (d) None of these
- Boiling / melting points of the following hydrides follow in order
 (a) $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 < \text{NH}_3$
 (b) $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$
 (c) $\text{SbH}_3 > \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$
 (d) $\text{SbH}_3 < \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$
- When PbO_2 reacts with conc. HNO_3 the gas evolved is
 (a) NO_2 (b) O_2
 (c) N_2 (d) N_2O
- When potassium ferrocyanide crystals are heated with conc. H_2SO_4 , the gas evolved is
 (a) SO_2 (b) NH_3
 (c) CO_2 (d) CO
- Hydrolysis of one mole of peroxodisulphuric acid produce
 (a) two moles of sulphuric acid
 (b) two moles of peroxomonosulphuric acid
 (c) one mole of sulphuric acid and one mole of peroxomonosulphuric acid
 (d) one mole of sulphuric acid, one mole of peroxomonosulphuric acid and one mole of hydrogen peroxide.
- $[\text{X}] + \text{H}_2\text{SO}_4 \longrightarrow [\text{Y}]$, a colourless gas with irritating smell
 $[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{Green solution}$
 The species $[\text{X}]$ and $[\text{Y}]$ are –
 (a) SO_3^{2-} , SO_2 (b) Cl^- , HCl
 (c) S^{2-} , H_2S (d) CO_3^{2-} , CO_2

9. In the reaction
 $\text{HNO}_3 + \text{HF} \longrightarrow \text{H}_2\text{NO}_3^+ + \text{F}^-$ base is
 (a) HF (b) HNO_3
 (c) HF and HNO_3 (d) None of these
10. HBr and HI reduce sulphuric acid, HCl can reduce KMnO_4 and HF reduces
 (a) H_2SO_4 (b) KMnO_4
 (c) $\text{K}_2\text{Cr}_2\text{O}_7$ (d) None of these
11. When Br_2 is treated with aqueous solutions of NaF, NaCl and NaI separately
 (a) F_2 , Cl_2 and I_2 are liberated
 (b) only F_2 and Cl_2 are liberated
 (c) only I_2 is liberated
 (d) only Cl_2 is liberated
12. The ease of liquefaction of noble gases increases in the order
 (a) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$
 (b) $\text{Xe} < \text{Kr} < \text{Ne} < \text{Ar} < \text{He}$
 (c) $\text{Kr} < \text{Xe} < \text{He} < \text{Ne} < \text{Ar}$
 (d) $\text{Ar} < \text{Kr} < \text{Xe} < \text{Ne} < \text{He}$
13. Which one of the following is correct pair with respect to molecular formula of Xenon compound and hybridization state of Xenon in it?
 (a) XeF_4 , sp^3 (b) XeF_2 , sp
 (c) XeF_2 , sp^3d (d) XeF_4 , sp^2
14. What are the products formed in the reaction of xenon hexafluoride with silicon dioxide?
 (a) $\text{XeSiO}_4 + \text{HF}$ (b) $\text{XeF}_2 + \text{SiF}_4$
 (c) $\text{XeOF}_4 + \text{SiF}_4$ (d) $\text{XeO}_3 + \text{SiF}_2$
15. Which of the following xenon-oxo compounds may not be obtained by hydrolysis of xenon fluorides?
 (a) XeO_2F_2 (b) XeOF_4
 (c) XeO_3 (d) XeO_4

Numeric Value Answer

16. How many bonding electron pairs are there in white phosphorus?
17. The number of S-S bonds in sulphur trioxide trimer (S_3O_9) is _____
18. The number of P-O-P bonds in cyclic metaphosphoric acid is _____
19. The number of P-O-P bonds in P_4O_{10} is _____
20. The number of Cl=O bonds in perchloric acid is, "_____."

ANSWER KEY

1	(a)	3	(c)	5	(b)	7	(c)	9	(b)	11	(c)	13	(c)	15	(d)	17	(0)	19	(6)
2	(a)	4	(a)	6	(d)	8	(a)	10	(d)	12	(a)	14	(c)	16	(6)	18	(3)	20	(3)

THE *d*-AND *f*-BLOCK ELEMENTS



MCQs with One Correct Answer

- What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid?
 - $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 - CrO_4^{2-} is reduced to +3 state of Cr
 - CrO_4^{2-} is oxidized to +7 state of Cr
 - Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
- The correct order of ionic radii of Y^{3+} , La^{3+} , Eu^{3+} and Lu^{3+} is
 - $\text{La}^{3+} < \text{Eu}^{3+} < \text{Lu}^{3+} < \text{Y}^{3+}$
 - $\text{Y}^{3+} < \text{La}^{3+} < \text{Eu}^{3+} < \text{Lu}^{3+}$
 - $\text{Y}^{3+} < \text{Lu}^{3+} < \text{Eu}^{3+} < \text{La}^{3+}$
 - $\text{Lu}^{3+} < \text{Eu}^{3+} < \text{La}^{3+} < \text{Y}^{3+}$
(Atomic nos. Y=39, La=57, Eu=63, Lu=71)
- Which of the following oxides of manganese is amphoteric?
 - MnO_2
 - Mn_2O_3
 - Mn_2O_7
 - MnO
- Which of the following compound is called Turnbull's blue?
 - Ferricyanide
 - Ferrous ferricyanide
 - Ferrous cyanide
 - Ferri-ferricyanide
- A compound of a metal ion M^{x+} ($Z = 24$) has a spin only magnetic moment of $\sqrt{15}$ Bohr Magnetons. The number of unpaired electrons in the compound are
 - 2
 - 4
 - 5
 - 3
- In which of the following compounds manganese has oxidation number equal to that of iodine in KIO_4 ?
 - Potassium manganate
 - Potassium permanganate
 - Manganous chloride
 - Manganese chloride
- When MnO_2 is fused with KOH , a coloured compound is formed, the product and its colour is:
 - K_2MnO_4 , purple green
 - KMnO_4 , purple
 - Mn_2O_3 , brown
 - Mn_3O_4 , black
- If an aqueous solution of KCN is added to a solution of ferrous salt then the complex formed is represented by
 - $[\text{Fe}(\text{CN})_6]^{4-}$
 - $[\text{Fe}(\text{CN})_6]^{3-}$
 - $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
 - $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

9. For the four successive transition elements (Cr, Mn, Fe and Co), the stability of +2 oxidation state will be there in which of the following order?
 (a) $\text{Mn} > \text{Fe} > \text{Cr} > \text{Co}$ (b) $\text{Fe} > \text{Mn} > \text{Co} > \text{Cr}$
 (c) $\text{Co} > \text{Mn} > \text{Fe} > \text{Cr}$ (d) $\text{Cr} > \text{Mn} > \text{Co} > \text{Fe}$
10. In neutral or faintly alkaline medium, thiosulphate is quantitatively oxidized by KMnO_4 to
 (a) SO_3^{2-} (b) SO_4^{2-}
 (c) SO_2 (d) SO_5^{2-}
11. An excess of $\text{Na}_2\text{S}_2\text{O}_3$ reacts with aqueous CuSO_4 to give
 (a) CuS_2O_3 (b) $\text{Cu}_2\text{S}_2\text{O}_3$
 (c) $\text{Na}_2[\text{Cu}(\text{S}_2\text{O}_3)_2]$ (d) $\text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$
12. Which of the following oxides of Cr is amphoteric
 (a) CrO_2 (b) Cr_2O_3
 (c) CrO_5 (d) CrO_3
13. The roasting of HgS in air produces
 (a) HgO (b) HgSO_3
 (c) HgSO_4 (d) Hg
14. Mercury (I) chloride sublimes, when this compound is heated and the vapours it gives off are cooled the substance collected consists of
 (a) Mercury (II) chloride
 (b) Mercury (I) and Mercury (II), chlorides
 (c) Mercury (II) chloride and mercury
 (d) Mercury
15. Consider the following statements
 (I) $\text{La}(\text{OH})_3$ is the least basic among hydroxides of lanthanoids.
 (II) Zr^{4+} and Hf^{4+} possess almost the same ionic radii.
 (III) Ce^{4+} can act as an oxidizing agent.
 Which of the above is/are true?
 (a) (I) and (III) (b) (II) and (III)
 (c) (II) only (d) (I) and (II)

Numeric Value Answer

16. The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is
17. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is
18. What are the oxidation states of Cr in butterfly structure.
19. Out of the following, how many oxides are acidic. MnO , Mn_2O_3 , MnO_2 , MnO_3 , Mn_2O_7
20. A light blue coloured compound (A) on heating gives a black compound (B) which reacts with glucose to give a red compound (C). After the completion of reaction, how many metal oxides are formed?

ANSWER KEY

1	(a)	3	(a)	5	(d)	7	(a)	9	(a)	11	(d)	13	(d)	15	(b)	17	(6)	19	(3)
2	(c)	4	(b)	6	(b)	8	(a)	10	(b)	12	(a)	14	(d)	16	(4)	18	(6)	20	(2)

COORDINATION COMPOUNDS



MCQs with One Correct Answer

- Which one is the most likely structure of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ if $1/3$ of total chlorine of the compound is precipitated by adding AgNO_3
 - $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$
 - $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot (\text{H}_2\text{O})_3$
 - $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$
 - $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- The octahedral complex $\text{CoSO}_4\text{Cl} \cdot 5\text{NH}_3$ exists in two isomeric forms **X** and **Y**. Isomer **X** reacts with AgNO_3 to give a white precipitate, but does not react with BaCl_2 . Isomer **Y** gives white precipitate with BaCl_2 but does not react with AgNO_3 .
Isomers **X** and **Y** are
 - ionization isomers
 - linkage isomers
 - coordination isomers
 - solvate isomers
- For an octahedral complex MX_4Y_2 (M = a transition metal, **X** and **Y** are monodentate achiral ligands), the correct statement, among the following, is
 - MX_4Y_2 has 2 geometrical isomers one of which is chiral.
 - MX_4Y_2 has 2 geometrical isomers both of which are achiral.
 - MX_4Y_2 has 4 geometrical isomers all of which are achiral.
 - MX_4Y_2 has 4 geometrical isomers two of which are chiral.
- Which of the following coordination compounds will give precipitate with an aqueous solution of AgNO_3 ?
 - $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
 - $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
 - $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$
 - $\text{Na}_3[\text{Cr}(\text{CN})_6]$
- Oxidation state of “V” in $\text{Rb}_4\text{K}[\text{HV}_{10}\text{O}_{28}]$ is
 - +5
 - +6
 - $+\frac{7}{5}$
 - +4
- The complex that can exhibit linkage isomerism is
 - $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_3$
 - $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$
 - $[\text{Co}(\text{NH}_3)_5(\text{NO}_3)](\text{NO}_3)_2$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$

7. The complex having the highest spin only magnetic moment is
 (a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (c) $[\text{MnF}_6]^{4-}$ (d) $[\text{NiCl}_4]^{2-}$
 8. Which of the following complex will show geometrical as well as optical isomerism (en=ethylenediamine)
 (a) $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ (b) $[\text{Pt}(\text{NH}_3)\text{Cl}_4]$
 (c) $[\text{Pt}(\text{en})_3]^{4+}$ (d) $[\text{Pt}(\text{en})_2\text{Cl}_2]$
 9. In which of the following pairs both the complexes show optical isomerism ?
 (a) $\text{cis}[\text{Cr}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3+}$, $\text{cis}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
 (b) $[\text{Co}(\text{en})_3]\text{Cl}_3$, $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (c) $[\text{PtCl}(\text{dien})]\text{Cl}$, $[\text{NiCl}_2\text{Br}_2]^{2-}$
 (d) $[\text{Co}(\text{NO}_3)_3(\text{NH}_3)_3]$, $\text{cis}[\text{Pt}(\text{en})_2\text{Cl}_2]$
 10. When 0.01 mole of a cobalt complex is treated with excess silver nitrate solution, 4.305 g silver chloride is precipitated. The formula of the complex is
 (a) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 (c) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (d) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_3$
 11. Which of the following complexes does not show the geometric isomerism ?
 (a) $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$
 (b) $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{NO}_2)(\text{C}_6\text{H}_5\text{N})]\text{NO}_2$
 (c) $\text{Pt}(\text{NH}_2\text{CH}_2\text{COO})_2$
 (d) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$
 12. $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]\text{Cl}$ exhibits
 (a) linkage isomerism, ionization isomerism and geometrical isomerism
 (b) ionization isomerism, geometrical isomerism and optical isomerism
 (c) linkage isomerism, geometrical isomerism and optical isomerism
 (d) linkage isomerism, ionization isomerism and optical isomerism
 13. Which one of the following has the highest molar conductivity?
 (a) Diamminedichloroplatinum (II)
 (b) Tetraamminedichlorocobalt (III) chloride
 (c) Potassium hexacyanoferrate (II)
 (d) Hexaaquachromium (III) chloride
 14. $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$ is called
 (a) Potassium aluminooxalate
 (b) Potassium trioxalatoaluminate (III)
 (c) Potassium aluminium (III) oxalate
 (d) Potassium trioxalatoaluminate (VI)
 15. When AgNO_3 is added to a solution of $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, the precipitate of AgCl shows two ionizable chloride ions. This means :
 (a) Two chlorine atoms satisfy primary valency and one secondary valency
 (b) One chlorine atom satisfies primary as well as secondary valency
 (c) Three chlorine atoms satisfy primary valency
 (d) Three chlorine atoms satisfy secondary valency
 16. Among the following, the pair of paramagnetic complexes is
 (a) $\text{K}_3[\text{Fe}(\text{CN})_6]$ and $\text{K}_3[\text{CoF}_6]$
 (b) $\text{K}_3[\text{Fe}(\text{CN})_6]$ and $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ and $\text{K}_3[\text{CoF}_6]$
 (d) $\text{K}_4[\text{Fe}(\text{CN})_6]$ and $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 17. The tetrahedral $[\text{CoI}_4]^{2-}$ and square planar $[\text{PdBr}_4]^{2-}$ complex ions are respectively
 (a) low spin, high spin
 (b) high spin, low spin
 (c) both low spin
 (d) both high spin
 18. Crystal field stabilization energy for high spin d^4 octahedral complex is:
 (a) $-1.8 \Delta_o$ (b) $-1.6 \Delta_o + P$
 (c) $-1.2 \Delta_o$ (d) $-0.6 \Delta_o$
 19. Low spin complex of d^6 -cation in an octahedral field will have the following energy :
 (a) $\frac{-12}{5} \Delta_o + P$ (b) $\frac{-12}{5} \Delta_o + 3P$
 (c) $\frac{-2}{5} \Delta_o + 2P$ (d) $\frac{-2}{5} \Delta_o + P$
- (Δ_o = Crystal Field Splitting Energy in an octahedral field, P = Electron pairing energy)

20. In which of the following octahedral complex species the magnitude of Δ_o will be maximum ?
 (a) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (b) $[\text{Co}(\text{CN})_6]^{3-}$
 (c) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$
21. The number of unpaired electrons in the complex ion $[\text{CoF}_6]^{3-}$ is (Atomic no.: Co = 27)
22. The possible number of isomers of the complex $[\text{MCl}_2\text{Br}_2]\text{SO}_4$ is _____.
23. The total number of coordination sites in ethylenediaminetetraacetate (EDTA^{4-}) is _____.
24. Considering that $\Delta_o > P$, the magnetic moment (in BM) of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ would be _____.
25. The volume (in mL) of 0.125 M AgNO_3 required to quantitatively precipitate chloride ions in 0.3 g of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is _____.
 $M_{[\text{Co}(\text{NH}_3)_6]\text{Cl}_3} = 267.46 \text{ g/mol}$
 $M_{\text{AgNO}_3} = 169.87 \text{ g/mol}$

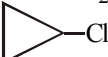
Numeric Value Answer

ANSWER KEY

1	(c)	4	(a)	7	(c)	10	(c)	13	(c)	16	(a)	19	(b)	22	(4)	25	(26.92)		
2	(a)	5	(a)	8	(d)	11	(d)	14	(b)	17	(b)	20	(b)	23	(6)				
3	(b)	6	(b)	9	(b)	12	(a)	15	(a)	18	(d)	21	(4)	24	(0.0)				

1. $(\text{CH}_3)_3\text{CMgCl}$ on reaction with D_2O produces :

(a) $(\text{CH}_3)_3\text{CD}$ (b) $(\text{CH}_3)_3\text{OD}$
(c) $(\text{CD}_3)_3\text{CD}$ (d) $(\text{CD}_3)_3\text{OD}$

2. $\text{CH}_3 - \text{CH}_2 - \underset{\text{Cl}}{\underset{|}{\text{CH}}} - \text{CH}_3$ obtained by chlorination of *n*-butane, will be
 (a) *l*-form (b) *d*-form
 (c) meso form (d) racemic mixture
3. Most reactive halide towards $\text{S}_{\text{N}}1$ reaction is
 (a) *n*-Butyl chloride (b) *sec*-Butyl chloride
 (c) *tert*-Butyl chloride (d) Allyl chloride
4. Arrange the following halides in the decreasing order of $\text{S}_{\text{N}}1$ reactivity :
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$, $\text{CH}_2 = \text{CHCH}(\text{Cl})\text{CH}_3$,
 I II
 $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$
 III
 (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{I} > \text{III}$
 (c) $\text{II} > \text{III} > \text{I}$ (d) $\text{III} > \text{II} > \text{I}$
5. The compound most reactive towards $\text{S}_{\text{N}}1$ reaction is
 (a) $\text{Me}_3\text{COCH}_2\text{Cl}$ (b) MeOCH_2Cl
 (c) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Cl}$ (d) 

(i) $(\text{CH}_3)_2\text{CH}-\text{CH}_2\text{Br} \xrightarrow{\text{C}_2\text{H}_5\text{OH}}$
 $(\text{CH}_3)_2\text{CH}-\text{CH}_2\text{OC}_2\text{H}_5 + \text{HBr}$

$$\text{(ii) } (\text{CH}_3)_2\text{CH}-\text{CH}_2\text{Br} \xrightarrow{\text{C}_2\text{H}_5\text{O}^-} (\text{CH}_3)_2\text{CH}-\text{CH}_2\text{OC}_2\text{H}_5 + \text{Br}^-$$

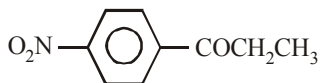
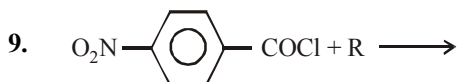
(a) S_{N1} and S_{N2} (b) S_{N1} and S_{N1}

(c) S_N2 and S_N2 (d) S_N2 and S_N1

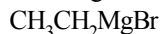
(a) *n*-Butyl chloride
(b) *iso*-Butyl chloride
(c) *sec*-Butyl chloride
(d) *tert*-Butyl chloride

8. Although hexafluoroethane (C_2F_6 , b.p. -79°C) and ethane (C_2H_6 , b.p. -89°C) differ very much in their molecular weights, their boiling points differ only by 10°C . This is due to

- (a) low polarizability of F
(b) nearly similar size of F and H
(c) both (a) and (b)
(d) Neither of the two



The reagent R may be



I



II



III



IV

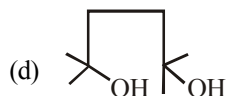
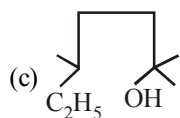
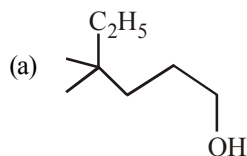
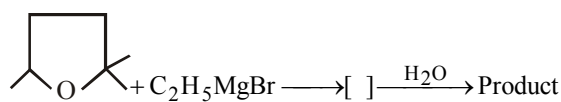
(a) I or II

(b) I or II or III

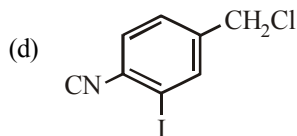
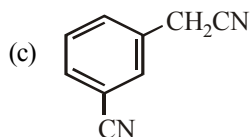
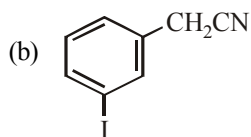
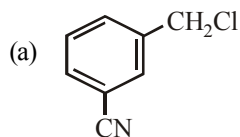
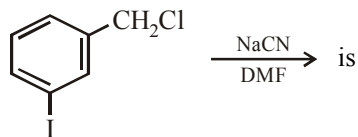
(c) III or IV

(d) Any of the four

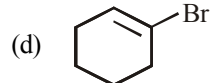
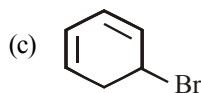
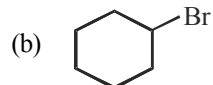
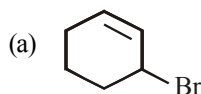
10. Pick up the final product in the following reaction.



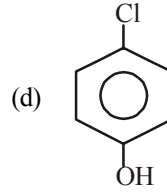
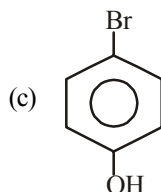
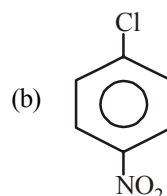
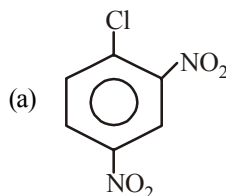
11. The structure of the major product formed in the following reaction



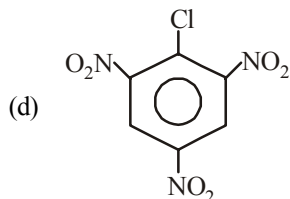
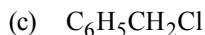
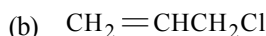
12. Which of the following is fast de-brominated ?



13. Which of the following is least reactive towards nucleophilic substitution with aq. KOH ?



14. Which of the following gives white precipitate when treated with silver nitrate solution ?

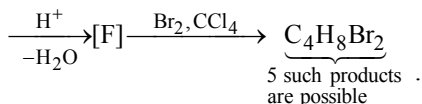
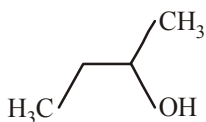


15. Chloropicrin is obtained by the reaction of

- steam on carbon tetrachloride
- nitric acid on chlorobenzene
- chlorine on picric acid
- nitric acid on chloroform

Numeric Value Answer

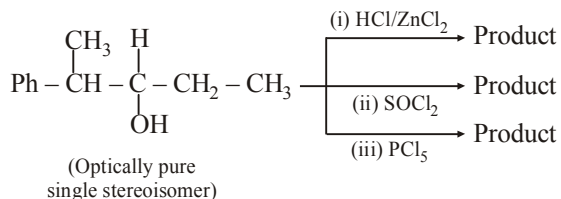
16.



How many structures for F are possible?

17. Find the number of total number of structural and configurational isomers of a bromo compound, C_5H_9Br , formed by the addition of HBr to 2-pentyne respectively.

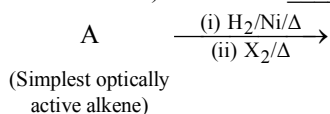
18.



Find the sum of total number of isomeric chlorides obtained in these reactions (consider only the major products)

19. How many possible monochloro structural isomers are expected to form on free radical chlorination of $(CH_3)_2CHCH_2CH_3$?

20. The total number of monohalogenated organic products in the following (including stereoisomers) reaction is _____.



ANSWER KEY

1	(a)	3	(c)	5	(b)	7	(d)	9	(c)	11	(b)	13	(d)	15	(d)	17	(6)	19	(4)
2	(d)	4	(c)	6	(a)	8	(c)	10	(b)	12	(c)	14	(d)	16	(3)	18	(4)	20	(8)

ALCOHOLS, PHENOLS AND ETHERS

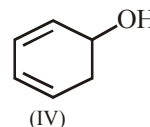
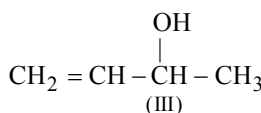
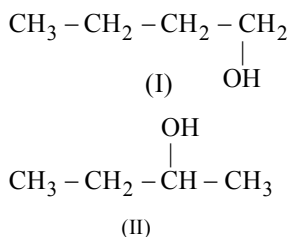
25

MCQs with One Correct Answer

- Which one of the following is not formed when glycerol reacts with HI ?
 (a) $\text{CH}_3 - \text{CHI} - \text{CH}_3$
 (b) $\text{CH}_3 - \text{CH} = \text{CH}_2$
 (c) $\text{CH}_2\text{OH} - \text{CHI} - \text{CH}_2\text{OH}$
 (d) $\text{CH}_2 = \text{CH} - \text{CH}_2\text{I}$
- The reaction of Lucas reagent is fastest with:
 (a) $(\text{CH}_3)_2\text{CHOH}$ (b) $\text{CH}_3(\text{CH}_2)_2\text{OH}$
 (c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) $(\text{CH}_3)_3\text{COH}$
- Following compounds are given:
 (i) $\text{CH}_3\text{CH}_2\text{OH}$ (ii) CH_3COCH_3
 (iii) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CHOH}}$ (iv) CH_3OH

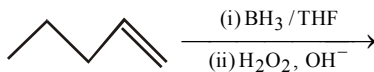
Which of the above compound(s), on being warmed with iodine solution and NaOH, will give iodoform?

- (i) and (ii)
 - (i), (iii) and (iv)
 - (i) only
 - (i), (ii) and (iii)
- Which of the following is the correct order for ease of dehydration—



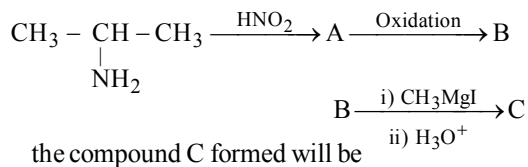
- $\text{I} > \text{III} > \text{II} > \text{IV}$
- $\text{IV} > \text{III} > \text{II} > \text{I}$
- $\text{IV} > \text{II} > \text{III} > \text{I}$
- $\text{III} > \text{IV} > \text{II} > \text{I}$

- The product of the following reaction is

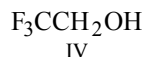
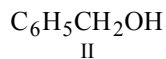
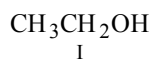


- 1-Pentanol
- 2-Pentanol
- Pentane
- 1,2-Pentanediol

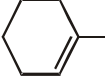
- In the following sequence of reactions,

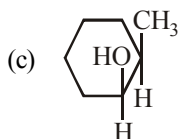
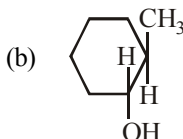
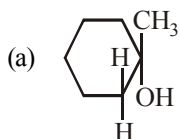


- (a) Butanol-1
(b) Butanol-2
(c) 2-Methylpropanol-1
(d) 2-Methyl-2-propanol
7. Ethanol when reacted with PCl_5 gives **A**, POCl_3 and HCl . **A** reacts with silver nitrite to form **B** (major product) and AgCl . **A** and **B** respectively are
- (a) $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
(b) C_2H_6 and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
(c) $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{NO}_2$
(d) C_2H_6 and $\text{C}_2\text{H}_5\text{NO}_2$
8. Arrange the following alcohols in order of increasing ease of dehydration



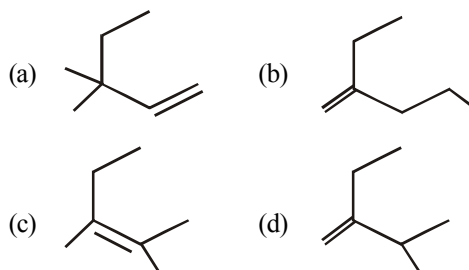
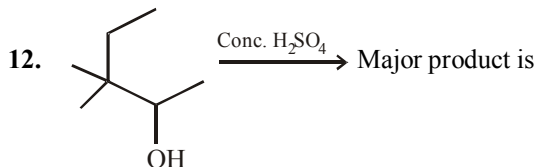
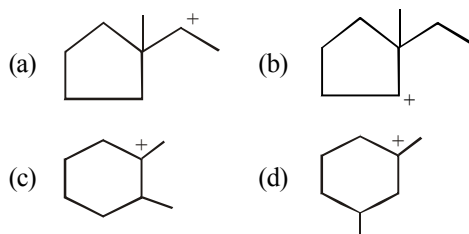
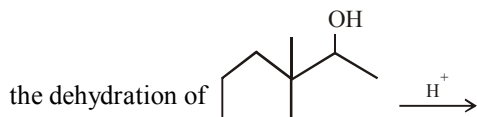
- (a) $\text{II} < \text{I} < \text{IV} < \text{III}$ (b) $\text{IV} < \text{III} < \text{II} < \text{I}$
(c) $\text{IV} < \text{III} < \text{I} < \text{II}$ (d) $\text{II} < \text{I} < \text{III} < \text{IV}$

9.  $\xrightarrow[\text{(ii) H}_2\text{O}_2, \text{OH}^-]{\text{(i) B}_2\text{H}_6}$ X. The compound X is

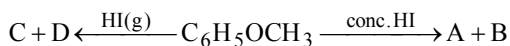


- (d) Both (b) and (c)

10. Which of the following is most reactive towards aqueous HBr ?
(a) 1-Phenyl - 2 - propanol
(b) 1- Phenyl - 1 - propanol
(c) 3-Phenyl - 1 - propanol
(d) 2- Phenyl - 1 - propanol
11. Which carbocation is more likely to be formed in

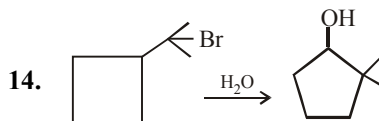


13. Anisole is treated with HI under two different conditions.



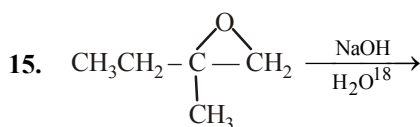
The nature of **A** to **D** will be

- (a) **A** and **B** are CH_3I and $\text{C}_6\text{H}_5\text{OH}$, while **C** and **D** are CH_3OH and $\text{C}_6\text{H}_5\text{I}$
(b) **A** and **B** are CH_3OH and $\text{C}_6\text{H}_5\text{I}$, while **C** and **D** are CH_3I and $\text{C}_6\text{H}_5\text{OH}$
(c) Both **A** and **B** as well as both **C** and **D** are CH_3I and $\text{C}_6\text{H}_5\text{OH}$ respectively.
(d) **A** and **B** are CH_3I and $\text{C}_6\text{H}_5\text{OH}$, while there is no reaction in the second case.

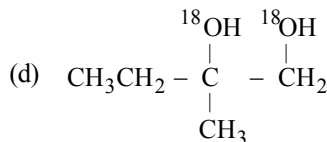
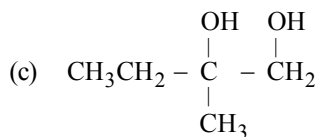
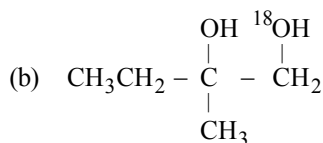
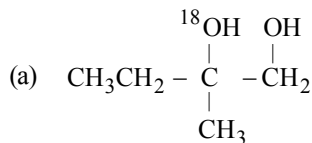


The driving force in the above reaction is

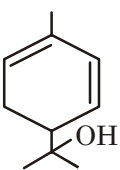
- (a) conversion of 1° carbocation to 2° carbocation
(b) conversion of 1° carbocation to 3° carbocation
(c) relief in steric strain due to expansion of ring
(d) both a & c

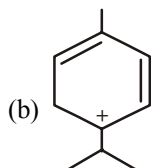
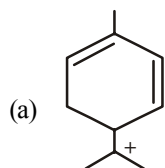


Major product is



16. Which of the carbocations is more likely to be

formed when  is heated with conc H_2SO_4 ?

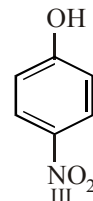
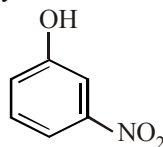
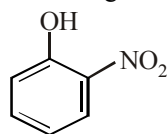


(c) both in equal amounts (d) None

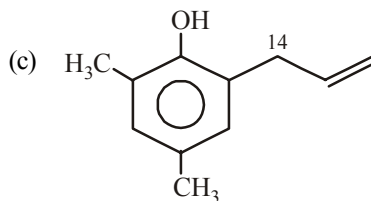
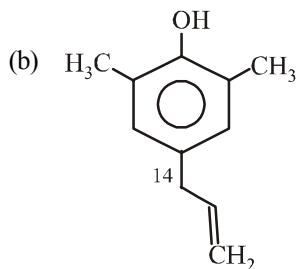
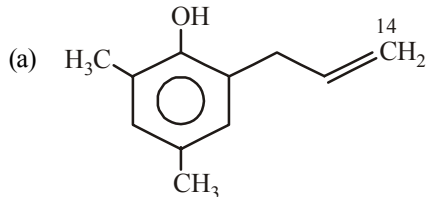
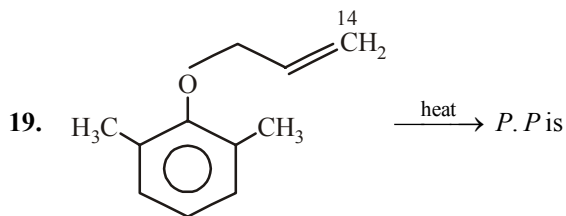
17. Phenol on reaction with Br_2 in non-polar aprotic solvent furnishes

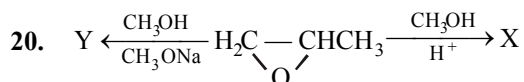
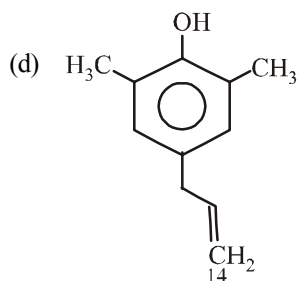
- (a) 2, 4, 6-tribromophenol
(b) *p*-bromophenol
(c) *o*- and *p*-bromophenol
(d) *m*-bromophenol

18. Arrange the following phenols in order of their increasing acidity.



- (a) $\text{I} > \text{III} > \text{II}$ (b) $\text{I} > \text{II} > \text{III}$
(c) $\text{III} > \text{I} > \text{II}$ (d) $\text{III} > \text{II} > \text{I}$

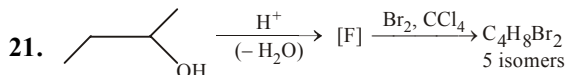




Here X and Y respectively are

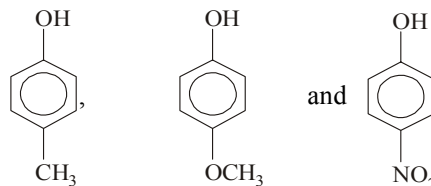
- (a) $\begin{array}{c} \text{CH}_2-\text{CHCH}_3 \\ | \quad | \\ \text{OCH}_3 \quad \text{OH} \end{array}$ and $\begin{array}{c} \text{HOCH}_2-\text{CHCH}_3 \\ | \\ \text{OCH}_3 \end{array}$
- (b) $\begin{array}{c} \text{HOCH}_2-\text{CHCH}_3 \\ | \\ \text{OCH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_2-\text{CHCH}_3 \\ | \quad | \\ \text{OCH}_3 \quad \text{OH} \end{array}$
- (c) $\begin{array}{c} \text{CH}_2-\text{CHCH}_3 \\ | \quad | \\ \text{OCH}_3 \quad \text{OH} \end{array}$ in both cases
- (d) $\begin{array}{c} \text{HOCH}_2-\text{CHCH}_3 \\ | \\ \text{OCH}_3 \end{array}$ in both cases

Numeric Value Answer

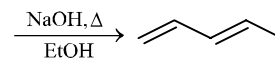
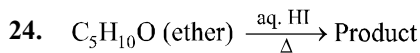
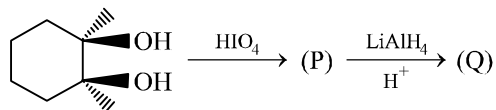


How many structures are possible for F?

22. How many of the following phenols will give lower yield during Kolbe-Schmidt reaction?

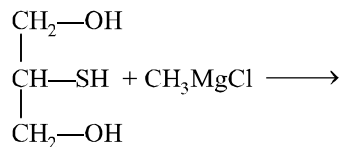


23. What is the total number of stereoisomers of the product 'Q'?



How many carbon atoms are present in the ring of reactant ether?

25. How many moles of CH_4 will evolve from the following reaction?



ANSWER KEY

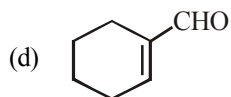
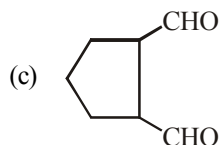
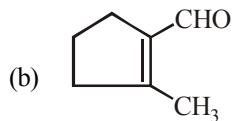
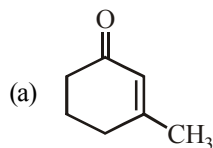
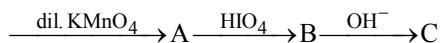
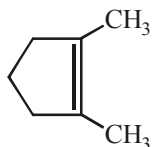
1	(c)	4	(b)	7	(c)	10	(b)	13	(c)	16	(b)	19	(d)	22	(1)	25	(3)		
2	(d)	5	(a)	8	(c)	11	(c)	14	(c)	17	(c)	20	(b)	23	(3)				
3	(d)	6	(d)	9	(b)	12	(c)	15	(b)	18	(c)	21	(3)	24	(4)				

ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

26

MCQs with One Correct Answer

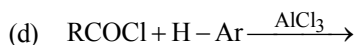
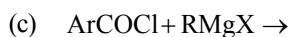
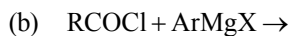
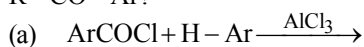
1. Suggest appropriate structures for the missing final compound. (The number of carbon atom remains the same throughout the reaction)



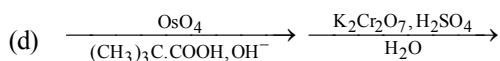
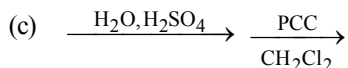
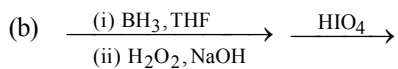
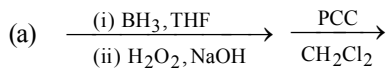
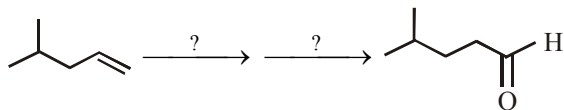
2. Identify the product C in the series

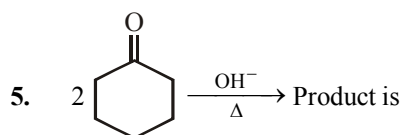


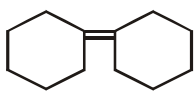
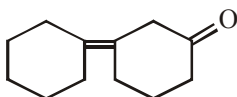
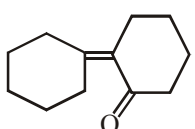
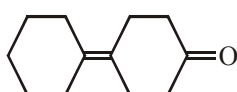
3. Which of the following reactions can produce $\text{R}-\text{CO}-\text{Ar}$?



4. The suitable reaction steps to carry out the following transformation



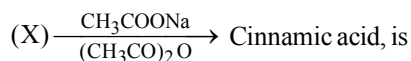


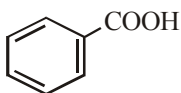
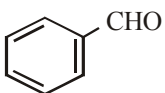
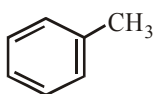
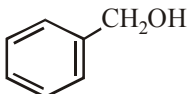
- (a) 
 (b) 
 (c) 
 (d) 

6. Which gives lactic acid on hydrolysis after reacting with HCN?

- (a) HCHO (b) CH_3CHO
 (c) $\text{C}_6\text{H}_5\text{CHO}$ (d) CH_3COCH_3

7. The reactant (X) in the reaction



- (a)  (b) 
 (c)  (d) 

8. The increasing order of the rate of HCN addition to compound A – D is

- (A) HCHO (B) CH_3COCH_3
 (C) PhCOCH_3 (D) PhCOPh
 (a) $\text{D} < \text{C} < \text{B} < \text{A}$ (b) $\text{C} < \text{D} < \text{B} < \text{A}$
 (c) $\text{A} < \text{B} < \text{C} < \text{D}$ (d) $\text{D} < \text{B} < \text{C} < \text{A}$

9. Oxidation of acetaldehyde with selenium dioxide produces

- (a) ethanoic acid (b) methanoic acid
 (c) glyoxal (d) oxalic acid

10. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$ with NaBH_4 gives

- (a) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (b) $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_2\text{OH}$
 (c) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CHO}$
 (d) $\text{C}_6\text{H}_5\text{CH}_2\text{CHOHCH}_3$

11. If 3-hexanone is reacted with NaBH_4 followed by hydrolysis with D_2O , the product will be

- (a) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
 (b) $\text{CH}_3\text{CH}_2\text{CD}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{CH}(\text{OD})\text{CH}_2\text{CH}_2\text{CH}_3$
 (d) $\text{CH}_3\text{CH}_2\text{CD}(\text{OD})\text{CH}_2\text{CH}_2\text{CH}_3$

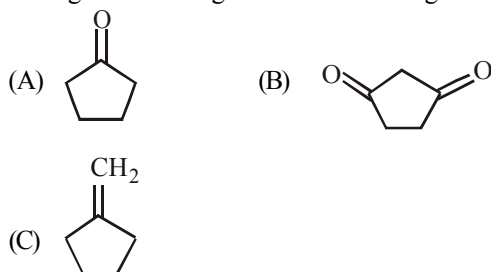
12. Consider the following statements:

- I. Acetophenone can be prepared by oxidation of 1-phenylethanol
 II. Acetophenone can be prepared by reaction of benzaldehyde with methyl magnesium bromide
 III. Acetophenone can be prepared by Friedel-Craft's reaction of benzene with acetyl chloride
 IV. Acetophenone can be prepared by distillation of calcium benzoate

Which of the above statements are correct?

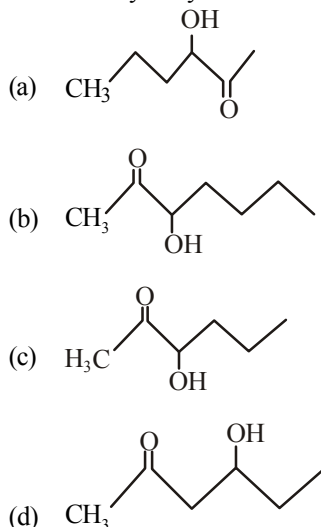
- (a) II and III (b) I and IV
 (c) I and III (d) III and IV

13. Arrange the following in order of decreasing acidity

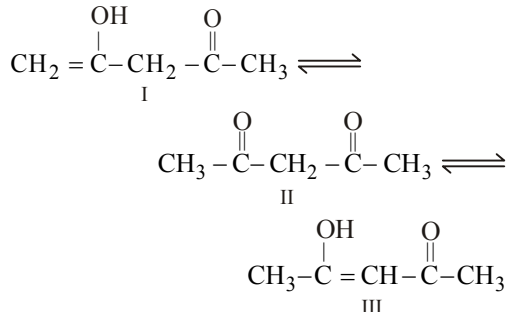


- (a) $\text{C} > \text{A} > \text{B}$ (b) $\text{C} > \text{B} > \text{A}$
 (c) $\text{A} > \text{C} > \text{B}$ (d) $\text{A} > \text{B} > \text{C}$

14. Which one of the following compounds will be most readily dehydrated?

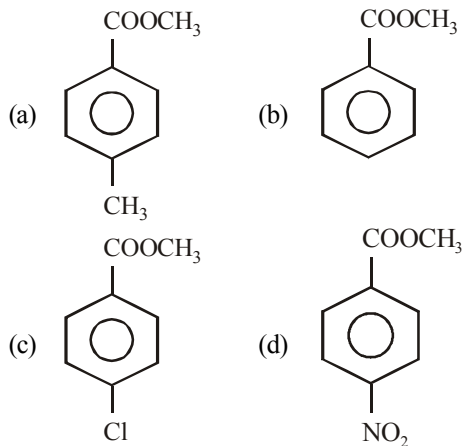


15. The order of stability of the following tautomeric compounds is :

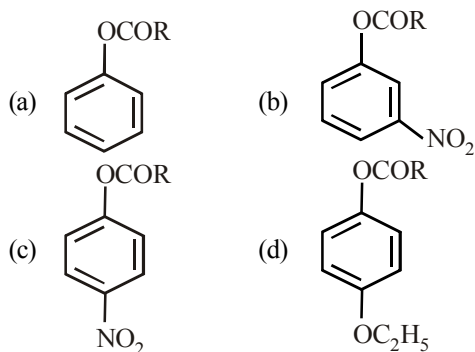


- (a) III > II > I (b) II > I > III
 (c) II > III > I (d) I > II > III
16. In the following reaction
- $$\text{RCH}_2\text{COOH} \xrightarrow{\text{Br}_2/\text{P}} \text{X} \xrightarrow{\text{excess NH}_3} \text{Y}$$
- The major compounds X and Y are
- (a) RCHBrCONH_2 ; $\text{RCH}(\text{NH}_2)\text{COOH}$
 (b) RCHBrCOOH ; $\text{RCH}(\text{NH}_2)\text{COOH}$
 (c) RCH_2COBr ; $\text{RCH}_2\text{COONH}_4$
 (d) RCHBrCOOH ; $\text{RCH}_2\text{CONH}_2$
17. The correct order of increasing acid strength of the compounds
- (A) $\text{CH}_3\text{CO}_2\text{H}$ (B) $\text{MeOCH}_2\text{CO}_2\text{H}$
 (C) $\text{CF}_3\text{CO}_2\text{H}$ (D) $\text{Me}_2\text{CHCO}_2\text{H}$
- is
- (a) $\text{D} < \text{A} < \text{B} < \text{C}$ (b) $\text{A} < \text{D} < \text{B} < \text{C}$
 (c) $\text{B} < \text{D} < \text{A} < \text{C}$ (d) $\text{D} < \text{A} < \text{C} < \text{B}$
18. Which of the following on oxidation followed by hydrolysis gives pyruvic acid ?
- (a) Acetaldehyde cyanohydrin
 (b) Acetone cyanohydrin
 (c) Formaldehyde cyanohydrin
 (d) None of these
19. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C', 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is:
- (a) CH_3COOH
 (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (c) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}}\text{COOH}$
 (d) $\text{CH}_3\text{CH}_2\text{COOH}$

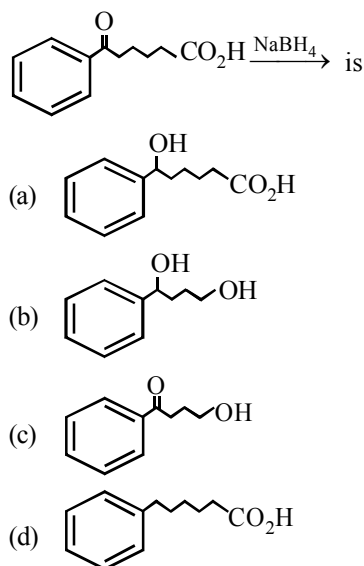
20. Which of the following will undergo alkaline hydrolysis most rapidly ?



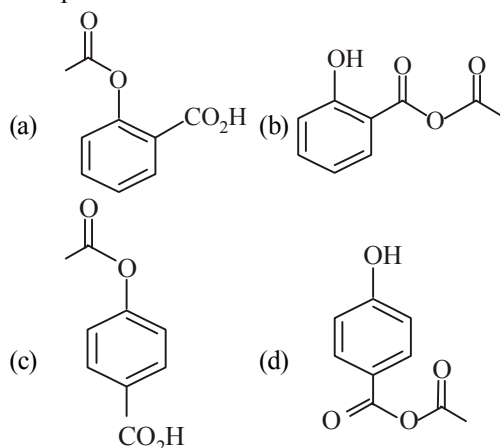
21. Acidic hydrolysis is minimum in



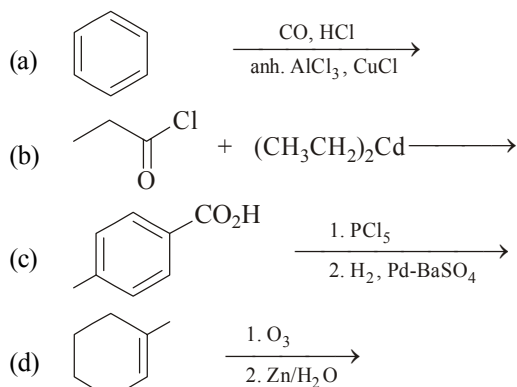
22. The major product of the following reaction,



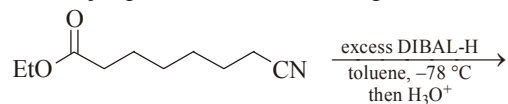
23. Reaction of phenol with NaOH followed by heating with CO_2 under high pressure, and subsequent acidification gives compound X as the major product, which can be purified by steam distillation. When reacted with acetic anhydride in the presence of a trace amount of conc. H_2SO_4 , compound X produces Y as the major product. Compound Y is



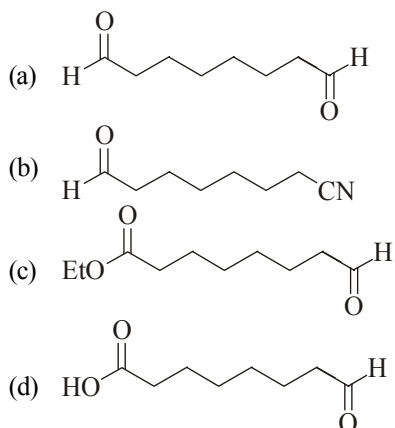
24. The product of which of the following reactions forms a reddish brown precipitate when subjected to Fehling's test?



25. The major product of the following reaction

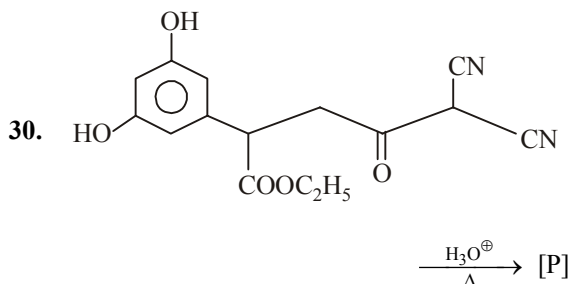
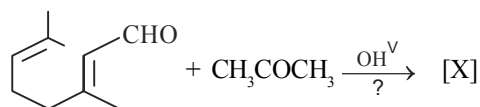


is



Numeric Value Answer

26. 1, 2-Diethylbenzene on ozonolysis gives how many different products?
27. How many products are formed when ethanal reacts with propanal in the presence of a base?
28. The amount (in mol) of bromoform (CHBr_3) produced when 1.0 mole of acetone reacts completely with 9.0 mole of bromine in the presence of aqueous NaOH is
29. Number of possible geometrical isomers for $[\text{X}]$



Number of moles of NaOH required to neutralize one mole of $[\text{P}]$

ANSWER KEY

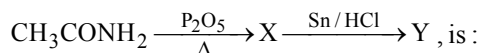
1	(a)	4	(a)	7	(b)	10	(a)	13	(a)	16	(b)	19	(d)	22	(a)	25	(a)	28	(11)
2	(d)	5	(c)	8	(a)	11	(c)	14	(d)	17	(a)	20	(d)	23	(a)	26	(3)	29	(4)
3	(a)	6	(b)	9	(c)	12	(c)	15	(a)	18	(a)	21	(d)	24	(d)	27	(4)	30	(3)

AMINES



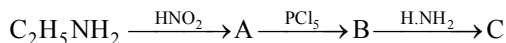
MCQs with One Correct Answer

1. The end product (Y) in the reaction sequence



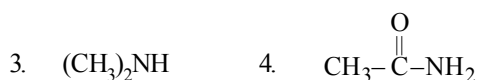
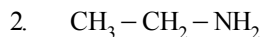
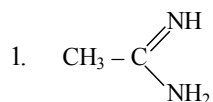
- (a) ethane nitrile (b) acetic acid
(c) ethanamine (d) chloroethane

2. The end product of the reactions is



- (a) ethyl cyanide (b) ethyl amine
(c) methyl amine (d) acetamide

3. The correct order of basicities of the following compounds is



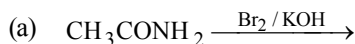
- (a) $2 > 1 > 3 > 4$ (b) $1 > 3 > 2 > 4$
(c) $3 > 1 > 2 > 4$ (d) $1 > 2 > 3 > 4$

4. *n*-Butylamine(I), diethylamine(II) and N, N-dimethylethyl amine(III) have the same molar mass.

The increasing order of their boiling point is

- (a) $\text{III} < \text{II} < \text{I}$ (b) $\text{I} < \text{II} < \text{III}$
(c) $\text{II} < \text{III} < \text{I}$ (d) $\text{II} < \text{I} < \text{III}$

5. Which of the following reactions will not give a primary amine?

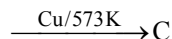
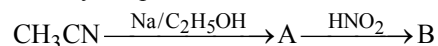


6. $\text{CH}_3\text{CH}_2\text{Cl} \xrightarrow{\text{NaCN}} \text{X} \xrightarrow[\text{acetic anhydride}]{\text{Ni/H}_2} \text{Y} \xrightarrow{\quad\quad\quad} \text{Z}$

Z in the above reacting sequence is

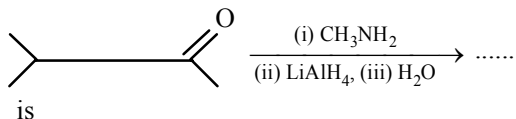
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NHCOCH}_3$
(b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
(c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCH}_3$
(d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCOCH}_3$

7. Identify the product C in the series



- (a) CH_3COOH (b) $\text{CH}_3\text{CH}_2\text{NHOH}$
(c) CH_3CONH_2 (d) CH_3CHO

8. The major organic product formed from the following reaction :



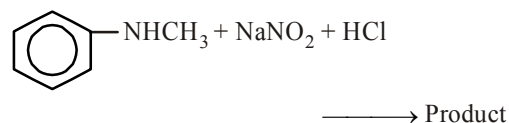
is

- (a)
- (b)
- (c)
- (d)

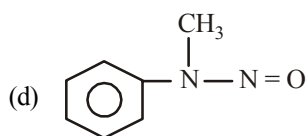
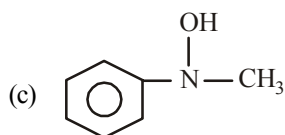
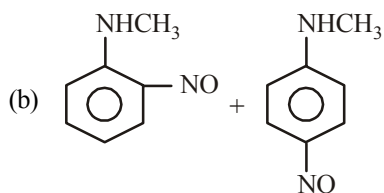
9. $\xrightarrow[\text{(ii) heat}]{\text{(i) H}_2\text{O}_2}$ Product P is

- (a)
- (b)
- (c)
- (d)

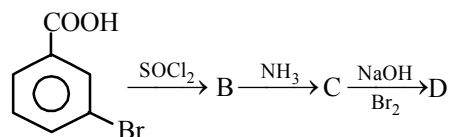
10. Predict the product:



- (a)



11. In a set of reactions *m*-bromobenzoic acid gave a product D. Identify the product D.



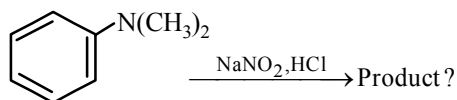
- (a)
- (b)
- (c)
- (d)

12.
$$\text{I} \xrightarrow{(\text{CH}_3\text{CO})_2\text{O, Pyridine}} \text{II} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) LiAlH}_4} \text{III}$$

The basicity order of I, II and III is –

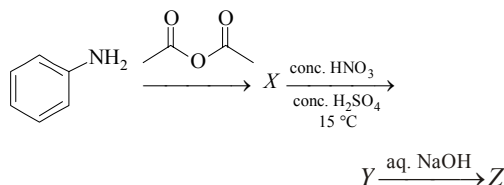
- (a) $\text{III} > \text{I} > \text{II}$ (b) $\text{I} > \text{II} > \text{III}$
 (c) $\text{III} > \text{II} > \text{I}$ (d) $\text{II} > \text{III} > \text{I}$

13. What could be the product for the following reaction?



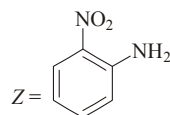
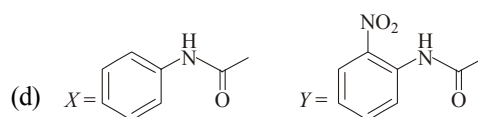
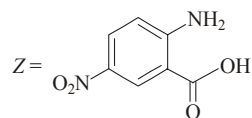
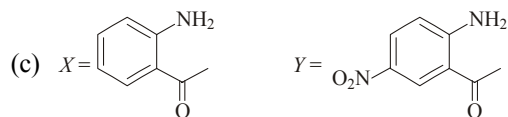
- (a)
- (b)
- (c)
- (d)

14. The major products X, Y and Z in the following sequence of transformations

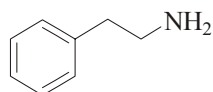


are

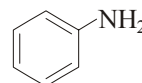
- (a) X = Y = Z =
- (b) X = Y = Z =



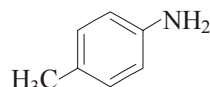
15. The correct order of basicity of the following amines



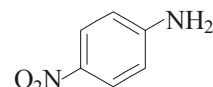
I



II



III



IV

is

- (a) I > II > III > IV (b) I > III > II > IV
 (c) III > II > I > IV (d) IV > III > II > I

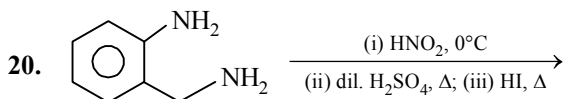
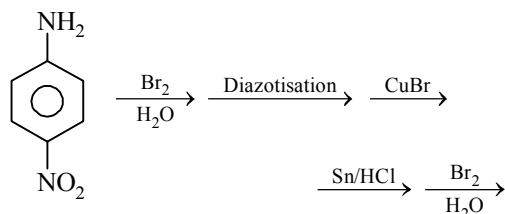
Numeric Value Answer

16. How many of the following method(s) is/ are used for eliminating nitrogen of an amine present outside the ring?
 Hofmann elimination, Cope elimination and Emde degradation
17. How many of the following compounds can be methylated by diazomethane?
 $\text{C}_2\text{H}_5\text{COOH}$, $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{OH}$ and $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$

18. Starting with nitrobenzene, what is the minimum number of following reagents required to convert it to P-dinitrobenzene?

- Conc. HNO_3 + Conc. H_2SO_4 ,
- NH_3
- $\text{C}_2\text{H}_5\text{OK}$
- $\text{Sn} + \text{HCl}$
- $\text{NaNO}_2 + \text{HBF}_4$
- $\text{NaNO}_2 + \text{Cu}$
- $(\text{CH}_3\text{CO})_2\text{O} + \text{Pyridine}$
- H_3O^+

19. Find the total number of bromine atoms present in the end product of following scheme of reactions.



How many hydrogen atoms are present in the final product of above reaction?

ANSWER KEY

1	(c)	3	(d)	5	(c)	7	(d)	9	(d)	11	(c)	13	(c)	15	(b)	17	(3)	19	(5)
2	(b)	4	(a)	6	(a)	8	(b)	10	(d)	12	(a)	14	(b)	16	(2)	18	(6)	20	(7)

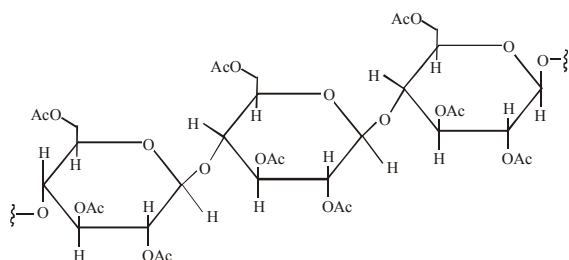
BIOMOLECULES



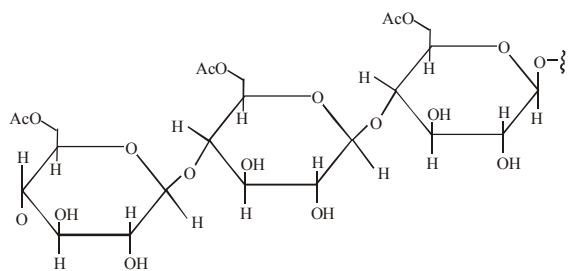
MCQs with One Correct Answer

- What will happen when D-(+)-glucose is treated with methanolic —HCl followed by Tollens' reagent ?
 - A black ppt. will be formed
 - A red ppt. will be formed
 - A green colour will appear
 - No characteristic colour or ppt. will be formed.
- Which one of the following statements is not true regarding (+) lactose ?
 - On hydrolysis (+) lactose gives equal amount of D(+) glucose and D(+) galactose.
 - (+) Lactose is a β -glycoside formed by the union of a molecule of D(+) glucose and a molecule of D(+) galactose.
 - (+) Lactose is a reducing sugar and does not exhibit mutarotation.
 - (+) Lactose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ contains 8-OH groups.
- The term anomers of glucose refers to
 - enantiomers of glucose
 - isomers of glucose that differ in configuration at carbon one (C-1)
 - isomers of glucose that differ in configurations at carbons one and four (C-1 and C-4)
 - a mixture of (D)-glucose and (L)-glucose
- The linkage between the two monosaccharide units in lactose is
 - C_1 of β -D-glucose and C_4 of β -D-galactose
 - C_1 of β -D-galactose and C_4 of β -D-glucose
 - C_1 of α -D-galactose and C_4 of β -D-glucose
 - C_1 of β -D-galactose and C_4 of α -D-glucose
- Natural glucose is termed D-glucose because :
 - —OH on the second carbon is on the right side in Fischer projection
 - —OH on the sixth carbon is on the right side in Fischer projection.
 - —OH on the fifth carbon is on the right side in Fischer projection.
 - It is dextrorotatory.
- Which of the following statement is not correct?
 - Amylopectin is a branched polymer of α -glucose.
 - Cellulose is a linear polymer of β -glucose
 - Glycogen is the food reserve of plants
 - All proteins are polymers of α -amino acids.

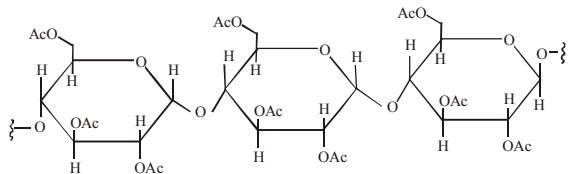
7. Fructose reduces Tollen's reagent due to:
- enolisation of fructose followed by conversion to glucose (having aldehydic group) by the base present in Tollen's reagent
 - asymmetric carbons
 - primary alcoholic group
 - secondary alcoholic group
8. Cellulose upon acetylation with excess acetic anhydride/ H_2SO_4 (catalytic) gives cellulose triacetate whose structure is
-



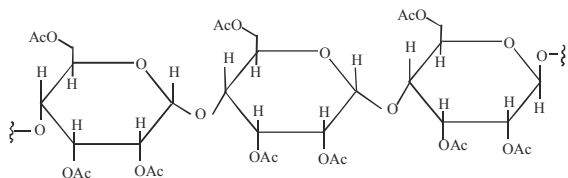
(b)



(c)



(d)



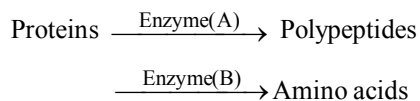
9. An organic compound with the formula $\text{C}_6\text{H}_{12}\text{O}_6$ forms a yellow crystalline solid with phenylhydrazine and gives a mixture of sorbitol and mannitol when reduced with sodium. Which among the following could be the compound?
- Fructose
 - Glucose
 - Mannose
 - Sucrose

10. **Statement-1** : Treatment of D-glucose with dilute alkali affords an equilibrium mixture consisting of D-mannose, D-fructose and starting substance D-glucose.

Statement-2 : The reaction involves an intermediate in which hybridisation of C_2 changes from sp^3 to sp^2 .

- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 - Statement -1 is True, Statement-2 is False.
 - Statement -1 is False, Statement-2 is True.
11. Which of the statements about "denaturation" given below are correct ?
- Denaturation of proteins causes loss of secondary and tertiary structures of the protein.
 - Denaturation leads to the conversion of double strand of DNA into single strand
 - Denaturation affects primary structure which gets distorted
- (B) and (C)
 - (A) and (C)
 - (A) and (B)
 - (A), (B) and (C)

12. During the process of digestion, the proteins present in food materials are hydrolysed to amino acids. The two enzymes involved in the process

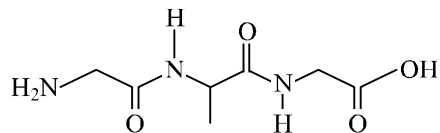


are respectively

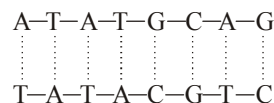
- Diastase and lipase
- Pepsin and trypsin

- (c) Invertase and zymase
(d) Amylase and maltase
13. In both DNA and RNA, heterocyclic base and phosphate ester linkages are at –
- (a) C_5' and C_1' respectively of the sugar molecule
(b) C_1' and C_5' respectively of the sugar molecule
(c) C_2' and C_5' respectively of the sugar molecule
(d) C_5' and C_2' respectively of the sugar molecule
14. Match List I (name of vitamin) with List II (deficiency result/disease) and select the correct answer using the codes given below the lists :
- | List I | List II |
|------------------|--------------------|
| I. Ascorbic acid | A. Beri-beri |
| II. Retinol | B. Cracked lips |
| III. Riboflavin | C. Scurvy |
| IV. Thiamine | D. Night blindness |
- (a) I - B, II - A, III - C, IV - D
(b) I - A, II - B, III - C, IV - D
(c) I - D, II - C, III - B, IV - A
(d) I - C, II - D, III - B, IV - A
15. Chargaff's rule states that in an organism
- (a) amount of adenine (A) is equal to that of guanine (G) and the amount of thymine (T) is equal to that of cytosine (C)
(b) amount of adenine (A) is equal to that of cytosine (C) and the amount of thymine (T) is equal to that of guanine (G)
(c) amount of adenine (A) is equal to that of thymine (T) and the amount of guanine (G) is equal to that of cytosine (C)
(d) amount of all bases are equal
16. Essential amino acids cannot be made by our body. Which of the following is not essential amino acid?
- (a) Leucine (b) Lysine
(c) Serine (d) Histidine

17. How many peptide bond(s) is/are present in the following structure?



- (a) One
(b) Two
(c) Three
(d) None of these
18. The average energy of each hydrogen bond in A-T pair is $x \text{ kcal mol}^{-1}$ and that in G-C pair is $y \text{ kcal mol}^{-1}$. Assuming that no other interaction exists between the nucleotides, the approximate energy required in kcal mol^{-1} to split the following double stranded DNA into two single strands is



[Each dashed line may represent more than one hydrogen bond between the base pair]

- (a) $10x + 9y$ (b) $5x + 3y$
(c) $15x + 6y$ (d) $5x + 45y$
19. A tetrapeptide is made of naturally occurring alanine, serine, glycine and valine. If the C-terminal amino acid is alanine and the N-terminal amino acid is chiral, the number of possible sequences of the tetrapeptide is
- (a) 12 (b) 8
(c) 6 (d) 4
20. Which one of the following is NOT correct?
- (a) D(–) Fructose exists in furanose structure
(b) D(+) Glucose exists in pyranose structure
(c) In sucrose the two monosaccharides are held together by peptide linkage
(d) Maltose is a reducing sugar

Numeric Value Answer

21. A strongly alkaline solution of a monoaminodicarboxylic acid contains how many basic groups ?
22. How many dipeptides are possible from two molecules of a typical α -amino acid ?

23. The optical rotation of α -D fructose is -21° and that of its β -form is -133° . The equilibrium mixture of these anomers has an optical rotation of -92° . What is the % of α -form (upto one decimal place) in its equilibrium mixture?
24. Consider all possible optical isomers of glucose individually. How many moles of periodic acid will be consumed by reaction of one mole of each optical isomer with it?
25. Starch is a polymer of two components: amylose and amylopectin. In the structure of amylopectin straight chain is formed by $Ca - Cb$ glycosidic linkage and branching occurs by $Cc - Cd$ glycosidic linkage. What is the value of $a + b + c + d$?

ANSWER KEY

1	(d)	4	(b)	7	(a)	10	(a)	13	(b)	16	(c)	19	(d)	22	(1)	25	(12)		
2	(c)	5	(c)	8	(a)	11	(c)	14	(d)	17	(b)	20	(c)	23	(36.6)				
3	(b)	6	(c)	9	(a)	12	(b)	15	(c)	18	(a)	21	(3)	24	(80)				

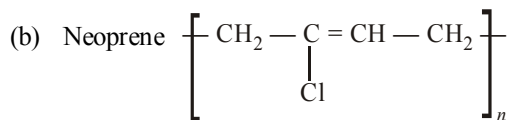
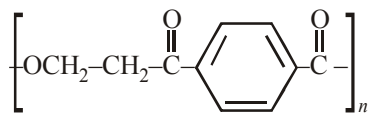
POLYMERS



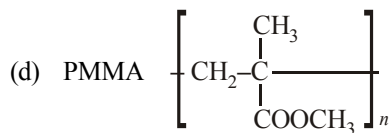
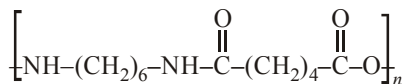
MCQs with One Correct Answer

- The turbidity of a polymer solution measures
 - the light scattered by a solution
 - the light absorbed by a solution
 - the light transmitted by a solution
 - none of the above
- On the basis of mode of formation, polymers can be classified?
 - as addition polymers only
 - as condensation polymers only
 - as copolymers
 - both as addition and condensation polymers
- Which of the following polymers do not involve cross linkages?
 - Melmac
 - Bakelite
 - Polythene
 - Vulcanised rubber
- Which one of the following is not an example of chain growth polymer?
 - Neoprene
 - Buna-S
 - PMMA
 - Glyptal
- $-\text{[NH(CH}_2\text{)}_6\text{NHCO(CH}_2\text{)}_4\text{CO]}_n-$ is a
 - addition polymer
 - thermosetting polymer
 - homopolymer
 - copolymer
- Which one of the following sets forms the biodegradable polymer?
 - $\text{CH}_2=\text{CH}-\text{CN}$ and $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
 - $\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$ and $\text{H}_2\text{N}-(\text{CH}_2)_5-\text{COOH}$
 - $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$ and $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$
 - $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$ and $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$

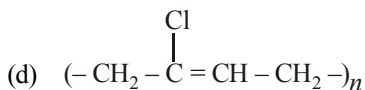
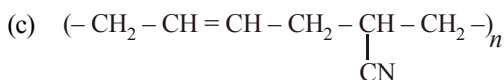
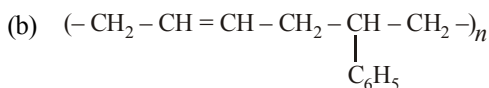
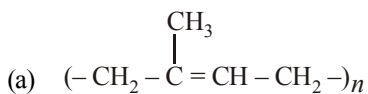
8. The synthetic polymer which resembles natural rubber is
 (a) neoprene (b) chloroprene
 (c) glyptal (d) nylon
9. Which of the following is not a biopolymer ?
 (a) Proteins (b) Rubber
 (c) Cellulose (d) RNA
10. The process of heat-softening, moulding and cooling to rigidity can be repeated for which plastics
 (a) thermoplastics
 (b) thermosetting plastics
 (c) both (a) and (b)
 (d) neither (a) nor (b)
11. Which one of the following statement is *not true*?
 (a) In vulcanization the formation of sulphur bridges between different chains make rubber harder and stronger.
 (b) Natural rubber has the *trans* -configuration at every double bond.
 (c) Buna-S is a copolymer of butadiene and styrene.
 (d) Natural rubber is a 1, 4 - polymer of isoprene.
12. Which of the following is not correctly matched?
 (a) Terylene



(c) Nylon-66



13. Which of the following statements is false?
 (a) Artificial silk is derived from cellulose.
 (b) Nylon-66 is an example of elastomer.
 (c) The repeat unit in natural rubber is isoprene.
 (d) Both starch and cellulose are polymers of glucose.
14. The condensation of hexamethylenediamine with sebacoyl chloride at 525 K gives
 (a) nylon-6,20 (b) nylon-6,01
 (c) nylon-6,10 (d) none of these
15. The monomer(s) used in the preparation of orlon, a substitute for wool is/are
 (a) caprolactam
 (b) tetrafluoroethene
 (c) styrene and 1, 3-butadiene
 (d) acrylonitrile
16. Polymer obtained by condensation polymerization is
 (a) Polythene
 (b) Teflon
 (c) Phenol-formaldehyde
 (d) Nitrile rubber
17. Polyvinyl alcohol can be prepared by
 (a) polymerization of vinyl alcohol
 (b) alkaline hydrolysis of polyvinyl acetate
 (c) polymerization of acetylene
 (d) reaction of acetylene with H_2SO_4 in presence of HgSO_4
18. Structure of some important polymers are given. Which one represents buna-S?



19. Match List-I (Monomer) with List II (Polymer) and select the correct answer using the codes given below the lists:

List I	List II
I. Hexamethylenediamine	A. Bakelite
II. Phenol	B. Dacron
III. Phthalic acid	C. Glyptal
IV. Terephthalic acid	D. Melamine
	E. Nylon

Codes:

- (a) I-E, II-A, III-B, IV-C
 (b) I-E, II-A, III-C, IV-B
 (c) I-D, II-C, III-A, IV-B
 (d) I-D, II-C, III-B, IV-A

20. **Statement-1 :** Vinylidene chloride forms isotactic, syndiotactic and atactic polymers.

Statement-2 : Vinylidene chloride does not contain chiral carbon.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (c) Statement -1 is True, Statement-2 is False.
 (d) Statement -1 is False, Statement-2 is True.

ANSWER KEY

1	(a)	3	(c)	5	(d)	7	(a)	9	(b)	11	(b)	13	(b)	15	(d)	17	(b)	19	(b)
2	(d)	4	(d)	6	(b)	8	(a)	10	(a)	12	(a)	14	(c)	16	(c)	18	(b)	20	(d)

CHEMISTRY IN EVERYDAY LIFE



MCQs with One Correct Answer

- Substances used for bringing down temperature in high fever are called
 - pyretics
 - antipyretics
 - antibiotics
 - antiseptics
- Amoxycillin is semi-synthetic modification of
 - penicillin
 - streptomycin
 - tetracyclin
 - chloroamphenicol
- Barbituric acid and its derivatives are well known
 - antipyretics
 - analgesics
 - antiseptics
 - traquillizers
- A medicine which promotes the secretion of urine is called
 - uretic
 - monuretic
 - diuretic
 - triuretic
- A drug effective in the treatment of pneumonia, bronchitis, etc, is
 - streptomycin
 - chloramphenicol
 - penicillin
 - sulphaguanidine
- Which of the following acts as an antioxidant in edible oils ?
 - Vitamin B
 - Vitamin C
 - Vitamin D
 - Vitamin E
- The drug which is effective in curing malaria is
 - quinine
 - aspirin
 - analgin
 - equanil
- Which of the following is an antidiabetic drug?
 - Insulin
 - Penicillin
 - Chloroquine
 - Aspirin
- Further growth of cancerous cells in the body is arrested by
 - physiotherapy
 - chemotherapy
 - electrotherapy
 - psychotherapy
- Sulphaguanidine is used for
 - dysentery
 - urinary infections
 - antiseptic
 - antipyretic
- H_1 – Receptor antagonists is a term associated with :
 - Antiseptics
 - Antihistamins
 - Antacids
 - Analgesics

12. Which one of the following can possibly be used as analgesic without causing addiction and mood modification?
- Diazepam
 - Morphine
 - N-acetyl-para-aminophenol
 - Tetrahydrocannabinol
13. Antiseptics and disinfectants either kill or prevent growth of microorganisms. Identify which of the following statements **is not true**:
- Chlorine and iodine are used as strong disinfectants.
 - Dilute solutions of boric acid and hydrogen Peroxide are strong antiseptics.
 - Disinfectants harm the living tissues.
 - A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant.
14. Structurally biodegradable detergents, should contain
- normal alkyl chain
 - branched alkyl chain
 - phenyl side chain
 - cyclohexyl side chain
15. Match the chemicals in Column I with their uses in Column II.

	Column I		Column II
(A)	Sodium perborate	(I)	Disinfectant
(B)	Chlorine	(II)	Antiseptic
(c)	Bithional	(III)	Milk bleaching agent
(D)	Potassium stearate	(IV)	Soap

- A - I, B - II, C - III, D - IV
- A - II, B - III, C - IV, D - I
- A - III, B - I, C - II, D - IV
- A - IV, B - I, C - II, D - III

ANSWER KEY

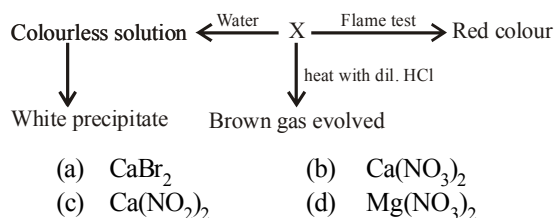
1	(b)	3	(d)	5	(c)	7	(a)	9	(b)	11	(b)	13	(b)	15	(c)				
2	(a)	4	(c)	6	(d)	8	(a)	10	(a)	12	(c)	14	(b)						

PRACTICAL CHEMISTRY



MCQs with One Correct Answer

- A solution of 10 mL $\frac{M}{10}$ FeSO_4 is titrated with KMnO_4 solution in acidic medium the amount of KMnO_4 used will be
 (a) 5 mL of 0.1 M (b) 10 mL of 0.1 M
 (c) 10 mL of 0.5 M (d) 10 mL of 0.02 M
- A red solid is insoluble in water. However, it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of metal appear on the cooler part of the test tube. The red solid is –
 (a) KI (b) Pb_3O_4
 (c) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (d) HgI_2
- The following figure shows a reaction scheme for a compound X, identify X.



- The prussian blue colour obtained in the Lassaigne's test for nitrogen is due to the formation of
 (a) Iron (II) hexacyanoferrate (III)
 (b) Iron (III) hexacyanoferrate (II)
 (c) Iron (III) hexacyanoferrate (III)
 (d) Iron (II) hexacyanoferrate (II)
- A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour, Y is
 (a) $\text{Mg}(\text{NO}_3)_2$ (b) Mg_3N_2
 (c) NH_3 (d) MgO
- Soda extract is prepared by –
 (a) fusing soda and mixture and then extracting with water
 (b) dissolving NaHCO_3 and mixture in dil. HCl
 (c) boiling Na_2CO_3 and mixture in dil. HCl
 (d) boiling Na_2CO_3 and mixture of salt in distilled water
- The salt used for performing bead test in qualitative inorganic analysis is :
 (a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (b) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$
 (c) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
 (d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

8. H_2S gas when passed through a solution of cations containing HCl precipitates the cations of second group in qualitative analysis but not those belonging to the fourth group. It is because
- presence of HCl decreases the sulphide ion concentration.
 - solubility product of group II sulphides is more than that of group IV sulphides.
 - presence of HCl increases the sulphide ion concentration.
 - sulphides of group IV cations are unstable in HCl.
9. An organic compound contains 46.78% of a halogen X. When 2.00 g of this compound is heated with fuming HNO_3 in the presence of AgNO_3 , 2.21 g AgX was formed. The halogen X is [Given: atomic weight of Ag = 108, F = 19, Cl = 35.5, Br = 80, I = 127]
- F
 - Cl
 - Br
 - I
10. In a titration experiment, 10 mL of an FeCl_2 solution consumed 25 mL of a standard $\text{K}_2\text{Cr}_2\text{O}_7$ solution to reach the equivalent point. The standard $\text{K}_2\text{Cr}_2\text{O}_7$ solution is prepared by dissolving 1.225 g of $\text{K}_2\text{Cr}_2\text{O}_7$ in 250 mL water. The concentration of the FeCl_2 solution is closest to [Given: molecular weight of $\text{K}_2\text{Cr}_2\text{O}_7 = 294 \text{ g mol}^{-1}$]
- 0.25 N
 - 0.50 N
 - 0.10 N
 - 0.04 N
11. Sodium nitroprusside, when added to an alkaline solution of sulphide ions, produces purple colour ion due to the formation of
- $\text{Na}[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]$
 - $\text{Na}_2[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]$
 - $\text{Na}_3[\text{Fe}(\text{CN})_5\text{NOS}]$
 - $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
12. In the estimation of sulphur by Carius method, 0.480 g of an organic compound gives 0.699 g of barium sulphate. The percentage of sulphur in this compound is (Atomic masses : Ba = 137, S = 32, O = 16)
- 20%
 - 15%
 - 35%
 - 30%
13. The colour of the solution/precipitate obtained in the elemental analysis of an organic compound and the molecule/ion responsible for the colour are given below. Choose the incorrectly matched pair
- Prussian blue - $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$
 - Black - PbS
 - Violet colour - $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$
 - Yellow - $(\text{NH}_4)_2\text{MoO}_4$
14. A is a lighter phenol and B is an aromatic carboxylic acid. Separation of mixture of A and B can be carried out easily by using a solution of
- sodium hydroxide
 - sodium sulphate
 - calcium chloride
 - sodium bicarbonate
15. To detect iodine in presence of bromine, the sodium extract is treated with NaNO_2 + glacial acetic acid + CCl_4 . Iodine is detected by the appearance of
- yellow colour of CCl_4 layer
 - purple colour of CCl_4
 - brown colour in the organic layer of CCl_4
 - deep blue colour in CCl_4
16. In Kjeldahl's method for the estimation of N_2 , potassium sulphate and copper sulphate are used. On the basis of their functions which of the following statement(s) is/are correct?
- Potassium sulphate raises the bpt. and ensures complete reaction.
 - Copper sulphate acts as catalyst.
 - Potassium sulphate acts as catalyst and copper sulphate raises the bpt.
- Only III is correct
 - I and II are correct
 - Only II is correct
 - None of the given option is correct
17. Which of the following compound will give white precipitate on heating with HNO_3 followed by addition of silver nitrate?
- $(\text{C}_2\text{H}_5)_3\text{NHCl}$
 - 2, 4, 6- Trinitrochlorobenzene
 - Both (a) and (b)
 - Neither (a) nor (b)
18. The ammonia evolved from the treatment of 0.30 g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralization. The organic compound is
- urea
 - benzamide
 - acetamide
 - thiourea

19. A sample of 0.5 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_2SO_4 . The remaining acid after neutralisation by ammonia consumed 80 mL of 0.5 M NaOH. The percentage of nitrogen in the organic compound is
 (a) 14 (b) 28
 (c) 42 (d) 56
20. Which one among the following is not correct?
 (a) Alumina can be used as adsorbent
 (b) Glycerol is purified by steam distillation
 (c) Water trapped in the chromatographic filter paper acts as stationary phase
 (d) In TLC the thickness of the adsorbent layer is about 0.2 mm thickness
21. Diazo compounds, sometimes do not respond Lassaigne's test for nitrogen because
 (a) these are quite stable compounds and do not decompose to elemental nitrogen
 (b) these contain very little carbon
 (c) these form organometallic compounds with sodium
 (d) their nitrogen contents is removed, during heating, in the form of nitrogen gas
22. One gram of an organic nitrogen-containing compound, on heating, decomposes to give 0.2 g of nitrogen. What should be nature of the compound ?
 (a) $\text{C}_3\text{H}_7\text{N}_2\text{Cl}$
 (b) $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$
 (c) $(\text{CH}_3)_3\text{N}$
 (d) Given data is insufficient.
23. In Lassaigne's test for the detection of halogens, the sodium fusion extract is first boiled with concentrated nitric acid. This is
 (a) to remove silver halides
 (b) to decompose Na_2S and NaCN , if present
 (c) to dissolve Ag_2S
 (d) to dissolve AgCN , if formed
24. In the test of basic radical which reagent is used in IVth group?
 (a) $\text{H}_2\text{S} + \text{HCl}$
 (b) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
 (c) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl} + \text{H}_2\text{S}$ gas
 (d) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl} + (\text{NH}_4)_2\text{CO}_3$
25. When excess of dilute NH_4OH is added to an aqueous solution of copper sulphate, an intense blue colour is obtained. This is due to the formation of
 (a) CuSO_4 (b) $\text{Cu}(\text{OH})_2$
 (c) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (d) $(\text{NH}_4)_2\text{SO}_4$

Numeric Value Answer

26. 3.92 g of ferrous ammonium sulphate react completely with 50 mL $\frac{\text{N}}{10}$ KMnO_4 solution. The percentage purity of the sample is
27. When 1.4 g of a nitrogen containing organic compound is heated with CuO in an atmosphere of CO_2 224 mL of free N_2 gas is liberated at STP. The percentage of nitrogen in the compound is
28. 0.25 g of an organic compound on Kjeldahl's analysis gave enough ammonia to just neutralize 10 cm^3 of 0.5 M H_2SO_4 . The percentage of nitrogen in the compound is
29. In Duma's method of estimation of nitrogen 0.35 g of an organic compound gave 55 mL of nitrogen collected at 300 K temperature and 715 mm pressure. The percentage composition of nitrogen in the compound would be : (Aqueous tension at 300 K = 15 mm)
30. 2.79 g of an organic compound when heated in Carius tube with conc. HNO_3 and H_3PO_4 formed converted into MgNH_4PO_4 ppt. The ppt. on heating gave 1.332 g of $\text{Mg}_2\text{P}_2\text{O}_7$. The percentage of P in the compound is

ANSWER KEY

1	(d)	4	(b)	7	(c)	10	(a)	13	(d)	16	(b)	19	(b)	22	(b)	25	(c)	28	(56)
2	(d)	5	(b)	8	(a)	11	(d)	14	(d)	17	(c)	20	(b)	23	(b)	26	(50)	29	(16.46)
3	(c)	6	(d)	9	(c)	12	(a)	15	(b)	18	(a)	21	(d)	24	(c)	27	(20)	30	(13.33)

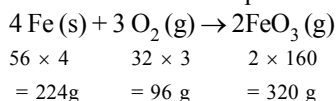
Hints & Solutions

CHAPTER

1

Some Basic Concepts of Chemistry

1. (a) Law of mass conservation states that matter can neither be created nor destroyed, or in a chemical reaction, the total mass of the reactants is equal to the total mass of the products.



2. (b) Statement (b) is correct according to the law of multiple proportions.

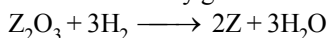
3. (a) Given: Density of water = 1.0 g mL^{-1}
Volume of water = 250 mL
 $\therefore$ Mass of water = Density $\times$ Volume
 $= 1.0 \times 250 = 250 \text{ g}$

18 g of water contains
 $= 6.023 \times 10^{23}$ molecules

$\therefore$ 250 g of water contains

$$= \frac{6.023 \times 10^{23}}{18} \times 250 = 83.65 \times 10^{23} \text{ molecules}$$

4. (d) The reaction may given as



0.1596 g of Z_2O_3 react with $\text{H}_2 = 6 \text{ mg} = 0.006 \text{ g}$

$\therefore$ 1 g of H_2 react with

$$= \frac{0.1596}{0.006} = 26.6 \text{ g of } \text{Z}_2\text{O}_3$$

$\therefore$ Eq. wt. of $\text{Z}_2\text{O}_3 = 26.6$ (from the definition of eq. wt.)

$$\text{Eq. wt. of Z} + \text{Eq. wt. of O} = 26.6$$

$$\Rightarrow \text{Eq. wt. of Z} = 26.6 - 8 = 18.6$$

$$\text{Valency of metal in } \text{Z}_2\text{O}_3 = 3$$

$$\text{Eq. wt. of metal} = \frac{\text{Atomic wt.}}{\text{valency}}$$

$$\therefore \text{At. wt. of Z} = 18.6 \times 3 = 55.8$$

5. (c) $\therefore$ Given of weight of hydrated crystal = 4 g
Weight of water = 0.8

$$\therefore \text{Weight of anhydrous salt} = 4 - 0.8 = 3.2$$

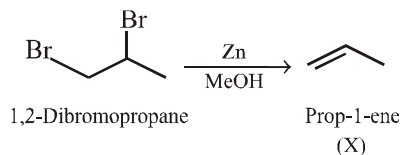
$$\therefore 3.2 \text{ g anhydrous salt} = 0.8 \text{ g of water}$$

$$\therefore 144 \text{ g anhydrous salt}$$

$$= \frac{0.8}{3.2} \times 144 = 36 \text{ g of water}$$

$$\therefore x = \frac{36}{18} = 2$$

6. (b)



Moles of 1, 2-dibromo propane

$$= \frac{20.2}{202} = 0.01 \text{ mole}$$

$$\text{Moles of prop-1-ene} = \frac{358}{42} = 0.085 \text{ mole}$$

$$\% \text{ yield} = \frac{0.085}{0.1} \times 100 = 85\%$$

7. (b) Using the relationship

$$\frac{\text{Mol. mass of oxide}}{\text{Mass of metal in molar mass}} = \frac{\text{mass of oxide}}{\text{mass of metal}}$$

$$\frac{4x+96}{4x} = \frac{10}{5} \Rightarrow x = 24$$

8. (a) No. of molecules in different cases

$$\begin{aligned} \text{(a)} \quad \therefore 22.4 \text{ litre at STP contains} \\ = 6.023 \times 10^{23} \text{ molecules of } \text{H}_2 \end{aligned}$$

$$\begin{aligned} \therefore 15 \text{ litre at STP contains} &= \frac{15}{22.4} \times 6.023 \times 10^{23} \\ &= 4.03 \times 10^{23} \text{ molecules of } \text{H}_2 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \therefore 22.4 \text{ litre at STP contains} \\ = 6.023 \times 10^{23} \text{ molecules of } \text{N}_2 \end{aligned}$$

$$\begin{aligned} \therefore 5 \text{ litre at STP contains} &= \frac{5}{22.4} \times 6.023 \times 10^{23} \\ &= 1.344 \times 10^{23} \text{ molecules of } \text{N}_2 \end{aligned}$$

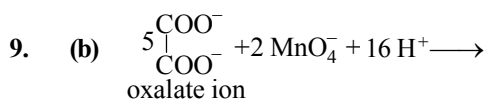
$$\text{(c)} \quad \therefore 2 \text{ g of } \text{H}_2 = 6.023 \times 10^{23} \text{ molecules of } \text{H}_2$$

$$\begin{aligned} \therefore 0.5 \text{ g of } \text{H}_2 &= \frac{0.5}{2} \times 6.023 \times 10^{23} \\ &= 1.505 \times 10^{23} \text{ molecules of } \text{H}_2 \end{aligned}$$

$$\text{(d)} \quad \text{Similarly } 10 \text{ g of } \text{O}_2 \text{ gas}$$

$$\begin{aligned} &= \frac{10}{32} \times 6.023 \times 10^{23} \text{ molecules of } \text{O}_2 \\ &= 1.88 \times 10^{23} \text{ molecules of } \text{O}_2 \end{aligned}$$

Thus (a) will have maximum number of molecules



From above equation 2 moles MnO_4^- required to oxidise 5 moles of oxalate.

Thus number of moles of MnO_4^- required to oxidise one mole of oxalate $= 2/5 = 0.4$

10. (b) Suppose weight of metallic chloride = 100 g

Then weight of metal = 47.22 g

Weight of chlorine = $100 - 47.22 = 52.78 \text{ g}$

$\therefore$ Equivalent weight of metal

$$= \left(\frac{\text{Mass of Metal}}{\text{Mass of chloride}} \right) \times 35.5$$

$$= \frac{47.22}{52.78} \times 35.5 = 31.76$$

11. (b) Volume of a spherical water droplet

$$= \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (1)^3$$

$$V = \frac{4}{3} \pi \text{ cm}^3$$

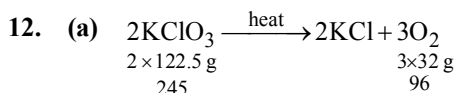
Mass of spherical water droplet

$$= P \times V = 1 \times \frac{4}{3} \pi = \frac{4}{3} \pi \text{ g}$$

18 g of water = 1 mol

no. of moles of water in a spherical water droplet

$$= \frac{4\pi}{3 \times 18} = \frac{2\pi}{27}$$



48 g of oxygen will be produced from 122.5 g of KClO_3

$\therefore$ Amount of 80% KClO_3 needed

$$= \frac{100}{80} \times 122.5 = 153.12 \text{ g}$$

13. (b) Meq. of $\text{CaCO}_3 = \frac{10}{50} \times 1000 = 200$

Meq. of $\text{HCl} = 250 \times 1 = 250$

Meq. of HCl left in the solution $= 250 - 200 = 50$

$\therefore$ Meq. of KOH required $V \times 2 = 50$

$\therefore V = 25 \text{ mL}$

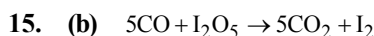
14. (c) The relation between molarity (M) and molality (m) is

$$d = M \left(\frac{1}{m} + \frac{M_2}{1000} \right), \quad M_2 = \text{Mol. mass of solute}$$

On putting value

$$1.252 = 3 \left(\frac{1}{m} + \frac{58.5}{1000} \right)$$

on solving $m = 2.79$



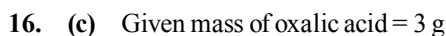
1 mol of $\text{I}_2 = 1$ mol of $\text{I}_2\text{O}_5 = 5$ mol of CO

$$\text{Hence, mol of CO} = 5 \times \frac{2.54}{254} = 0.05$$

$$\text{Mass of CO} = 0.05 \times 28 = 1.4 \text{ g}$$

$$\text{Mass of CO}_2 = 2 - 1.4 = 0.6 \text{ g}$$

$$\text{Mass \% of CO}_2 = \frac{0.6}{2} \times 100 = 30\%$$



Molar mass of oxalic acid,

$$(\text{COOH})_2 \cdot 2\text{H}_2\text{O} = 126$$

$$\text{We know that, density} = \frac{\text{Mass}}{\text{Volume}}$$

$$V = 250 \text{ mL}$$

$$\rho = \frac{M}{V} \Rightarrow M = \rho \times V$$

$$\therefore \text{Mass of solvent} = \text{Density} \times \text{Volume}$$

$$\text{Molality (m)} = \frac{n_{\text{solute}}}{m_{\text{solvent}}} = \frac{m/M}{\rho V}$$

$$= \frac{3/126}{1.9 \times 250 \times 10^{-3}}$$

$$= \frac{10^3}{42 \times 19 \times 25}$$

$$m = 0.05$$

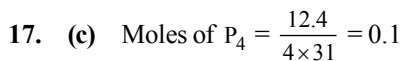
$$n_f \text{ of oxalic acid} = 2$$

$$\text{Normality (N)} = \text{Molality} \times n_f$$

$$N = \frac{n}{V(\text{in L})} \times n_f$$

$$= \frac{3/126}{250 \times 10^{-3}} \times 2$$

$$= \frac{40}{42} \times 2 = 0.19$$

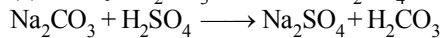
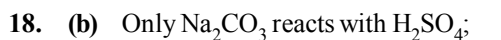


$$\text{Moles of O}_2 = \frac{12.8}{32} = 0.4$$

Let x moles of P_4 form P_4O_6 .

Then, moles of O_2 required $= x \times 3 + (0.1 - x) \times$

$$5 = 0.4 (\text{given}) \Rightarrow x = 0.05$$



$$\text{m-moles of Na}_2\text{CO}_3 = \text{m-moles of H}_2\text{SO}_4$$

$$= 20 \times 0.1 = 2$$

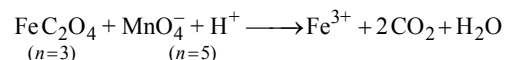
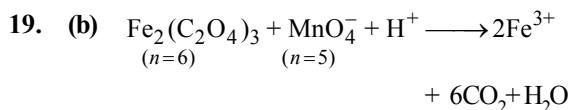
$$\text{m-moles of Na}_2\text{CO}_3 \text{ in 250 mL solution}$$

$$= \frac{250}{25} \times 2 = 20$$

$$\text{wt. of Na}_2\text{CO}_3 = 20 \times 106 \times 10^{-3} = 2.12 \text{ g}$$

$$\% \text{ of Na}_2\text{CO}_3 = \frac{2.12}{5} \times 100 = 42.4$$

$$\therefore \% \text{ Na}_2\text{SO}_4 = 100 - 42.4 = 57.6$$

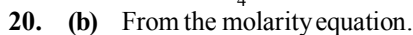


Total equivalents of $(\text{Fe}_2(\text{C}_2\text{O}_4)_3 + \text{FeC}_2\text{O}_4)$

$$= \text{equivalents of KMnO}_4$$

$$\therefore 0.5 \times 6 + 0.5 \times 3 = x \times 5$$

$$\text{moles of KMnO}_4 = x = 0.9$$

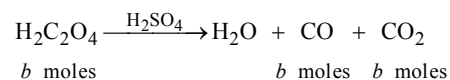
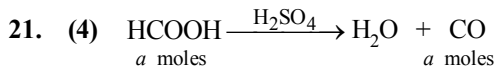


$$M_1V_1 + M_2V_2 = MV$$

Let M be the molarity of final mixture,

$$M = \frac{M_1V_1 + M_2V_2}{V} \text{ where } V = V_1 + V_2$$

$$M = \frac{480 \times 1.5 + 520 \times 1.2}{480 + 520} = 1.344 \text{ M}$$



Total number of moles of gases formed

$$= a + 2b$$

Moles of gas (CO_2) absorbed by $\text{KOH} = b$

$$\text{Hence, } b = \frac{1}{6}(a + 2b) [\text{Volume} \propto \text{No. of moles}]$$

$$a/b = 4$$



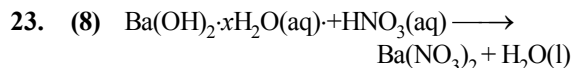
$$\text{Molar mass of } \text{M}_x\text{Cl}_y = 85.5 \times 2 = 171 \text{ g/mol}$$

$$(M)(x) + (35.5)(y) = 171 \quad \dots (i)$$

$$\frac{\text{Mass of chlorine}}{\text{Mass of metallic chloride}} = \frac{0.835}{1}$$

$$= \frac{35.5y}{M(x) + (35.5)(y)}$$

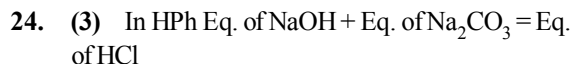
On solving $y = 4$ and $x = 1$
 $x + y = 5$



Eq. of $\text{Ba(OH)}_2 \cdot x\text{H}_2\text{O} = \text{eq. of HNO}_3$

$$\frac{0.7875}{(171 + 18x)} \times 2 = \frac{1}{4} \times \frac{20}{1000}$$

$$x = 8$$



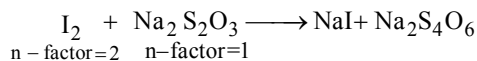
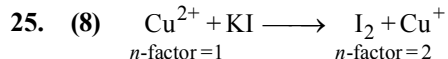
$$(0.5)(1) + (0.5)(1) = (x)(1) \quad x = 1$$

In MeOH, Eq. of $\text{NaOH} + \text{Eq. of Na}_2\text{CO}_3 + \text{Eq. of NaHCO}_3 = \text{Eq. of HCl}$

$$(0.5)(1) + (0.5)(2) + (0.5)(1) = (y)(1)$$

$$y = 2$$

$$x + y = 3$$



Eq. of $\text{Cu}^{2+} = \text{Eq. of I}_2 \text{ produced} = \text{Eq. of I}_2 \text{ reacted} = \text{Eq. of Na}_2\text{S}_2\text{O}_3$.

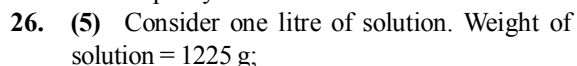
$$(x)(1) = \left(\frac{40}{1000}\right)(0.1)(1)$$

$$x = 4 \times 10^{-3} \text{ mol}$$

$$\text{Mass of pure Cu} = 4 \times 10^{-3} \times 63.5 = 0.254 \text{ g}$$

$$\% \text{ Purity} = \frac{0.254}{0.276} \times 100 = 92\%$$

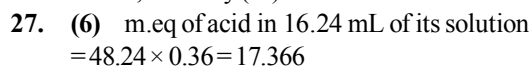
$$\% \text{ Impurity} = 8\%$$



$$\text{Weight of solute} = 1225 \times 0.4 = 490 \text{ g}$$

$$\text{Moles of solute} = \frac{490}{98} = 5.0$$

Hence, molarity (M) = 5.0



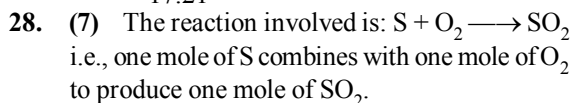
$\Rightarrow$ Mass of acetic acid

$$= 17.366 \times 10^{-3} \times 60 = 1.042 \text{ g}$$

$\Rightarrow$ Mass of solution = 17.21 g

$\Rightarrow$ Mass percentage of acid

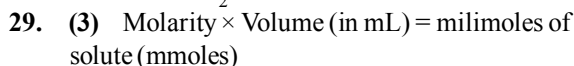
$$= \frac{1.042}{17.21} \times 100 = 6$$



$$\text{Moles of S} = \frac{10}{32} = 0.3125$$

$\Rightarrow$ Moles of $\text{SO}_2 = 0.3125$

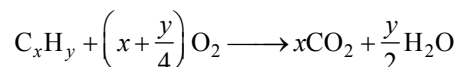
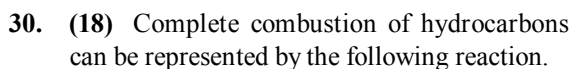
$$\Rightarrow \text{Vol of SO}_2 = 0.3125 \times 22.4 = 7.0 \text{ L}$$



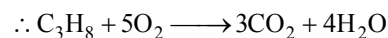
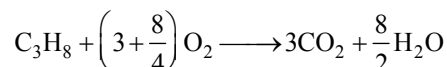
Therefore, mmoles of HNO_3 required

$$= 60 \times 0.2 = 12$$

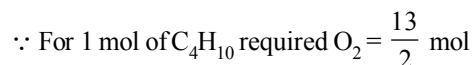
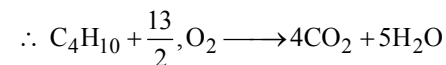
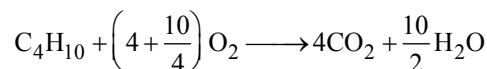
$$\Rightarrow V(\text{mL}) = \frac{\text{m mole}}{\text{molarity}} = \frac{12}{4} = 3 \text{ mL}$$



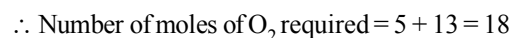
For propane combustion reaction is



Similarly, for butane is



$$= \frac{13}{2} \times 2 = 13 \text{ mol}$$



CHAPTER

2

Structure of Atom

1. (d) In S.I. units the P.E. = $\frac{-Ze^2}{4\pi\epsilon_0 r}$. For Li^{2+} , $Z=3$.

$$\therefore \text{P.E.} = \frac{-3e^2}{4\pi\epsilon_0 r}$$

2. (a) ${}^{20}_{10}\text{Ne}$ contains 10 protons and 10 neutrons

$$\therefore M_1 = 10m_p + 10m_n$$

${}^{40}_{20}\text{Ca}$ contains 20 protons and 20 neutrons

$$\therefore M_2 = 20m_p + 20m_n$$

$$\therefore M_2 = 2M_1$$

3. (d) Isodiaphers have same difference of number of neutrons and protons or $(A - 2Z)$ must be same.
4. (c) Energy of an electron at infinite distance from the nucleus is zero. As an electron approaches the nucleus, the electron attraction increases and hence the energy of electron decreases and thus becomes negative. Thus as the value of n decreases, i.e. lower the orbit is, more negative is the energy of the electron in it.
5. (b) The energy associated with radiation of wave length $975 \text{ \AA}$ is given by

$$E = \frac{12400}{975} \text{ eV} = 12.72 \text{ eV}$$

[For a radiation of wavelength $x \text{ \AA}$, we

$$\text{know } E = \frac{12400}{x} \text{ eV}]$$

$$E = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ eV}$$

$$\text{or } E = 13.6 \left[\frac{1}{1} - \frac{1}{n_2^2} \right] \text{ eV} \quad [\because n_1 = 1]$$

$$\text{or } 12.72 = 13.6 \left[1 - \frac{1}{n_2^2} \right] \quad [\because E = 12.72 \text{ eV}]$$

$$\text{or } 1 - \frac{1}{n_2^2} = \frac{12.72}{13.6}$$

$$\text{or } \frac{1}{n_2^2} = \frac{13.6 - 12.72}{13.6} = \frac{0.88}{13.6} \approx \frac{1}{16}$$

$$\text{or } n_2^2 \approx 16$$

$$\text{or } n_2 \approx 4$$

The transition $n_4 \longrightarrow n_3$ will give the **longest wave length**.

6. (d) Total energy of a revolving electron is the sum of its kinetic and potential energy.

$$\text{Total energy} = \text{K.E.} + \text{P.E.}$$

$$= \frac{e^2}{2r} + \left(-\frac{e^2}{r} \right) = \frac{e^2 - 2e^2}{2r}$$

$$= -\frac{e^2}{2r}$$

7. (a) The frequency distribution of the emitted radiation from black body depends only on its temperature. The maximum value of intensity of radiation emitted will be more for higher temperature. Thus, correct representation is shown in option (a).
8. (b) The maximum number of electrons in the shell with principle quantum number (n) is $2n^2$.

9. (b) For H atom, first Balmer line in series is

$$E_3 - E_2 = \frac{-E_1(\text{H})}{(3)^2} + \frac{E_1(\text{H})}{(2)^2} = \frac{5E_1(\text{H})}{36}$$

For He^+ ion ($Z = 2$)

$$E_6 - E_4 = -\frac{E_1(\text{H}) \times (2)^2}{6^2} - \frac{E_1(\text{H}) \times (2)^2}{4^2}$$

$$= -E_1(\text{H}) \times 2^2 \left| \frac{16 - 36}{16 \times 36} \right|$$

$$= \frac{4 \times 20}{36 \times 16} E_1(\text{H}) = \frac{5E_1(\text{H})}{36}$$

10. (c) $v_n = 2.186 \times 10^6$

$$\Rightarrow 1.093 \times 10^6 = 2.186 \times 10^6 \times \frac{1}{n}; n = 2$$

from Bohr theory we know $2\pi r = n\lambda$

$$= 2\lambda, \text{ where } \lambda = \frac{h}{mv}$$

$$\text{or } r = 0.529 \frac{n^2}{Z} \Rightarrow 0.529 \times 4 \text{ \AA}$$

$\therefore$ Circumference of the orbit

$$\Rightarrow 2 \times 0.529 \times 4 \times 10^{-10}$$

$$\Rightarrow 13.30 \times 10^{-10} \text{ m}$$

11. (b) Angular momentum = $\frac{nh}{2\pi}$

$$3.1652 \times 10^{-34} = \frac{n \times 6.626 \times 10^{-34}}{2\pi};$$

$$n = 3$$

$$\therefore \bar{v} = R \cdot Z^2 \cdot \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right);$$

$$\bar{v} = R \cdot 2^2 \cdot \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \Rightarrow \frac{5R}{9}$$

12. (a) Energy absorbed = $13.6 \times 1.5 = 20.4 \text{ eV}$ out of this 6.8 eV is converted to K.E.

$$6.8 \text{ eV} \Rightarrow 6.8 \times 1.6 \times 10^{-19} \text{ J};$$

$$6.8 \times 1.6 \times 10^{-19} = \text{K.E.} \Rightarrow \left(\frac{1}{2} \right) mv^2$$

$$v = \sqrt{\frac{2\text{KE}}{m}} = \sqrt{\frac{2 \times 1.088 \times 10^{-18}}{9.1 \times 10^{-31}}}$$

$$= 1.54 \times 10^6 \text{ m/s}$$

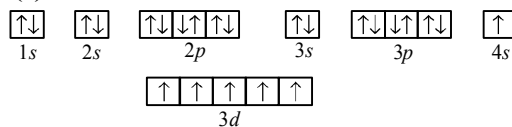
13. (d) According to $(n + l)$ rule, maximum is the $(n + l)$ value higher is the energy level. If two level has same $(n + l)$ value then the level which has higher value of n will have higher energy.

14. (c) (I) Beyond a certain wavelength the line spectrum becomes band spectrum.

(II) For Balmer series $n_1 = 2$

(IV) For calculation of longest wavelength use nearest value of n_2 . Hence for longest wavelength in Balmer series of hydrogen spectrum, $n_1 = 2$ & $n_2 = 3$.

15. (d)



Out of 6 electrons in $2p$ and $3p$ must have one electron with $m = +1$ and $s = -\frac{1}{2}$ but in $3d$ -subshell an orbital having $m = +1$ may have spin quantum no.

Therefore, minimum and maximum possible values are 2 and 3 respectively.

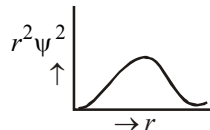
16. (a) As we know that,

$$E_n \propto \frac{-Z^2}{n^2}$$

$$\text{or } E_{2s} \propto \frac{-Z}{(2)^2}$$

Therefore, as the atomic number increases, the energy of orbital decreases. The atomic number of H, Li, Na and K respectively, are 1, 3, 11 and 19. Thus, the correct order of energy of $2s$ -orbitals is $K < Na < Li < H$.

17. (c) $\ell = 2$ represent d orbital for which



18. (b) Given $E_1 = 25 \text{ eV}$ $E_2 = 50 \text{ eV}$

$$E_1 = \frac{hc}{\lambda_1} \quad E_2 = \frac{hc}{\lambda_2} \quad \therefore \frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{25}{50} = \frac{1}{2} \quad \therefore \lambda_1 = 2\lambda_2$$

19. (b) Cathode rays are independent of the nature of the gas and electrodes placed in discharge tube. They travel in a straight line in the absence of external electrical or magnetic field.

20. (c) Energy emitted by the bulb = 600 W = 600 J s⁻¹ (1W = 1 J s⁻¹)
 $\lambda = 331.3 \times 10^{-9} \text{ m}$

$$\text{Energy of one photon} = h\nu = \frac{hc}{\lambda}$$

$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{331.3 \times 10^{-9}}$$

$$= 0.059 \times 10^{-17} \approx 0.06 \times 10^{-17} \text{ J}$$

No. of photon emitted from the lamp per second

$$= \frac{600}{0.06 \times 10^{-17}} = 1.0 \times 10^{21}$$

21. (2) $\Delta\nu = \frac{0.001}{100} \times 30,000 = 0.3 \text{ cm sec}^{-1}$

According to uncertainty principle,

$$\Delta x \cdot \Delta p \approx \frac{h}{4\pi}; \quad \Delta x \cdot \Delta\nu \approx \frac{h}{4\pi m}$$

$$\Delta x \times 9.1 \times 10^{-28} \times 0.3 \approx \frac{6.625 \times 10^{-27} \times 7}{4 \times 22}$$

$$\Delta x \approx 1.93 \text{ cm.} \approx 2$$

22. (2) Energy of photon, $E = h\nu = \frac{hc}{\lambda}$

Here, $c = 3.0 \times 10^8 \text{ ms}^{-1}$

In first case, $\lambda = 800 \text{ nm} = 800 \times 10^{-9} \text{ m}$

$$\therefore E_1 = \frac{(6.626 \times 10^{-34} \text{ Js}) \times (3 \times 10^8 \text{ ms}^{-1})}{800 \times 10^{-9} \text{ m}}$$

$$= 2.48 \times 10^{-19} \text{ J}$$

In second case, $\lambda = 400 \text{ nm} = 400 \times 10^{-9} \text{ m}$

$$\therefore E_2 = \frac{(6.626 \times 10^{-34} \text{ Js}) \times (3 \times 10^8 \text{ ms}^{-1})}{400 \times 10^{-9} \text{ m}}$$

$$= 4.91 \times 10^{-19} \text{ J}$$

Ratio of energy of first and second radiations,

$$\frac{E_1}{E_2} = \frac{2.48 \times 10^{-19} \text{ J}}{4.97 \times 10^{-19} \text{ J}} = \frac{1}{2}$$

$$E_1 : E_2 = 1 : 2 \text{ or } E_2 = 2E_1$$

Thus, energy of the radiation with wavelength 400 nm is twice that of the radiation of wavelength 800 nm.

23. (3) In the 1st Bohr orbit of H:

$$v = 2.18 \times 10^6 \text{ ms}^{-1}.$$

Now, let us consider that in Li²⁺ the electron is in nth orbit. Speed of electron in nth Bohr orbit of Li²⁺ is

$$v = (Li^{2+}) = 2.18 \times 10^6 \times \frac{3}{n}$$

Now, applying the condition of equal speed :

$$2.18 \times 10^6 \times \frac{3}{n} = 2.18 \times 10^6 \Rightarrow n = 3$$

24. (3) $\Delta E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.03 \times 10^{-7}}$
 $= 1.93 \times 10^{-18} \text{ J}$

$$\text{Also: } \Delta E = kZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 2.18 \times 10^{-18}$$

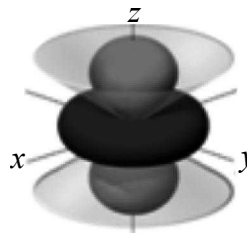
$$\left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Rightarrow 1.93 \times 10^{-18} = 2.18 \times 10^{-18} \left(1 - \frac{1}{n^2} \right)$$

$$\Rightarrow n = 3.$$

25. (1) For 4d_{z²}-orbital, $n = 4$ and $l = 2$.

$$\text{Number of radial nodes} = n - l - 1 = 4 - 2 - 1 = 1$$



Nodal planes are planes having zero probability of finding electron. Due to unusual shape of d_{z²}-orbital, it does not have a nodal plane but it does possess 2 nodal cones or 2 angular nodes.

$$26. \quad (576) \quad \lambda = \frac{h}{mv} \Rightarrow v_1 = \frac{h}{m\lambda_1} \text{ and } v_2 = \frac{h}{m\lambda_2}$$

$$E = \frac{1}{2}mv^2 \Rightarrow E_1 - E_2 = \frac{1}{2}m(v_1^2 - v_2^2)$$

$$\Rightarrow \frac{hc}{\lambda} = \frac{h^2}{2m} \left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2} \right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{h}{2mc} \left(\frac{1}{4^2} - \frac{1}{6^2} \right) \frac{1}{(10^{-6})^2}$$

$$\Rightarrow \frac{1}{\lambda} = \frac{h}{2mc} \left(\frac{20}{36 \times 16} \right) \frac{1}{10^{-12}}$$

$$\lambda = \frac{mc}{h} \left(\frac{36 \times 16}{10} \right) 10^{-12} \text{ m} = \frac{mc}{h} \times 576 \times 10^{-11} \text{ m}$$

$$\Rightarrow x = 576$$

$$27. \quad (653.4) \quad \bar{\nu} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda_H} = R \times 1^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5}{36} R$$

$$\frac{1}{\lambda_{Na^+}} = R \times 11^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{363}{4} R$$

$$\text{Hence } \frac{\lambda_H}{\lambda_{Na^+}} = \frac{\frac{363R}{4}}{\frac{5R}{36}} = 653.4$$

$$28. \quad (7.45)$$

$$\text{K.E. of photoelectron} = e^- \times \text{stopping potential}$$

$$= 1.602 \times 10^{-19} \times 300$$

$$= 4.806 \times 10^{-17} \text{ J}$$

$$\text{K.E. of photoelectron} = h(\nu - \nu_0) = h \left(\nu - \frac{c}{\lambda_0} \right)$$

$$(\nu_0 = \text{Threshold frequency})$$

$$4.806 \times 10^{-17} = 6.626 \times 10^{-34} \left(\nu - \frac{3 \times 10^8}{1500 \times 10^{-10}} \right)$$

$$= 6.626 \times 10^{-34} (\nu - 2 \times 10^{15})$$

$$\nu - 2 \times 10^{15} = \frac{4.806 \times 10^{-17}}{6.626 \times 10^{-34}} = 72.53 \times 10^{15}$$

$$\Rightarrow \nu = 7.45 \times 10^{16} \text{ Hz}$$

$$29. \quad (6.604) \quad \Delta E = E_3 - E_2 = h\nu = \frac{hc}{\lambda} \text{ or}$$

$$\lambda = \frac{hc}{E_3 - E_2}$$

$$\text{Given } E_2 = -5.42 \times 10^{-12} \text{ erg}$$

$$E_3 = -2.41 \times 10^{-12} \text{ erg}$$

$$\therefore \lambda = \frac{6.626 \times 10^{-27} \times 3 \times 10^{10}}{-2.41 \times 10^{-12} - (-5.42 \times 10^{-12})}$$

$$= \frac{19.878 \times 10^{-17}}{3.01 \times 10^{-12}} = 6.604 \times 10^{-5} \text{ cm} = 6.604 \text{ \AA}$$

$$30. \quad (80) \quad \text{Let the \% of isotope with At. wt. 10.01} = x$$

$$\therefore \% \text{ of isotope with At. wt. 11.01} = (100 - x)$$

$$\text{At. wt. of boron} = \frac{x \times 10.01 + (100 - x) \times 11.01}{100}$$

$$\Rightarrow 10.81 = \frac{x \times 10.01 + (100 - x) \times 11.01}{100}$$

$$\therefore x = 20$$

$$\text{Hence \% of isotope with At. wt. 10.01} = 20\%$$

$$\therefore \% \text{ of isotope with At. wt. 11.01} = 100 - 20 = 80\%$$

CHAPTER

3

Classification of Elements & Periodicity in Properties

- (d) Most of the transition metal compounds (ionic as well as covalent) are coloured both in the solid state and in aqueous solution in contrast to the compounds of *s* and *p*-block elements due to the presence of incomplete *d*-subshell.
- (c) In *iso*-electronic species, cation with greater positive charge will have a smaller radius and anion with greater negative charge will have the largest radius.
Thus, the order of ionic radii will be,
$$\text{N}^{3-} > \text{O}^{2-} > \text{F}^{-} > \text{Na}^{+}$$
- (c) Elements (a), (b) and (d) belong to the same group since each one of them has two electrons in the *s* sub shell. In contrast, element (c) has seven electrons in the valence shell and hence does not lie in the same group in which elements (a), (b) and (d) lie.
- (c) atomic number 9 is for F so its halogen series.
- (a) There are four elements which is lighter than the rare-earth elements that are:
 - eka – boron (Eb) – scandium
 - eka – aluminium (Ea) – Gallium
 - eka – manganese (Em) – Technetium
 - eka – Silicon (Es) – Germanium
- (b) The atomic number 85 lies in the sixth period and since the next halogen, if discovered, will lie in the 7th period. Hence, its atomic number must be $85+32=117$.
- (c) Atomic number of the given element = 15; group no. = 10 + 5 (valence electrons) = 15, period = 4th Therefore atomic number of the element below the above element = $15 + 18 = 33$.
- (a) O^{2-} and F^{-} are isoelectronic. Hence have same number of shells, therefore greater the nuclear charge smaller will be the size i.e.,
$$\text{O}^{2-} > \text{F}^{-}$$
 further Li^{+} and B^{3+} are isoelectronic, therefore
$$\text{Li}^{+} > \text{B}^{3+}$$
 Hence the correct order of atomic size is.
$$\text{O}^{2-} > \text{F}^{-} > \text{Li}^{+} > \text{B}^{3+}$$
- (a) Ionisation potential of nitrogen is more than that of oxygen. This is because nitrogen has more stable **half-filled** *p*-orbitals. ($\text{N} = 1s^2, 2s^2, 2p^3$, $\text{O} = 1s^2, 2s^2, 2p^4$)
- (a) In Cu it has completely filled *d*-orbital so highest energy is absorbed when it convert in Cu^{+} ion.
- (b) $\text{K}^{+} \rightarrow \text{K}^{2+} + e^{-}$. Since e^{-} is to be removed from stable configuration.
- (b) Along the period, I.P. generally increases but not regularly. Be and B are exceptions. First I.P. increases in moving from left to right in a period, but I.P. of B is lower than Be.
- (b) IE_2 of Mg is lower than that of Na because in case of Mg^{+} , 3*s*-electron has to be removed whereas in case of Na^{+} , an electron is removed from the stable inert gas configuration which is difficult.

14. (b) In case of Ne, the electron is being removed from a species with a charge of +1 whereas in case of Na^+ , the electron is being removed from a species with charge of +2. Hence IE_1 of $\text{Na}^+ > \text{IE}_1$ of Ne.
15. (a) Zero, because of the stable electronic configuration the noble gases do not show any force of attraction towards the incoming electron.
16. (a) Acidic character of oxide $\propto$ Non-metallic nature of element.
Non-metallic character increases along the period. Hence order of acidic character is

$$\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$$
17. (a) O^- ion exerts a force of repulsion on the incoming electron. The energy is required to overcome it.
18. (c) Basicity of oxides decreases in a period and increases in a group.
 $\therefore \text{SnO}_2, \text{Al}_2\text{O}_3$ and ZnO are amphoteric oxides.
19. (a) (i) < (iii) < (ii) < (iv). In case of half filled or completely filled subshell, not easy to add electron so (i) & (iii) has less tendency to accept electron. Thus option (a) is correct.
20. (c) The basic character of metal oxides decreases from left to right in a period and increases down the group.
21. (106) $Unh = Un - \text{nil} - \text{hexium} = 106$
22. (731) The I.E. of Ag will be less than Cu, due to bigger size of Ag. From Ag to Au, $14f$ electrons are added, which provides very poor shielding effect. Therefore, due to high nuclear charge, the I.E. of $\text{Au} > \text{Ag}$. Correct order: $\text{Au} (890) > \text{Cu} (745) > \text{Ag} (731)$.
23. (7) Long form of periodic table contains 18 groups and 7 periods (completely filled)
24. (7) Third period contains total 8 elements but 'Si' is metalloid.
25. (2) (iv) $\text{Li} < \text{Li}^+$ (vii) $\text{Ba} < \text{Sr}$

CHAPTER

4

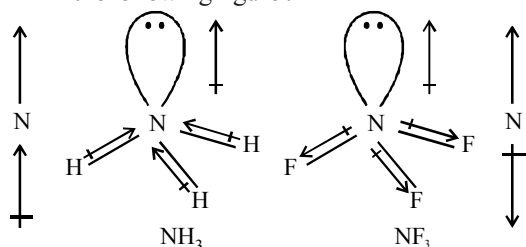
Chemical Bonding and Molecular Structure

1. (c) Percent ionic character is given by following equation.

$$\% \text{ of ionic character} = 16(X_A - X_B) + 3.5(X_A - X_B)^2$$

From the above relation, it is clear that as soon as $(X_A - X_B)$ increases, % ionic character will also increase. Therefore, curve C show a correct path.

2. (a) In NH_3 the atomic dipole (orbital dipole due to lone pair) and bond dipole are in the same direction whereas in NF_3 these are in opposite direction so in the former case they are added up whereas in the latter case net result is reduction of dipole moment. It has been shown in the following figure :



3. (c) $\text{B.O.} = \frac{1}{2}(N_b - N_a)$

where, N_b = electrons in bonding orbitals N_a = electrons in antibonding orbitals.

The electronic configuration of CO (14) is

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\therefore \text{B.O.} = \frac{1}{2}(10 - 4) = \frac{6}{2} = 3$$

The electronic configuration of O_2^{2-} (18) is

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^2 = \pi^* 2p_y^2$$

$$\text{B.O.} = \frac{1}{2}(10 - 8) = 1$$

The electronic configuration of O_2^- (17) is

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi^* 2p_x^2 = \pi^* 2p_y^1$$

$$\text{B.O.} = \frac{1}{2}(10 - 7) = \frac{3}{2} = 1.5$$

The electronic configuration of B_2 (10) is

$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^1 = \pi 2p_y^1$$

$$\text{B.O.} = \frac{1}{2}[6 - 4] = \frac{2}{2} = 1$$

Electronic configuration of N_2^+ (13) is

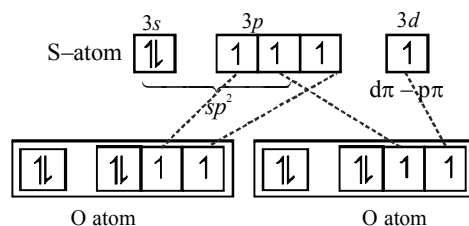
$$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^1$$

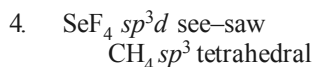
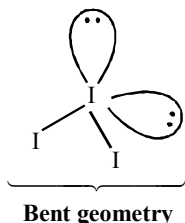
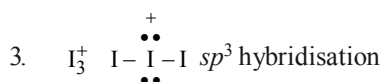
$$\text{B.O.} = \frac{1}{2}[9 - 4] = \frac{5}{2} = 2.5$$

Thus, option (c) is correct.

4. (a) 1. PH_5 doesn't exist because d -orbital of P interact with s -orbital of H. Bond formed is not stable and not energetically favorable and in case of BiCl_5 + 5 oxidation state is not stable due to inert pair effect.

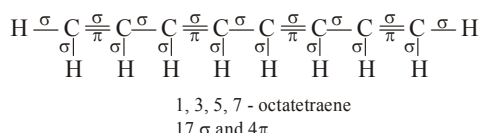
2.





Thus, option (a) is correct

5. (b)



6. (a) $\text{R}-\text{N}=\text{C}=\text{O}$

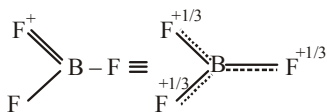
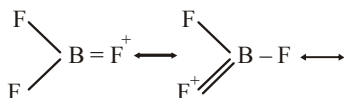
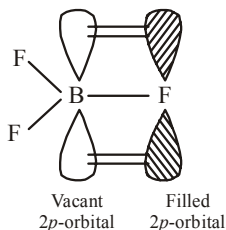
Nitrogen has 2 sigma bond and a lone pair of electron

$\therefore$ hybridisation is sp^2 . Carbon has 2 sigma bond

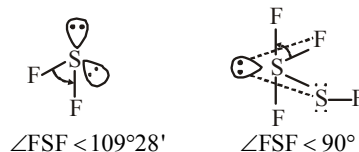
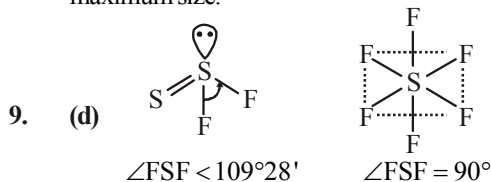
$\therefore$ hybridisation is sp . Oxygen has 1 sigma bond and two lone pair of electron $\therefore$ hybridisation is sp^2 .

NOTE: π -bonding electron does not take part in hybridisation.

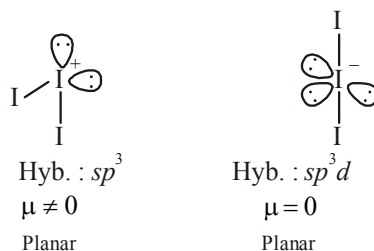
7. (b) The delocalised $p\pi-p\pi$ bonding between filled p -orbital of F and vacant p -orbital of B leads to shortening of B-F bond length which results in higher bond dissociation energy of the B-F bond.



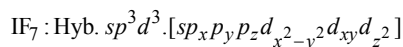
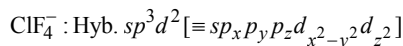
8. (b) As the size of anion increases polarizability of anion increase. Among given anions I^- has maximum size.



10. (d)



11. (c) IO_2F_2^- : Hyb. sp^3d , [$\equiv sp_x p_y p_z d_{z^2}$]



d_{xy} : orbital has two nodal planes xz and yz .

12. (b) SO_3

$$\left. \begin{array}{l} \sigma \text{ bond} = 3 \\ \pi \text{ bond} = 3 \end{array} \right\} \text{number of covalent bond} = 6$$

$$\text{Lone pair} = \frac{1}{2} (6 - 6) = 0$$

$$\text{Hybridization} = 3 + 0 = 3 (sp^2)$$

$$\text{Unhybridized } p\text{-orbitals left} = 1 \text{ so } p\pi-p\pi = 1$$

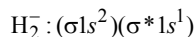
$$\pi \text{ bond} = 3$$

$$p\pi-d\pi = 3 - 1 = 2$$

Thus, SO_3 has 3 σ , one $p\pi-p\pi$ and two $p\pi-d\pi$ bonds.

13. (c) H_2^+ : ($\sigma 1s^1$)

$$\therefore \text{B.O.} = \frac{1}{2} (1 - 0) = \frac{1}{2}$$



$$\therefore \text{B.O.} = \frac{1}{2} (2 - 1) = \frac{1}{2}$$

Even though the bond order of H_2^+ and H_2^- are equal but H_2^+ is more stable than H_2^- as in the latter, an electron is present in the antibonding (σ^*1s) orbital of higher energy, which results in repulsion and decrease the stability.

14. (c) Bond energy $\propto$ Bond order
Bond order can be determined by MO configuration.

NO : No. of electrons = 7 + 8 = 15

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^1$$

$$\therefore \text{Bond order} = \frac{10-5}{2} = 2.5$$

NO^+ : No. of electrons = 15 - 1 = 14

Delete $\pi^* 2p_x^1$ from NO configuration

$$\therefore \text{Bond order} = \frac{10-4}{2} = 3$$

NO^- : No. of electrons = 15 + 1 = 16

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2,$$

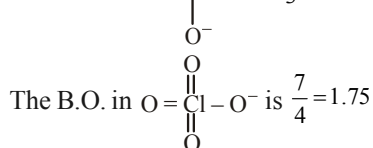
$$\pi^* 2p_x^1 = \pi^* 2p_y^1$$

$$\therefore \text{Bond order} = \frac{10-6}{2} = 2$$

$\therefore$ Bond order; hence bond strength
 $NO^+ > NO > NO^-$

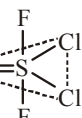
15. (c) The B.O in $Cl-O^-$ is 1
The B.O in $O=Cl-O^-$ is 1.5

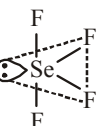
The B.O. in $O=Cl=O$ is $\frac{5}{3} = 1.66$

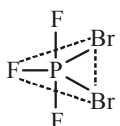


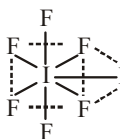
The B.O. in $O=Cl-O^-$ is $\frac{7}{4} = 1.75$

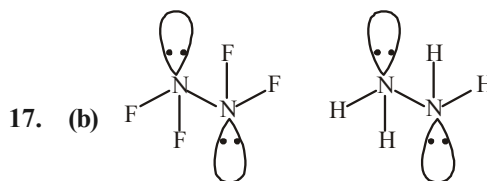
The bond length increases as B.O. decreases.

16. (a) $\Rightarrow$  $\Rightarrow$ both S—F bonds are of equal length.

$\Rightarrow$  $\Rightarrow$ Equatorial Se—F bond is shorter than axial Se—F bond.

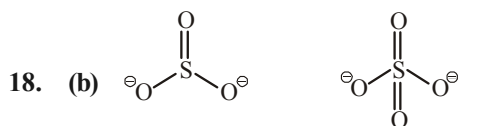
$\Rightarrow$  $\Rightarrow$ Equatorial P—F bond is shorter than axial P—F bond.

$\Rightarrow$  $\Rightarrow$ Equatorial I—F bond is longer than axial I—F bond.



It can be explained on the basis of Bent's rule. In N_2F_4 , N—N bond has more s -character hence bond length decreases.

While in N_2H_4 , N—N bond has less s -character (i.e., more p -character), hence bond lengths increases.



$$B.O. = \frac{4}{3} = 1.33 \quad B.O. = \frac{6}{4} = 1.5$$

$$\left[B.L. \propto \frac{1}{B.O.} \right]$$

$$\therefore SO_3^{2-} > SO_4^{2-} > SO_2 > SO_3$$

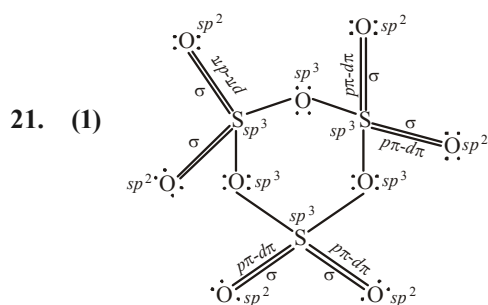
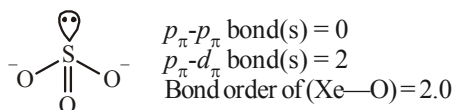
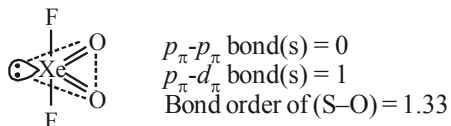
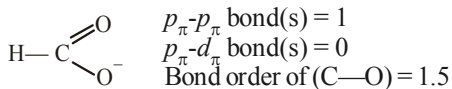
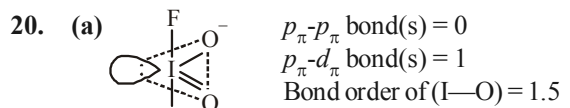
Bent's Rule

19. (c)

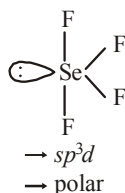
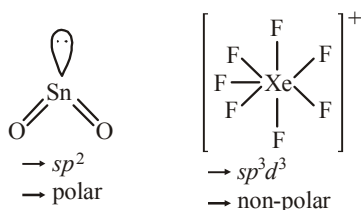
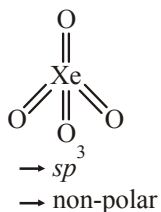
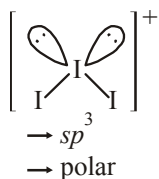
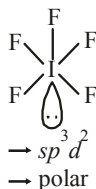
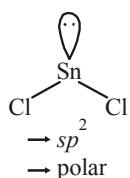
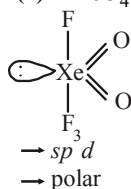
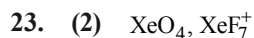
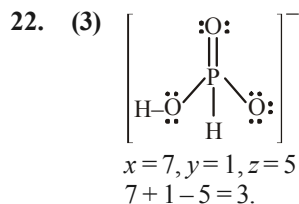
Compounds Hybridization of central atom

CO_2	sp
$HCOOH$	sp^2
BF_3	sp^2
BF_4^-	sp^3
NH_3	sp^3
NH_4^+	sp^3
PCl_3	sp^3
PCl_5	sp^3d

Hybridization of NH_3 and NH_4^+ remains unchanged after transformation i.e. sp^3 .

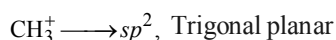
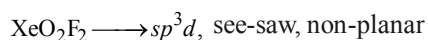
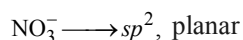
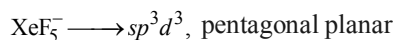
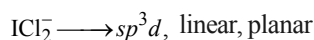
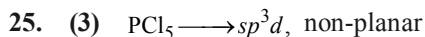


$$p = 6, q = 6; \frac{p}{q} = 1$$



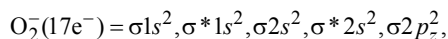
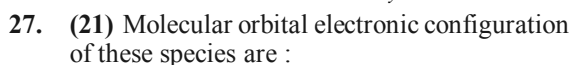
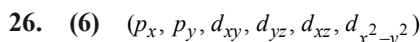
24. (5)

Species	Nodal Plane
3s	0
4d _{z²}	0
4s	0
2p _x	1
3d _{x²-y²}	2
4p _z	1
4d _{xy}	2
3p _y	1
$\therefore$ Total no. of orbitals	
$= 3s, 4d_{xy}, 4d_{z^2}, 3d_{x^2-y^2}, 4s$ (Five)	



$$a = 4, b = 5, c = 3$$

$$\text{so, } \frac{a+b}{c} = 3$$



$$\pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^2$$

$$O_2(16e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2,$$

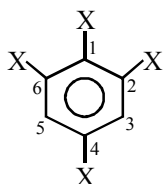
$$\pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$$

$$O_2^{2-}(18e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2,$$

$$\pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^2$$

Therefore number of antibonding electrons are 7, 6 and 8 respectively and the sum is 21.

28. (1.6) The substituent in *o*- and *p*- position, will cancel each others dipole moment.



The angle between 2 and 6 positions is 120° .
The resultant dipole moment generated by these two is:

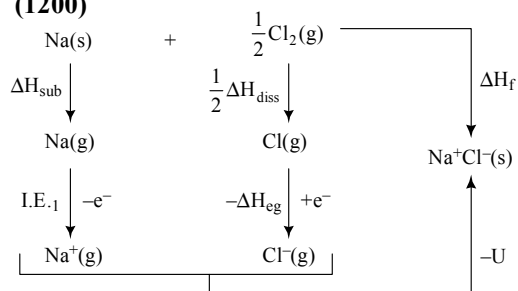
$$\mu = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos \theta}$$

$$= \sqrt{1.6^2 + 1.6^2 + 2 \times 1.6 \times 1.6 \times \cos(120)} = 1.6 \text{ D}$$

29. (50) The hybridization of S in SF_6 is sp^3d^2 .

$$\% p \text{ character} = \frac{3}{6} \times 100 = 50$$

30. (1200)



$$\Delta H_f = \Delta H_{\text{sub}} + \frac{1}{2} \Delta H_{\text{diss}} + \text{I.E.}_1 - \Delta H_{\text{eg}} - U$$

$$\Rightarrow -200 = 650 + \frac{1}{2} \times 400 + 500 - 350 - U$$

$$\Rightarrow U = 1200 \text{ kJ/mol}$$

CHAPTER

5

States of Matter

1. (a) Given initial volume (V_1) = 600 c.c.; Initial pressure (P_1) = 750 mm of Hg and final volume (V_2) = 500 c.c. according to Boyle's law,

$$P_1 V_1 = P_2 V_2$$

$$\text{or } 750 \times 600 = P_2 \times 500$$

$$\text{or } P_2 = \frac{750 \times 600}{500} = 900 \text{ mm of Hg}$$

Therefore increase of pressure = (900 – 750)
= 150 mm of Hg

2. (d) n of $O_2 = \frac{16}{32} = \frac{1}{2}$

$$n \text{ of } H_2 = \frac{3}{2}$$

$$\text{Total no. of moles} = \frac{3}{2} + \frac{1}{2} = 2$$

$$\text{The total volume occupied} = 2 \times 22.444.8L \\ = 44800mL$$

3. (b) Pressure exerted by hydrogen will be proportional to its mole fraction.

$$\text{Mole fraction of } H_2 = \frac{\frac{w}{2}}{\frac{w}{16} + \frac{w}{2}} = \frac{8}{9}$$

4. (b) Under identical conditions, $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$
As rate of diffusion is also inversely proportional

$$\text{to time, we will have, } \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

$$(a) \text{ Thus, For He, } t_2 = \sqrt{\frac{4}{2}}(5s) = 5\sqrt{2}s \neq 10s$$

$$(b) \text{ For } O_2, t_2 = \sqrt{\frac{32}{2}}(5s) = 20s$$

$$(c) \text{ For CO, } t_2 = \sqrt{\frac{28}{2}}(5s) \neq 25s$$

$$(d) \text{ For CO}_2, t_2 = \sqrt{\frac{44}{2}}(5s) \neq 55s$$

5. (c) According to question :

For initial stage of a gas,

$$P_1 = 1 \text{ atm, } V_1 = 24.6 \text{ litres,}$$

$$T_1 = 273 + 27 = 300 \text{ K}$$

For final stage of a gas,

$$P_2 = 10 \text{ atm, } V_2 = ?$$

$$T_2 = 327 + 273 = 600 \text{ K}$$

By gas equation,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1 \times 24.6}{300} = \frac{10 \times V_2}{600}$$

$$\Rightarrow V_2 = \frac{24.6 \times 600}{300 \times 10}$$

$$= 4.92 \text{ litres.}$$

6. (b) At the same conditions of T and P , $V \propto n$

$$n_{N_2} = \frac{14}{28} = \frac{1}{2}; V_{N_2} \propto \frac{1}{2}$$

$$n_{O_3} = \frac{36}{48} = \frac{3}{4}; V_{O_3} \propto \frac{3}{4}$$

$$\text{Hence, } V_{N_2}/V_{O_3} = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$$

$$\therefore 3V_{N_2} = 2V_{O_3}$$

7. (b) The correct graph which represents the van der Waal's gas under the given condition is option (b).

8. (c) Total pressure; $P = \frac{nRT}{V} = \frac{4 \times 0.0821 \times 300}{100}$
 $= 0.985 \text{ atm.}$

$$P_A = \frac{n_A}{n_A + n_B} \cdot P = \frac{1}{4} \times 0.985 = 0.246 \text{ atm.}$$

$$P_B = \frac{n_B}{n_A + n_B} \cdot P = \frac{3}{4} \times 0.985 = 0.739 \text{ atm.}$$

Total partial pressure ($P_A + P_B$) = 0.985 atm.

9. (b) First we should calculate the number of moles of the gas under the given conditions by the relation $PV = nRT$

Here $P = 7.6 \times 10^{-10} \text{ mm Hg}$

$$= \frac{7.6 \times 10^{-10}}{760} \text{ atm.} = 1 \times 10^{-12} \text{ atm.}$$

$V = 1 \text{ litre}$

$T = 273 + 0 = 273 \text{ K}$

$R = 0.082 \text{ litre atm./K/mol}$

Putting the values in equation

$$n = \frac{PV}{RT} = \frac{1 \times 10^{-12} \times 1}{0.082 \times 273} \text{ moles}$$

Now since 1 mole contains $= 6.023 \times 10^{23}$ molecules

$$\frac{10^{-12}}{0.082 \times 273} \text{ moles will have}$$

$$= \frac{6.023 \times 10^{23} \times 10^{-12}}{0.082 \times 273} \text{ molecules}$$

$$= 2.7 \times 10^{10} \text{ molecules}$$

10. (d) No. of balloons that can be filled

$$= \frac{V \text{ of } H_2 \text{ available}}{V \text{ of one balloon}}$$

Calculation of total volume of hydrogen in the cylinder at N.T.P.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$P_1 = 1 \text{ atm}$$

$$P_2 = 20 \text{ atm}$$

$$V_1 = ?$$

$$V_2 = 2.82 \text{ l}$$

$$T_1 = 273 \text{ K}$$

$$T_2 = 273 + 27 = 300 \text{ K}$$

$$\therefore V_1 = \frac{20 \times 2.82 \times 273}{300 \times 1} = 51.3241 = 51324 \text{ mL}$$

Actual volume to be transferred into balloons
 $= 51324 - 2820 \text{ mL} = 48504 \text{ mL}$

[$\because$ 2820 mL of H_2 will remain in cylinder]

$$\text{Volume of one balloon} = \frac{4}{3} \pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{21}{2}\right)^3$$

$$\left[\because r = \frac{\text{diameter}}{2} \right]$$

$$= 4851 \text{ ml} = 4.851 \text{ L}$$

$$\text{No. of balloons that can be filled up} = \frac{48504}{4.851}$$

$$= 9.999 = 10$$

11. (d) Let $V \text{ mL}$ of gas is effused

$$\frac{V / 220}{V / 200} = \sqrt{\frac{d_{O_2}}{d_{\text{mix}}}} \Rightarrow d_{\text{mix}} = 1.936 \text{ g/L}$$

Let density of ozone is d . In 100 volume ozonised oxygen, 60% O_2 and 40% by volume O_3 is present

$\therefore$ Mass of mixture = mass of ozone + mass of oxygen

$$100 \times 1.936 = 40 \times d + 60 \times 1.6$$

$\therefore$ Density of O_3 is 2.44 g/L

12. (d) From ideal gas equation

$$PV = nRT$$

Maximum number of moles in container,

$$n = \frac{PV}{RT} = \frac{2 \times 2.4}{0.0821 \times 298} = 0.18 \text{ moles}$$

Maximum weight of N_2 in container

$$= 0.183 \times 28 = 5.127 \text{ g}$$

Hence, at 5.127 g exploding can occur. Thus, the maximum amount of nitrogen that can be safely put in this container at 298 K temperature and exert pressure less than 2 atm will be closest to 4.2 g.

13. (c) Density (ρ) = $\frac{PM}{RT}$ (1 bar = 0.987 atm)

$$\rho_{N_2} = \frac{4 \times 0.987 \times 28}{R \times 300}$$

Let the molar mass of gas be x

$$\rho_{\text{gas}} = \frac{2 \times 0.987 \times x}{R \times 300}$$

$$\text{Given } \rho_{\text{gas}} = \rho_{N_2} \times 2$$

$$\frac{2 \times 0.987 \times x}{R \times 300} = \frac{4 \times 0.987 \times 28}{R \times 300} \times 2$$

$$\therefore x = 112 \text{ g/mol.}$$

14. (a) Given $T = 27^\circ\text{C} = 27 + 273 = 300\text{ K}$

$$V = 10.0\text{ L}$$

$$\text{Mass of He} = 0.4\text{ g}$$

$$\text{Mass of oxygen} = 1.6\text{ g}$$

$$\text{Mass of nitrogen} = 1.4\text{ g}$$

$$n_{\text{He}} = 0.4/4 = 0.1$$

$$n_{\text{O}_2} = 1.6/32 = 0.05$$

$$n_{\text{N}_2} = 1.4/28 = 0.05$$

$$n_{\text{total}} = n_{\text{He}} + n_{\text{O}_2} + n_{\text{N}_2} = 0.1 + 0.05 + 0.05 = 0.2$$

$$P = \frac{nRT}{V} = \frac{0.2 \times 0.082 \times 300}{10} = 0.492\text{ atm}$$

15. (c) Draw a line at constant P parallel to volume axis. Take volume corresponding to each temperature.

$$\text{From volume axis, } V_1 > V_2 > V_3$$

$$\text{Hence, } T_1 > T_2 > T_3$$

16. (b) $PV = \frac{1}{3} mNu^2 = \frac{1}{3} \mu u^2$

$$\text{or } u = \sqrt{3PV/M}$$

$$\text{at STP, } u \propto \sqrt{\frac{1}{M}}$$

i.e., higher the molar mass lesser will be the value of u_{rms} .

Molecular masses of H_2 , N_2 , O_2 and HBr are 2, 28, 32 and 81. Hence the correct order is $\text{HBr} < \text{O}_2 < \text{N}_2 < \text{H}_2$.

17. (c) $\text{K.E.} = \frac{3}{2} nRT$; $\text{K.E.}(\text{H}_2) = \frac{3}{2} \times \frac{w}{2} \times R \times 300$;

$$\text{K.E.}(\text{CH}_4) = \frac{3}{2} \times \frac{w}{16} \times R \times 600$$

$$\text{Hence } \frac{\text{K.E.}(\text{H}_2)}{\text{K.E.}(\text{CH}_4)} = 4$$

18. (c) $U_{\text{rms}} \sqrt{\frac{3RT}{M}}$

$$= \frac{(U_{\text{rms}})_{\text{H}_2}}{(U_{\text{rms}})_{\text{N}_2}} = \sqrt{\frac{T_{\text{H}_2}}{M_{\text{H}_2}} \times \frac{M_{\text{N}_2}}{T_{\text{N}_2}}}$$

$$(U_{\text{rms}})_{\text{H}_2} = \sqrt{5} (U_{\text{rms}})_{\text{N}_2}$$

$$\therefore \frac{(U_{\text{rms}})_{\text{H}_2}}{(U_{\text{rms}})_{\text{H}_2}} \times \sqrt{5} = \sqrt{\frac{T_{\text{H}_2}}{T_{\text{N}_2}} \times \frac{28}{2}}$$

$$= \frac{\sqrt{5}}{1} = \sqrt{\frac{T_{\text{H}_2}}{T_{\text{N}_2}} \times 14}$$

$$= 5 = \frac{T_{\text{H}_2}}{T_{\text{N}_2}} \times 14$$

$$T_{\text{N}_2} \times 5 = T_{\text{H}_2} \times 14$$

$$\therefore T_{\text{N}_2} > T_{\text{H}_2}$$

19. (b) Compressibility factor $Z = \frac{PV_m}{RT}$

$$Z = \frac{PV_m}{RT} = \left(\frac{V_m}{V_m - b} \right) - \left(\frac{a}{RTV_m} \right)$$

$$= \frac{1}{1 - \frac{b}{V_m}} - \frac{a}{RTV_m} \quad \dots (i)$$

Considering the equation (i) the value of compressibility factor Z decreases as the value of ' b ' decreases and the value of compressibility factor Z decreases as the value of ' a ' increases at constant temperature and fixed volume.

20. (d) $K = \frac{3}{2} n_T RT \Rightarrow n_T = \frac{3000 \times 2}{3 \times 250 \times 8.314}$

$$3 \times 10^3 = \frac{3}{2} n_T \times 8.314 \times 250$$

$$\frac{x}{20} + \frac{30-x}{40} = \frac{2}{8.314}$$

$$\% \text{Ne} = 28.3$$

21. (5) $r_g = \frac{1}{5} \cdot r_{\text{H}_2}$

$$\frac{M_g}{M_{\text{H}_2}} = \left[\frac{r_{\text{H}_2}}{r_g} \right]^2 = (5)^2 = 25; M_g = 2 \times 25 = 50$$

$$10y = 50 \Rightarrow y = 5$$

$$22. (8) \frac{r_{CH_4}}{r_x} = 2 = \sqrt{\frac{M_x}{M_{CH_4}}} = \sqrt{\frac{M_x}{16}}, \text{ or } M_x = 64$$

$$8y = 64 \Rightarrow y = 8$$

$$23. (4) v_{rms} \text{ of } X = \sqrt{\frac{3RT_x}{M_x}}; v_{mp} \text{ of } Y = \sqrt{\frac{2RT_y}{M_y}}$$

$$\text{Given } v_{rms} = v_{mp}$$

$$\Rightarrow \sqrt{\frac{3RT_x}{M_x}} = \sqrt{\frac{2RT_y}{M_y}}$$

$$\Rightarrow M_y = \frac{2RT_y M_x}{3RT_x} = \frac{2 \times 60 \times 40}{3 \times 400} = 4$$

$$24. (7) \text{ Equation of line is =}$$

$$p - 2 = \frac{10 - 2}{4 - 15}(V - 15)$$

$$p - 2 = -\frac{8}{11}(V - 15)$$

$$p = 2 - \frac{8V}{11} + \frac{15 \times 8}{11}$$

$$p = \left(\frac{142}{11} - \frac{8V}{11} \right)$$

$$f(T) = \frac{1}{nR} \left(\frac{142V}{11} - \frac{8V^2}{11} \right)$$

$$\frac{d(f(T))}{dV} = \frac{1}{nR} \left(\frac{142}{11} - \frac{8V}{11} \right) = 0$$

$$V = \frac{142}{11 \times 16} = 8.875$$

$$p = \frac{142}{11} - \frac{8}{11} \times \frac{8.875}{16} = \frac{71}{11}$$

$$T_{Max} = \frac{PV}{nR} = \frac{71}{11} \times \frac{142}{16 \times 0.0821} = 700$$

$$\Rightarrow 7 \times 10^2 \text{ K}$$

$$25. (4) P_i = 722 \text{ mm} \quad P_f = 760 \text{ mm}$$

$$T_i = 107 + 273 = 380 \text{ K} \quad P_f = 100 \text{ K}$$

$$d_i = 1 \text{ g/cm}^3 \quad d_f = ?$$

$$\frac{d_i T_i R}{P_i} = \frac{d_f T_f R}{P_f}$$

$$\Rightarrow d_f = \left(\frac{P_f}{P_i} \right) \left(\frac{T_i}{T_f} \right) d_i = \left(\frac{760}{722} \right) \left(\frac{380}{100} \right) \times 1 \text{ g/cm}^3$$

$$= 4 \text{ g/cm}^3$$

$$26. (9) V, n \text{ constant.}$$

$$\frac{P_i}{T_i} = \frac{P_f}{T_f} \Rightarrow P_f = \frac{T_f}{T_i} P_i = \left(\frac{109}{100} \right) P_i$$

$$\Rightarrow P_{\text{increases}} = \Delta P = P_f - P_i = \frac{9}{100} P_i$$

$$\therefore \% \text{ Pressure increases}$$

$$= \frac{\Delta P_f}{P_i} \times 100 = \frac{9P_i}{100P_i} \times 100\%$$

$$X\% = 9\%$$

$$X = 9$$

$$27. (3) u_{rms} = \sqrt{\frac{u_1^2 N_1 + u_2^2 N_2}{N_1 + N_2}}$$

$$\Rightarrow u_{rms}^2 = \frac{u_1^2 N_1 + u_2^2 N_2}{N_1 + N_2}$$

$$25 = \frac{4 \times 7^2 + 6u_2^2}{10} \Rightarrow u_2 \sqrt{\frac{54}{6}} = 3 \text{ ms}^{-1}$$

$$28. (1) \text{ Average kinetic energy depends only on the temperature. Thus, } n = 1$$

$$29. (405) \mu_{rms} = \sqrt{\frac{3RT}{M}}$$

$$450 = \sqrt{\frac{3RT}{4 \times 10^{-3}}}$$

$$(\text{molecular mass of He} = 4 \times 10^{-3} \text{ kg})$$

$$RT = 270$$

$$\text{Total K.E. of He gas} = \frac{3}{2} nRT = \frac{3}{2} \times \frac{4}{4} \times 270 = 405 \text{ J}$$

$$30. (8) T_c = \frac{8a}{27Rb} \text{ and } P_c = \frac{a}{27b^2}$$

$$\Rightarrow \frac{T_c}{P_c} = \frac{8b}{R} \quad \left[\because \frac{T_c}{P_c} = \frac{mb}{R} \right]$$

$$\Rightarrow \frac{mb}{R} = \frac{8b}{R}$$

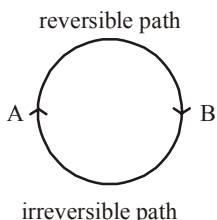
$$\therefore m = 8$$

CHAPTER

6

Thermodynamics

1. (c)



For a cyclic process the net change in the internal energy is zero because the change in internal energy does not depend on the path.

2. (b) $\Delta U = q + w = q - P\Delta V$

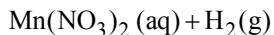
$$= 200\text{J} - 10^5 \text{ Pa} \times (5 - 4) \times 10^{-3} \text{ m}^3$$

$$(\because 1\text{L} = 10^{-3} \text{ m}^3)$$

$$= 200\text{J} - 100 \text{ Pa m}^3$$

$$= 200\text{J} - 100\text{J} = 100\text{J} \quad (\because 1\text{J} = \text{Pa m}^3)$$

3. (c) $\text{Mn(s)} + 2\text{HNO}_3(\text{aq}) \longrightarrow ;$



$$\text{Moles of H}_2 = \text{Moles of Mn} = \frac{110}{55} = 2$$

Work done due to expansion

$$= -P\Delta V = -\Delta nRT = -2 \times 8.314 \times 300 = -4988.4\text{J}$$

4. (a) Volume of 0.5 mole of steam at 1 atm pressure

$$= \frac{nRT}{P} = \frac{0.5 \times 0.0821 \times 373}{1.0} = 15.3\text{L}$$

Change in volume = vol. of steam – vol. of water
 $= 15.3 - \text{negligible} = 15.3\text{L}$

Work done by the system,

$$w = P_{\text{ext}} \times \text{volume change}$$

$$= 1 \times 15.3 = 15.3 \text{ litre-atm}$$

$$= 15.3 \times 101.3 \text{ J} = 1549.89 \text{ J}$$

‘w’ should be negative as the work has been done by the system on the surroundings.

$$w = -1549.89 \text{ J} = -1.54 \text{ kJ}$$

5. (b) Amount of work done in reversible isothermal expansion

$$w = -2.303nRT \log \frac{V_2}{V_1}$$

$$w = -2.303 \times 2 \times 8.314 \times 298 \log \frac{20}{10}$$

$$= -2.303 \times 2 \times 8.314 \times 298 \times 0.3010$$

$$= -3434.9 \text{ J} = -3.4 \text{ kJ}$$

i.e., work is done by the system.

6. (b) $w = \text{Area of P-V curve}$. As area under graph 3 is maximum and area under graph 1 is minimum, so w_3 is maximum and w_1 is minimum $\therefore w_1 < w_2 < w_3$

7. (b) $w = -nRT \ln \frac{P_1}{P_2}$

$$= -10 \times 8.314 \times 300 \ln \frac{10}{1}$$

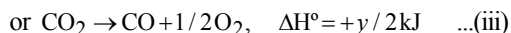
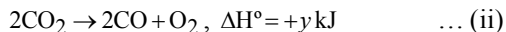
$$= -57441.43 \text{ J}$$

$$w = -mgh$$

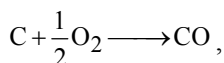
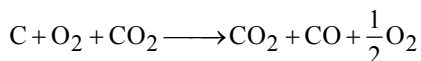
$$\therefore m \times 9.81 \times 100 = 57441.42$$

$$m = 58.55 \text{ kg}$$

8. (b) Given $C + O_2 \rightarrow CO_2$, $\Delta H^\circ = -x \text{ kJ}$ (i)



By adding no. (i) and (iii) eq.



$$\Delta H^\circ = y/2 - x = \frac{y - 2x}{2} \text{ kJ}$$

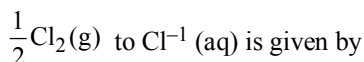
9. (a) $I_2(s) + Cl_2(g) \longrightarrow 2ICl(g)$

$$\begin{aligned} \Delta_r H &= [\Delta H(I_2(s) \rightarrow I_2(g)) + \Delta H_{I-I} + \Delta H_{Cl-Cl}] \\ &\quad - [\Delta H_{I-Cl}] \\ &= 151.0 + 242.3 + 62.76 - 2 \times 211.3 = 33.46 \end{aligned}$$

$$\Delta_f H^\circ(ICl) = \frac{33.46}{2} = 16.73 \text{ kJ/mol}$$

10. (d) Given, mass of ethanol = 1 kg = 1000 g
Latent heat of vaporisation of ethanol = 855 Jg⁻¹
Specific heat capacity of ethanol = 2.44 J/gK⁻¹
Heat, $q = mc\Delta T + \text{heat of vaporisation}$
 $= 1000 \times 2.44(351.45 - 293.45) + 855 \times 1000 \text{ J}$
 $= 9.97 \times 10^5 \text{ J}$

11. (c) The energy involved in the conversion of



$$\Delta H = \frac{1}{2} \Delta_{\text{diss}} H_{Cl_2}^{(-)} + \Delta_{\text{eg}} H_{Cl}^{(-)} + \Delta_{\text{hyd}} H_{Cl}^{(-)}$$

Substituting the values from given data, we get

$$\begin{aligned} \Delta H &= \left(\frac{1}{2} \times 240 \right) + (-349) + (-381) \text{ kJ mol}^{-1} \\ &= (120 - 349 - 381) \text{ kJ mol}^{-1} \\ &= -610 \text{ kJ mol}^{-1} \end{aligned}$$

12. (a) Molar heat capacity at constant pressure,
 $C_p = 248.2 \times M \text{ J.kg}^{-1}$

where M is the molar mass of the gas.

Similarly, $C_v = 149 \times M \text{ J.kg}^{-1}$

$$C_p - C_v = R$$

$$\therefore 248.2 \times M - 149M = 8.314$$

$$M = \frac{8.314}{248.2 - 149} = 0.0838 \text{ kg/mol}$$

Molar mass of the gas = 83.8 g/mol

13. (a) Given $S(s) + 3F_2(g) \rightarrow SF_6(g)$;

$$\Delta H = -1100 \text{ kJ} \quad \dots (i)$$

$$S(s) \rightarrow S(g); \Delta H = 275 \text{ kJ} \quad \dots (ii)$$

$$1/2 F_2(g) \rightarrow F(g); \Delta H = 80 \text{ kJ} \quad \dots (iii)$$

To get $SF_6(g) \rightarrow S(g) + 6F(g)$ we can proceed as (ii) + 6 × (iii) - (i)

$$\therefore SF_6(g) \rightarrow S(g) + 6F(g); \Delta H = 1855 \text{ kJ}$$

Thus average bond energy for S-F bond

$$= \frac{1855}{6} = 309.16 \text{ kJ}$$

14. (d) Entropy change

$$\Delta S = \Delta S_{\text{product}} - \Delta S_{\text{reactant}}$$

$$= 2(186.7) - (223 + 130.6)$$

$$= 373.4 - 353.6$$

$$= 19.8 \text{ JK}^{-1} \text{ mol}^{-1}$$

15. (a) $2CH_3COOH(g) \rightleftharpoons (CH_3COOH)_2(g)$

$$\begin{array}{cc} 1 & 0 \\ 1 - 0.982 & \frac{0.982}{2} \end{array}$$

$$K = \frac{(CH_3COOH)_2}{(CH_3COOH)^2} = \frac{0.982}{2 \times (0.018)^2} = 1515.4$$

Now, ΔH° for dimerization = $-2 \times 33 \text{ kJ} = -66 \text{ kJ}$

$$\text{Thus, } \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$-2.303 RT \log K^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$-2.303 \times 8.314 \times 400 \times \log(1515.4)$$

$$= -66 \times 10^3 - 400 \times \Delta S^\circ$$

$$-242359.2 = -66000 - 400 \Delta S^\circ$$

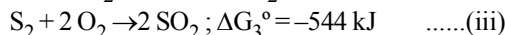
$$\Delta S^\circ = -\frac{41640.8}{400} = -104 \text{ JK}^{-1} \text{ mol}^{-1}$$

16. (d) For a spontaneous process,

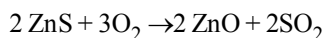
$$\Delta S = (\Delta S_{\text{system}} + \Delta S_{\text{surrounding}}) > 0$$

17. (d) Since the process is at equilibrium $\Delta G = 0$ and for $\Delta G = 0$, the conditions should be $\Delta H > 0$, $\Delta S > 0$.

18. (b) For the reaction



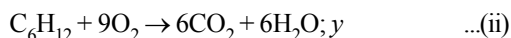
ΔG° for the reaction



can be obtained by adding eqn. (1), (2) and (3)

$$\Rightarrow \Delta G^\circ = 293 - 480 - 544 = -731 \text{ kJ}$$

19. (b) $\text{C}_6\text{H}_6 + \frac{15}{2} \text{O}_2 \rightarrow 6\text{CO}_2 + 3\text{H}_2\text{O}; x \quad \dots(i)$



To get equation (iv), (i) + 3 × (iii) – (ii)



Molar enthalpy of hydrogenation of benzene to cyclohexane = $x - y + 3z$

20. (d) $\text{H}_2\text{O}(\ell) \xrightleftharpoons{1\text{atm}} \text{H}_2\text{O}(\text{g})$

$$\Delta H = 40630 \text{ J mol}^{-1}$$

$$\Delta S = 108.8 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\Delta G = \Delta H - T\Delta S$$

$$\text{When } \Delta G = 0, \quad \Delta H - T\Delta S = 0$$

$$T = \frac{\Delta H}{\Delta S} = \frac{40630 \text{ J mol}^{-1}}{108.8 \text{ J mol}^{-1}} = 373.4 \text{ K.}$$

21. (– 13538)

$$\text{From } \Delta H^\circ = \Delta U^\circ + \Delta n_g RT$$

$$\Delta H^\circ = -20 \times 1000 - 1 \times 8.314 \text{ J / mol.K} \times 298 \text{ K}$$

$$= -22477.57 \text{ J}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -22477.57 - (298 \times -30)$$

$$= -13538 \text{ J}$$

22. (48.00)

Work done is given by the area under the trapezium.

$$\therefore |w| = \frac{1}{2} (6 + 10) \times 6 = 48 \text{ J}$$

23. (6.25) $\Delta U = nC_v \Delta T$

$$5000 = 4 \times C_v (500 - 300)$$

$$C_v = 6.25 \text{ JK}^{-1} \text{ mol}^{-1}$$

24. (1) For reversible adiabatic process

$$\left(\frac{T_1}{T_2} \right)^\gamma = \left(\frac{P_2}{P_1} \right)^{1-\gamma} \quad \text{or } \gamma \log \frac{T_1}{T_2} = 1 - \gamma \log \frac{P_2}{P_1}$$

$$\gamma \log \frac{300}{90} = 1 - \gamma \log \frac{10}{200}$$

$$\gamma (\log 10 - \log 3) = 1 - \gamma (\log 10 - \log 200)$$

$$\gamma = 1.66$$

$$\text{Atomicity} = 1$$

25. (4) In $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$

$$2 \times 56 + 48 = 160 \quad 2 \times 27 = 54$$

Heat of reaction = $399 - 199 = 200 \text{ kcal}$ [Al and Fe are in their standard states]

$$\text{Volume of Al} = \frac{54}{2.7} = 20 \text{ cc}$$

$$\text{Volume of Fe}_2\text{O}_3 = \frac{160}{5.2} = 30.77 \text{ cc}$$

$$\text{Total volume} = 20 + 30.77 = 50.77 \text{ cc}$$

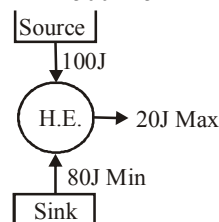
$$\therefore \text{Fuel value per cc} = \frac{200}{50.77} = 3.94 \approx 4 \text{ kcal/cc}$$

26. (1) $Q_v = 1.25 \times 4 = 5$

$$Q_v = 5 = \frac{x}{80} \times 400$$

$$\Rightarrow x = 1$$

27. (4) $\eta_{\text{max}} = 1 - \frac{400}{500} = \frac{1}{5} = 20\%$



$$T = \frac{80}{20}$$

$$T = 4$$

28. (76) $\Delta H = \Delta E + W$

$$= \Delta E + (P_2 V_2 - P_1 V_1)$$

$$= 50 + (4 \times 7 - 1 \times 2)$$

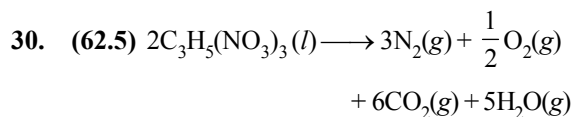
$$= 76 \text{ L-atm}$$

29. (31) At equilibrium

Heat lost by iron = Heat gain by water

$$20 \times 0.5 \times (473 - T) = 100 \times 4.2 \times (T - 300)$$

$$\Rightarrow T = 304 \text{ K} = 31^\circ\text{C}$$



$$\Delta H^\circ_{\text{reaction}} = \sum H^\circ_{\text{f(P)}} - \sum H^\circ_{\text{f(R)}}$$

$$\Delta H^\circ_{\text{reaction}} = 3 \times 0 + \frac{1}{2} \times 0 + 6 \times -393.5$$

$$+ 5 \times -241.8 - (2 \times -364)$$

$$= -2361 - 1209 + 728$$

$$= -2842 \text{ kJ/mol}$$

For 2 moles of nitroglycerine enthalpy

$$= -2842 \text{ kJ/mol.}$$

$\therefore$ For 1 mole of nitroglycerine enthalpy

$$= \frac{-2842}{2} = -1421 \text{ kJ/mol.}$$

227.1 g of $\text{C}_3\text{H}_5(\text{NO}_3)_3$ has enthalpy

$$= -1421 \text{ kJ/mol.}$$

$\therefore$ 10 g of $\text{C}_3\text{H}_5(\text{NO}_3)_3$ has enthalpy

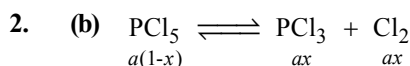
$$= -\frac{1421}{227.1} \times 10 = -62.5 \text{ kJ/mol}$$

CHAPTER

7

Equilibrium

1. (a)



$$a=2, x=0.4, V=2\text{ L}$$

$$\therefore [\text{PCl}_5] = \frac{2(1-0.4)}{2} = 0.6 \text{ mol L}^{-1}$$

$$[\text{PCl}_3] = [\text{Cl}_2] = \frac{2 \times 0.4}{2} = 0.4 \text{ mol L}^{-1}$$

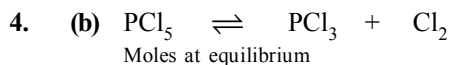
$$\therefore K_c = \frac{0.4 \times 0.4}{0.6} = 0.267$$

3. (a) For reation (i)

$$K_1 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

and for reaction (ii)

$$K_2 = \frac{[\text{N}_2]^{1/2}[\text{O}_2]^{1/2}}{[\text{NO}]}; \therefore K_1 = \frac{1}{K_2^2}$$



$$\frac{1}{2} \qquad \frac{1}{2} \qquad \frac{1}{2}$$

Mole fraction at equilibrium

$$\frac{1}{3} \qquad \frac{1}{3} \qquad \frac{1}{3}$$

Partial pressure at equilibrium

$$\frac{P}{3} \qquad \frac{P}{3} \qquad \frac{P}{3}$$

$$\therefore K_p = \frac{\frac{P}{3} \times \frac{P}{3}}{\frac{P}{3}} = \frac{P}{3}$$



$$1 - \alpha \qquad \alpha \qquad 2\alpha$$

Total moles at equilibrium = $1 + 2\alpha$

$$= 1 + \frac{20 \times 2}{100} = 1.4$$

$$\frac{P_1}{P_2} = \frac{n_1 T_1}{n_2 T_2}; \quad \frac{1}{P_2} = \frac{1 \times 300}{1.4 \times 600}$$

$$\therefore P_2 = 2.8 \text{ atm}$$



Apply law of mass action,

$$K_p = \frac{(P_{\text{CO}})^2}{P_{\text{CO}_2}} \text{ or } 63 = \frac{(10P_{\text{CO}_2})^2}{P_{\text{CO}_2}}$$

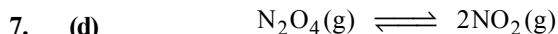
(Given $K_p = 63$ and $P_{\text{CO}} = 10P_{\text{CO}_2}$)

$$\text{or } 63 = \frac{100(P_{\text{CO}_2})^2}{P_{\text{CO}_2}} \text{ or } 63 = 100 P_{\text{CO}_2}$$

$$P_{\text{CO}_2} = \frac{63}{100} = 0.63 \text{ atm}$$

$$P_{\text{CO}} = 10P_{\text{CO}_2} = 10 \times 0.63 = 6.3 \text{ atm}$$

$$P_{\text{total}} = P_{\text{CO}_2} + P_{\text{CO}} = 0.63 + 6.3 = 6.93 \text{ atm}$$



$$\text{Initial moles} \qquad 1 \qquad 0$$

$$\text{Moles at eqm.} \qquad (1 - \alpha) \qquad 2\alpha$$

(α = degree of dissociation)

Total number of moles at equil.

$$= (1 - \alpha) + 2\alpha$$

$$= (1 + \alpha)$$

$$P_{\text{N}_2\text{O}_4} = \frac{(1 - \alpha)}{(1 + \alpha)} \times P$$

$$P_{\text{NO}_2} = \frac{2\alpha}{(1 + \alpha)} \times P$$

$$K_p = \frac{(P_{\text{NO}_2})^2}{P_{\text{N}_2\text{O}_4}} = \frac{\left(\frac{2\alpha}{(1 + \alpha)} \times P\right)^2}{\left(\frac{1 - \alpha}{(1 + \alpha)} \times P\right)} = \frac{4\alpha^2 P}{1 - \alpha^2}$$

Given, $K_p = 2$, $P = 0.5$ atm

$$\therefore K_p = \frac{4\alpha^2 P}{1 - \alpha^2}$$

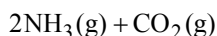
$$2 = \frac{4\alpha^2 \times 0.5}{1 - \alpha^2}$$

$$\alpha = 0.707 \approx 0.71$$

$\therefore$ Percentage dissociation

$$= 0.71 \times 100 = 71$$

8. (b) $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons$



$$K_p = \frac{(P_{\text{NH}_3})^2 \times (P_{\text{CO}_2})}{P_{\text{NH}_2\text{COONH}_4(\text{s})}} = (P_{\text{NH}_3})^2 \times (P_{\text{CO}_2})$$

As evident by the reaction, NH_3 and CO_2 are formed in molar ratio of 2 : 1. Thus if P is the total pressure of the system at equilibrium, then

$$P_{\text{NH}_3} = \frac{2 \times P}{3}; \quad P_{\text{CO}_2} = \frac{1 \times P}{3}$$

$$K_p = \left(\frac{2P}{3}\right)^2 \times \frac{P}{3} = \frac{4P^3}{27}$$

$$\text{Given } K_p = 2.9 \times 10^{-5}$$

$$\therefore 2.9 \times 10^{-5} = \frac{4P^3}{27}$$

$$P^3 = \frac{2.9 \times 10^{-5} \times 27}{4}$$

$$P = \left(\frac{2.9 \times 10^{-5} \times 27}{4}\right)^{1/3} = 5.82 \times 10^{-2} \text{ atm}$$

9. (c) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$

At equilibrium, we have $[\text{N}_2] = 0.79(1 - \alpha)$

$[\text{O}_2] = 0.21(1 - \alpha)$, $[\text{NO}] = 2\alpha$

Total no. of moles

$$= 0.79(1 - \alpha) + 0.21(1 - \alpha) + 2\alpha = 1 + \alpha$$

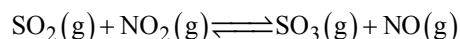
$$P_{\text{N}_2} = \frac{0.79(1 - \alpha)}{1 + \alpha} \times 1; \quad P_{\text{O}_2} = \frac{0.21(1 - \alpha)}{1 + \alpha} \times 1$$

$$P_{\text{NO}} = \frac{2\alpha}{1 + \alpha} \cdot 1$$

$$K_p = \frac{P_{\text{NO}}^2}{P_{\text{N}_2} \cdot P_{\text{O}_2}} = \frac{4\alpha^2}{0.79 \times 0.21(1 - \alpha)^2} = 1.1 \times 10^{-3}$$

$$\alpha = 0.0067 \quad \text{vol \% of NO} = 1.33$$

10. (c)



Initial moles	2	2	2	2
at eqm.	2 - x	2 - x	2 + x	2 + x

$$(\because Q_p < K_p)$$

Total no. of moles of gases at equilibrium

$$= 8 + 2 = 10$$

$$K_p = \frac{P_{\text{SO}_3} \cdot P_{\text{NO}}}{P_{\text{SO}_2} \cdot P_{\text{NO}_2}}$$

$$\Rightarrow 25 = \frac{\left(\frac{2+x}{10} \times P\right)^2}{\left(\frac{2-x}{10} \times P\right)^2}$$

$$\Rightarrow 5 = \frac{2+x}{2-x}$$

$$\Rightarrow x = 1.33$$

$$\text{Partial Pressure of NO}_2 = \frac{2-x}{10} \times P_{\text{total}}$$

$$= \frac{0.666}{10} \times 2 = 0.133 \text{ atm}$$

11. (c) Catalyst only changes time to achieve equilibrium, it does not affect the value of equilibrium constant.
12. (b) The most favourable conditions are :
- (i) high pressure ($\Delta n < 0$)
 - (ii) low temperature (Exothermic reaction)
 - (iii) catalyst Fe in presence of Mo.
13. (d) Moles of CO_2 present at equilibrium

$$= \frac{1.642 \times 50}{0.0821 \times 1000} = 1$$

Mole % of XCO_3 decomposed

$$= \frac{1}{4} \times 100 = 25\%$$

Hence, 75% remains undecomposed.

14. (a) From graph, $[\text{A}]_{\text{eq}} = 0.1 \text{ M}$, $[\text{B}]_{\text{eq}} = 0.4 \text{ M}$

$$K_c = \frac{[\text{B}]^2}{[\text{A}]} = 1.6$$

15. (c) $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$
- | | | |
|-----|---|---|
| 0.1 | 0 | 0 |
|-----|---|---|

Initial concentration

$$\begin{array}{ccc} 0.1(1-0.0132) & 0.1 \times 0.0132 & 0.1 \times 0.0132 \\ \text{at eqb.} & & \end{array}$$

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

$$= \frac{0.1 \times 0.0132 \times 0.1 \times 0.0132}{0.1(1-0.0132)} = 1.76 \times 10^{-5}$$

16. (c) Here $\text{Hg}_2\text{Cl}_2 \longrightarrow \text{Hg}_2^{2+} + 2\text{Cl}^-$

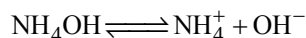
$$K_{\text{sp}} = [\text{Hg}_2^{2+}][\text{Cl}^-]^2 = s \times (2s)^2 = 4s^3$$

$$4s^3 = 3.2 \times 10^{-17}$$

$$\Rightarrow s^3 = \frac{32}{4} \times 10^{-18} \Rightarrow s^3 = 8 \times 10^{-18}$$

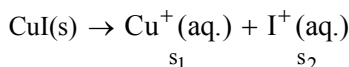
$$\therefore s = 2 \times 10^{-6} \text{ M}$$

17. (c) $K_{\text{sp}}(\text{AgBr}) < K_{\text{sp}}(\text{AgCl})$
Therefore, AgBr will precipitate first and at that time all Cl^- will be present.
18. (a) Buffer solution can be prepared by mixing equal volumes of 0.2 M NH_4OH (weak base) and 0.1 M HCl (strong acid). It results in the formation of NH_4OH (weak base) + NH_4Cl (salt of weak base NH_4OH), which is basic buffer mixture.
19. (a) Dil. aqueous solutions of NH_3 is NH_4OH .



On adding solid ammonium chloride, the reaction moves backward due to common ion effect. The concentration of OH^- decreases and, hence, the pH decreases.

20. (b) $K_{\text{sp}}(\text{CuI}) = 4 \times 10^{-12}$
 $K_{\text{sp}}(\text{Ag}_2\text{CrO}_4) = 4 \times 10^{-12}$

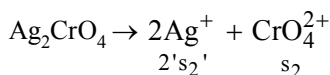


$$K_{\text{sp}_1} = s_1 \times s_1$$

$$\Rightarrow 4 \times 10^{-12} = s_1^2$$

$$s_1 = 2 \times 10^{-6} \text{ mol/L}$$

$$K_{\text{sp}}(\text{Ag}_2\text{CrO}_4) = 4 \times 10^{-12}$$



$$K_{\text{sp}_2} = (2s_2)^2(s_2)$$

$$4 \times 10^{-12} = 4s_2^3$$

$$s_2^3 = 10^{-12}$$

$$\Rightarrow s_2 = 10^{-4} \text{ mol/L}$$

$$\frac{\text{Solubility of CuI}}{\text{Solubility of Ag}_2\text{CrO}_4} = \frac{s_1}{s_2}$$

$$= \frac{2 \times 10^{-6}}{10^{-4}} = 2 \times 10^{-2} = 0.02$$



$$\begin{array}{ccc} \text{Initial mole} & 4 & 0 & 0 \\ \text{Mole at equilibrium} & 4 - 2 = 2 & 2 & 2 \end{array}$$

equilibrium

Total mole of equilibrium = 2 + 2 + 2 = 6 mole
and total Pressure (P) = 3 atm

$$K_p = \frac{P_{\text{PCl}_3} P_{\text{Cl}_2}}{P_{\text{PCl}_5}}$$

Partial pressure = mole fraction $\times$ total pressure

$$= \frac{\text{mole of element present at equilibrium}}{\text{total mole at equilibrium}} \times \text{total pressure}$$

$$(P_{\text{PCl}_3}) = \frac{2}{6} \times 3 = 1 \quad ; \quad (P_{\text{Cl}_2}) = \frac{2}{6} \times 3 = 1$$

$$(P_{\text{PCl}_5}) = \frac{2}{6} \times 3 = 1 \quad ; \quad K_p = \frac{1 \times 1}{1} = 1 \text{ atm}$$

22. (3) $K_p = \frac{(P_{\text{H}_2\text{O}})^2}{(P_{\text{H}_2})^2}$

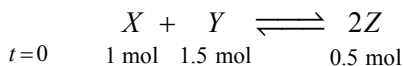
Given H_2 is 45% by volume at constant temperature in closed vessel ($P \propto V$).

So $P_{\text{H}_2\text{O}} = 0.55$ and $P_{\text{H}_2} = 0.45$

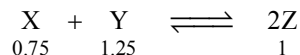
$$K_p = \left(\frac{0.55}{0.45} \right)^2 = 1.5$$

$$\Rightarrow 2K_p = 1.5 \times 2 = 3$$

23. (16)



$$\begin{array}{ccc} t = t_{eq} & 1-x & 1.5-x & & 0.5+2x \\ 0.5+2x=1; x=0.25 \end{array}$$

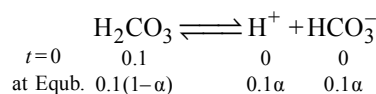


$$K_{eq} = \frac{[Z]^2}{[X][Y]} = \frac{1}{0.75 \times [1.25]} = \frac{x}{15}$$

$$\Rightarrow x = \frac{15}{(0.75 \times 1.25)} = 16$$



$$\begin{array}{ccc} 30 \text{ bar} & \dots\dots\dots & 1 \text{ mol/L} \\ 3 \text{ bar} & \dots\dots\dots & 0.1 \text{ mol/L} \end{array}$$



$$4.0 \times 10^{-7} = \frac{0.1\alpha^2}{1-\alpha}$$

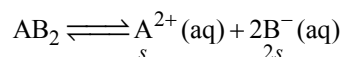
$$\Rightarrow (1-\alpha) = 1$$

$$\alpha^2 = 4 \times 10^{-6} \Rightarrow \alpha = 2 \times 10^{-3}$$

$$[\text{H}^+] = 2 \times 10^{-4} \text{ M}$$

$$\text{pH} = 4 \times \log 2 = 3.7 \approx 37 \times 10^{-1}$$

25. (2.0)

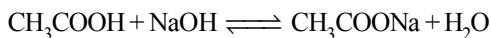


$$K_{sp} = 4s^3 = 3.2 \times 10^{-11}$$

$$\Rightarrow s^3 = 8 \times 10^{-12}$$

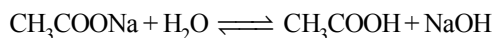
$$\Rightarrow s = 2 \times 10^{-4}$$

26. (8.22)



Let acid be = V mL

V mL of 0.01 M CH_3COOH will require V mL of 0.01 M NaOH. But CH_3COONa formed will make solution alkaline due to hydrolysis.



$$[\text{CH}_3\text{COONa}] = \frac{0.01}{2} = 0.005 \text{ M}$$

Using equation for pH of salt of weak acid and strong base.

$$\begin{aligned} \text{pH} &= 7 + \frac{\text{pK}_a}{2} + \frac{\log C}{2} = 7 + \frac{4.74}{2} + \frac{\log 0.005}{2} \\ &= 8.22 \end{aligned}$$

27. (8) The chemical reaction of NH_4OH (NH_3) and HCl follows the eqn: $\text{NH}_4\text{OH} + \text{HCl} \rightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}$. Henderson Hasselbalch equation can be used to calculate the pOH of resulted basic buffer solutions.

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$= \text{pK}_b + \log \frac{[\text{NH}_4\text{Cl}]}{[\text{NH}_4\text{OH}]}$$

$$= 5 + \log \frac{[1.0]}{[0.1]}$$

$$\text{pOH} = 6$$

$$\text{pH} + \text{pOH} = 14$$

$$\text{pH} = 14 - 6 = 8.0$$

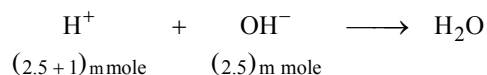
28. (4) Given $[\text{H}_3\text{O}^+] = 1 \times 10^{-10} \text{ M}$
at 25°C $[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$

$$\therefore [\text{OH}^-] = \frac{10^{-14}}{10^{-10}} = 10^{-4}$$

$$\text{Now, } [\text{OH}^-] = 10^{-\text{pOH}} = 10^{-4} = 10^{-\text{pOH}}$$

$$\therefore \text{pOH} = 4$$

29. (2)



$$[\text{H}^+]_{\text{Final}} = \frac{(3.5 - 2.5)}{25 + 50 + 25} = 10^{-2} \text{ M}$$

$$\text{pH} = -\log [\text{H}^+] = -\log (10^{-2}) = 2$$

30. (8.98) $\text{HA} + \text{NaOH} \longrightarrow \text{NaA} + \text{H}_2\text{O}$

At the end point, the solution contains only NaA whose concentration is $0.1/2 = 0.05 \text{ M}$

Since the salt NaA is formed by strong alkali (NaOH) and weak acid HA (indicated by its low K_a value), its pH can be evaluated by the following relation.

$$\text{pH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$$

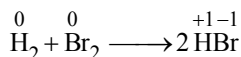
$$= \frac{1}{2}(14 + 5.2518 + (-1.3010)) = 8.98$$

CHAPTER

8

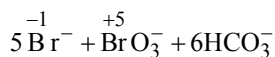
Redox Reactions

- (a) Ox. no. of Cr on both side is + 6.
- (d) $2\text{I}^- \rightarrow \text{I}_2$ is oxidation (loss of electrons) ; Cr (+6) changes to Cr (+3) by gain of electrons. Hence Cr is reduced.
- (a) In a redox reaction, one molecule is oxidised and other molecule is reduced i.e. oxidation number of reactants are changed.



Here H_2 is oxidised and Br_2 is reduced, thus it is oxidation-reduction reaction.

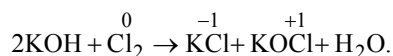
- (c) CaOCl_2 or $\text{Ca}(\text{OCl})\text{Cl}$ is the mixed salt of $\text{Ca}(\text{OH})_2$ with HCl and HOCl .
- (d) $\overset{0}{3\text{Br}_2} + \overset{0}{6\text{CO}_3^{2-}} + \overset{0}{3\text{H}_2\text{O}} \rightarrow$



O.N. of Br_2 changes from 0 to -1 and +5 hence it is reduced as well as oxidised.

- (b) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
The most electronegative element O is present in BaSO_4 and H_2O_2 , where its O.N. are -2 and -1 respectively.
- (d) In Na_2O , SnCl_2 and Na_2O_2 central atom is either in lowest or highest oxidation state, so it can function either as an oxidising or a reducing agent but not both. However, the oxidation state of N in NaNO_2 is +3 which lies between its highest (+5) and lowest (-3) values.

- (b) SO_2 oxidises H_2S to S, since the O.N. of S change from -2 to 0.
- (d) In H_2SO_4 , sulphur is in highest oxidation state (+6), hence H_2SO_4 will be strongest oxidising agent.
- (c) A reaction in which a substance undergoes simultaneous oxidation and reduction is called disproportionation reaction. In these reactions the same substance simultaneously acts as an oxidising agent and as a reducing agent. Here Cl undergoes simultaneous oxidation and reduction.



- (b) The reaction involve 6 electrons, i.e. $\text{Cr}_2\text{O}_7^{2-}$ accepts 6 electrons for its reduction to Cr^{3+}
 $2\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{I}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{I}_2$
- (a) (i) $\text{Mn}^{n+} + n\text{e}^- \rightleftharpoons \text{M}$, for this reaction, high negative value of E° indicates lower reduction potential, that means M will be a good reducing agent.

Stronger reducing agent $\Rightarrow$ Easy to oxidise

$\Downarrow$

Lower reduction potential $\Leftarrow$ higher oxidation potential

(ii) Element	F	Cl	Br	I
Reduction potential (E° volt)	+2.87	+1.36	+1.06	+0.54

As reduction potential decreases from fluorine to iodine, oxidising nature also decreases from fluorine to iodine.

(iii) The size of halide ions increases from F^- to I^- . The bigger ion can lose electron easily. Hence the reducing nature increases from HF to HI.

13. (d)

14. (d)	A	B	C
	+0.5C	-3.0V	-1.2V

The higher the negative value of reduction potential, the more is the reducing power. Hence the correct order is $B > C > A$.

15. (c) In $Cr_2O_7^{3-}$, let the oxidation state of Cr be x .

$$\therefore 2(x) + 7(-2) = -2$$

$$2x - 14 = -2$$

$$2x = 12$$

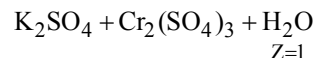
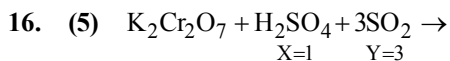
$$x = +6$$

In ClO_3^- , let the oxidation state of Cl be x .

$$\therefore 1(x) + 3(-2) = -1$$

$$x - 6 = -1$$

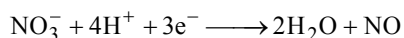
$$x = +5$$



The sum of x, y and $z = 1 + 3 + 1 = 5$

17. (4) The element is Ti (At. no. 22). Electronic configuration is $1s^2, 2s^2p^6, 3s^2p^6d^2, 4s^2$. the energy level of $3d$ and $4s$ is very close. It can have Ti^{4+} O.S.

18. (3) In the equation all the atoms are balanced, to balance charge add $3e^-$ to L.H.S.



19. (5) NO_3^-

Let x be the oxidation number of nitrogen

$$[x + 3 \times (-2)] = -1 \Rightarrow x = -1 + 6 = +5$$

20. (7) $H_4IO_6^- : 4 + x - 12 = -1$

$$\Rightarrow x = 7$$

Oxidation state of I = +7

CHAPTER

9

Hydrogen

- (a) $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
 $\text{Zn} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
 $\therefore$ Ratio of volumes of H_2 evolved is 1 : 1
- (b) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$
 $2\text{Al} + 2\text{NaOH} + 2\text{H}_2\text{O} \rightarrow 2\text{NaAlO}_2 + 3\text{H}_2$
 The ratio of volumes of hydrogen evolved is 1 : 1.
- (c) Pure hydrogen is evolved by reacting absolute alcohol and Na
 $\text{C}_2\text{H}_5\text{OH} + \text{Na} \rightarrow \text{C}_2\text{H}_5\text{ONa} + \frac{1}{2}\text{H}_2$
 other statements are correct.
- (d) All metal hydrides are ionic in nature.
- (d) It forms calcium and magnesium complex with Ca^{2+} and Mg^{2+} ions present in hard water.
- (c) Sea water is purified by reverse osmosis.
- (c) $\text{Ca}(\text{OH})_2$ is used for the softening of temporary hard water.
 $\text{Ca}(\text{OH})_2(\text{aq}) \xrightarrow{\text{Cloudiness}} \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$
- (b) Water has high dielectric constant i.e., 78.39 C^2/Nm^2 , high liquid range and can dissolve maximum number of compounds. That is why it is used as universal solvent.
- (a) Molecular weight of $\text{H}_2\text{O}_2 = 34$
 Equivalent weight of $\text{H}_2\text{O}_2 = 17$
 $\therefore$ 1 L of 1 N H_2O_2 has = 17 g of H_2O_2
 $\therefore$ 1 L of 1.5 N H_2O_2 has = $1.5 \times 17 = 25.5$ g of H_2O_2
- (a) Normality of $\text{H}_2\text{O}_2 = \frac{\text{Vol. strength}}{5.6}$
 Volume of normal (1N) H_2O_2 solution = 5.6 volumes.
 $\therefore$ Volume of strength of 1.5 N H_2O_2
 = $1.5 \times 5.6 = 8.4$ volumes.
- (a) $\text{H}_2\text{SO}_4 + \text{BaO}_2 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
- (a) Equivalent weight = $\frac{\text{Molar mass}}{\text{Valence factor}}$

$$= \frac{1 \times 2 + 16 \times 2}{2}$$

 Eq. wt. of $\text{H}_2\text{O}_2 = 17$

$$N = \frac{30.36}{17} = 1.78 \text{ N}$$

 Volume strength = $5.6 \times \text{normality}$

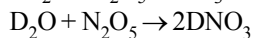
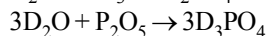
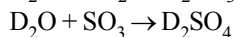
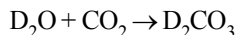
$$= 5.6 \times 1.78 = 10 \text{ V}$$
- (b) $\therefore$ 22.4L O_2 at S.T.P. is obtained from = 68 g H_2O_2
 $\therefore$ 1L O_2 at S.T.P. is obtained from = $\frac{68}{22.4}$ g H_2O_2
 $\therefore$ 20L O_2 at S.T.P. is obtained from

$$= \frac{68}{22.4} \times 20 \text{ g } \text{H}_2\text{O}_2 = 60.71 \text{ g of } \text{H}_2\text{O}_2$$

 Therefore, strength of H_2O_2 in 20 volume H_2O_2
 = 60.71 g/L
 $\therefore$ 1000 mL O_2 at S.T.P. is obtained from = 60.71 g H_2O_2
 $\therefore$ 100 mL O_2 at S.T.P. is obtained from

$$= \frac{60.71}{1000} \times 100 = 6.071\% \text{ } \text{H}_2\text{O}_2$$
- (a) $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + [\text{O}]$
 weak acid
- (a) As H_2O_2 is losing electrons so it is acting as reducing agent.

16. (d) The D_2O reacts with given compounds as follows:

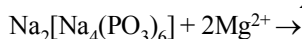
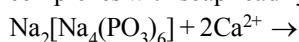


17. (c) 6 volume H_2O_2 would give 6 volumes of O_2 per unit volume of H_2O_2

18. (c) In water gas shift reaction carbon monoxide reacts with steam at high temperature to produce carbondioxide and hydrogen.

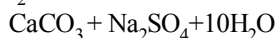
19. (c) Lime can remove temporary hardness of water which is caused due to bicarbonate salt of Ca and Mg.

20. (d) When sodium hexametaphosphate is added to hard water it combines with Ca^{2+} , Mg^{2+} ions present in hard water and forms a complex. These complexes with soap readily produce lather.



21. (3) The ratio of ortho to para hydrogen can never be more than 3 : 1.

22. (100) Molecular mass of $Na_2CO_3 \cdot 10H_2O = 286$
Molecular mass of $CaCO_3 = 100$



$$0.286 \text{ g } Na_2CO_3 \cdot 10H_2O \equiv \frac{0.286}{286} = 10^{-3} \text{ moles}$$

$$\text{Moles of } CaCO_3 = 10^{-3} \text{ moles}$$

$$\text{Mass of } CaCO_3 = 10^{-3} \times 100 = 10^{-1} \text{ g}$$

$$1 \text{ in } 1000 \text{ g of hard water} \equiv 10^{-1} \text{ g of } CaCO_3$$

$$10^6 \text{ g of hard water} \equiv \frac{10^{-1}}{1000} \times 10^6$$

$$\equiv 100 \text{ g. of } CaCO_3$$

$$\text{ppm hardness} = 100$$

23. (15) Volume strength = $5.6 \times N$

$$\text{Thus } N_1 = \frac{10}{5.6}, N_2 = \frac{15}{5.6}, N_3 = \frac{20}{5.6}$$

$$\text{After mixing total volume} = 300 \text{ mL}$$

$$N_1V_1 + N_2V_2 + N_3V_3 = NV$$

$$\frac{10}{5.6} \times 0.1 + \frac{15}{5.6} \times 0.1 + \frac{20}{5.6} \times 0.1 = 0.3 \times N$$

$$N = \frac{10 + 15 + 20}{5.6} \times \frac{0.1}{0.3} = \frac{15}{5.6}$$

$$\text{Volume strength} = 5.6 \times \frac{15}{5.6} = 15$$

24. (6) 6000 parts of ordinary water (H_2O) contains 1 part of heavy water (D_2O).

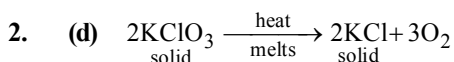
25. (143) 1 g of H_2 gas, release 143 kJ of energy.

CHAPTER

10

The s-Block Element

1. (a) Lithium does not form peroxide.



3. (b) Sodium metal on burning in air gives sodium peroxide.

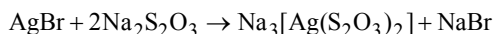
4. (d) Stability of a compound depends upon its enthalpy of formation ΔH_f . The more negative value of ΔH_f shows more stability of a compound. Thus KCl is more stable than LiCl,

$$\Delta H_f \text{ for LiCl} = -408 \text{ kJ/mol}$$

$$\Delta H_f \text{ for KCl} = -436 \text{ kJ/mol}$$

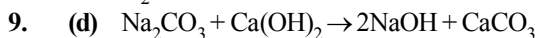
5. (a) If lattice energy is less than hydration energy, the ionic compound will be soluble in water.

6. (c) The unreacted AgBr is removed by soluble complex formation



7. (a) The ionisation potential value of lithium is maximum among alkali metals i.e., its tendency to ionise to give Li^+ ions should be the minimum i.e. Li should be the poorest reducing agent. But, lithium is the strongest reducing agent in aq. solution. This is due to the largest value of hydration energy of Li^+ ions.

8. (d) Na_2O_2 is peroxide of sodium not superoxide. The formula of sodium superoxide is NaO_2 .



10. (b)

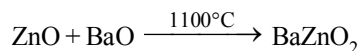
11. (c) The property of sodium atom to emit photons in yellow region of visible spectrum, due to electrically stimulated electron transitions is used in street lights.

12. (a)

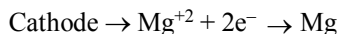
13. (d) Halides of group I and II impart characteristic colour to the flame due to low IE of the central atom. However, ionization energy of Be and Mg atoms is high due to their small size, hence they can't be excited to higher levels by Bunsen burner flame. Thus, their halides don't impart colour to flame.

14. (c) $\text{ZnS} + \text{BaSO}_4$ is lithopone. It is used as white pigment.

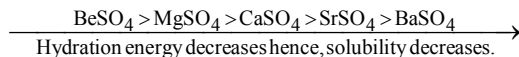
15. (c) Philosopher's wool is ZnO



16. (d) $\text{MgCl}_2 \xrightarrow[\text{(Molten)}]{\text{Electrolysis}} \text{Mg}^{2+} + 2\text{Cl}^-$
Cation Anion



17. (d) BeSO_4 is most soluble because hydration energy is more than lattice energy.

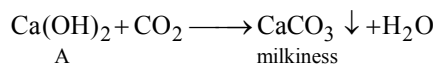
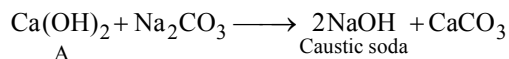
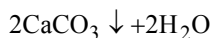
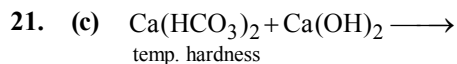


18. (a) $\text{MgCO}_3 \xrightarrow{\text{Heat}} \text{MgO} + \text{CO}_2$

The metal whose oxide is stable, has its carbonate unstable

19. (b) The basic character of oxides increases down the group.

20. (a)



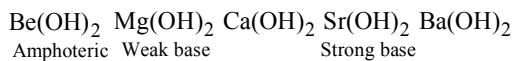
22. (c)

23. (d) Hasenclever and Batchmann's methods are used for the manufacture of bleaching powder.

24. (a) Metal halide on hydrolysis with water form corresponding hydroxides.

The basic strength of hydroxide increases as we move down in a group. This is because of the

increase in size which results in decrease of ionization energy which weakens the strength of M – O bonds in MOH and thus increases the basic strength.



Hence, $\text{Be}(\text{OH})_2$ will have lowest pH.

25. (b) $30 \text{ mL N/50 HCl} \equiv 30 \text{ mL N/50 Ca}(\text{HCO}_3)_2$
 $\equiv 30 \text{ mL N/50 CaCO}_3 \equiv 100 \text{ mL tap water}$

Mass of CaCO_3 in 100 mL tap water

$$= \frac{E \times N \times V}{1000} = \frac{50 \times 30}{50 \times 1000} = 0.03 \text{ g}$$

$$\Rightarrow \text{hardness} = 300 \text{ ppm}$$

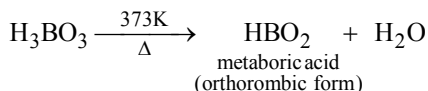
CHAPTER

11

The p-Block Elements

(Groups 13 and 14)

1. (c) H_3BO_3 on heating at 373K yields metaboric acid (HBO_2)



2. (b) $\text{Al}_2\text{Cl}_6 + 12\text{H}_2\text{O} \rightleftharpoons 2[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 6\text{Cl}^-$
3. (b)
4. (b) CaF_2 when added to fused cryolite, lowers the m.p. and increases the conductivity.

5. (b) $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^-$
- $$\xrightarrow{2\text{H}_2\text{O}} [\text{Al}(\text{OH})_4(\text{H}_2\text{O})_2]^-$$

6. (d) Lewis acidic strength: $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$
As BI_3 is strongest Lewis acid among all boron halides therefore, heat of adduct formation will be the maximum numerically, for BI_3 .

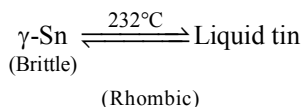
7. (a) $2\text{Al} + \text{N}_2 \xrightarrow{\Delta} 2\text{AlN}$

8. (a)

9. (a)
- $$\text{SiCl}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \xrightarrow[\text{-4HCl}]{\Delta} \underset{\substack{\text{Silicic acid} \\ \text{Silica gel}}}{\text{Si}(\text{OH})_4(\text{aq})}$$
- $$\xrightarrow{\Delta} \text{SiO}_2 \cdot x\text{H}_2\text{O}(\text{s})$$

10. (d) It is hydrolysed with water to form a $\text{Si}(\text{OH})_4$.

11. (b) $\alpha\text{-tin} \xrightleftharpoons[15.2^\circ\text{C}]{\text{(Grey)}} \beta\text{-Sn} \xrightleftharpoons[164^\circ\text{C}]{\text{(White)}}$
(most common and stable)



Although transitions of white to grey tin occurs at any temperature below 15.2°C , it becomes rapid only at -50°C , unless a catalyst is present. During the conversion of white tin to grey tin (in cold countries) volume increases. Grey Sn is very brittle easily crumbles down to powder. This phenomenon is called tin diseases, tin pest or tin plague.

12. (a) Lesser electronegativity of X and similar size of valence shell of X and B favours back bonding.

Cl and F are more electronegative than OMe and NMe group and Cl has large size of $3p$ -orbital.

Among $\text{B}(\text{OMe})_3$ and $\text{B}(\text{NMe}_2)_3$, N in NMe is less electronegative than O in OMe group. Thus, the tendency to form π -bond with boron follows the order.



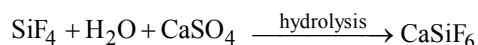
13. (c) $\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \rightarrow 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + 2\text{H}_2\text{O}$

14. (d) Fullerene contains both single and double bond with C-C at a distance of 143.5 pm and 138.3 pm.

15. (b) SnCl_4 is colourless volatile liquid and SnCl_2 is colourless solid Sn conducts electricity and it belongs to 14 group.

16. (a) The stability of $+2$ O.S. follows the order $\text{Pb}^{2+} > \text{Sn}^{2+} > \text{Ge}^{2+}$
Hence reducing power $\text{Ge} > \text{Sn} > \text{Pb}$

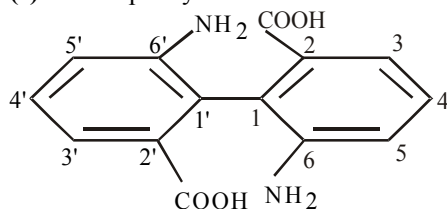
17. (d) $2\text{CaF}_2 + \text{SiO}_2 + \text{H}_2\text{SO}_4 \longrightarrow$



18. (c) $\text{Pb}(\text{NO}_3)_2 \xrightarrow{\Delta} \text{PbO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$
19. (c) $\text{SiO}_2 + 3\text{C} \longrightarrow \text{SiC} + 2\text{CO}$
20. (d) CO molecule is linear ($\text{C} \equiv \text{O}$) and it is supposed that carbon atom in it is sp -hybridised. CO is electron deficient molecule and has π acceptor properties. Due to presence of pair of electrons, it acts as ligand or Lewis base. In metallurgy it is used as reducing agent.
21. (6) In aqueous solution the probable aluminate species is $[\text{Al}(\text{H}_2\text{O})_2(\text{OH})_4]$, hence, co-ordination number of Al is 6.
22. (3) CO_2 , SiO_2 acidic CaO basic and SnO_2 amphoteric.
23. (4) Boron in B_2H_6 is sp^3 hybridised and hence four orbitals are involved.
24. (32) It contains 20 six-membered rings and 12 five-membered rings.
Total no. of rings = $20 + 12 = 32$
25. (56) Producer gas is $\text{CO} + \text{N}_2$.
Sum of molecular mass = $(16 + 12) + (14 + 14)$
= 56

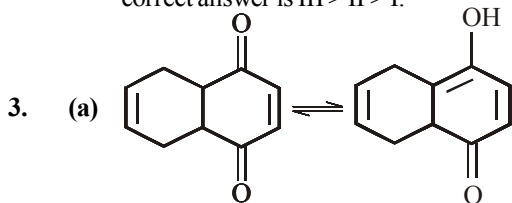
Organic Chemistry- Some Basic Principles and Techniques

1. (b) It is biphenyl derivative

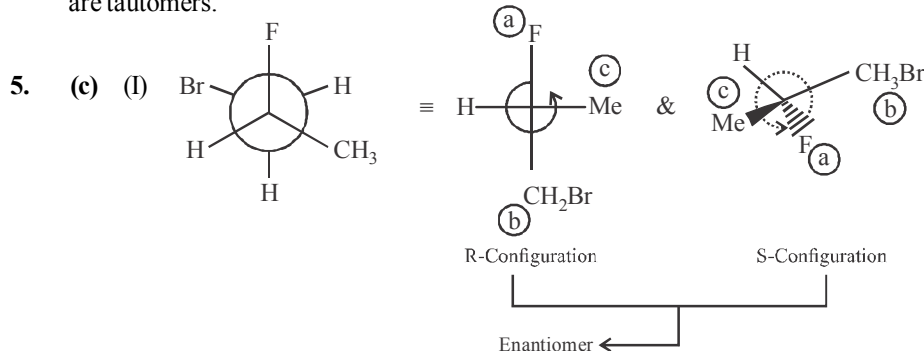


2. (a) Enolic form predominates in compounds containing two carbonyl groups separated by a $-\text{CH}_2$ group. This is due to following two factors.

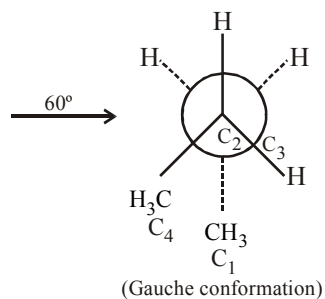
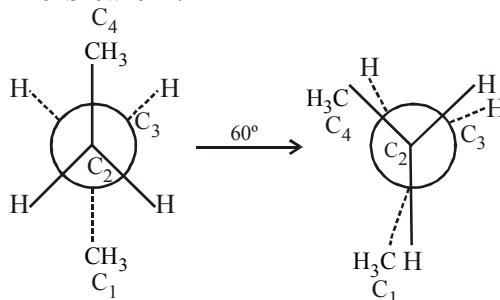
- Presence of conjugation which increases stability.
- Formation of intramolecular hydrogen bond between enolic hydroxyl group and second carbonyl group which leads to stabilisation of the molecule. Hence the correct answer is $\text{III} > \text{II} > \text{I}$.

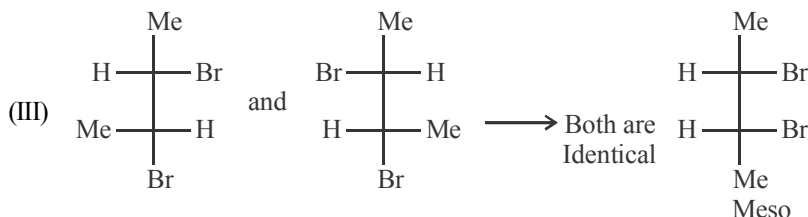
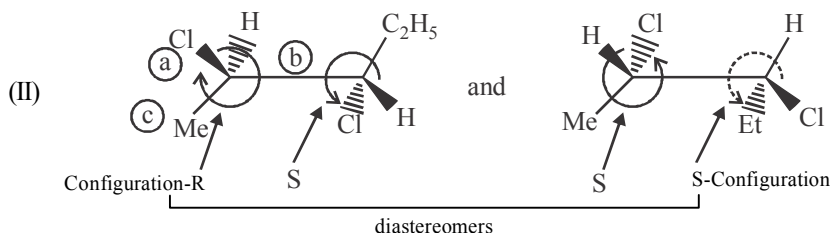


are tautomers.



4. (c) Any conformation between two extreme positions i.e. eclipsed and staggered is known as Gauche or Skew form.

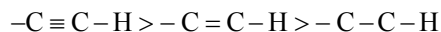




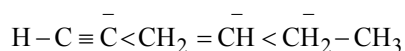
6. (d) The amount of s -character in various hybrid orbitals is as follows.

$$sp = 50\%, sp^2 = 33\% \text{ and } sp^3 = 25\%$$

Therefore s -character of the C – H bond in acetylene (sp) is greater than that of the C – H bond in alkene (sp^2 hybridized) which in turn has greater s character of the C–H bond than in alkanes. Thus owing to a high s -character of the C – H bond in alkynes, the electrons constituting this bond are more strongly held by the carbon nucleus with the result the hydrogen present on such a carbon atom can be easily removed as proton. The acidic nature of three types of C – H bonds follows the following order



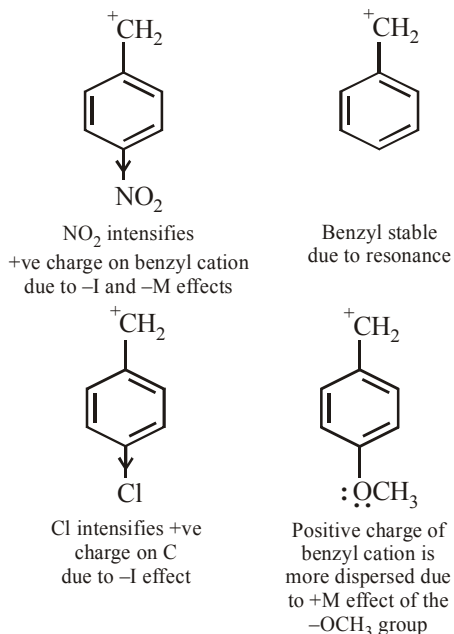
Further, as we know that conjugate base of a strong acid is a weak base, hence the correct order of basicity is



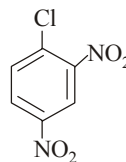
7. (c) $CH_2 = CH - \ddot{Cl} : \leftrightarrow \ddot{CH}_2 - CH = \overset{+}{Cl} :$

a double bond is formed between C and Cl. Hence it is less reactive due to resonance.

8. (d) The stability of carbonium ion is influenced by both resonance and inductive effect.



9. (c) Cl^- is the best leaving group among the given option.
10. (d) $-NO_2$ group is electron attractive group, so it is able to deactivate the benzene ring.



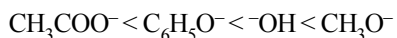
hence withdrawal of electrons from ortho and para position cause easy removal of $-Cl$ atom due to development of positive charge on o - and p -positions.

11. (c) The stronger the acid, the weaker the conjugate base formed.

The acid character follows the order :

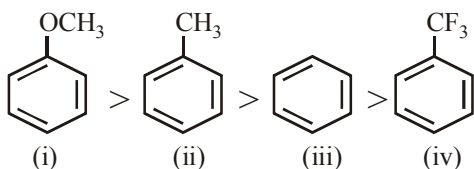


The basic character will follow the order

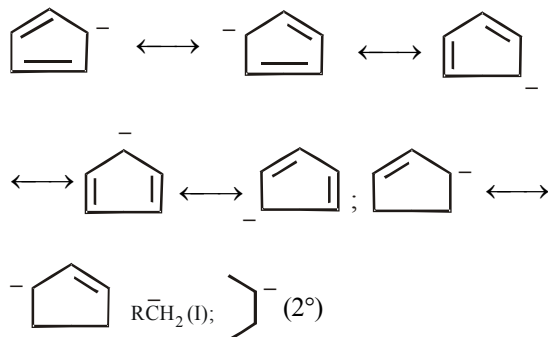


Hence, decreasing order of rate of given reactions with Nu^- is $D > C > A > B$

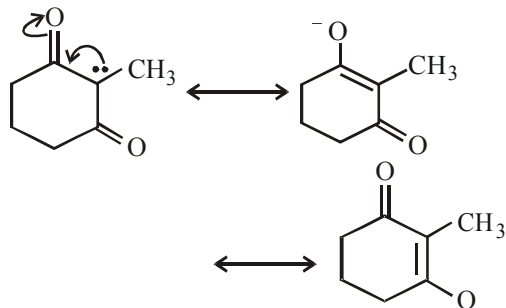
12. (d) $-OCH_3$ and $-CH_3$ groups are activating group while $-CF_3$ is a deactivating group.



13. (a)

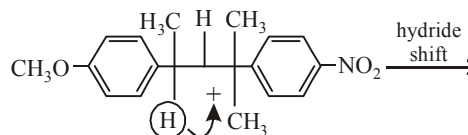
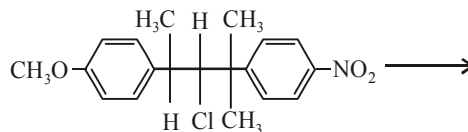


14. (b) (a), (c) and (d) are resonating structures.

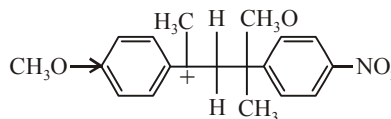


15. (c) 16. (a)

17. (a) This is an example of S_N1 reaction involving carbocation as intermediate.

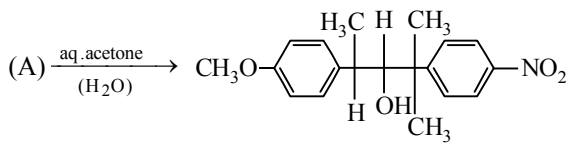


(A) 2° Carbocation

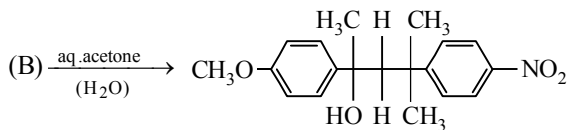


(B) 3° Carbocation; more stable;
(positive charge is dispersed
due to +M effect of $-OCH_3$)

This carbocation is especially stabilised through resonance in which $-\ddot{O}-CH_3$ group acts as a good electron donor.

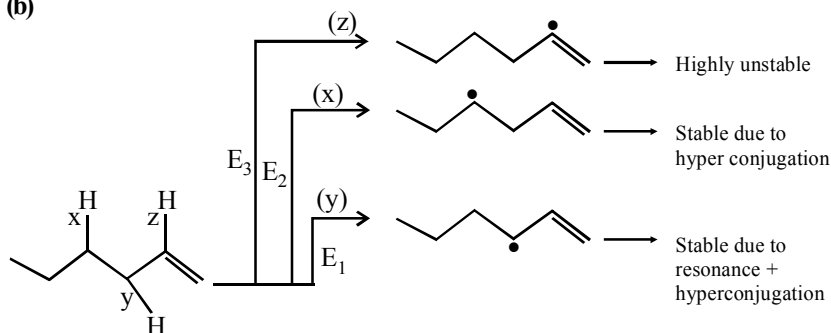


(i)



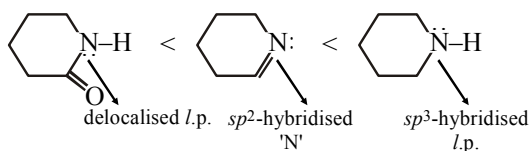
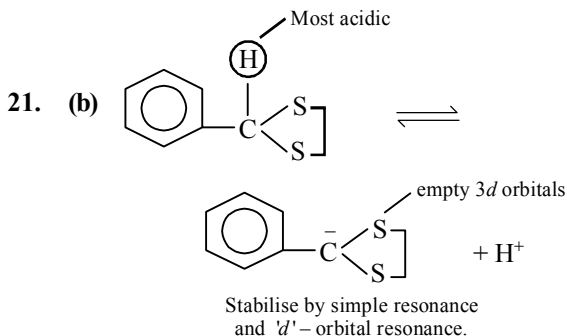
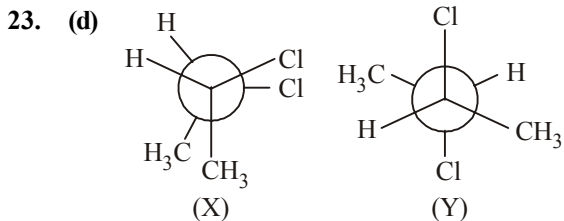
(ii)

18. (b)

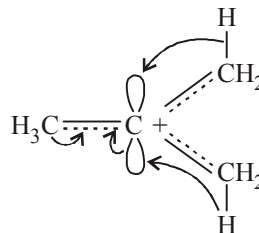


so bond dissociation energy order is $E_1 < E_2 < E_3$ i.e., $(z) > (x) > (y)$

19. (b) Increasing order of basic strength

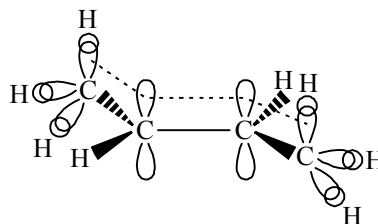
20. (d) B.P. $\propto$ extent of H bonding $\propto$ surface area of molecule.22. (b) Structure III and I has complete octet for all atoms III is more stable than I because $N^+ > O^+$ is the stability order.

Both X and Y are conformers of each other, where X is eclipsed and Y is staggered form.

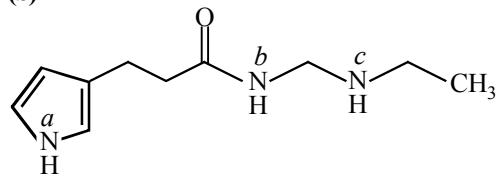
24. (d) *Tert*-butyl cation is more stable than isopropyl cation, because of better hyper conjugation between σ and vacant p -orbital.

(Hyper-conjugation in *tert*-butyl carbocation)

Whereas, *trans*-2-butene is more stable than propene because of hyper-conjugation between σ and π^* -orbitals.



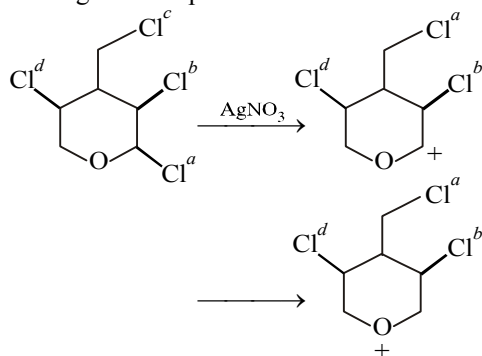
25. (b)



In the given compound most acidic proton will be $N^b - H$. This is because its conjugate base will be most stable and the most nucleophilic nitrogen will be N^c , because the lone pair of electrons present on this N is localised.

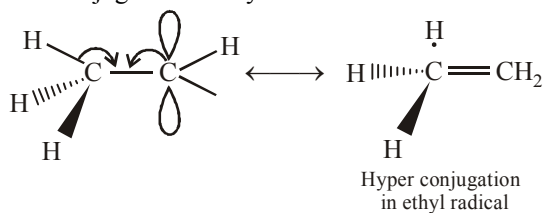
26. (a) The reaction between haloalkane and AgNO_3 gives carbocation intermediate. So, more easily it will be formed, more readily it will react to give precipitate.

In the given compound



O being electronegative, will attract electron density and cleave $\text{C}-\text{Cl}^a$ bond density easily.

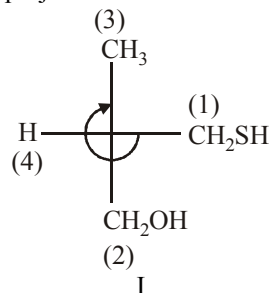
27. (a) The lower stability of ethyl anion (CH_3CH_2^-) compared to methyl anion (CH_3^-) is because of +I-effect of methyl group of ethyl anion. The higher stability of ethyl radical compared to methyl radical is due to σ and p -orbital conjugation in ethyl radical.



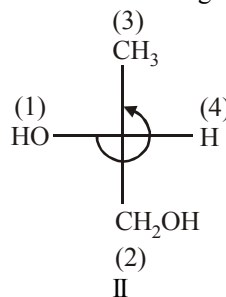
28. (d)
-
- 5-Hydroxycyclohex-2-en-1-one

29. (d)
-

Fischer projection of the above compounds are:



So I should be 'R' but here, lower priority group is on horizontal line. Therefore configuration will be reversed i.e. correct configuration 'S'.



II should be 'S', lower priority group is on horizontal line i.e. 'R'

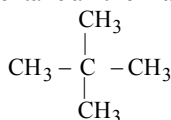
30. (a)
-

CHAPTER

13

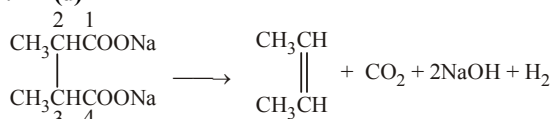
Hydrocarbons

1. (c) In neopentane all the H atoms are same (1°).



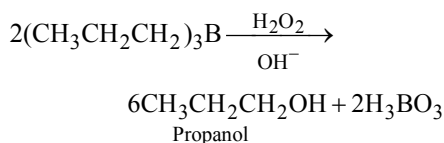
2. (d) In *trans*-isomer polarity developed by one $-\text{CH}_3$ group is cancelled by polarity of the other $-\text{CH}_3$ group placed in opposite direction while the *cis*-isomer is polar although weak causing high boiling point than the *trans*-isomer.

3. (d)

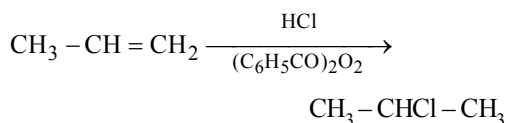


4. (b) The relative reactivities of alkenes towards HX are directly related to the stabilities of the intermediate carbocations. Isobutene (IV) forms the most stable 3° carbocation followed by butene-1 (II) which forms 2° carbocation. Ethylene (I) and vinyl chloride (III), both form 1° carbocation, but the carbocation from vinyl chloride will be less stable because electron-withdrawing halogen intensifies the positive charge and thus destabilizes the carbocation. Hence vinyl chloride will be less reactive than ethylene.

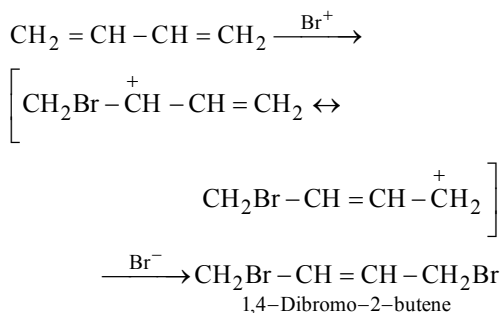
5. (d) $6\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{ether, } 0^\circ\text{C}]{\text{B}_2\text{H}_6}$
1-Propene



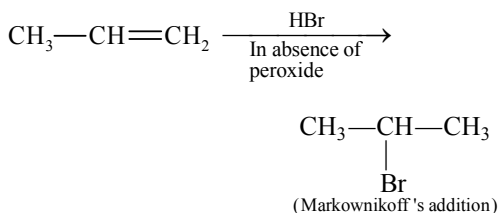
6. (a) Peroxide effect is observed only in case of HBr. Therefore, addition of HCl to propene even in the presence of benzoyl peroxide occurs according to Markovnikoff's rule :



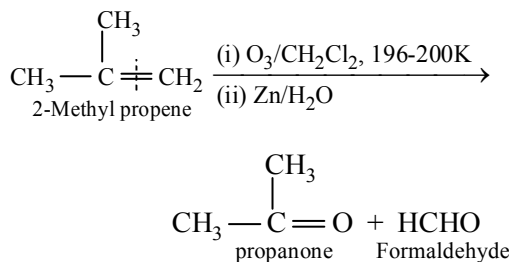
7. (a) The intermediate 2° carbocation shows resonance



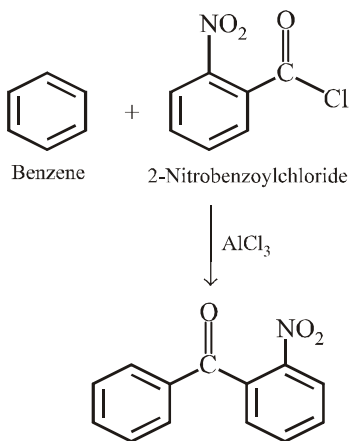
8. (b) Always remember *cis*-2-butene + $\text{Br}_2 \rightarrow$ Racemic-2, 3-dibromobutane
trans-2-Butene + $\text{Br}_2 \rightarrow$ Meso-2, 3-dibromobutane.
9. (d) Addition of hydrogen halide to alkene is an example of electrophilic addition involving carbocations as intermediates.



10. (a)

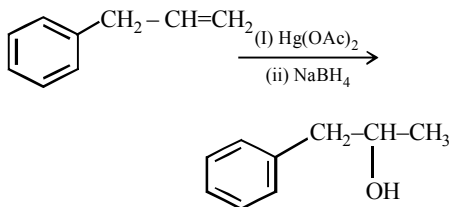


11. (a)



This reaction is Friedel-Craft acylation.

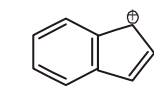
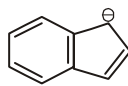
12. (a)



13. (a) Anti-Markownikoff addition is possible only in case of HBr and not in HCl and HI. In HBr both the chain initiation and propagation steps are exothermic, while in HCl, first step is exothermic, and second step is endothermic and in HI, no step is exothermic. Hence HCl and HI do not undergo anti-Markownikoff's addition.

14. (b) The intermediate carbocation is more stable, because positive charge is in conjugation with the benzene nucleus, due to possibility of resonance.

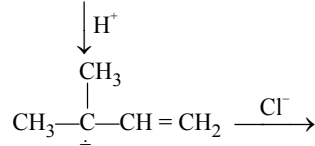
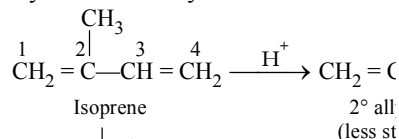
15. (d) The species which follows Huckel's rule ($4n + 2$) π will be most stable species.

It has $8\pi e^-$ s, doesn't follow Huckel's ruleIt has $4\pi e^-$ s, doesn't follow Huckel's ruleIt has $10\pi e^-$ s, follows Huckel's ruleIt has $2\pi e^-$ s, follows Huckel's rule

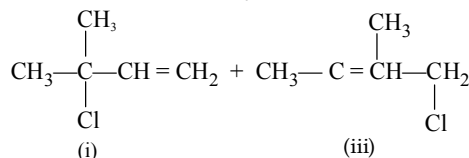
As both the species in option (d) follow Huckel's rule. Thus, it is correct option.

16. (a) H^+ adds to C^1 to form a more stable allylic 3° carbocation rather than to C^2 or C^3 to form a nonallylic 1° carbocation ($\text{CH}_2 - \text{CH}(\text{CH}_3)\text{CH} = \text{CH}_2$

and $\text{H}_2\text{C} = \text{C}(\text{CH}_3)\text{CH}_2 - \text{CH}_2^+$, respectively) or to C^4 to yield to a 2° allylic carbocation.



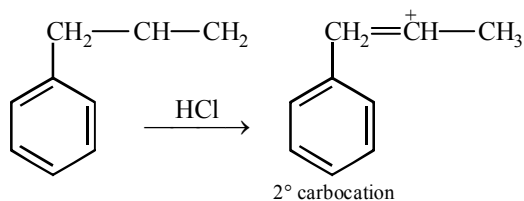
3° allylic carbocation (more stable)
(also favoured by $-\text{CH}_3$ group)

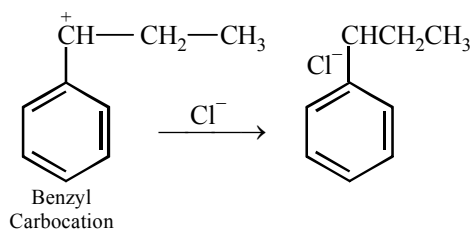


17. (c) Hot $\text{KMnO}_4/\text{NaIO}_4$ are strong oxidising agents which will give CO_2 and $\text{COOH}-\text{COOH}$.

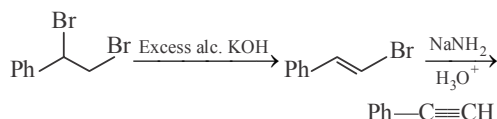
18. (d) Rate of electrophilic addition is proportional to nucleophilicity of alkene and stability of carbocation, so order will be. $\text{IV} > \text{I} > \text{II} > \text{III}$.

19. (c)

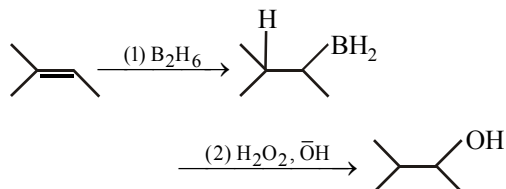




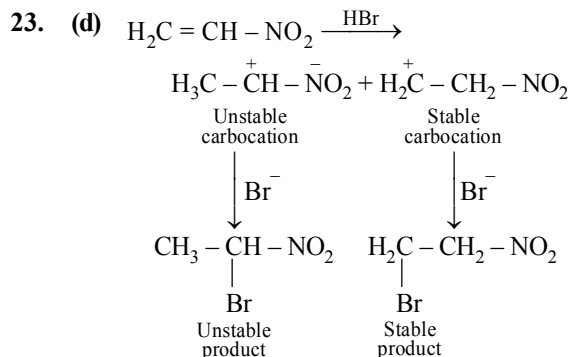
20. (a)



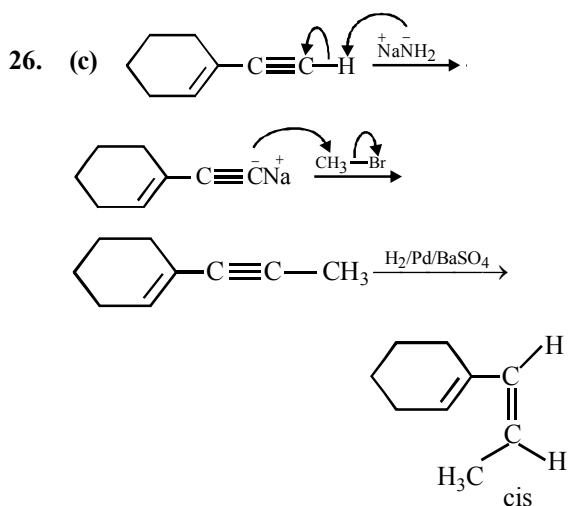
21. (b) Hydroboration oxidation reaction.



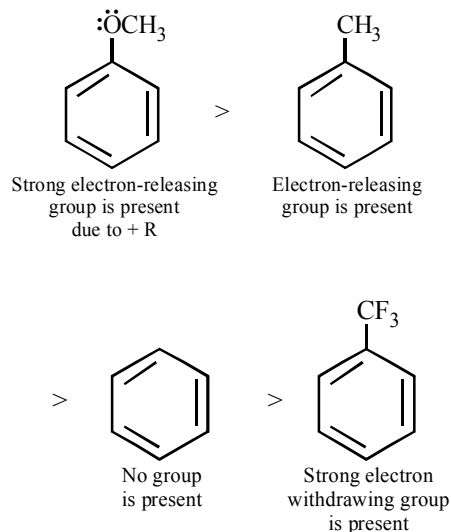
22. (d) Corresponding alkene is not possible.

24. (b) Addition on triple bond takes place by the *syn*-addition of hydrogen in presence of poisoned palladium catalyst.

Since the configuration of the double bond already present is *cis*, the compound formed will have a plane of symmetry and hence optically inactive.

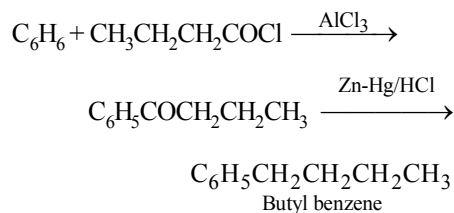
25. (c) Alkynes add hydrogen in *cis*-manner the *cis*-alkene adds bromine in *anti* manner forming racemic product.

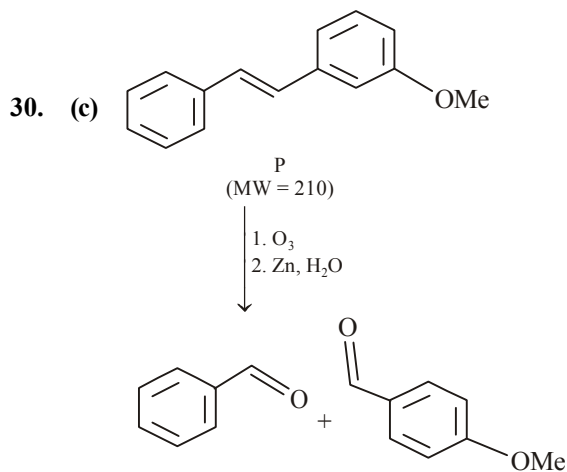
27. (a)



28. (c) Bromination in presence of light leads to side chain substitution. For nuclear substitution we require halogen carrier.

29. (c) The Friedel-Crafts alkylation reaction will give propyl phenyl ketone which further on Clemmensen's reduction will give butyl benzene.





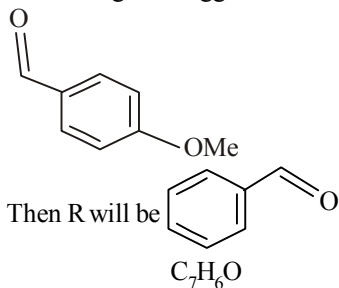
$$\text{No. of moles of P} = \frac{\text{Given weight}}{\text{M.W.}} = \frac{420}{210} = 2 \text{ milimol}$$

$$\text{No. of milimoles of Q formed} = 2 \times \frac{40}{100} = \frac{4}{5} \text{ milimol}$$

$$\text{Now, calculating M.W. of Q} = \frac{\text{mass}}{\text{mol}}$$

$$= \frac{10.8.8\text{mg}}{\frac{4}{5} \text{ m mol}} = 136 \text{ g/mol}$$

M.W. 136 g/mol suggests that compound Q will

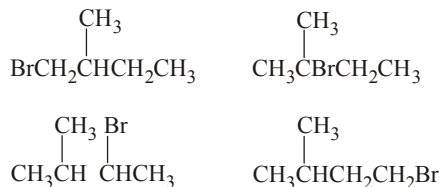


$$\text{Mol. wt} = 106 \text{ g/mol}$$

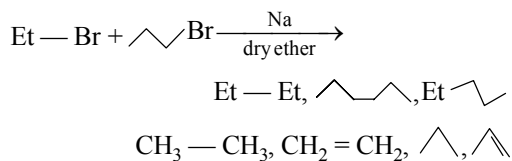
$$\text{Moles of R} = 2 \times \frac{40}{100} = \frac{4}{5} \text{ milimol}$$

$$\text{Mass of R} = \frac{4}{5} \times 10^{-3} \text{ mol} \times 106 \times 10^3 \text{ mg/mol} = 84.8 \text{ mg.}$$

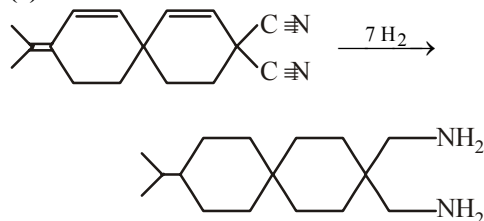
31. (4) All alkyl bromides having carbon skeleton of isopentane (2-methylbutane $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$) will give isopentane via Grignard reagent.



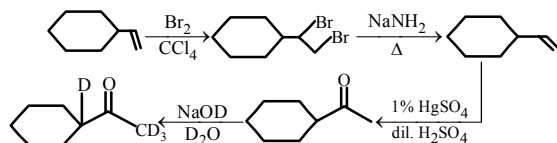
32. (7)



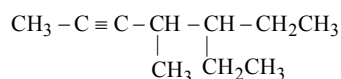
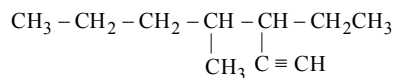
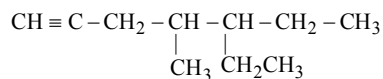
33. (7)



34. (4) '4'-Deuterium exchange occur.



35. (3) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{CH}_2 - \text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{CH}_3$
- 3-Ethyl-4-methylheptane



CHAPTER

14

Environment Chemistry

1. (a) In Antarctica ozone depletion is due to formation of acrolein.
2. (b) 3. (b) 4. (b) 5. (c)
6. (b) The coldest region is mesosphere (temp. – 27°C to – 92°C)
7. (a) Radiation coming from sun or outerspace have high energy or short wavelength, which are allowed to enter by green house gases. However, radiation emitted by earth is in infrared region, having long wavelength, are reflected back by the envelope of green house gases.
8. (d) London smog is formed in morning during winter.
9. (a) The average residence time of NO is 4 days.
10. (a) Normal rain water has pH 5.6. Thunderstorm results in the formation of NO and HNO₃ which lowers the pH.
11. (d) Decrease in D.O causes death of fish.
12. (a) 13. (c) 14. (c)
15. (c) Because they are very sensitive to sulphur dioxide and in cities the amount of SO₂ is high so lichen do not grow in cities.

The Solid State

- (b) Liquid crystals on heating first become turbid and then clear.
- (d) Orthorhombic system has 4 bravais lattice:
(i) primitive (ii) *bcc* (iii) *fcc* (iv) end centred
- (d) Since in NaCl type of structure 4 formula units form a cell.

58.5 gm. of NaCl = 6.023×10^{23} atoms

$$1 \text{ gm of NaCl} = \frac{6.023 \times 10^{23}}{58.5} \text{ atoms}$$

4 atoms constitute 1 unit cell

$$\therefore \frac{6.023 \times 10^{23}}{58.5} \text{ atoms constitute}$$

$$= \frac{6.023 \times 10^{23}}{58.5 \times 4} = 2.57 \times 10^{21} \text{ unit cells.}$$

- (d) Order of Bragg diffraction (n) = 2;
Wavelength (λ) = $1 \text{ \AA}$ and angle (θ) = 60° .

We know from the Bragg's equation

$$n\lambda = 2d \sin \theta$$

$$\text{or } 2 \times 1 = 2d \sin 60^\circ$$

$$\Rightarrow 2 \times 1 = 2d \cdot \frac{\sqrt{3}}{2} \Rightarrow d = \frac{2}{\sqrt{3}} = 1.15 \text{ \AA}$$

(where d = Distance between the scattering planes)

- (a) Atoms are present in the corners of cube = A and atom present at body centre = B.
We know that a cubic unit cell has 8 corners.

Therefore contribution of each atom A at the corner = $\frac{1}{8}$. Since number of atoms per unit cell

is 8, therefore total contribution = $8 \times \frac{1}{8} = 1$. We

also know that atoms in the body centre, therefore number of atoms of B per unit cell = 1.
Thus formula of the compound is AB.

- (b) The face centered cubic unit cell contains 4 atom

$$\therefore \text{Total volume of atoms} = 4 \times \frac{4}{3} \pi r^3 = \frac{16}{3} \pi r^3$$

- (b) For bcc lattice, $\sqrt{3}a = 4R \Rightarrow a = \frac{4 \times 96}{\sqrt{3}} \text{ pm}$
= 221.7 pm

(where R is the radius of Ca atom)

Volume of unit

$$\text{cell} = a^3 = (211.7 \times 10^{-12})^3 \text{ m}^3 = 10.9 \times 10^{-30} \text{ m}^3$$

- (b) Volume of crystal = $(1.0 \text{ cm})^3 = (10^{-2} \text{ m})^3 = 10^{-6} \text{ m}^3$
Number of unit cells

$$= \frac{10^{-6} \text{ m}^3}{10.9 \times 10^{-30} \text{ m}^3} = 9.17 \times 10^{22}$$

- (c) In a *fcc* lattice, the distance between the cation and anion is equal to the sum of their radii, which is equal to half of the edge length of unit cell,

i.e. $r^+ + r^- = \frac{a}{2}$ (where a = edge length)

$$r^+ = 95 \text{ pm}, r^- = 181 \text{ pm}$$

$$\begin{aligned}\text{Edge length} &= 2r^+ + 2r^- = (2 \times 95 + 2 \times 181) \text{ pm} \\ &= (190 + 362) \text{ pm} = 552 \text{ pm}.\end{aligned}$$

10. (c) Number of Cu-atoms per unit cell

$$= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4.$$

$$\text{Number of Ag-atoms per unit cell} = \frac{1}{4} \times 12 = 3$$

Number of Au-atoms per unit cell = 1 (at body centre)

Formula : $\text{Cu}_4\text{Ag}_3\text{Au}$

11. (c) Let the number of Z atoms in the *ccp* arrangement = 100

Thus the number of tetrahedral sites = 200

Since all the tetrahedral sites are occupied by X atoms, the number of X atoms = 200

Hence ratio of X : Z = 2 : 1

Thus the formula is X_2Z

12. (a) Fraction of unoccupied sites in NaCl crystal

$$\begin{aligned}&= 1 - \frac{2.165 \times 10^3}{2.178 \times 10^3} \\ &= \frac{2.178 \times 10^3 - 2.165 \times 10^3}{2.178 \times 10^3} = \frac{0.13 \times 10^3}{2.178} \\ &= \frac{130}{2178} = 5.96 \times 10^{-3}\end{aligned}$$

13. (a) $\rho = \frac{ZM}{N_A V}$

$$Z = \frac{\rho N_A V}{M} = \frac{2 \times 6 \times 10^{23} \times (5 \times 10^{-8})^3}{75}$$

$Z = 2$, which represents *bcc* structure

$$\begin{aligned}\therefore r &= \frac{\sqrt{3}}{4} a = \frac{\sqrt{3}}{4} \times 5 = 2.165 \text{ \AA} = 216.5 \text{ pm} \\ &\approx 217 \text{ pm}\end{aligned}$$

14. (d) $M = \frac{\rho \times a^3 \times N_A \times 10^{-30}}{Z}$

$$= \frac{10 \times (100)^3 \times 6.02 \times 10^{23} \times 10^{-30}}{4} = 15.05$$

$\therefore$ Number of atoms in 100 g

$$= \frac{6.02 \times 10^{23}}{15.05} \times 100 = 4 \times 10^{25}$$

15. (b) $\rho = \frac{ZM}{N_A V}$

$$Z = \frac{\rho N_A V}{M}$$

$$= \frac{8.92 \times 6.02 \times 10^{23} \times (362)^3 \times 10^{-30}}{63.55} = 4$$

$\therefore$ It has *fcc* unit cell

16. (b) $r_+ / r_- = \frac{180}{187} = 0.962$ which lies in the

range of 0.732 – 1.000, hence co-ordination number = 8 i.e., the structure is CsCl type.

17. (a) ZnS has cubic close packed (*ccp*) structure. The S^{2-} ions are present at the corners of the cube and at the centre of each face. Zinc ions occupy half of the tetrahedral sites. Each zinc ion is surrounded by four sulphide ions which are disposed towards the corners of a regular tetrahedron. Similarly, S^{2-} ion is surrounded by four Zn^{2+} ions.

18. (c) Number of O-atoms per unit cell

$$= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4.$$

Number of octahedral holes per unit cell

$$= 1 \times 4 = 4$$

$$\text{Number of } \text{Fe}^{3+} \text{ ions per unit cell} = \frac{50 \times 4}{100} = 2$$

$$\begin{aligned}\text{Number of tetrahedral voids per unit cell} \\ &= 2 \times 4 = 8.\end{aligned}$$

$$\text{Number of Zn}^{2+} \text{ ions per unit cell} = \frac{1}{8} \times 8 = 1$$

Hence, formula : ZnFe_2O_4

$$\begin{aligned} 19. \quad (b) \quad \text{Volume of unit cell} &= a \times b \times c \\ &= 5 \times 10^{-8} \times 8 \times 10^{-8} \times 4 \times 10^{-8} \\ &= 1.6 \times 10^{-22} \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Mass of unit cell} &= 1.6 \times 10^{-22} \times 5.2 \\ &= 8.32 \times 10^{-22} \text{ g} \end{aligned}$$

Number of molecules in one unit cell

$$= \frac{8.32 \times 10^{-22} \text{ g}}{166.4 \text{ g mol}^{-1}} = 3$$

$$\begin{aligned} 20. \quad (a) \quad \text{Number of A cations per unit cell} &= 4 \\ \text{Number of B anions per unit cell must be} &= 2 \times 4 = 8 \end{aligned}$$

$$\begin{aligned} \text{Number of tetrahedral voids per unit cell} &= 2 \times 4 = 8 \end{aligned}$$

$$\begin{aligned} \text{Number of octahedral voids per unit cell} &= 1 \times 4 = 4. \end{aligned}$$

Hence, occupancy of tetrahedral voids = 100%

$$21. \quad (d) \quad \text{If in an ionic crystal of the type } A^+, B^-, \text{ equal number of cations and anions are missing from their lattice sites so that the electrical neutrality is maintained. The defect is called Schottky defect.}$$

$$22. \quad (d) \quad \text{For } n\text{-type, impurity added to silicon should have more than 4 valence electrons.}$$

$$23. \quad (d) \quad \text{In non stoichiometric defects ratio of cation and anion is not same that is represented by chemical ideal formula, this is due to variable oxidation nos of cation. Transition elements show variable oxidation nos.}$$

$$\begin{aligned} 24. \quad (b) \quad \text{Since each Sr}^{++} \text{ ion provides one cation vacancy, hence} \\ \text{Concentration of cation vacancies} &= \text{mole \% of SrCl}_2 \text{ added} \end{aligned}$$

$$= 10^{-4} \text{ mole \%} = \frac{10^{-4}}{100} \times 6.023 \times 10^{23} = 6.023 \times 10^{17}$$

$$25. \quad (c) \quad \text{Frenkel defect.}$$

$$26. \quad (910) \quad \text{For } bcc,$$

$$\begin{aligned} d &= \frac{\sqrt{3}}{2} a \text{ or } a = \frac{2d}{\sqrt{3}} = \frac{2 \times 4.52}{1.732} = 5.219 \text{ \AA} \\ &= 522 \text{ pm} \end{aligned}$$

$$\begin{aligned} \rho &= \frac{z \times M}{a^3 \times N_A \times 10^{-30}} \\ &= \frac{2 \times 39}{(522)^3 \times (6.023 \times 10^{23}) \times 10^{-30}} \\ &= 0.91 \text{ g/cm}^3 = 910 \text{ kg m}^{-3} \end{aligned}$$

$$27. \quad (1.123) \quad \frac{r_{\text{Na}^+}}{r_{\text{Cl}^-}} = 0.55 \text{ and } \frac{r_{\text{K}^+}}{r_{\text{Cl}^-}} = 0.74$$

$$\frac{r_{\text{Na}^+}}{r_{\text{Cl}^-}} + 1 = 0.55 + 1 \text{ and } \frac{r_{\text{K}^+}}{r_{\text{Cl}^-}} + 1 = 0.74 + 1$$

$$\frac{r_{\text{Na}^+} + r_{\text{Cl}^-}}{r_{\text{Cl}^-}} = 1.55 \text{ and } \frac{r_{\text{K}^+} + r_{\text{Cl}^-}}{r_{\text{Cl}^-}} = 1.74$$

Now edge length ratio of KCl and NaCl is

$$\frac{1.74}{1.55} = \frac{r_{\text{K}^+} + r_{\text{Cl}^-}}{r_{\text{Cl}^-}} \times \frac{r_{\text{Cl}^-}}{r_{\text{Na}^+} + r_{\text{Cl}^-}} = 1.123$$

$$28. \quad (27) \quad \text{Density is given by}$$

$$d = \frac{Z \times M}{N_A a^3}; \text{ where } Z = \text{number of formula units}$$

present in unit cell, which is 4 for fcc

a = edge length of unit cell. M = Molecular mass

$$\begin{aligned} 2.72 &= \frac{4 \times M}{6.02 \times 10^{23} \times (404 \times 10^{-10})^3} \\ &(\because 1 \text{ pm} = 10^{-10} \text{ cm}) \end{aligned}$$

$$M = \frac{2.72 \times 6.02 \times (404)^3}{4 \times 10^7} = 26.99 = 27 \text{ g mole}^{-1}$$

29. (5) For bcc ; $r = \frac{\sqrt{3}}{2} a$;

$$d = \frac{n \times M}{N_{Av} \times a^3} \quad \text{or} \quad n = \frac{d \times N_{Av} \times a^3}{M}$$

$$\Rightarrow n = \frac{2 \times 6 \times 10^{23} (5 \times 10^{-8})^3}{75} = 2$$

Therefore Metal crystallizes in BCC structure

and for a BCC lattice $\sqrt{3}a = 4r$

$$r = \frac{\sqrt{3}}{4} a = \frac{\sqrt{3} \times 5}{4} = 2.165 \text{ \AA} = 216.5 \text{ pm}$$

$$x = \frac{216.5}{43.3}$$

$$x = 5$$

30. (4) For an octahedral void $a = 2(r + R)$

In fcc lattice the largest void present is octahedral void. If the radius of void sphere is R and of lattice sphere is r,

$$\text{Then, } r = \frac{\sqrt{2} \times 400}{4} = 141.42 \text{ pm} \quad (a = 400 \text{ pm})$$

Applying condition for octahedral void,

$$2(r + R) = a$$

$$\therefore 2R = a - 2r = 400 - 2 \times 141.42$$

$$\therefore \text{Diameter of greatest sphere} = \mathbf{117.16 \text{ pm}}$$

$$d = \frac{117.16}{29.29} = 4$$

Solutions

1. (c) Applying the law of equivalence,

$$N_1 V_1 + N_2 + V_2 + N_3 + V_3 = N_R V_R$$

$$\frac{N}{10} \times 50 + \frac{N}{3} \times 30 + \frac{N}{2} \times 10 = N_R \times 1000$$

$$5N + 10N + 5N = 1000 \times N_R$$

$$\Rightarrow N_R = \frac{N}{50}$$

2. (a)
- $K_H = 100 \text{ kbar} = 10^5 \text{ bar} = 1 \text{ bar}$

$$P = K_H \times x_A$$

$$x_A = \frac{P}{K_H} = \frac{1}{100 \times 10^3} = 10^{-5}$$

$$\text{Moles of water} = \frac{1000}{18} = 55.5$$

$$\text{Weight of water} = 1000 \text{ g } (\because 1000 \text{ mL} = 1000 \text{ g})$$

$$\text{Mole fraction} = 10^{-5} = \frac{x}{55.5 + x}$$

As $55.5 \gg x$, thus neglecting x from denominator

$$10^{-5} = \frac{x}{55.5} \Rightarrow x = 55.5 \times 10^{-5} \text{ moles}$$

or 0.555 millimoles.

3. (b)
- $P_{\text{total}} = P_A^\circ X_A + P_B^\circ X_B$

$$550 = P_A^\circ \times \frac{1}{4} + P_B^\circ \times \frac{3}{4}$$

$$P_A^\circ + 3P_B^\circ = 550 \times 4 \quad \dots(i)$$

In second case

$$P_{\text{total}} = P_A^\circ \times \frac{1}{5} + P_B^\circ \times \frac{4}{5}$$

$$P_A^\circ + 4P_B^\circ = 560 \times 5 \quad \dots(ii)$$

Subtract (i) from (ii)

$$\therefore P_B^\circ = 560 \times 5 - 550 \times 4 = 600$$

$$\therefore P_A^\circ = 400$$

4. (c) For dilute solution,
- $\frac{\Delta P}{P^\circ} \Rightarrow \frac{n_{\text{solute}}}{n_{\text{solvent}}}$

For solution in A,

$$\frac{\Delta P_A}{P_A^\circ} = \frac{W/M}{W_A/M_A} = \frac{W}{M} \times \frac{M_A}{W_A} \quad \dots(i)$$

$$\text{For solution in B, } \frac{\Delta P_B}{P_B^\circ} = \frac{W}{M} \times \frac{M_B}{W_B} \quad \dots(ii)$$

$$\text{From (i) and (ii), } \frac{\Delta P_A/P_A^\circ}{\Delta P_B/P_B^\circ} = 2 = \frac{M_A W_B}{M_B W_A} = \frac{M_A}{M_B}$$

$$(W_A = W_B) M_A = M_B$$

5. (d) From the given data

In case I

3 mole of A + 1 mole of B, V.P. = 550 mm

Using the equation

$P = P_A^\circ x_A + P_B^\circ x_B$, we have

$$550 = P_A^\circ \times \frac{3}{4} + P_B^\circ \times \frac{1}{4} \quad \left[= \frac{3}{4}, = \frac{1}{4} \right]$$

$$\text{or } 550 = 0.75 P_A^\circ + 0.25 P_B^\circ$$

In case II

4 mole of A + 1 mole of B; V.P. = 560 mm

$$\therefore 560 = P_A^\circ \times 0.80 + P_B^\circ \times 0.20$$

$$\left[x_A = \frac{4}{5} \text{ or } 0.8 \text{ and } x_B = \frac{1}{5} = 0.20 \right]$$

Solving the above two equations, we get

$$P_A^\circ = 600 \text{ mm of Hg}$$

6. (b) $\therefore P_A = P_A^\circ X_A$,

$$P_B = P_B^\circ X_B$$

$$(P_A^\circ \text{ and } P_B^\circ = \text{v.p. of pure A and B})$$

$$Y_A = \frac{P_A}{P_A + P_B}$$

$$\Rightarrow \frac{P_A^\circ X_A}{P_A^\circ X_A + P_B^\circ (1 - X_A)}$$

$$\Rightarrow Y_A = \frac{P_A^\circ X_A}{X_A (P_A^\circ - P_B^\circ) + P_B^\circ}$$

$$\Rightarrow \frac{1}{Y_A} = \left(\frac{P_A^\circ - P_B^\circ}{P_A^\circ} \right) + \frac{P_B^\circ}{P_A^\circ} \cdot \frac{1}{X_A}$$

$$\text{So, slope is } \frac{P_B^\circ}{P_A^\circ} \text{ and intercept} = \frac{P_A^\circ - P_B^\circ}{P_A^\circ}$$

7. (a) The given data are

$$P_{\text{water}} = 17.0 \text{ torr;}$$

$$P_{\text{total}} (4 \text{ mole \% solution})$$

$$= P_{\text{NH}_3} + P_{\text{water}} = 50.0 \text{ torr}$$

$$X_{\text{NH}_3} = 0.04 \text{ and } X_{\text{water}} = 0.96$$

Now according to Raoult's law;

$$P_{\text{water}} = X_{\text{water}} P_{\text{water}}^\circ$$

$$= 0.96 \times 17.0 \text{ torr} = 16.32 \text{ torr}$$

Now Henry's law constant for ammonia is

$$K_H(\text{NH}_3) = \frac{P_{\text{NH}_3}}{X_{\text{NH}_3}} = \frac{33.68 \text{ torr}}{0.04} = 842 \text{ torr}$$

Hence, for 5 mole % solution, we have

$$P_{\text{NH}_3} = K_H(\text{NH}_3) X_{\text{NH}_3}$$

$$= (842 \text{ torr})(0.05) = 16.15 \text{ torr}$$

$$\text{Thus, } P_{\text{total}} (5 \text{ mole \% solution})$$

$$= P_{\text{NH}_3} + P_{\text{water}} = 42.1 + 16.15 = 58.25 \text{ torr}$$

8. (a) Given vapour pressure of pure solvent (P°) = 121.8 mm Hg;

$$\text{Weight of solute (w)} = 15 \text{ g}$$

Weight of solvent (W) = 250 g; vapour pressure of solution (P) = 120.2 mm Hg and Molecular weight of solvent (M) = 78 from Raoult's law.

$$= \frac{P^\circ - P}{P^\circ} = \frac{w}{m} \times \frac{M}{W}$$

$$\frac{121.8 - 120.2}{121.8} = \frac{15}{m} \times \frac{78}{250}$$

$$\text{or } m = \frac{15 \times 78}{250} \times \frac{121.8}{1.6} = 356.2$$

9. (b) As $\Delta T_f = K_f m$
 $\Delta T_b = K_b m$

$$\text{Hence, we have } m = \frac{\Delta T_f}{K_f} = \frac{\Delta T_b}{K_b}$$

$$\text{or } \Delta T_f = \Delta T_b \frac{K_f}{K_b}$$

$$\Rightarrow [\Delta T_b = 100.18 - 100 = 0.18^\circ\text{C}]$$

$$= 0.18 \times \frac{1.86}{0.512} = 0.654^\circ\text{C}$$

As the freezing point of pure water is 0°C ,

$$\Delta T_f = 0 - T_f$$

$$0.654 = 0 - T_f$$

$$\therefore T_f = -0.654$$

Thus the freezing point of solution will be -0.654°C .

10. (c) Moles of urea = $\frac{5}{60}$; moles of fructose

$$= \frac{5}{180}; C_{\text{urea}} = \frac{0.08 \times 1000}{100} \Rightarrow 0.833$$

$$\text{moles of sucrose} = \frac{5}{342};$$

$$C_{\text{sucrose}} = 0.0146 \times \frac{1000}{100} = 0.146$$

moles of KCl(effective)

$$= 2 \times \frac{5}{74.5} = \frac{5}{37.25}; \frac{0.134 \times 1000}{100} = 1.34$$

$\pi \propto C$

this order is $\pi_4 > \pi_1 > \pi_2 > \pi_3$

11. (c) $\Delta T_f(\text{normal}) = K_f m = 1.86 \times 0.01 = 0.0186$;

$$i = \frac{\Delta T_{f(\text{obs})}}{\Delta T_{f(\text{nor})}} = \frac{0.0205}{0.0186} = 1.10 = 1 + \alpha; \alpha = 0.1$$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.01 \times 0.1^2}{1-0.1} = \frac{1}{9} \times 10^{-3};$$

$$K_b = \frac{K_w}{K_a} = 1.0 \times 10^{-14} \times 9 \times 10^3 = 9 \times 10^{-11}$$

12. (c) Let x and y be the molar masses of A and B respectively.

$$\text{Then, } \frac{\Delta T_f}{K_f} = m = 1 = \frac{8 \times 1000}{(x+2y) \times 100} = \frac{80}{x+2y} \dots (i)$$

$$\text{And } \frac{\Delta T_f}{K_f} = m = 1 = \frac{10 \times 1000}{(2x+y) \times 100} = \frac{100}{2x+y} \dots (ii)$$

Solving (i) and (ii), $x = 40$; $y = 20$

13. (b) According to Raoult's law

$$\frac{\Delta P}{P^\circ} = \frac{n}{n+N} \quad (\text{mole fraction of solute})$$

$$\frac{10}{P^\circ} = 0.2 \quad \therefore P^\circ = 50 \text{ mm of Hg}$$

For other solution of same solvent

$$\frac{20}{P^\circ} = \frac{n}{n+N} \quad (\text{Mole fraction of solute})$$

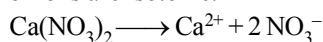
$$\Rightarrow \frac{20}{50} = \text{Mole fraction of solute}$$

$$\Rightarrow \text{Mole fraction of solute} = 0.4$$

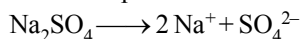
As mole fraction of solute + mole fraction of solvent = 1

Hence, mole fraction of solvent = $1 - 0.4 = 0.6$

14. (a) The solution which provide same number of ions are isotonic.



Total ions produced = 3



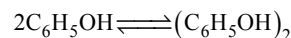
Total ions produced = 3

$\therefore 0.1 \text{ M Ca}(\text{NO}_3)_2$ and 0.1 M

Na_2SO_4 are isotonic.

15. (c) Molality of solution,

$$= \frac{23.5 \times 1000}{94 \times 500} = 0.5 \quad (\text{mol. mass} = 94)$$



(α = degree of dimerization)

Van't Hoff factor of phenol,

$$= \left[1 - \left(1 - \frac{1}{n} \right) \alpha \right] = \left[1 - \left(1 - \frac{1}{2} \right) \alpha \right] = 1 - \frac{\alpha}{2}$$

Hence

$$\Delta T_f = i K_f m = \left(1 - \frac{\alpha}{2} \right) \times 12.0 \times 0.5 = \left(1 - \frac{\alpha}{2} \right) \times 6.0$$

$$3.60 = \left(1 - \frac{\alpha}{2} \right) \times 6.0 \Rightarrow \alpha = 0.80$$

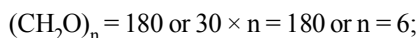
$\therefore$ 80% of phenol is present in dimeric form.

16. (d) Mole ratio of C : H : O is 1 : 2 : 1 so empirical formula is CH_2O

$$m = \frac{\Delta T_b}{K_b} \Rightarrow \frac{0.15}{0.51} \Rightarrow 0.294;$$

$$0.294 = \frac{5}{M} \times \frac{1000}{95}; M \approx 180$$

M = molar mass of solute



$\therefore$ molecular formula is $\text{C}_6\text{H}_{12}\text{O}_6$.

17. (d) Given,

$$W_{\text{H}_2\text{SO}_4} = 38 \text{ g}, \quad W_{\text{H}_2\text{O}} = 100 - 38 = 62 \text{ g}$$

$$\therefore \Delta T_f = \frac{1000 \times K'_f \times w}{mW}$$

$$\Delta T_f = \frac{1000 \times 1.86 \times 38}{98 \times 62} \quad \therefore \Delta T_{f \text{ Normal}} = 11.633$$

$$\text{Now, } \frac{\Delta T_{f \text{ exp}}}{\Delta T_{f \text{ N}}} = i = 2.50$$

$$\therefore \Delta T_{\text{exp}} = 11.633 \times 2.50 = 29.08$$

$$\therefore \text{f. pt.} = 273 - 29.08 = 243.92 \text{ K}$$

18. (c) For Na_3PO_4 , $i = 1 + 3\alpha = 1 + 3 \times 0.5 = 2.5$; for MgSO_4 , $i = 1 \propto = 1 + 0.6 = 1.6$
 100 g solution contains 8.2 g Na_3PO_4 and 12 g MgSO_4
 $\Delta T_b = K_b \cdot m \cdot i =$

$$K_b \cdot \left[\frac{\text{effective no. of moles of } (\text{Na}_3\text{PO}_4 + \text{MgSO}_4) \times 1000}{\text{mass of solvent (in g)}} \right]$$

$$\Delta T_b = 0.50 \left[\frac{\frac{8.2}{164} \times 2.5 + \frac{12}{120} \times 1.6}{79.8} \right] \times 1000 = 1.78^\circ\text{C}$$

$$T_b = 100 + 1.785 \Rightarrow 101.78^\circ\text{C}$$

19. (b) Total vapour pressure of solution varies linearly with mole fraction of component χ .
 Thus, correct graph is given in option (b).
 20. (c) Henry's law; $P = K_H \cdot X_{\text{gas}}$

$$\frac{1}{\text{Henry's law constant } (K_H)} \propto$$

Solubility of gas $\propto$ Partial pressure of gas

$$K_H : \text{Ar} > \text{O}_2 > \text{CO}_2 > \text{CH}_4$$

$$4.30 \quad 34.86 \quad 1.67 \quad 0.41$$

Solubility: $\text{Ar} < \text{O}_2 < \text{CO}_2 < \text{CH}_4$

21. (0.857) Mole fraction of A, $x_A = \frac{60}{100} = 0.60$;

$$P_A^\circ = 24.0 \text{ kPa}$$

$$\text{Mole fraction of B, } x_B = \frac{40}{100} = 0.40;$$

$$P_B^\circ = 12.0 \text{ kPa}$$

P_A = Partial vapour pressure of A at 350 K

$$= P_A^\circ x_A = 24 \times 0.60 = 14.4 \text{ kPa}$$

P_B = Partial vapour pressure of B at 350 K

$$= P_B^\circ x_B = 12 \times 0.40 = 4.88 \text{ kPa}$$

Mole fraction of A in vapour, i.e., first distillate

$$x'_A = \frac{P_A}{P_A + P_B} = \frac{14.4}{14.4 + 4.8} = \frac{14.4}{19.2} = 0.75$$

Mole fraction of B in first distillate $x'_B = 1 - 0.75 = 0.25$

This distillate is redistilled.

P'_A = Partial vapour pressure of

$$A = P_A^\circ x'_A = 24 \times 0.75 = 18 \text{ kPa}$$

P'_B = Partial vapour pressure of

$$B = P_B^\circ x'_B = 12 \times 0.25 = 3 \text{ kPa}$$

Mole fraction of A in second distillate = mole fraction of A in the vapour above first distillate

$$= \frac{P'_A}{P'_A + P'_B} = \frac{18}{18 + 3} = 0.857$$

Hence, mole % of A in 2nd distillate = 85.7

22. (14.49) Mole fraction of urea in its solution

$$= \frac{\frac{12}{60}}{\frac{12}{60} + \frac{140.4}{18}} \Rightarrow 0.025$$

Mole fraction of glucose

$$= \frac{\frac{18}{180}}{\frac{18}{180} + \frac{178.2}{18}} \Rightarrow 0.01$$

$\therefore$ Mole fraction of glucose is less so vapour pressure above the glucose solution will be higher than the pressure above urea solution, so some H_2O molecules will transfer from glucose to urea side in order to make the solutions of equal mole fraction to attain equilibrium, let x moles H_2O transferred

$$\therefore \frac{0.2}{0.2 + 7.8 + x} = \frac{0.1}{0.1 + 9.9 - x} \Rightarrow x = 4$$

now mass of glucose solution

$$\Rightarrow 196.2 - 4 \times 18 \Rightarrow 124.2$$

$$\text{wt. \% of glucose} = \frac{18}{124.2} \times 100 \Rightarrow 14.49$$

23. (300) Osmotic pressure of urea from the formula

$$\pi V = nRT \Rightarrow \pi = \frac{nRT}{V}$$

$$\Rightarrow \frac{10}{60} \times 0.0821 \times T \quad [1 \text{ dm}^3 = 1 \text{ litre}]$$

5% solution means

$$100 \text{ mL} \equiv 5 \text{ g}$$

$$1000 \text{ mL} \equiv 50 \text{ g/L}$$

Osmotic pressure of solution having non-volatile solute

$$\pi_2 = \frac{50}{M} \times .0821 \times T$$

For isotonic solution, $\pi_1 = \pi_2$

$$\frac{10}{60} \times .0821 \times T = \frac{50}{M} \times .0821 \times T$$

$$M = 300 \text{ g mol}^{-1}$$



$$\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \end{array}$$



$$\begin{array}{ccc} 1 & 0 & 0 \\ 1-h & h & h \end{array}$$

$\therefore$ Total particles after dissolution of

$$\text{NaHSO}_4 = 1 + 1 - h + h + h = 2 + h; \quad i = \frac{2+h}{1}$$

$$= 2 + h$$

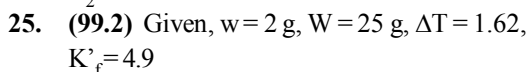
$$\text{Now, } \Delta T_f = K_f \times \text{molality} \times i$$

$$0.345 = 1.86 \times 0.08 \times (2 + h)$$

$$\therefore 2 + h = 2.319$$

$$\therefore h = 0.319$$

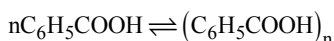
i.e., 31.9% of HSO_4^- shows proton transfer to H_2O .



$$\therefore \Delta T = \frac{1000 \times K_f' \times w}{m \times W}$$

$$1.62 = \frac{1000 \times 4.9 \times 2}{25 \times m}$$

$$\therefore m_{\text{exp}} = 241.98$$



$$\text{Before association} \quad 1 \quad 0$$

$$\text{After association} \quad (1 - \alpha) \quad \frac{\alpha}{n}$$

$\therefore$ Total number of mole at equilibrium

$$= 1 - \alpha + \frac{\alpha}{n}$$

$$\frac{m_N}{m_{\text{exp}}} = 1 - \alpha + \frac{\alpha}{n}$$

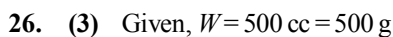
For dimer formation $n = 2$

$$i = \frac{1 - \frac{\alpha}{2}}{1} = 1 - \frac{\alpha}{2}$$

$$\frac{122.0}{241.98} = 1 - \frac{\alpha}{2} \quad (m_N = 122.0 \text{ for } \text{C}_6\text{H}_5\text{COOH})$$

$$\text{or } 1 - \frac{\alpha}{2} = 0.504$$

$$\therefore \alpha = 0.992 \text{ or } 99.2\%$$



($\therefore$ density of water = 1 g/cc)

$$\text{As, Density} = \frac{\text{Weight}}{\text{Volume}}$$

$$\text{Weight} = \text{Density} \times \text{Volume} = \frac{1 \text{ g}}{\text{cc}} \times 500 \text{ cc} = 500 \text{ g}$$

$w =$ amount of urea = ?

$$K_f = 1.86^\circ\text{C}; \Delta T_f = 0.186^\circ\text{C}$$

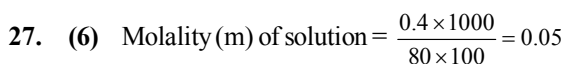
$m =$ Molecular mass of urea = 60

$$m = \frac{1000 \times K_f \times w}{W \times \Delta T_f}$$

$$60 = \frac{1000 \times 1.86 \times w}{500 \times 0.186}$$

$$60 = 20 \times w$$

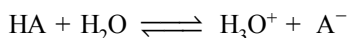
$$\therefore w = \frac{60}{20} = 3 \text{ g}$$



$$\Delta T_f(\text{normal}) = K_f \times m = 1.86 \times 0.05 = 0.093 \text{ K}$$

Van't Hoff factor,

$$i = \frac{\Delta T_f(\text{observed})}{\Delta T_f(\text{normal})} = \frac{0.12}{0.093} = 1.290$$



$$\alpha = \frac{i - 1}{n - 1} = \frac{1.290 - 1}{2 - 1} = 0.29$$

$$K_a = \frac{C\alpha^2}{1 - \alpha} = \frac{0.05 \times 0.29^2}{1 - 0.29} = 5.92 \times 10^{-3}$$

28. (2) Let x g be the mass of element in 51.0 g of saturated solution.

Mass of benzene in 51.0 g of saturated solution
 $= 51.0 - x$ g

Total mass of benzene containing x g of solute
 $= 50 + 51 - x = (101 - x)$ g

$$\Delta T_f = \frac{1000 K_f W_B}{M_B W_A} = \frac{1000 \times 5.5 \times x}{4 \times 25 \times (101 - x)} = 0.55 \text{ (given)}$$

$$\Rightarrow x = 1.0 \text{ g}$$

Hence, solubility

$$= \frac{W_B \times 100}{W_A} = \frac{1}{(51 - 1)} \times 100 = 2.0 \text{ g}$$

29. (5) For initial solution,

$$\therefore \pi = \frac{500}{760} \text{ atm, } T = 283 \text{ K}$$

$$\frac{500}{760} \times V_1 = n \times R \times 283 \quad \dots(i)$$

After dilution, let volume becomes V_2 and temperature is raised to 25°C , i.e., 298 K.

$$\pi = \frac{105.3}{760} \text{ atm}$$

$$\frac{105.3}{760} \times V_2 = n \times R \times 298 \quad \dots(ii)$$

$$\therefore \text{By Eqs. (i) and (ii), we get } \frac{V_1}{V_2} = \frac{283}{298} \times \frac{105.3}{500}$$

$$\frac{V_1}{V_2} = \frac{1}{5}$$

$$\therefore V_2 = 5V_1$$

i.e., Solution was diluted to 5 times.

$$30. (4) \Delta T = \frac{1000 \times K'_b \times w}{m \times W}$$

$$1.68 = \frac{1000 \times 2.34 \times 28}{m \times 315}$$

$$\therefore m_{\text{exp}} = 123.80$$

$$\frac{m_N}{m_{\text{exp}}} = 1 - \alpha + \frac{\alpha}{n}$$

$$\therefore \alpha = 1 \quad \therefore \frac{m_N}{m_{\text{exp}}} = \frac{1}{n} \quad (m_N \text{ of P} = 31)$$

$$\therefore \frac{31}{123.80} = \frac{1}{n}$$

$$\therefore n \approx 4$$

1. (a) Oxidising tendency $\propto \frac{1}{\text{Electrode potential}}$

TX $\longrightarrow$ No reaction

TY $\longrightarrow$ X, Z

TZ $\longrightarrow$ X

$\Rightarrow$ order of electrode potential is

TY < TZ < TX

$\Rightarrow$ Order of oxidation of the anion is

$Y^- > Z^- > X^-$

2. (a) $2\text{Ag}^+ + \text{H}_2 \longrightarrow 2\text{H}^+ + 2\text{Ag}$

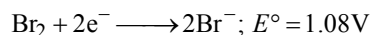
$$E = E^\circ - \frac{0.0591}{2} \log \frac{[\text{H}^+]^2}{P_{\text{H}_2} \times [\text{Ag}^+]^2}$$

$$0.222 = 0.7995 - \frac{0.0591}{2} \log \frac{1}{[\text{Ag}^+]^2}$$

$$[\text{Ag}^+] = 10^{-9.8}$$

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = (10^{-9.8}) \times (1) = 10^{-9.8}$$

3. (b) $\text{Sn}^{4+} + 2\text{e}^- \longrightarrow \text{Sn}^{2+}; E^\circ = 0.13\text{V}$



E° value shows Br_2 has higher reduction potential.

Hence

$$\begin{aligned} E_{\text{cell}} &= E_{\text{Br}_2/\text{Br}^-} - E_{\text{Sn}^{4+}/\text{Sn}^{2+}} \\ &= 1.08 - 0.13 = 0.95\text{V} \end{aligned}$$

$$\text{Now } -\Delta G = nF E_{\text{cell}}$$

$$n = 2, F = 96500$$

$$-\Delta G = 2 \times 96500 \times 0.95 \text{ kJ/mol.}$$

$$\text{Also, } \Delta G = -2.303RT \log K_{\text{eq}}$$

$$\log K_{\text{eq}} = -\frac{\Delta G}{2.303 \times R \times T}$$

$$= \frac{-(-2 \times 96500 \times 0.95)}{2.303 \times 8.314 \times 298} = 32.13$$

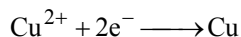
$$K_{\text{eq}} = \text{antilog } 32.682 \approx 10^{32}$$

4. (a) When pH = 14 $[\text{H}^+] = 10^{-14}$ and $[\text{OH}^-] = 1\text{M}$

$$K_{sp} = [\text{Cu}^{2+}][\text{OH}^-]^2 = 10^{-19}$$

$$\therefore [\text{Cu}^{2+}] = \frac{10^{-19}}{[\text{OH}^-]^2} = 10^{-19}$$

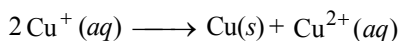
The half cell reaction



$$E = E^\circ - \frac{0.059}{2} \log \frac{1}{[\text{Cu}^{2+}]}$$

$$= 0.34 - \frac{0.059}{2} \log \frac{1}{10^{-19}} = -0.22\text{V}$$

5. (c) The reaction



$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{1} \log \frac{[\text{Cu}^{2+}]}{[\text{Cu}^+]^2}$$

At equilibrium $E_{\text{cell}} = 0$

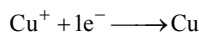
$$\therefore E_{\text{cell}}^\circ = 0.059 \log K_{\text{eq}}$$

$$\text{or, } \log K_{\text{eq}} = \frac{0.52 - 0.16}{0.059}$$

$$\therefore K_{\text{eq}} = 1.2 \times 10^6$$

6. (b) $\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$

$$E_1^\circ = 0.15\text{V}; \Delta G_1^\circ = -n_1 E_1^\circ F$$



$$E_2^\circ = 0.50\text{V}; \Delta G_2^\circ = -n_2 E_2^\circ F$$



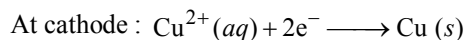
$$-nE^\circ F = -1 n_1 E_1^\circ F + (-1) n_2 E_2^\circ F$$

$$-nE^\circ F = -1 (n_1 E_1^\circ F + n_2 E_2^\circ F)$$

$$E^\circ = \frac{n_1 E_1^\circ + n_2 E_2^\circ}{n} = \frac{0.15 \times 1 + 0.50 \times 1}{2}$$

$$\Rightarrow 0.325$$

7. (d) The cell reactions are :



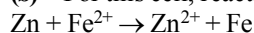
We know that :

$$\Delta G^\circ = -nF E_{\text{cell}}^\circ ; n = 2 \text{ mol}$$

$$\begin{aligned} E_{\text{cell}}^\circ &= [E_{(\text{Cu}^{2+}/\text{Cu})}^\circ - E_{(\text{Fe}^{2+}/\text{Fe})}^\circ] \\ &= (+0.34 \text{ V}) - (-0.44 \text{ V}) = +0.78 \text{ V} \\ F &= 96500 \text{ C mol}^{-1} \end{aligned}$$

$$\begin{aligned} \therefore \Delta G^\circ &= -nF E_{\text{cell}}^\circ \\ &= -(2 \text{ mol}) \times (96500 \text{ C mol}^{-1}) \times (+0.78 \text{ V}) \\ &= -150540 \text{ CV} = -150540 \text{ J } (\because 1 \text{ CV} = 1 \text{ J}) \\ &= -150.54 \text{ kJ} \end{aligned}$$

8. (b) For this cell, reaction is;



$$E = E^\circ - \frac{0.0591}{n} \log \frac{c_1}{c_2}; E^\circ = E + \frac{0.0591}{n} \log \frac{c_1}{c_2}$$

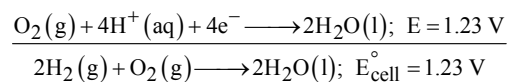
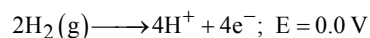
$$E^\circ = 0.2905 + \frac{0.0591}{2} \log \frac{10^{-2}}{10^{-3}} = 0.32 \text{ V.}$$

$$E^\circ = \frac{0.0591}{2} \log K_{\text{eq}}$$

$$\log K_{\text{eq}} = \frac{0.32 \times 2}{0.0591} = \frac{0.32}{0.0295}$$

$$\therefore K_{\text{eq}} = 10^{0.0295}$$

9. (a) We know



$$\Delta G_{298}^\circ = -nF E^\circ = -4 \times 96500 \times 1.23 = -474.78 \text{ kJ}$$

$$\Delta_f H_{298}^\circ = \Delta_f G_{298}^\circ + T \Delta_r S_{298}^\circ$$

$$= (-474.78) + 298 \times (-0.32)$$

$$= -570.14 \text{ kJ/mol}$$

$$\therefore \Delta_f H_{298}^\circ = -285.07 \text{ kJ/mol}$$

10. (d) Molarity = 0.01 M ; Resistance = 40 ohm;

$$\text{Cell constant } \frac{l}{A} = 0.4 \text{ cm}^{-1}$$

Specific conductivity (κ)

$$= \frac{\text{cell constant}}{\text{resistance}} = \frac{0.4}{40} = 0.01 \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\text{Molar conductance } (\Lambda_m) = \frac{1000\kappa}{\text{Molarity}}$$

$$= \frac{1000 \times 0.01}{0.01} = 10^3 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

11. (a) $2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$

$$E_{\text{H}}(\text{Eq. wt}) = \frac{2}{2} = 1 \text{ g}$$

$$= \frac{22400}{2} = 11200 \text{ mL (STP)}$$

$$\text{Total charge passed} = \frac{96500 \times 112}{11200} = 965$$

$$Q = It = 965$$

$$I = \frac{965}{965} = 1 \text{ amp.}$$

12. (c) $\text{ClCH}_2\text{COONa} + \text{HCl} \rightarrow$



$$\Lambda_{\text{ClCH}_2\text{COONa}} + \Lambda_{\text{HCl}} = \Lambda_{\text{ClCH}_2\text{COOH}} + \Lambda_{\text{NaCl}}$$

$$224 + 203 = \Lambda_{\text{ClCH}_2\text{COOH}} + 38.2$$

$$\Lambda_{\text{ClCH}_2\text{COOH}} = 427 - 38.2$$

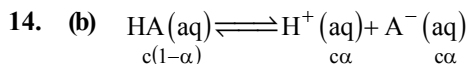
$$= 388.8 \text{ ohm}^{-1} \text{ cm}^2 \text{ g eq}^{-1}$$

13. (a) $\Lambda_{\text{eq}}^\infty(\text{NH}_4\text{OH}) = \Lambda_{\text{eq}}^\infty(\text{NH}_4\text{Cl}) +$

$$\Lambda_{\text{eq}}^\infty(\text{NaOH}) - \Lambda_{\text{eq}}^\infty(\text{NaCl})$$

$$= 129.8 + 217.8 - 109.3 = 238.3 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

$$\alpha = \frac{\Lambda_{\text{eq}}}{\Lambda_{\text{aq}}^\infty} = \frac{9.30}{238.3} = 0.04$$



$$K_a = \frac{c\alpha^2}{1-\alpha}; \text{ where } \alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$$

$$\therefore K_a = \frac{c \left(\frac{\Lambda_m}{\Lambda_m^\infty} \right)^2}{\left(1 - \frac{\Lambda_m}{\Lambda_m^\infty} \right)} = \frac{c\Lambda_m^2}{\Lambda_m^\infty (\Lambda_m^\infty - \Lambda_m)}$$

15. (a) (Eq. = Charge on the ion/Total charge)

$$[\text{K}^+] = \frac{1}{8} \text{ mole} \times 2 = \frac{1}{4} \text{ mole} = \frac{1}{4} \text{ Eq.}$$

$$[\text{Al}^{3+}] = \frac{6}{8} = \frac{3}{4} \text{ Eq.}$$

$$[\text{SO}_4^{2-}] = \frac{8}{8} = 1 \text{ Eq.}$$

$$\begin{aligned} \Lambda_{\text{eq}}^\circ \text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O} \\ = \lambda_{\text{eq}}^\circ(\text{K}^+) + \lambda_{\text{eq}}^\circ(\text{Al}^{3+}) + \lambda_{\text{eq}}^\circ(\text{SO}_4^{2-}) \\ = \frac{1}{4} \times 73.5 + 149 \times \frac{3}{4} + 85.8 \times 1 \\ = 18.375 + 111.75 + 85.8 = 215.92 \end{aligned}$$

16. (a) By Faraday's 1st law of electrolysis,

$$\frac{W}{E} = \frac{Q}{96500} \quad (\text{where } Q = it = \text{charge of ion})$$

We know that no. of gram equivalent

$$= \frac{W}{E} = \frac{it}{96500} = \frac{1 \times 965}{96500} = \frac{1}{100}$$

(where $i = 1 \text{ A}$, $t = 16 \times 60 + 5 = 965 \text{ sec.}$)

Since, we know that

$$\text{Normality} = \frac{\text{No. of gram equivalent}}{\text{Volume (in litre)}} = \frac{\frac{1}{100}}{1}$$

$$= 0.01 \text{ N}$$

17. (b) The molar conductivity of the dissociated form of crotonic acid is

$$\Lambda_m(\text{HC}) = \Lambda_m(\text{HCl}) + \Lambda_m(\text{NaC}) - \Lambda_m(\text{NaCl})$$

$$= (426 + 83 - 126) \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$= 383 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

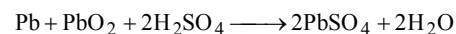
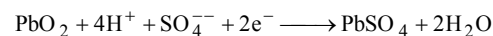
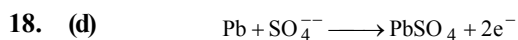
The molar conductivity of HC,

$$\begin{aligned} \Lambda_m(\text{HC}) &= \frac{\kappa}{C} = \frac{3.83 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}}{0.001} \times 1000 \\ &= 38.3 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1} \end{aligned}$$

The degree of dissociation,

$$\alpha = \frac{\Lambda_m(\text{HC})}{\Lambda_m^\infty(\text{HC})} = \frac{(38.3 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1})}{(383 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1})} = 0.1$$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{(10^{-3})(0.1)^2}{1-0.1} = 1.11 \times 10^{-5}$$



The reaction indicates that 2 moles of H_2SO_4 corresponds to $2 \times 96500 \text{ C}$ and 2 mol $\text{H}_2\text{SO}_4 \equiv 4 \text{ equiv. of } \text{H}_2\text{SO}_4$.

$2 \times 96500 \text{ C}$ consumed 4 equiv. of H_2SO_4

and $100 \times 10^{-3} \times 9.65 \times 10^5 \text{ C}$ consumed

$$= \frac{4 \times 100 \times 10^{-3} \times 9.65 \times 10^5}{2 \times 96500} = 2 \text{ equiv. } \text{H}_2\text{SO}_4$$

$$\therefore \text{Decrease in normality} = \frac{2}{5} = 0.40$$

19. (c) Gold deposited in the first cell = 9.85 g
At. wt. of Gold = 197, Oxidation number of gold = +3

$$\text{Eq. Wt. of Gold} = \frac{197}{3}$$

$W = Zit$

(where W stands for the weight of ions deposited, i for current and t for time and Z for electro-chemical equivalent of the electrolyte.)

$\therefore$ Charge required to deposit 1 g eq. of gold = $1F = 96,500 \text{ C}$

$\therefore$ Charge required to deposit 9.85 g of gold or

$$\frac{9.85}{197/3} \text{ g eq. of gold} = \frac{96,500 \times 9.85 \times 3}{197} \text{ C}$$

$$= 14475 \text{ C}$$

According to Faraday's second law,

$$\frac{\text{Wt. of Cu}}{\text{Eq. wt. of Cu}} = \frac{\text{Wt. of Gold}}{\text{Eq. wt. of Gold}}$$

$\Rightarrow$ Wt. of Cu deposited

$$= \frac{9.85 \times 3}{197} \times \frac{63.5}{2} = 4.76 \text{ g}$$

$$\text{Current} = \frac{Q}{t} = \frac{14475}{5 \times 3600} \text{ A} = \frac{193}{240} \text{ A} = 0.80 \text{ A}$$

20. (c) $\text{Cu}^{2+}(\text{aq}) + \text{H}_2 \rightleftharpoons 2\text{H}^+ + \text{Cu}(\text{s})$
For the given electrochemical cell
 $\text{Pt} | \text{H}_2 (p = 1 \text{ atm}) |$
 $\text{H}^+ (\text{aq. } x\text{M}) || \text{Cu}^{2+} (\text{aq. } 1.0 \text{ M}) | \text{Cu}(\text{s})$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{H}^+]^2}{[\text{Cu}^{2+}]}$$

$$0.49 = 0.37 - \frac{0.0591}{2} \log \frac{x^2}{1}$$

$$0.49 = 0.37 - \frac{0.0591}{2} \times 2 \log x$$

$$0.15 = -0.0591 \times \log x$$

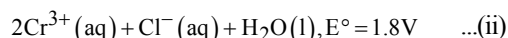
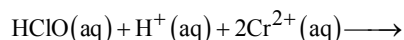
$$2.54 = -\log x$$

$$\text{Also, } -\log [\text{H}^+] = \text{pH}$$

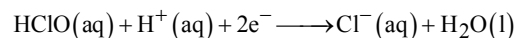
$$\therefore \text{pH} = -\log x$$

$$\Rightarrow \text{pH} = 2.54 \approx 2.5$$

21. (1.39) $\text{Cr}^{2+}(\text{aq}) \longrightarrow \text{Cr}^{3+}(\text{aq}) + \text{e}^-$, $E^{\circ} = 0.41 \text{ V}$
... (i)



[Reaction (i)] $\times -2$ + [Reaction (ii)] we get.

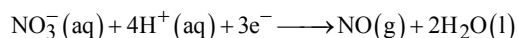


$$\Delta G^{\circ} = -2FE^{\circ} - 2(-1 \times FE^{\circ})$$

$$-2FE^{\circ} = -2F(1.8) + 2F(0.41)$$

$$E^{\circ} = +1.8 - 0.41 = 1.39 \text{ V}$$

22. (0.84) After addition of Cd and its oxidation into Cd^{2+} .



$$0.1 - x \quad 0.4 - 4x;$$

$$\text{where } x = 0.08$$

$$[\text{NO}_3^-]_{\text{remaining}} = 0.02 \text{ M}; [\text{H}^+]_{\text{remaining}} = 0.08 \text{ M}$$

$$E_{\text{NO}_3^-/\text{NO}} - E_{\text{NO}_3^-/\text{NO}}^{\circ} - \frac{0.0591}{3} \log \frac{1}{[\text{NO}_3^-][\text{H}^+]^4}$$

$$= 0.96 - \frac{0.0591}{3} \log \frac{1}{(0.02)(0.08)^4} = 0.84 \text{ V}$$

23. (0.0529) Using Faraday's second law of electrolysis,

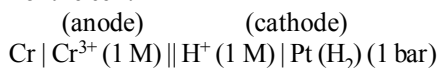
$$\frac{\text{Weight of Cu deposited}}{\text{Weight of Ag deposited}} = \frac{\text{Equ. wt. of Cu}}{\text{Equ. wt. of Ag}}$$

$$\Rightarrow \frac{w_{\text{Cu}}}{0.18} = \frac{63.5}{2} \times \frac{1}{108}$$

$$\Rightarrow w_{\text{Cu}} = \frac{63.5 \times 18}{2 \times 108 \times 100} = 0.0529 \text{ g.}$$

24. (1.5)

For the cell:

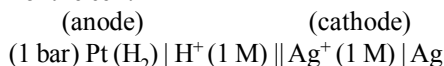


$$E_{\text{cell}}^{\circ} = E_{\text{H}^+/\text{H}_2}^{\circ} - E_{\text{Cr}^{3+}/\text{Cr}}^{\circ}$$

$$0.74 = 0 - E_{\text{Cr}^{3+}/\text{Cr}}^{\circ}$$

$$E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.74 \text{ V}$$

For the cell:



$$E_{\text{cell}}^{\circ} = E_{\text{Ag}^+/\text{Ag}}^{\circ} - E_{\text{H}^+/\text{H}_2}^{\circ}$$

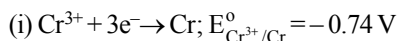
$$0.80 = E_{\text{Ag}^+/\text{Ag}}^{\circ} - 0$$

$$E_{\text{Ag}^+/\text{Ag}}^{\circ} = +0.80 \text{ V}$$

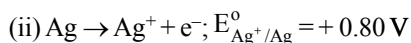
For the cell:



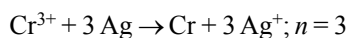
Reduction at cathode:



Oxidation at anode:



$$\text{eq. (i)} + 3 \times \text{eq. (ii)}$$



$$E_{\text{cell}}^{\circ} = E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} - E_{\text{Ag}^+/\text{Ag}}^{\circ}$$

$$E_{\text{cell}}^{\circ} = -0.74 - 0.80 = -1.54 \text{ V}$$

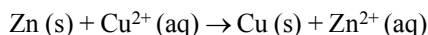
$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{Ag}^+]^3}{[\text{Cr}^{3+}]}$$

$$= -1.54 - \frac{0.06}{3} \log \frac{(0.1)^3}{(0.1)} = -1.54 - \frac{0.06}{3} \times (-2)$$

$$= -1.5 \text{ V}$$

25. (0.15)

Cell reaction for Daniell's cell:



$$\text{Reaction quotient } Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

At start, $Q = 1$ as $[\text{Zn}^{2+}] = [\text{Cu}^{2+}]$

also $\log Q = 0$ at point O for $Q = 1$

This is the maximum EMF, the cell can generate.

As the reaction proceeds, $[\text{Zn}^{2+}]$ will increase and $[\text{Cu}^{2+}]$ will decrease. Thus, the value of Q will increase and the cell EMF will decrease.

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

$$E_{\text{cell}} = 1.07 \text{ V}, E_{\text{cell}}^{\circ} = 1.10 \text{ V}, n = 2$$

$$1.07 = 1.10 - \frac{0.06}{2} \log \frac{1.5}{[\text{Cu}^{2+}]}$$

$$0.03 = 0.03 \log \frac{1.5}{[\text{Cu}^{2+}]}$$

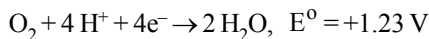
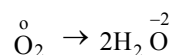
$$\log \frac{1.5}{[\text{Cu}^{2+}]} = 1$$

$$\Rightarrow \frac{1.5}{[\text{Cu}^{2+}]} = 10$$

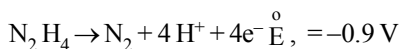
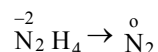
$$\Rightarrow [\text{Cu}^{2+}] = 0.15 \text{ mol/L}$$

26. (76.43)

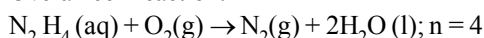
Reduction at cathode:



Oxidation at anode:



Overall cell reaction:



$$E_{\text{cell}}^{\circ} = E_{\frac{1}{2}\text{O}_2/\text{H}_2\text{O}}^{\circ} - E_{\text{N}_2/\text{N}_2\text{H}_4}^{\circ}$$

$$= (1.23 - 0.9) \text{ V} = +0.33 \text{ V}$$

Maximum available potential $E_{\text{cell}}^{\circ} = 0.33 \text{ V}$

Maximum amount of energy $= -\Delta G$

$$= nF E^{\circ} = 4 \times 96500 \times 0.33 \text{ J} = 1,27,380 \text{ J}$$

Assuming 60% efficiency

$$W_{\text{max}} = 127380 \times \frac{60}{100} \text{ J} = 76.43 \text{ KJ}$$

27. (6) Molecular mass of $\text{H}_2\text{SO}_4 = 98 \text{ g/mol}$

Molarity of initial solution

$$= \frac{\% \text{W/W} \times 10 \times d}{\text{molecular mass}} = \frac{38 \times 10 \times 1.274}{98} = 4.94$$

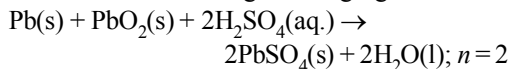
Molarity of final solution

$$= \frac{19.6 \times 10 \times 1.2}{98} = 2.4 \text{ M}$$

Decrease in amount of H_2SO_4 during this discharging process

$$= (4.94 - 2.4) \times 98 \times 2.36 \text{ g} = 6 \times 98 \text{ g}$$

Overall reaction during discharging



$\therefore$ For 1 mole of H_2SO_4 +1 F is required

$6 \times 98 \text{ g H}_2\text{SO}_4 = 6 \text{ moles of H}_2\text{SO}_4$

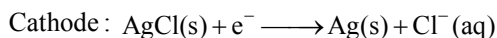
Thus, number of Faradays required = 6

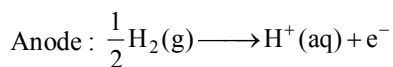
28. (142)

Sodium metal :

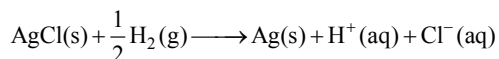
$$E = E_0 + (\text{KE})_{\text{max}}; E_{\text{cell}}^{\circ} = 0.22 \text{ V}$$

Cell reaction





Overall:



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log[\text{H}^+][\text{Cl}^-]$$

$$E_{\text{cell}} = 0.22 - \frac{0.06}{1} \log[10^{-1}][10^{-1}]$$

$$= 0.22 + 0.12 = 0.34 \text{ V}$$

$$(\text{KE})_{\text{max}} = E_{\text{cell}} = 0.34 \text{ eV}$$

So, $E = 2.3 + 0.34 = 2.64 \text{ eV} = \text{Energy of photon incident}$

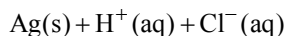
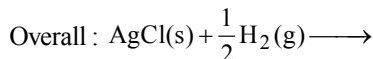
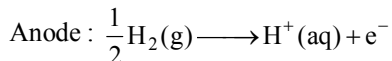
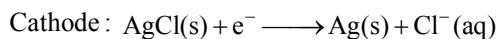
For potassium metal:

$$E = E_0 + (\text{KE})_{\text{max}}$$

$$2.64 = 2.25 + (\text{KE})_{\text{max}}$$

$$(\text{KE})_{\text{max}} = 0.39 = E_{\text{cell}}$$

Cell reaction



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log[\text{H}^+][\text{Cl}^-]$$

$$0.39 = 0.22 - 0.06 \log[\text{H}^+]^2$$

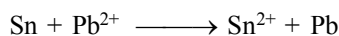
$$0.39 = 0.22 - 0.12 \log[\text{H}^+]$$

$$0.17 = 0.12 \times \text{pH}$$

$$\text{pH} = 17/12 = 1.4166 \approx 1.42 = 142 \times 10^{-2}$$

29. (2.15)

At equilibrium state $E_{\text{cell}} = 0$; $E_{\text{cell}}^0 = 0.01 \text{ V}$



$$E = E_{\text{cell}}^0 - \frac{0.06}{n} \log \frac{[\text{P}]}{[\text{R}]}$$

$$0 = 0.01 - \frac{0.06}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

$$-0.01 = -\frac{0.06}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

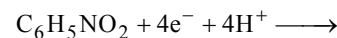
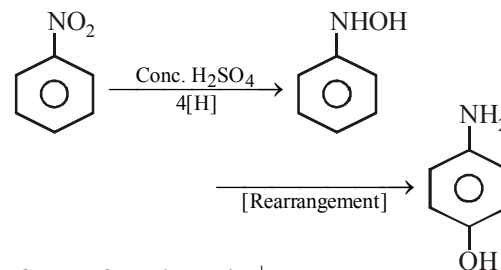
$$\frac{1}{3} = \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

$$\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} = 10^{1/3} = 2.15$$

30. (9.81) 9.65 ampere current was passed for 1.0 hour (3600 seconds)

Number of moles of electrons passed

$$= \frac{I(\text{A}) \times t(\text{s})}{96500} = \frac{9.65 \text{ A} \times 3600 \text{ s}}{96500} = 0.36 \text{ moles}$$



p-Aminophenol + H₂O

$\therefore$ 4 moles of electrons reduces 1 mole of nitrobenzene to *p*-aminophenol.

$$\therefore 0.36 \text{ moles of electrons will reduce } \frac{0.36}{4} = 0.09$$

moles of nitrobenzene to *p*-aminophenol.

p-aminophenol molar mass = 109.14 g/mol

Mass of *p*-aminophenol obtained

$$= 109.14 \text{ g/mol} \times 0.09 \text{ mol} = 9.81 \text{ g}$$

Chemical Kinetics

1. (b) Rate of disappearance of reactants = Rate of appearance of products

$$-\frac{1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{1}{4} \frac{d[\text{NO}_2]}{dt} = \frac{d[\text{O}_2]}{dt}$$

$$\frac{1}{2} k[\text{N}_2\text{O}_5] = \frac{1}{4} k'[\text{N}_2\text{O}_5] = k''[\text{N}_2\text{O}_5]$$

$$\frac{k}{2} = \frac{k'}{4} = k''$$

$$k' = 2k, \quad k'' = \frac{k}{2}$$

2. (b) Given: 75% reaction gets completed in 32 min

$$\begin{aligned} \text{Thus, } k &= \frac{2.303}{t} \log \frac{a}{(a-x)} \\ &= \frac{2.303}{32} \log \frac{100}{(100-75)} \\ &= \frac{2.303}{32} \log 4 = 0.0433 \text{ min}^{-1} \end{aligned}$$

Now we can use this value of k to get the value of time required for 50% completion of reaction

$$\begin{aligned} t &= \frac{2.303}{k} \log \frac{a}{(a-x)} = \frac{2.303}{0.0433} \log \frac{100}{50} \\ &= \frac{2.303}{0.0433} \log 2 = 16 \text{ min} \end{aligned}$$

3. (b) $k = \frac{2.303}{t} \log \frac{a}{(a-x)}$

$(a-x)$ is the concentration left after 100 sec.

$$2.7 \times 10^{-3} = \frac{2.303}{100} \log \frac{0.29}{(a-x)}$$

$$\Rightarrow \frac{0.27}{2.303} = \log \frac{0.29}{(a-x)} \Rightarrow 0.117 = \log \frac{0.29}{(a-x)}$$

$$\Rightarrow (a-x) = 0.22 \text{ M.}$$

4. (c) $t_{90\%} = \frac{2.303}{k} \log \frac{100}{100-90} \quad \text{(I)}$

$$t_{50\%} = \frac{2.303}{k} \log \frac{100}{100-50} \quad \text{(II)}$$

$$\text{Dividing } \frac{t_{90\%}}{t_{50\%}} = \frac{\log 10}{\log 2}$$

$$\therefore t_{90\%} = 3.3 t_{50\%}$$

5. (d) If rate = $k[\text{A}]^x[\text{B}]^y[\text{C}]^z$
 $5.0 \times 10^{-3} = k[0.010]^x[0.005]^y[0.010]^z \quad \dots\text{(i)}$
 $5.0 \times 10^{-3} = k[0.010]^x[0.005]^y[0.015]^z \quad \dots\text{(ii)}$
 $1.0 \times 10^{-2} = k[0.010]^x[0.010]^y[0.010]^z \quad \dots\text{(iii)}$
 $1.25 = k[0.005]^x[0.005]^y[0.010]^z \quad \dots\text{(iv)}$

From 1st and 2nd sets of data - no change in rate is observed with the change in concentration of 'C'. So the order with respect to 'C' is zero.

From 1st and 4th sets of data

Dividing eq. (iv) by eq. (i)

$$\frac{1.25 \times 10^{-3}}{5.0 \times 10^{-3}} = \left[\frac{0.005}{0.010} \right]^x$$

$$\text{or } 0.25 = (0.5)^x \text{ or } (0.5)^2 = (0.5)^x$$

$$\therefore x = 2$$

The order with respect to 'A' is 2 from the 1st and 3rd sets of data Dividing eq. (i) by eq. (iii)

$$\frac{5.0 \times 10^{-3}}{1.0 \times 10^{-2}} = \left[\frac{0.005}{0.010} \right]^y$$

$$\text{or } (0.5)^1 = (0.5)^y \Rightarrow y = 1$$

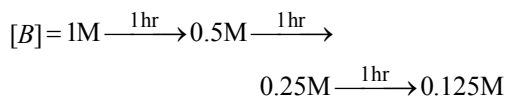
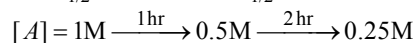
The order with respect to 'B' is 1

So the order with respect to the reactants A, B and C is 2, 1 and 0.

6. (d) Units of k indicate that reaction I is of second order and reaction II is first order.

For I reaction, $t_{1/2} \propto 1/a$,

first $t_{1/2} = 1$ hr, second $t_{1/2} = 2$ hr



$$\frac{[A]}{[B]} = \frac{0.25M}{0.125M} = 2$$

7. (d) $A \longrightarrow 3B$; $P_T = P_0 + 2x$
After time t , $P_0 - x \quad 3x$

$$x = \frac{P_T - P_0}{2} \quad k = \frac{1}{t} \ln \left(\frac{P_0}{P_0 - x} \right)$$

After long time, $0 \quad 3P_0$

$$x = \frac{P_T - \frac{P_\infty}{3}}{2} \quad k = \frac{1}{t} \ln \left(\frac{\frac{P_\infty}{3}}{\frac{P_\infty}{3} - \left(\frac{3P_T - P_\infty}{6} \right)} \right)$$

$$P_\infty = 3P_0 \quad x = \frac{3P_T - P_\infty}{6} \quad k = \frac{1}{t} \ln \left(\frac{P_\infty / 3}{\frac{P_\infty}{2} - \frac{P_T}{2}} \right)$$

$$k = \frac{1}{t} \ln \left(\frac{2P_\infty}{3(P_\infty - P_T)} \right)$$

8. (a) $A \longrightarrow 2B + C$
 $t = 0 \quad P_i \quad 0 \quad 0$
 $t = t \quad P_i - y \quad 2y \quad y$

Total pressure $P_t = P_i - y + 2y + y = P_i + 2y$

$$\frac{P_t - P_i}{2} = y$$

$$k = \frac{2.303}{t} \log \frac{2P_i}{3P_i - P_t}$$

9. (a) $t_{1/2} = \frac{0.693}{k}$

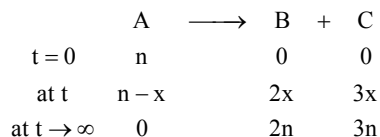
$$= \frac{0.693}{2.772 \times 10^{-3}} = 250 \text{ sec}$$

Since volume is changing therefore half-life should be defined on the basis of moles, so moles of A remaining after 250 sec. Also final volume of the container

$$V_f = \frac{n_f}{n_i} \times V_i = \frac{0.05 + 0.15}{0.1} \times 2 = 4 \text{ litre}$$

$$\therefore \text{Final concentration of A} = \frac{0.05}{4} = 0.0125 M$$

10. (d) Let n is the moles of reagent 'R' when R is reacted with A at time $t = 0$



$$\therefore 5n = n_2 \Rightarrow n = \frac{n_2}{5}$$

$$n + 4x = n_1 \Rightarrow x = \frac{n_1 - n}{4}$$

$$k = \frac{2.303}{t} \log \left(\frac{n}{n - x} \right)$$

$$\text{so } k = \frac{1}{t} \ln \left(\frac{4n_2}{5(n_2 - n_1)} \right)$$

11. (b) $A(g) \longrightarrow 2B(g) + C(g)$

Let initial pressure $P_0 \quad 0 \quad 0$
After 10 min. $(P_0 - x) \quad 2x \quad x$
After long time ($t \rightarrow \infty$) $0 \quad 2P_0 \quad P_0$

as per given $(P_0 - x) + 2x + x = \text{vapour pressure of H}_2\text{O} = 188$

$$P_0 + 2x = 160 \text{ and } 3P_0 + 28 = 388$$

so, $P_0 = 120$ and $x = 20$ torr

$$k = \frac{1}{t} \ln \left(\frac{P_0}{P_0 - x} \right)$$

$$\Rightarrow \frac{1}{10} \ln \left(\frac{120}{100} \right) = \frac{1}{10} (\ln 4 + \ln 3 - \ln 10) \\ = 0.02 \text{ min}^{-1} = 1.2 \text{ hr}^{-1}$$

12. (c) $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
2 mol of gaseous nitrogen pentoxide on complete decomposition gives 5 mol of gaseous products.

Therefore, initial pressure of $N_2O_5 = 584.5 \times \frac{2}{5} \\ = 233.8 \text{ mm Hg.}$

Let x be the amount of N_2O_5 decomposed after 30 min.

∴ After 30 min.

Pressure due to $\text{N}_2\text{O}_5 = 233.8 - x$

Pressure due to $\text{NO}_2 = 2x$

and pressure due to $\text{O}_2 = \frac{x}{2}$

Total pressure after 30 min

$$= 284.5 \text{ mm Hg} = 233.8 - x + 2x + \frac{x}{2}$$

$$\text{or } 233.8 + \frac{3x}{2} = 284.5$$

or $x = 33.8 \text{ mm Hg}$

Hence pressure of N_2O_5 after 30 min.

$$= 233.8 - 33.8 = 200 \text{ mm Hg}$$

$$k = \frac{2.303}{t} \log \frac{a}{(a-x)} = \frac{2.303}{30} \log \frac{233.8}{200}$$

$$k = \frac{2.303}{30} \times 0.0679 = 5.2 \times 10^{-3} \text{ min}^{-1}$$

13. (d) The activation energy of reverse reaction will depend upon whether the forward reaction is exothermic or endothermic.

As $\Delta H = E_a(\text{forward reaction}) - E_a(\text{backward reaction})$

For exothermic reaction

$$\Delta H = -ve$$

$$\therefore -\Delta H = E_a(f) - E_a(b)$$

$$\text{or } E_a(f) = E_a(b) - \Delta H$$

$$\therefore E_a(f) < E_a(b)$$

for endothermic reaction

$$\Delta H = +ve$$

$$\therefore \Delta H = E_a(f) - E_a(b) \text{ or } E_a(f) = \Delta H + E_a(b)$$

$$\therefore E_a(f) > E_a(b).$$

14. (b) $T_2 = T$ (say), $T_1 = 25^\circ\text{C} = 298\text{K}$,
 $E_a = 104.4 \text{ kJ mol}^{-1} = 104.4 \times 10^3 \text{ J mol}^{-1}$
 $k_1 = 3 \times 10^{-4}$, $k_2 = ?$,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \frac{k_2}{3 \times 10^{-4}} = \frac{104.4 \times 10^3 \text{ J mol}^{-1}}{2.303 \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1})}$$

$$\left[\frac{1}{298} - \frac{1}{T} \right]$$

$$\text{As } T \rightarrow \infty, \frac{1}{T} \rightarrow 0$$

$$\therefore \log \frac{k_2}{3 \times 10^{-4}} = \frac{104.4 \times 10^3 \text{ J mol}^{-1}}{2.303 \times 8.314 \times 298}$$

$$\log \frac{k_2}{3 \times 10^{-4}} = 18.297, \frac{k_2}{3 \times 10^{-4}} = 1.98 \times 10^{18}$$

$$k_2 = (1.98 \times 10^{18}) \times (3 \times 10^{-4}) = 6 \times 10^{14} \text{ s}^{-1}$$

$$15. \text{ (a) } 2.303 \log \frac{k_2}{k_1} = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

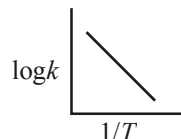
$$\log \frac{k_2}{k_1} = \frac{9.0 \times 10^3}{2.303 \times 2} \left[\frac{308 - 298}{308 \times 298} \right]$$

$$\frac{k_2}{k_1} = 1.63; k_2 = 1.63k_1;$$

$$\begin{aligned} \text{Increase in } k_1 &= \frac{k_2 - k_1}{k_1} \times 100 \\ &= \frac{1.63k_1 - k_1}{k_1} \times 100 = 63.0\% \end{aligned}$$

$$16. \text{ (c) } k = 1.2 \times 10^{14} e^{-25000/RT} \text{ sec}^{-1} \text{ or}$$

$$\log k = \log 1.2 \times 10^{14} - \frac{25000}{R} \cdot \frac{1}{T}$$



Equation of straight line

$$\text{slope} = -\frac{2500}{R}$$

$$17. \text{ (b) } \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log 2 = \frac{E_a}{2.303 \times 8.314} \left[\frac{1}{293} - \frac{1}{308} \right]$$

$$0.3 = \frac{E_a}{2.303 \times 8.314} \times \frac{15}{293 \times 308}$$

$$E_a = \frac{0.3 \times 2.303 \times 8.314 \times 293 \times 308}{15}$$

$$= 34558 \text{ J mol}^{-1} = 34.5 \text{ kJ mol}^{-1}$$

18. (c) $k = Ae^{-E_a/RT}$

For methyl nitrite $k_1 = 10^{13} \times e^{[-152300/(8.314 \times T)]}$

For ethyl nitrite $k_2 = 10^{14} \times e^{[-157700/(8.314 \times T)]}$

if $k_1 = k_2$, then

$$10^{13} e^{[-152300/(8.314 \times T)]} = 10^{14} e^{[-155700/(8.314 \times T)]}$$

$$10 = e^{(+157700-152300)/(8.314 \times T)}$$

$$\text{or } 2.303 \log 10 = \frac{157700 - 152300}{8.314 \times T}$$

$$\therefore T = 282 \text{ K}$$

19. (a) Given, $A \longrightarrow B$

and 20% solution of A decomposes 25% in 20 minute at 25°C

$$\therefore \text{Initial amount, } a = 20$$

$$\therefore \text{Amount left, } (a - x) = 20 \times \frac{75}{100} = 15$$

$$\therefore k_{25} = \frac{2.303}{20} \log_{10} \frac{20}{15} \quad (\because t = 20 \text{ minute})$$

$$= 0.0144 \text{ minute}^{-1}$$

$$\text{Now, } 2.303 \log_{10} \frac{k_{40}}{k_{25}} = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\therefore 2.303 \log_{10} \frac{k_{40}}{k_{25}} = \frac{70 \times 10^3}{8.314} \left[\frac{313 - 298}{313 \times 298} \right]$$

$$\frac{k_{40}}{k_{25}} = 3.872$$

$$\therefore k_{40} = 0.05575 \text{ min}^{-1} \quad (\because k_{25} = 0.0144)$$

Now suppose amount 'm' is left in 30% solution in 20 minute at 40°C.

$$k_{40} = \frac{2.303}{t} \log_{10} \frac{a}{(a - x)}$$

$$0.05575 = \frac{2.303}{20} \log_{10} \frac{30}{m}$$

$$\therefore m = 9.838$$

$\therefore$ % decomposition

$$= \frac{(a - m)}{a} \times 100 = \frac{30 - 9.838}{30} \times 100 = 67.21\%$$

20. (d) An activity of 6.023 curie means

$$6.023 \times 3.7 \times 10^{10} \text{ dps.}$$

$$\text{Activity} = \lambda \times N$$

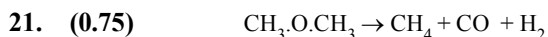
$$\therefore 6.023 \times 3.7 \times 10^{10} = 3.7 \times 10^4 \times N$$

$$\text{or } N = \frac{6.023 \times 3.7 \times 10^{10}}{3.7 \times 10^4}$$

$$\therefore \text{Mass of } X^{100}$$

$$= \left[\frac{6.023 \times 3.7 \times 10^{10}}{3.7 \times 10^4} \right] \times \frac{1}{6.023 \times 10^{23}} \times 100$$

$$= 10^{-15} \text{ g}$$



Total conc.

Initial state	a	0	0	0	a
---------------	-----	-----	-----	-----	-----

After 12 mts.	$a - x$	x	x	x	$a + 2x$
---------------	---------	-----	-----	-----	----------

Calculation of initial concentration (a) of CH_3OCH_3

$$PV = nRT$$

$$\text{or } \frac{n}{V} = \frac{P}{RT}$$

$$a = \frac{P}{RT} = \frac{0.4}{0.082 \times 773} = 6.31 \times 10^{-3} \text{ moles L}^{-1}$$

Calculation of k

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{14.5} = 4.78 \times 10^{-2} \text{ min}^{-1}$$

Substituting the values in the first order equation.

$$k = \frac{2.303}{t} \log \frac{a}{a - x}$$

$$4.78 \times 10^{-2} = \frac{2.303}{12} \log \frac{a}{a - x}$$

$$\frac{a}{a - x} = 1.77446$$

$$a - x = \frac{6.31 \times 10^{-3}}{1.77446} \text{ moles L}^{-1} = 3.556 \times 10^{-3}$$

moles L⁻¹

$$\therefore x = (6.310 - 3.556) \times 10^{-3} \text{ moles/l}$$

$$= 2.754 \times 10^{-3} \text{ moles L}$$

$\therefore$ After 12 min.

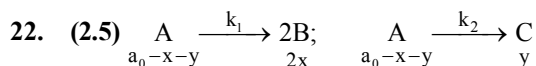
Total no. of moles L⁻¹

$$= a + 2x = 6.31 \times 10^{-3} + 2 \times 2.754 \times 10^{-3}$$

$$= 11.818 \times 10^{-3}$$

$$\therefore P = \frac{n}{V} RT = 11.818 \times 10^{-3} \times 0.082 \times 773$$

$$= 0.75 \text{ atm}$$



$$\frac{d[A]}{dt} = (k_1 + k_2)[A]; \quad \frac{[B]}{[C]} = \frac{2k_1}{k_2} = \frac{2x}{y}$$

$$= \frac{k_1}{k_2} = \frac{x}{y}$$

$$\therefore k_1 = k_2 = x = y$$

$$(k_1 + k_2)t = \ln \left[\frac{A_0}{A_t} \right]$$

$$2 \times 6.93 \times 10^{-3} \times 50 = \ln \left[\frac{A_0}{A_t} \right]$$

$$= [A]_t = \frac{[A_0]}{2} = a_0 - x - y = a_0/2 \quad (\because x = y)$$

$$\therefore x = a_0/4. \text{ Total moles in container}$$

$$a_0 - x - y + 2x + y = a_0 + x = 1.25 a_0$$

$$\% \text{ increase in moles} \Rightarrow 25, \text{ So final pressure}$$

$$= 2 \times 1.25 = 2.5 \text{ atm}$$

23. (18.39) Let the initial concentration (A) = 100
 Final concentration at 298 K = 100 - 10 = 90
 Final concentration at 308 K = 100 - 25 = 75
 Substituting the values in the 1st order rate reaction

$$t = \frac{2.303}{k_{298}} \log \frac{100}{90} \quad \dots(i)$$

$$t = \frac{2.303}{k_{308}} \log \frac{100}{75} \quad \dots(ii)$$

$$\text{From (i) and (ii)} \quad \frac{k_{308}}{k_{298}} = 2.73$$

Substituting the value in the following relation

$$E_a = \frac{2.303 R \times T_1 \times T_2}{T_2 - T_1} \log \frac{k_2}{k_1}$$

$$= \frac{2.303 \times 8.314 \times 298 \times 308}{308 - 298} \log 2.73$$

$$E_a = 76.62 \text{ kJ mol}^{-1} = 18.39 \text{ kcal mol}^{-1}$$

24. (378.74) $\ln \frac{k'_1}{k_1} = \frac{E_1}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \quad \dots(i)$

$$\ln \frac{k'_2}{k_2} = \frac{E_2}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \quad \dots(ii)$$

Solving we get (ii) - (i),

$$\ln \frac{k'_2}{k_2} - \ln \frac{k'_1}{k_1} = \left(\frac{E_2 - E_1}{R} \right) \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

(for equimolar formation of B and C, $k'_2 = k'_1$)

$$\ln \left(\frac{k_1}{k_2} \right) = \left(\frac{8314}{8.314} \right) \left(\frac{T_2 - 300}{300 \times T_2} \right)$$

$$\ln 2 = (1000) \left(\frac{T_2 - 300}{300 \times T_2} \right)$$

$$T_2 = 378.74 \text{ K}$$

25. (1.8) Rate constant at 300 K = k

$\therefore$ Rate constant at

$$310 \text{ K} = k + \left[k \times \frac{7}{100} \right] = 1.07 k$$

$$\text{Thus, } 2.303 \log \frac{k_2}{k_1} = \frac{E_a(f)}{R} \left[\frac{T_2 - T_1}{T_2 T_1} \right]$$

$$2.303 \log \frac{1.07k}{k} = \frac{E_a(f)}{2} \left[\frac{310 - 300}{310 \times 300} \right]$$

$$\therefore E_a(f) = 1258.68 \text{ cal}$$

Now, equilibrium constant at 300 K = K'

Equilibrium constant at 310 K

$$= K' + \frac{3}{100} \times K' = 1.03 K'$$

$$\text{Using } 2.303 \log \frac{K_2}{K_1} = \frac{\Delta H}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$2.303 \log \frac{1.03K'}{K'} = \frac{\Delta H}{2} \left[\frac{310 - 300}{310 \times 300} \right]$$

$$\therefore \Delta H = 549.89 \text{ cal}$$

Since, $\Delta H = E_a(f) - E_a(b)$

$$\therefore 549.89 = 1258.68 - E_a(b)$$

$$\therefore E_a(b) = 708.79 \text{ cal} \quad \frac{E_a(f)}{E_a(b)} = 1.8$$

26. (8.42) $t = 4.5$ billion years and $T_{1/2}$ of $^{40}\text{K} = 1.26$ billion years.

Then, percentage of ^{40}K remaining $\left[\frac{a-x}{a} \right] \times 100$

$$t = \frac{2.303}{\lambda} \log \frac{a}{a-x}$$

$$\lambda = 0.693 / T_{1/2} = 0.693 / 1.26 = 0.55$$

$$\frac{a}{a-x} = 11.86 \quad \therefore \frac{a-x}{a} = \frac{1}{11.86} = 0.0843$$

$$\therefore \frac{a-x}{a} \times 100 = 0.0843 \times 100 = 8.42\%$$

27. (8) Since $[B] \gg [A]$, $[B]$ remains practically constant and the rate is dependent only on $[A]$. The reaction is pseudo first order w.r.t. A .
Rate $= k[A][B] = k[A] \times 5 = 5k[A] = k'[A]$

$$k' = \frac{2.303}{t} \log \frac{C_o}{C_t} = \frac{2.303}{100} \log \frac{C_o}{C_t}$$

$$\log \frac{C_o}{C_t} = \frac{100 k'}{2.303} = \frac{100 \times 5k}{2.303}$$

$$= \frac{100 \times 5 \times 5.052 \times 10^{-3}}{2.303} = 1.0968$$

$$\frac{C_o}{C_t} = 12.5, \quad C_t = \frac{C_o}{12.5} = \frac{0.1}{12.5} = 0.008M$$

Hence, concentration of the reactant A remaining after $100 \text{ s} = 0.008 \text{ M} = 8 \text{ mM}$

28. (5) Rate of reaction $= k[\text{CO}]^x[\text{O}_2]^y$
 $4 \times 10^{-5} = k(0.02)^x(0.02)^y \quad \dots(i)$
 $1.6 \times 10^{-4} = k(0.04)^x(0.02)^y \quad \dots(ii)$
 $8 \times 10^{-5} = k(0.02)^x(0.04)^y \quad \dots(iii)$

$$\frac{(i)}{(ii)} \frac{1}{4} = \left(\frac{1}{2} \right)^x$$

$$x = 2$$

$$\frac{(i)}{(iii)} \frac{1}{2} = \left(\frac{1}{2} \right)^y$$

$$y = 1$$

From eq. (1)

$$4 \times 10^{-5} = k(0.02)^2(0.02)^1$$

$$k = \frac{4 \times 10^{-5}}{4 \times 10^{-4} \times 2 \times 10^{-2}} = \frac{10}{2} = 5$$

29. (8) For I order reaction $r_1 = k[\text{C}]^1$

$$\therefore \frac{R_1}{R_2} = k_1 / k_2 = \text{temperature coefficient}$$

$$\text{Let temperature coefficient be } a \quad \frac{r_{35}}{r_{25}} = \frac{k_{35}}{k_{25}} = a$$

$$\frac{r_{45}}{r_{35}} = \frac{k_{45}}{k_{35}} = a \quad \therefore \frac{r_{45}}{r_{25}} = a \times a = a^2$$

$$\text{Similarly, } \frac{r_{75}}{r_{25}} = a^5$$

$$\text{For I reaction } (r_{75})_I = 2^5 \times (r_{25})_I$$

$$\text{For II reaction } (r_{75})_{II} = 3^5 \times (r_{25})_{II}$$

$$\therefore \frac{(r_{75})_{II}}{(r_{75})_I} = \frac{3^5}{2^5} = 7.594 \approx 8 \quad [\because (r_{25})_I = (r_{25})_{II}]$$

30. (3) Rate of reaction $= -\frac{1}{2} \frac{dP_{\text{NO}}}{dt} = -\frac{dP_{\text{H}_2}}{dt}$

For P_{H_2} constant :

$$-\frac{1}{2} \frac{dP_{\text{NO}}}{dt} = 1.5 = k(359)^m (P_{\text{H}_2})^n$$

$$-\frac{1}{2} \frac{dP_{\text{NO}}}{dt} = 0.25 = k(152)^m (P_{\text{H}_2})^n$$

$$\therefore \frac{1.5}{0.25} = \left(\frac{359}{152} \right)^m \quad \therefore m = 2$$

$$\text{For } P_{\text{NO}} \text{ constant : } -\frac{dP_{\text{H}_2}}{dt} = 1.60 = (P_{\text{NO}})^m (289)^n$$

$$-\frac{dP_{\text{H}_2}}{dt} = 0.79 = (P_{\text{NO}})^m (147)^n$$

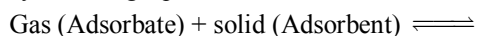
$$\therefore \frac{1.60}{0.79} = \left(\frac{289}{147} \right)^n \quad \therefore n = 1$$

$$\therefore \text{Rate} = k[P_{\text{NO}}]^2 [P_{\text{H}_2}]^1$$

$$\therefore \text{Order of reaction} = 2 + 1 = 3$$

Surface Chemistry

1. (a) Adsorption of a gas on solid is represented by following equilibria,



Gas adsorbed on solid + Heat

Initially adsorption increases with increase in pressure at a particular temperature then got slow. After attaining equilibrium adsorption become independent of pressure.

2. (b) The Langmuir isotherm assumes that the gas attaches to the surface of adsorbent. The free gas and adsorbed gas are in dynamic equilibrium. The surface of the adsorbent is uniform that is, all the adsorption sites are equivalent. The ability of a molecule to get adsorbed at a given site is independent of the occupation of neighbouring sites. Thus, option (b) is correct.
3. (d) Amount of CH_3COOH adsorbed on 1g of charcoal

$$= (0.5 - 0.49 \text{ mol L}^{-1}) \times \frac{1000 \text{ mL}}{1000 \text{ mL/L}} = 1 \times 10^{-3} \text{ mol}$$

So, No. of CH_3COOH molecules in $1 \times 10^{-3} \text{ mol}$ of acetic acid

$$= 6.02 \times 10^{23} \text{ mol}^{-1} \times 10^{-3} \text{ mol} = 6.02 \times 10^{20}$$

Area occupied by one CH_3COOH molecule

$$= \frac{3.01 \times 10^2 \text{ m}^2}{6.02 \times 10^{20}} = 5 \times 10^{-19} \text{ m}^2$$

Thus the surface area of the charcoal covered by each CH_3COOH molecule is $5 \times 10^{-19} \text{ m}^2$.

4. (c) Mass of O_2 per gram of adsorbent = $\frac{3.6}{1.2} = 3$
 No. of moles of O_2 per gram of adsorbent = $\frac{3}{32}$

Volume of O_2 per gram of adsorbent

$$= \frac{3}{32} \times \frac{0.0821 \times 273}{1} = 2.10$$

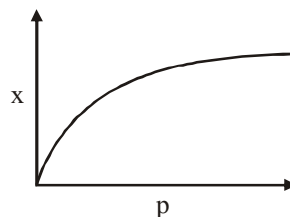
5. (b) $\Delta H < 0, \Delta S < 0, \Delta G < 0$
6. (c) Mass of oxalic acid adsorbed by 10 g charcoal

$$= 200 \times 10^{-3} (0.5 - 0.4) \times 90 = 1.8 \text{ g}$$

$$\frac{x}{m} = \frac{1.8}{10} = 0.18$$

7. (c) The slope = $\frac{1}{n} = \frac{1}{5} = 0.2$
8. (b) We know that, $x/m = K \cdot p^{1/n} (n > 1)$
 $x = K p^{1/n}$
 $m = 1 \text{ g}$ (given)

The above relation between x and p will be correctly represented by Freundlich absorption isotherm curve.

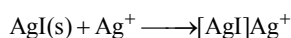
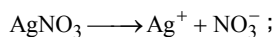
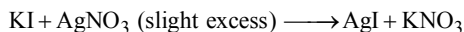


9. (d) In acid hydrolysis of methyl acetate all are present in one phase (liquid)
10. (a)
11. (b) In first case the given compounds have same anion but different cations having different charge hence they will precipitate negatively charged sol i.e. 'A'.

In second case the given compounds have similar cation but different anion with different charge. Hence they will precipitate positively charged sol. i.e. 'B'.

12. (c) $\text{SnO}_2 + 4\text{HCl} \longrightarrow \text{Sn}^{4+} + 4\text{H}_2\text{O}$; Sn^{4+} ions are preferentially adsorbed by SnO_2 particles.

13. (b)



14. (c) The sols obtained in the two cases will be oppositely charged.
15. (a) Since the sol particles migrate towards cathode, they are positively charged. Hence, anions would be effective in coagulation. Greater is the valence of effective ion, smaller will be its coagulating value.
16. (c) O/W type emulsion is stabilized by soaps of alkali metals whereas W/O type emulsions are stabilized by soaps of bi and trivalent cations.
17. (d) Volume of gold dispersed in 1 L water

$$= \frac{\text{Mass}}{\text{Density}} = \frac{1.9 \times 10^{-4} \text{ g}}{19 \text{ g cm}^{-3}} = 1 \times 10^{-5} \text{ cm}^3$$

Radius of gold sol particle

$$= 10 \text{ nm} = 10 \times 10^{-7} \text{ cm} = 10^{-6} \text{ cm}$$

$$\text{Volume of gold sol particle} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (10^{-6})^3 = 4.19 \times 10^{-18} \text{ cm}^3$$

$$\therefore \text{No. of gold sol particles in } 1 \times 10^{-5} \text{ cm}^3$$

$$= \frac{1 \times 10^{-5}}{4.19 \times 10^{-18}} = 2.38 \times 10^{12}$$

$$\therefore \text{No. of gold sol particles in one mm}^3$$

$$= \frac{2.38 \times 10^{12}}{10^6} = 2.4 \times 10^6$$

18. (a) Particle size of colloidal particle = $1 \text{ m}\mu$ to $100 \text{ m}\mu$ (suppose $10 \text{ m}\mu$)

$$V_C = \frac{4}{3} \pi r^3 = V_C = \frac{4}{3} \pi (10)^3$$

Particle size of true solution particle = $1 \text{ m}\mu$

$$V_S = \frac{4}{3} \pi (1)^3 \text{ hence now } \frac{V_C}{V_S} = 10^3$$

19. (c) As_2S_3 sol is negatively charged owing to preferential adsorption of S^{2-} ions. Cation would be the effective ion in coagulation.

Flocculating value = minimum milli mol of the

$$\text{effective ion per litre of sol} = \frac{4 \times 0.005 \times 10^3}{4 + 16} = 1.0$$

20. (a) Preparation of egg albumin sol. is done in following two steps.

(i) Prepare 100 mL of 5% (w/V) solution of NaCl in water in a 250 mL beaker.

(ii) Break one egg in a porcelain dish and pipette out the albumin and pour it in sodium chloride solution. Stir well to ensure the preparation of sol.

21. (6) $\log \frac{x}{m} = \log K + \frac{1}{n} \log P$

$\therefore$ Plot of $\log \frac{x}{m}$ versus $\log P$ is linear with slope

$$= \frac{1}{n} \text{ and intercept} = \log K$$

$$\text{Thus } \frac{1}{n} = \tan \theta = \tan 45^\circ = 1 \text{ or } n = 1$$

$$\log K = 0.301 \text{ or } K = \text{antilog } 0.301 = 2$$

At $P = 3 \text{ atm}$

$$\frac{x}{m} = KP^{1/n} = 2 \times (3)^1 = 6$$

22. (3) Adsorbed moles of $\text{H}_2 = \frac{0.03 \times 2.46 \times 10^{-3}}{0.0821 \times 300}$

$$= 3 \times 10^{-6}$$

$\therefore$ No. of adsorbed molecules of

$$\text{H}_2 = 3 \times 10^{-6} \times 6 \times 10^{23}$$

$$\Rightarrow = 18 \times 10^{17}$$

Total no. of surface sites available

$$= 5.4 \times 10^{16} \times 1000$$

$$\Rightarrow 5.4 \times 10^{19} \text{ cm}^2$$

No. of surface sites that is occupied by adsorption of

$$H_2 = \frac{10}{100} \times 5.4 \times 10^{19}$$

$$\Rightarrow 5.4 \times 10^{18}$$

No. of surface sites occupied by one molecule

$$H_2 = \frac{5.4 \times 10^{18}}{18 \times 10^{17}} = 3$$

$$\begin{aligned} 23. \quad (8) \quad \frac{x}{m} &= K \times P^{1/n} \\ \log \frac{x}{m} &= \frac{1}{n} \log P + \log K \\ \frac{1}{n} &= \tan 45^\circ = 1 \end{aligned}$$

$$\Rightarrow n = 1$$

$$\log K = 0.3$$

at pressure $P = 4$ atm.

$$\log \frac{x}{m} = 1 \times \log 4 + 0.3$$

$$\Rightarrow \log \frac{x}{m} = 2 \log 2 + \log 2 \quad (\log 2 = 0.3)$$

$$\Rightarrow \log \frac{x}{m} = 3 \log 2 = \log 8$$

$$\Rightarrow \frac{x}{m} = 8.$$

$$24. \quad (7) \quad \text{Volume of gold} = \frac{1.9 \times 10^{-3} \text{ g}}{19 \text{ g/cm}^3} = 10^{-4} \text{ cm}^3$$

Radius of gold particle = 10.5 nm

$$= 10.5 \times 10^{-7} \text{ cm.}$$

Volume of gold-sol particle

$$= \frac{4}{3} \pi r^3 = \frac{4}{3} \times \frac{22}{7} \times (10.5 \times 10^{-7})^3 \text{ cm.}$$

$$= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} \times 10^{-21} \text{ cm}^3$$

$$= 4.851 \times 10^{-18} \text{ cm}^3$$

As gold sol occupying 50% of the space, the

$$\text{available volume} = \frac{1}{2} \times 10^{-4} \text{ cm}^3$$

$$\text{Number of gold particles} = \frac{0.5 \times 10^{-4}}{4.851 \times 10^{-18}}$$

$$= 1.03 \times 10^{13}$$

Gold particles are dispersed in a litre of water i.e. 10^6 mm^3 .

Number of gold particles in one mm^3 of sol

$$= \frac{1.03 \times 10^{13}}{10^6 \text{ mm}^3} = 1.03 \times 10^7 \approx 10^7 \text{ per mm}^3$$

Hence, $n = 7$.

25. (30) Before micelle formation, number of particles of cationic colloidal electrolyte per litre

$$= 10^{-3} \times 6 \times 10^{23} = 6 \times 10^{20} / \text{L}$$

$$= 6 \times 10^{20} / 10^6 \text{ mm}^3 = 6 \times 10^{14} / \text{mm}^3$$

Number of micelles in $1 \text{ mm}^3 = 2 \times 10^{13}$

Thus, number of cations in one micelle

$$= \frac{6 \times 10^{14}}{2 \times 10^{13}} = 3 \times 10 = 30.$$

26. (100)

$$400 \text{ mL of colloid requires} = \frac{1.46 \text{ g}}{36.5 \text{ g/mol}}$$

$$= 0.04 \text{ mol} = 40 \text{ m-mol}$$

$$1000 \text{ mL of colloid requires} = \frac{40}{400} \times 1000$$

$$= 100 \text{ m-mol}$$

Flocculation value of HCl = 100

27. (25) 10 mL of 1% NaCl solution =

1 mL of 10% NaCl solution

For 100 mL of gold sol = 250 mg. of substance required

For 10 mL of gold sol = 25 mg. of substance required

Hence, gold number = 25.

28. (16.25) Final volume of gas, at 608 torr pressure

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{760 \times 1}{608}$$

$$\Rightarrow 1.25 \text{ L or } V_2 = 1250 \text{ mL}$$

Volume occupied by gas = Volume of vessel
 – Volume occupied by charcoal

$$= 1000 - \frac{16}{1.6} = 990 \text{ mL}$$

Difference of volume is due to adsorption of gas by charcoal.

$$\therefore \text{Volume of gas adsorbed by charcoal} \\ = 1250 - 990 \Rightarrow 260 \text{ mL}$$

Volume of the gas adsorbed per gram of charcoal

$$= \frac{260}{16} = 16.25 \text{ mL/g at 608 torr and } 27^\circ\text{C}.$$

29. (7.46) Total surface area of carbon

$$= \frac{44}{7} \times 10^7 \text{ cm}^2$$

$$r = 10^{-8} \text{ cm}$$

$$\text{Surface area of } \text{NH}_3 = \pi r^2 = \frac{22}{7} \times 10^{-16} \text{ cm}^2$$

$$\text{No. of } \text{NH}_3 \text{ molecules adsorbed} = \frac{\frac{44}{7} \times 10^7}{\frac{22}{7} \times 10^{-16}}$$

Vol. of NH_3 adsorbed at 1 atm 273 K

$$= \frac{2 \times 10^{23}}{6 \times 10^{23}} \times 22.4 = 7.46 \text{ L}$$

30. (60) No. of sodium lauryl sulphate molecules

$(\text{CH}_3(\text{CH}_2)_{11}\text{SO}_4^-\text{Na}^+)$ in 1 litre solution

$$= 10^{-3} \times 6 \times 10^{23} = 6 \times 10^{20}$$

No. of sodium lauryl sulphate molecules per $\text{mm}^3 = 6 \times 10^{14}$

No. of colloidal particles per $\text{mm}^3 = 10^{13}$

No. of molecules per colloidal particle

$$= \frac{6 \times 10^{14}}{10^{13}} = 60$$

General Principles and Processes of Isolation of Elements

- (b) Calcination is used for removal of volatile impurities and decompose carbonates into their respective oxides.

$$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$$

Bauxite Alumina

$$\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$$

Limestone Calcium oxide
- (a) For example, Ag_2S is converted into $\text{Na}[\text{Ag}(\text{CN})_2]$. When Zn is added, Ag is displaced.
- (a) Zone refining is based on the difference in solubility of impurities in molten and solid state of the metal. This method is used for obtaining metals of very high purity.
- (c) Froth reduces the surface tension of water and the solution forms froth.
- (c) Tricalcium phosphate and calcium silicate is Thomas slag
- (b) Zr and Ti are purified by van Arkel method.

$$\text{Zr(s)} + 2\text{I}_2(\text{g}) \longrightarrow \text{ZrI}_4(\text{g})$$

$$\text{ZrI}_4(\text{g}) \xrightarrow[\text{filament}]{\text{On the hot}} \text{Zr(s)} + 2\text{I}_2(\text{g})$$
- (d) Calcination is a process of heating a substance to a high temperature but below the melting or fusion point, causing loss of moisture, reduction or oxidation and dissociation into simpler substances.
- (a) The more stable CO_2 has higher enthalpy of formation than CS_2
- (d) Ni is purified by Mond's process by decomposition of $\text{Ni}(\text{CO})_4$.
- (d) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$ is calcination.
- (c) Fe_2O_3 (3 parts) + Al powder (1 part) is known as thermite

$$\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 3\text{Fe}$$
- (a) Chalcopryite : CuFeS_2
 Fool's gold : FeS_2
 Carnalite : $\text{KMgCl}_3 \cdot 6\text{H}_2\text{O}$
 Bauxite : $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
- (c) Azurite is a basic carbonate ore of copper. $2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
- (b) Cassiterite contains the magnetic impurities of FeSO_4 and thus concentrated by electro-magnetic separation.
- (c) Lead pyromorphite is a phosphate ore. It can be represented as $3\text{Pb}_3(\text{PO}_4)_2 \cdot \text{PbCl}_2$.
- (d) (A) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{+3} \text{Al}_2\text{O}_3 + 2\text{Cr}$
 (B) $\text{Au} + 2\text{CN}^- + \text{H}_2\text{O} + \frac{1}{2}\text{O}_2 \longrightarrow [\text{Au}(\text{CN})_2]^- + 2\text{OH}^-$
 $2[\text{Au}(\text{CN})_2]^- + \text{Zn} \rightarrow [\text{Zn}(\text{CN})_4]^{2-} + 2\text{Au}$
 (C) Statement is true
 (D) $\text{Ag}_2\text{S} + 4\text{NaCN} \xrightarrow{\text{O}_2} 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$
 $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \longrightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}$
- (c) The enthalpy of formation of Al_2O_3 is very high and therefore, it cannot be reduced by carbon. It is reduced by electrolytic method.
- (a) Generally tin is obtained by carbon reduction and lead by self-reduction.

$$\text{SnO}_2 \longrightarrow \text{Sn} + \text{CO}_2 \quad (\text{carbon reduction})$$

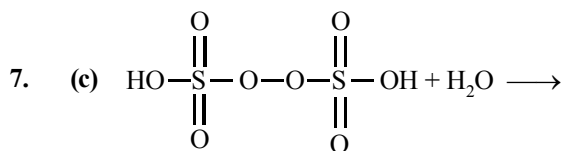
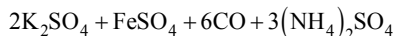
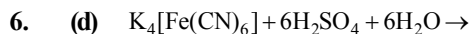
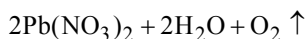
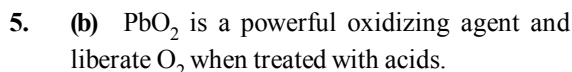
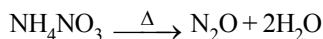
$$2\text{PbO} + \text{PbS} \longrightarrow 3\text{Pb} + \text{SO}_2 \quad (\text{self-reduction})$$
- (d) $2\text{Cu}_2\text{S} + 3\text{O}_2 \longrightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$
 $3\text{Cu}_2\text{O} + \text{CH}_4 \longrightarrow 6\text{Cu} + 2\text{H}_2\text{O} + \text{CO}$

(From green logs of wood)
- (a) It is used in metallurgy of Zn.

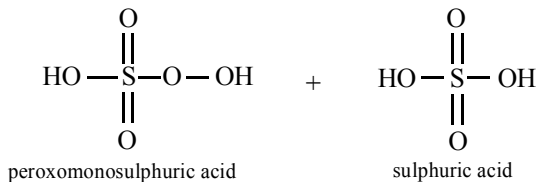
CHAPTER

21

The p-Block Elements (Group 15, 16, 17 and 18)

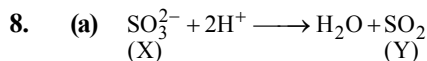


peroxodisulphuric acid

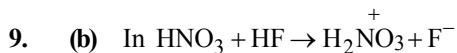
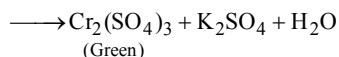
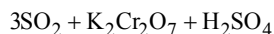


peroxomonosulphuric acid

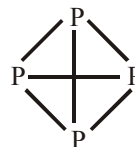
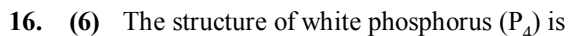
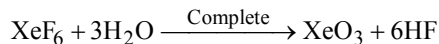
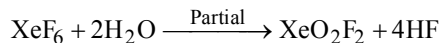
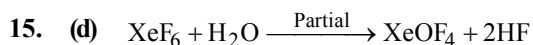
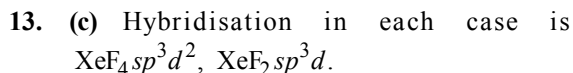
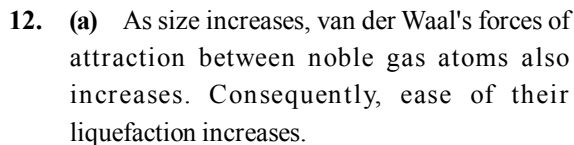
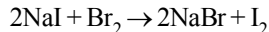
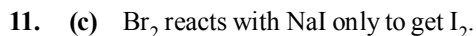
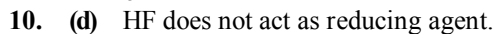
sulphuric acid



SO_2 is a colourless gas with irritating smell.

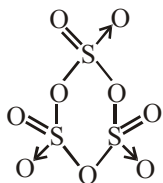


HNO_3 accepts H^+ hence it is a base.



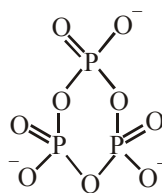
Hence bonding electron pairs are 6.

17. (0) S_3O_9 (also called α -sulphur trioxide) is an ice like solid with the following molecular structure.



There is no S-S bond in S_3O_9

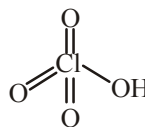
18. (3)



No. of P-O-P bonds = 3. It is the most stable among all the cyclic metaphosphates

19. (6)

20. (3) The structure of perchloric acid is



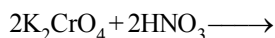
The number Cl = O bond in HClO_4 is 3.

CHAPTER

22

The *d*- and *f*-Block Elements

1. (a) When a solution of potassium chromate is treated with an excess of dilute nitric acid. Potassium dichromate and H_2O are formed.



Hence $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed.

2. (c) In lanthanoid series there is a regular decrease in the atomic as well as ionic radii of trivalent ions (M^{3+}) as the atomic number increases. Although the atomic radii do show some irregularities but ionic radii decreases from La (103 pm) to Lu (86 pm).

3. (a) The more the oxidation state, the more is the acid character, MnO , Mn_2O_3 are basic, MnO_2 is amphoteric and Mn_2O_7 is acidic.

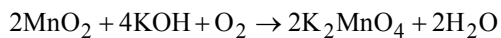
4. (b) Ferrous ferricyanide is known as Turnbull's blue

5. (d) Magnetic moment $\mu = \sqrt{n(n+2)}$ where n = number of unpaired electrons $\sqrt{15} = \sqrt{n(n+2)}$

$$\therefore n = 3$$

6. (b) In KIO_4 O.S. of I is +7 and in KMnO_4 O.S. of Mn is +7

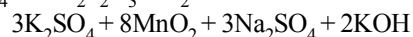
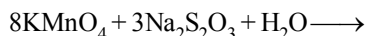
7. (a) Stable oxidation state of Mn in alkaline medium is +6. So, MnO_2 is oxidised to K_2MnO_4 by atmospheric oxygen in KOH medium.



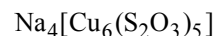
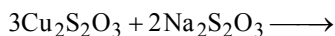
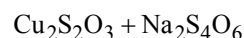
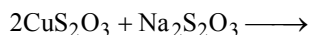
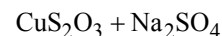
8. (a) $\text{Fe}^{2+} + 6\text{KCN} \rightarrow [\text{Fe}(\text{CN})_6]^{4-} + 6\text{K}^+$

9. (a)

10. (b) In neutral or faintly alkaline medium thiosulphate is quantitatively oxidized by KMnO_4 to SO_4^{2-}



11. (d) $\text{CuSO}_4 + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow$



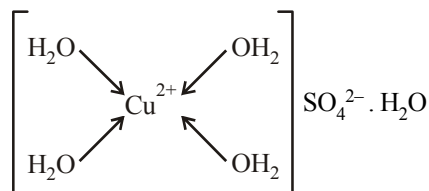
12. (a) Acidic nature of oxides increases with the increase in O.S. of the metal atom e.g., Cr_2O_3 is basic. CrO_2 is acidic, CrO_5 is also acidic in nature.

13. (d) $\text{HgS} + \text{O}_2 \xrightarrow{773-873\text{K}} \text{Hg} + \text{SO}_2$

14. (d) $\text{Hg}_2\text{Cl}_2 \xrightarrow{\Delta} \text{Hg} + \text{HgCl}_2$

15. (b) As a result of lanthanoid contraction Zr^{4+} and Hf^{4+} possess almost the same ionic radii. Ce^{4+} is an oxidising agent. Ce^{4+} gains electron to acquire more stable Ce^{3+} state. $\text{La}(\text{OH})_3$ is the most basic among lanthanoid hydroxides.

16. (4) The number of water molecules directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 4.



Coordination Compounds

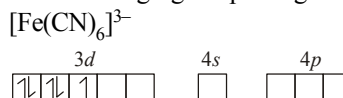
- (c) The ions present in the ionisation sphere are precipitated. Hence $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl} \cdot 2\text{H}_2\text{O}$ contains $1/3$ Cl in ionisation sphere to be precipitated by AgNO_3 as AgCl
- (a) $\text{CoSO}_4 \cdot \text{Cl} \cdot 5\text{NH}_3$ has two ionization isomers i.e. $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. On reacting them with AgNO_3 and BaCl_2 following reactions give white precipitate.
 - $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl} + \text{AgNO}_3 \rightarrow \text{AgCl} + [\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
[White ppt]
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + [\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
[White ppt]
- (b) MX_4Y_2 is an octahedral complex and it has two geometrical isomers i.e. *cis* and *trans* forms both of them are achiral.
- (a) In $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$, the chloride ions are ionisable. As they are present outside the coordination sphere.

$$[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3 + 3\text{AgNO}_3 \longrightarrow [\text{Cr}(\text{NH}_3)_6]\text{NO}_3 + 3\text{AgCl} \downarrow$$

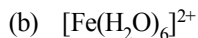
white ppt.
- (a)
- (b) Linkage isomerism arises in a coordination compound containing ambidentate ligand. Since NO_2 is an ambidentate ligand which can join to the metal atom through N or O-atom. Thus, $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ show linkage isomerism.
- (a) $[\text{Fe}(\text{CN})_6]^{3-}$
The oxidation number of Fe in $[\text{Fe}(\text{CN})_6]^{3-}$ is +3.
 $\text{Fe}^{3+} : [\text{Ar}] 3d^5 4s^0$



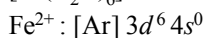
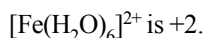
As CN^- is a strong ligand pairing will occur.



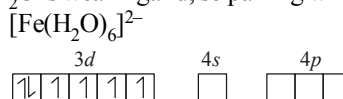
$$\therefore n = 1 \Rightarrow \mu = \sqrt{1(1+2)} = \sqrt{3}$$



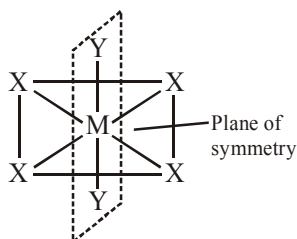
The oxidation number of Fe in



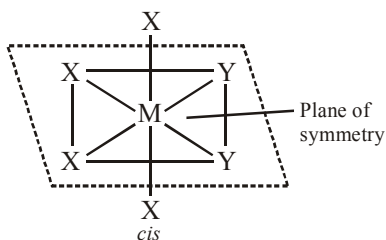
As H_2O is weak ligand, so pairing will not occur.



$$\therefore n = 4 \Rightarrow \mu = \sqrt{4(4+2)} = \sqrt{24}$$



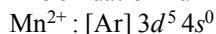
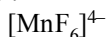
trans
(Achiral)



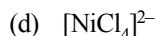
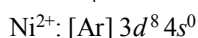
cis
(Achiral)



The oxidation number of Mn is +2.


 As F^- is a weak field ligand, so pairing will not occur.


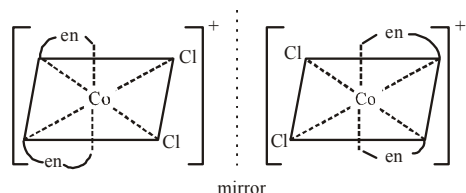
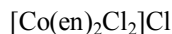
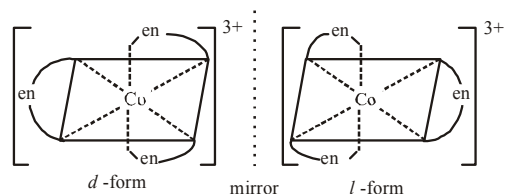
$$\therefore n = 5 \Rightarrow \mu = \sqrt{5(5+2)} = \sqrt{35}$$


 In $[\text{NiCl}_4]^{2-}$, the oxidation number of Ni will be +2.

 As Cl^- is a weak field ligand, so pairing will not occur.


$$n = 2 \Rightarrow \mu = \sqrt{2(2+2)} = \sqrt{8}$$

 Thus, $[\text{MnF}_6]^{4-}$ will have the highest spin only magnetic moment.

8. (d) $[\text{Pt}(\text{en})_2\text{Cl}_2]$ is a complex of the type $\text{M}(\text{AA})_2\text{B}_2$ which is octahedral such compounds exhibit optical and geometrical isomerism both.
9. (b) $[\text{Co}(\text{en})_3]\text{Cl}_3$



10. (c) Number of moles of AgCl

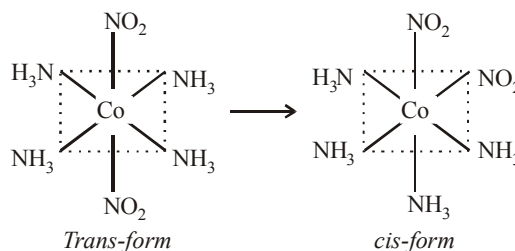
$$= \frac{4.305}{143.5} = 0.03 \text{ mol}$$

 0.01 mol. of complex $\rightarrow$ 0.03 mol of AgCl

 1 mol. of complex $\rightarrow$ 3 moles of AgCl

 Hence, the compound has three ionisable Cl atoms. Thus formula of the complex must be $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$.

11. (d) (a) square planar - *cis* and *trans* isomers
 (b) square planar - 3 isomers (while NH_3 at position 1, groups NH_2OH , NO_2 and $\text{C}_6\text{H}_5\text{N}$ can occupy position 3).
 (c) 2 isomers (*cis* and *trans*)
12. (a) The given compound may have linkage isomerism due to presence of NO_2 group which may be in the form $-\text{NO}_2$ or $-\text{ONO}$. It may have ionisation isomerism due to presence of two ionisable group $-\text{NO}_2$ and $-\text{Cl}$. It may have geometrical isomerism in the form of *cis-trans* form as follows :
- $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]\text{NO}_2$ and $[\text{Co}(\text{NH}_3)(\text{NO}_2)_2]\text{Cl}$
 Ionisation isomers.
- $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)_2]\text{Cl}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})_2]\text{Cl}$
 Linkage isomers



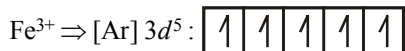
Geometrical isomers

13. (c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ produces maximum number of ions (5) in the solution.
 $\text{K}_4[\text{Fe}(\text{CN})_6] \rightleftharpoons 4\text{K}^+ + [\text{Fe}(\text{CN})_6]^{4-}$
14. (b) IUPAC name is potassium trioxalatoaluminate (III).
15. (a) Since the precipitate of AgCl shows two ionisable chloride ion the complex must have the structure.
 $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 + 2\text{AgNO}_3 \rightarrow$
 $[\text{Co}(\text{NH}_3)_5\text{Cl}](\text{NO}_3)_2 + 2\text{AgCl}$
- Hence two chlorine atoms satisfy the primary valency and one secondary valency.
16. (a) $\text{K}_3[\text{Fe}(\text{CN})_6]$

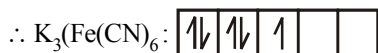
Let 'x' be the oxidation state of Fe

$$x + 6(-1) = -3$$

$$x = +3$$



CN^- strong field ligand, hence pairing occurs.



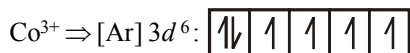
It is paramagnetic as it has 1 unpaired electron.



Let 'x' be the oxidation state of Co

$$x + 6(-1) = -3$$

$$x = +3$$

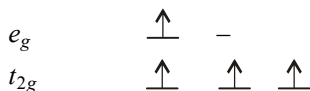


F^- weak field ligand hence, no pairing occurs.

It is paramagnetic as it has 4 unpaired electron.

17. (b)

18. (d) d^4 in high spin octahedral complex

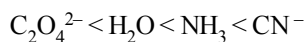


$$\text{CFSE} = [0.6 \times 1] + [-0.4 \times 3] = -0.6 \Delta_0$$

19. (b) $d^6 - t_{2g}^{2,2,2} e_g^{0,0}$ (in low spin)

$$\text{C.F.S.E} = -0.4 \times 6\Delta_0 + 3P = -\frac{12}{5} \Delta_0 + 3P$$

20. (b) Crystal field splitting depends upon the field strength of ligand. The field strength of ligand Δ decreases as shown below

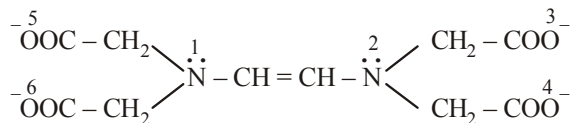


hence the crystal field splitting will be maximum for $[\text{Co}(\text{CN})_6]^{3-}$

21. (4)

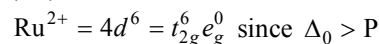
22. (4)

23. (6)



EDTA^{4-} is a hexadentate ligand, since hexa means six and the ligand attaches six times.

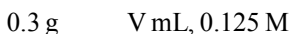
24. (0.0)



No. of unpaired electrons = 0

$\therefore$ Magnetic moment = 0 B.M.

25. (26.92)



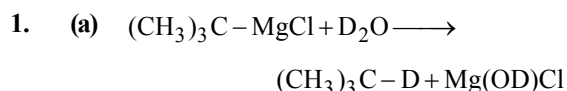
$$\text{Number of moles of the complex} = \frac{0.3}{267.46}$$

$$\text{Number of moles of AgNO}_3 = 0.125 \times \text{V} \times 10^{-3}$$

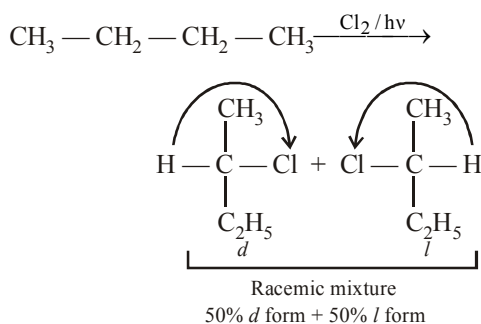
$$\frac{0.3}{267.46} \times 3 = 0.125 \times \text{V} \times 10^{-3}$$

$$\text{Or, V} = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 \text{ mL}$$

Halolalkanes and Haloarenes

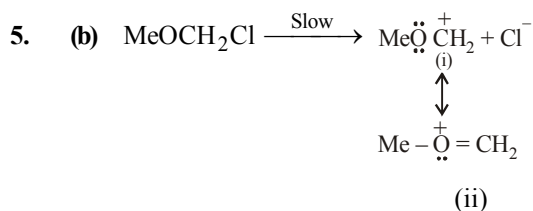
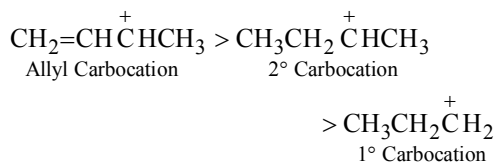


2. (d)



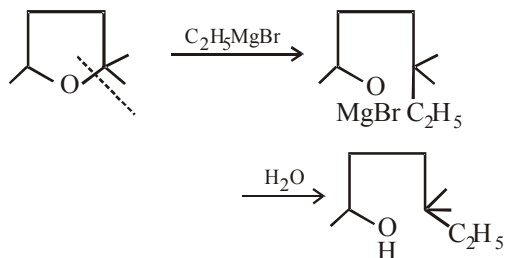
$\text{Cl}^\bullet$ may attack on either side and give a racemic mixture of 2-chloro butane which contain 50% *d* form and 50% *l*-form.

3. (c) More stable the carbocation, more reactive will be the parent alkyl halide towards $\text{S}_{\text{N}}1$ reaction.
 $3^\circ > \text{Benzyl} > \text{Allyl} > 2^\circ > 1^\circ > \text{methyl}$
4. (c) Stability of the three corresponding carbocations

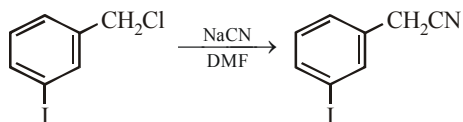


Though (ii) contains +ve charge on oxygen. Since octet around each atom is complete the structure II is more stable than I.

6. (a) A strong nucleophile favours the $\text{S}_{\text{N}}2$ reaction and a weak nucleophile favours the $\text{S}_{\text{N}}1$ reaction. First reaction is $\text{S}_{\text{N}}1$ reaction because $\text{C}_2\text{H}_5\text{OH}$ used as solvent which is a weak nucleophile. Second reaction is $\text{S}_{\text{N}}2$ reaction because $\text{C}_2\text{H}_5\text{O}^-$ is strong nucleophile.
7. (d) The boiling points of alkyl chlorides increases with increase in molecular weight. In case of isomeric alkyl chloride, the order of boiling point is
 primary > secondary > tertiary
8. (c) Small difference in boiling points of C_2F_6 and C_2H_6 is due to the fact that (i) the F atom is only slightly larger than H, and (ii) F has low polarizability
9. (c) Organomagnesium and organolithium compounds can't be prepared from the alkyl (or aryl) halide having $-\text{NO}_2$ group. On the other hand, organocopper and organocadmium compounds, do not react with the $-\text{NO}_2$ group.
10. (b) When Grignard reagent reacts with an epoxide, the alkyl group of the Grignard reagent goes to more substituted carbon atom.

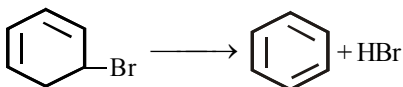


11. (b)



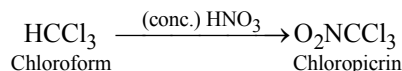
Nuclear substitution will not take place.

12. (c)

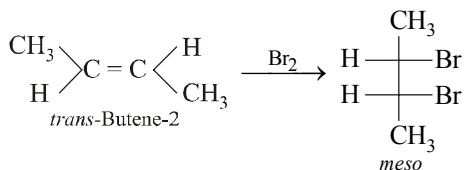
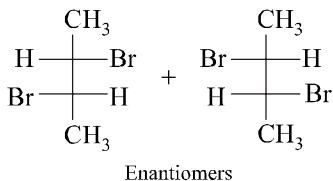
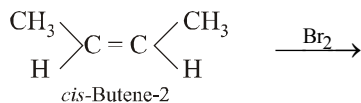
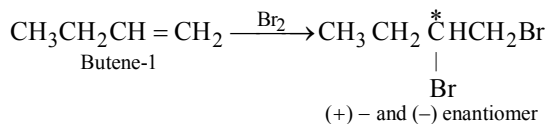
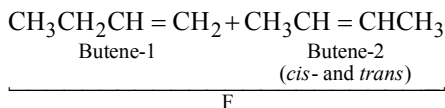
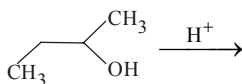


It results in formation of more stable aromatic compound which is benzene.

13. (d) 14. (d)

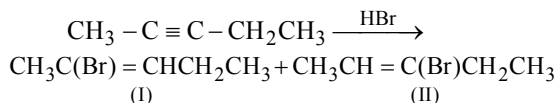
15. (d) Chloropicrin is nitrochloroform. It is obtained by the nitration of chloroform with HNO_3 .

16. (3)



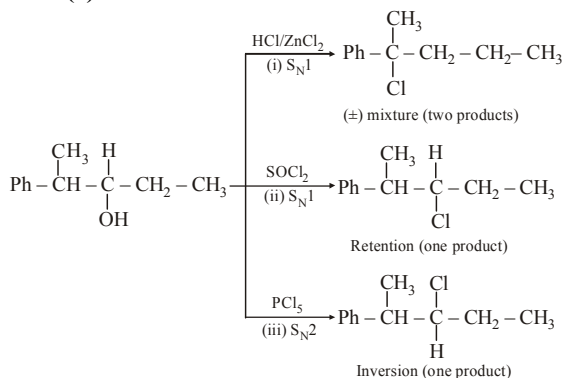
So F can have three possible structures

17. (6) Addition of HBr to 2-pentyne gives two structural isomers (I) and (II)



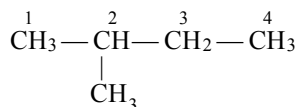
Each one of these will exist as a pair of geometrical isomers. Thus, there are two structural and four configurational isomers. (total 6)

18. (4)

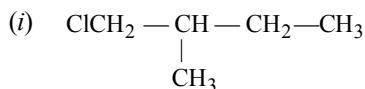


Total no. of products is 4.

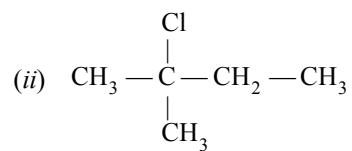
19. (4) In the given compound, there are four different types of hydrogen atoms as shown below:



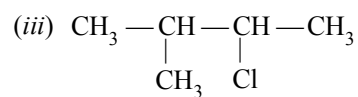
The replacement of each of these hydrogen atoms will give the following four monochloro derivatives.



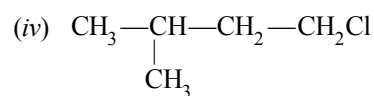
1-chloro-2-methylbutane



2-chloro-2-methylbutane

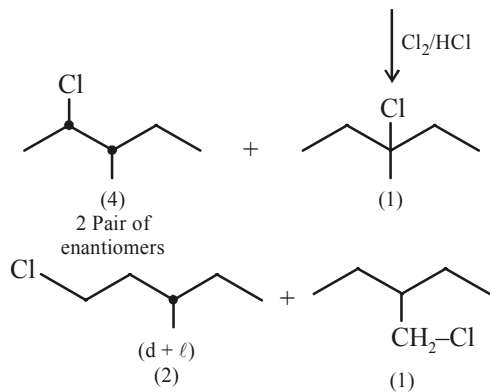
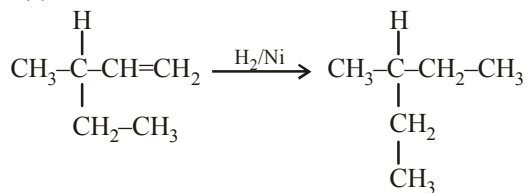


2-chloro-3-methylbutane

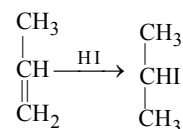
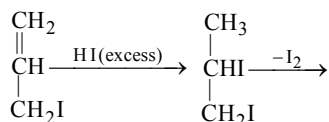
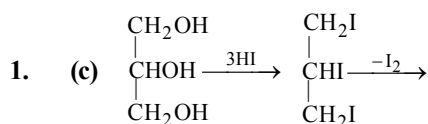


1-chloro-3-methylbutane

20. (8)



Alcohols, Phenols and Ethers



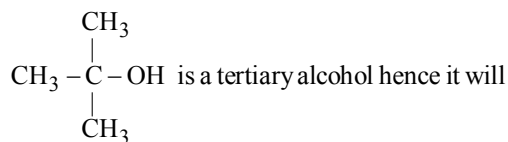
∴ All the compounds except (c), $\text{CH}_2\text{OH}-\text{CHI}-\text{CH}_2\text{OH}$ are formed during reaction of glycerol with excess HI.

2. (d) Lucas reagent is anhydrous ZnCl_2 and conc. HCl . It is used to distinguish between 1° , 2° and 3° alcohols.

3° alcohols → immediate turbidity

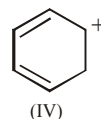
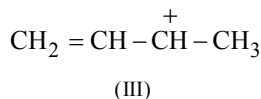
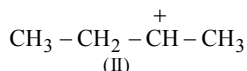
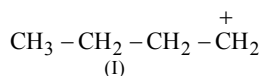
2° alcohols → turbidity after 5 minutes

1° alcohols → No turbidity at room temp.



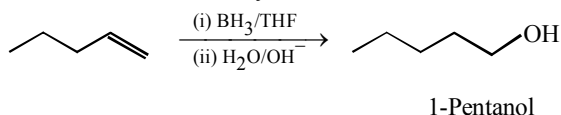
give fastest reaction with Lucas reagent.

3. (d) Among the given compounds only CH_3OH does not give iodoform reaction.
4. (b) The ease of dehydration can be determined by seeing the stability of carbocation as intermediate.

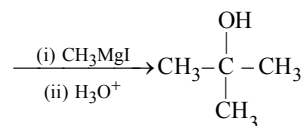
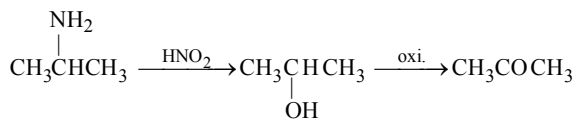


(IV) is most stable due to resonance followed by (III) which is allylic carbocation. Next is (II) as it is a secondary carbocation and least stable is (I) as it is a primary carbocation.

5. (a) Hydroboration-oxidation leads to *anti*-Markownikoff's hydration, thus

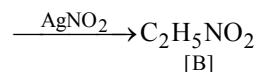
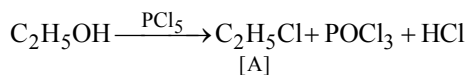


6. (d)

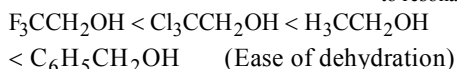
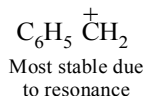
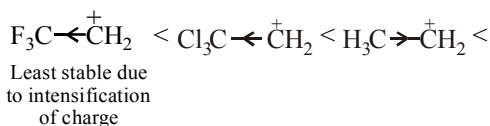


2-Methyl-2-propanol

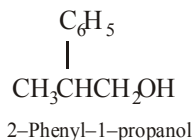
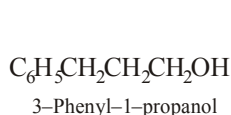
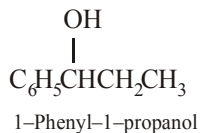
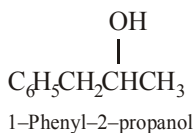
7. (c)



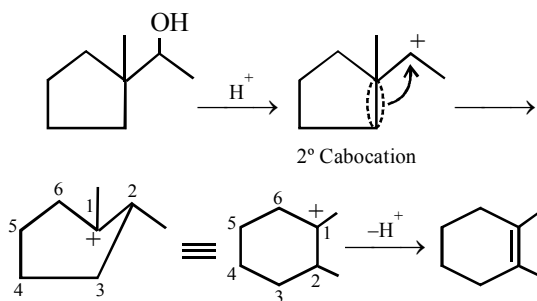
8. (c) Higher the stability of the carbocation, more easily it is formed and more will be the ease of its dehydration. Stability of the carbocations of the corresponding alcohols is



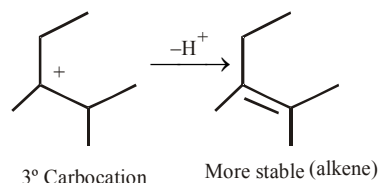
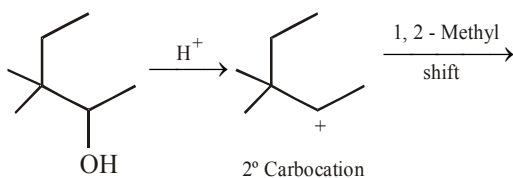
9. (b) Hydroboration – oxidation reaction of alkenes leads to anti-Markovnikov's hydration. Further addition of water adds in *syn*-manner, i.e., H and OH are added to the same face of the double bond leading *trans*-product. In short, hydroboration-oxidation of alkenes is regioselective as well as stereoselective.
10. (b) It is a nucleophilic substitution reaction, 1-phenyl-1-propanol will be most reactive because of formation of more stable benzyl type of carbocation.



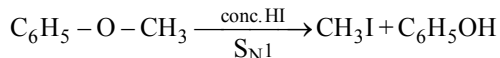
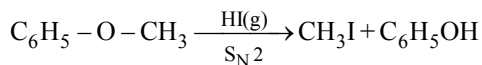
11. (c)



12. (c) It is an example of dehydration of alcohol. The 2° carbocation formed at first stage rearranges to the more stable 3°.

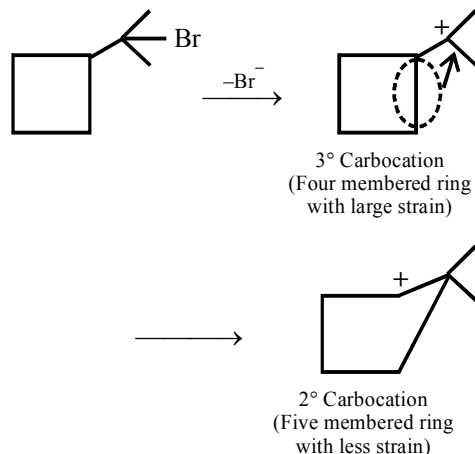


13. (c) Although in both cases products are CH_3I and $\text{C}_6\text{H}_5\text{OH}$; the two reactions follow different mechanism.



Remember that during $\text{S}_\text{N}1$ reaction, CH_3^+ is formed because it is more stable than C_6H_5^+ .

14. (c)

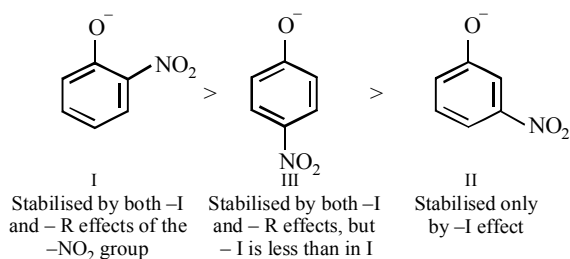


15. (b) Base catalysed epoxide opening is a typical $\text{S}_\text{N}2$ reaction in which attack of the nucleophile takes place at the less hindered epoxide.

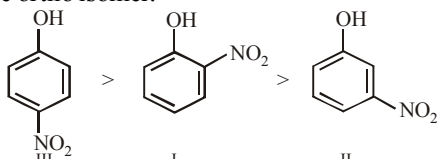
16. (b)

17. (c) In presence of non-protic solvent such as CHCl_3 or CCl_4 , concentration of electrophile (Br^+) is less, hence reaction stops at the monobromo stage

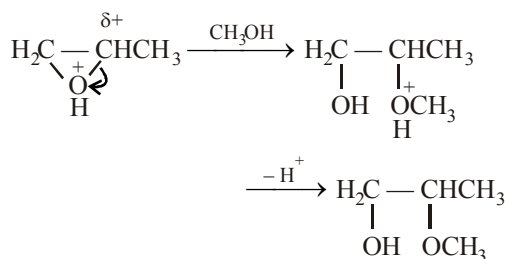
18. (c) Observe the relative stability of their corresponding conjugate bases.



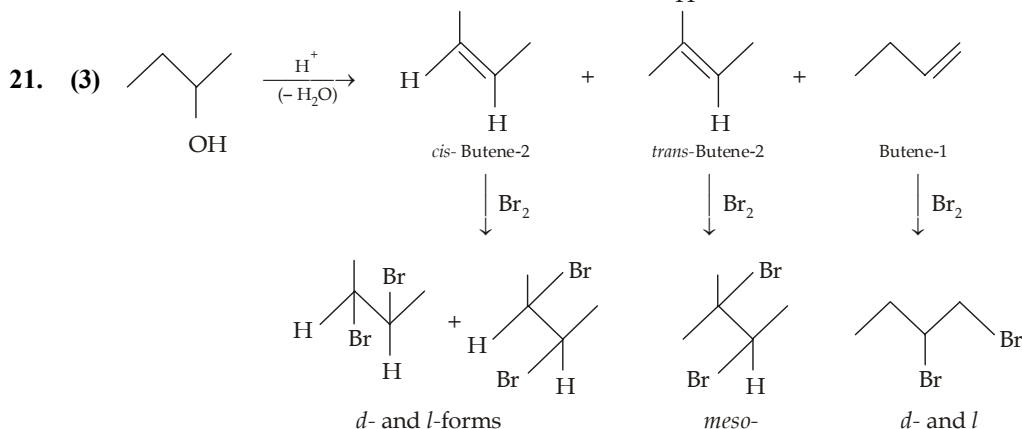
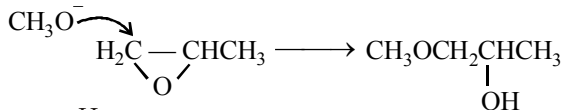
However, the acidity of the corresponding phenols will be different because of H-bonding in the ortho isomer.



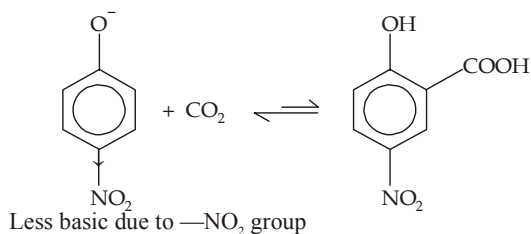
19. (d) It is an example of Claisen rearrangement.
20. (b) In the acid-catalysed ring opening of an unsymmetrical ethoxide, the nucleophile attacks primarily at the more substituted carbon atom because such atom of the protonated epoxide acquires a considerable positive charge. This resembles like a more stable 2° or 3° carbocation and hence the reaction is S_N1 like.



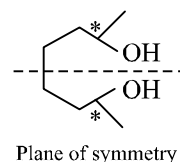
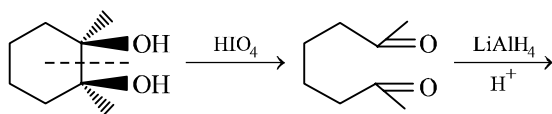
In case of CH_3ONa , $^-\text{OCH}_3$, being a strong nucleophile, opens the strained epoxide ring in a direct S_N2 reaction, i.e. by attacking at the least hindered carbon atom.



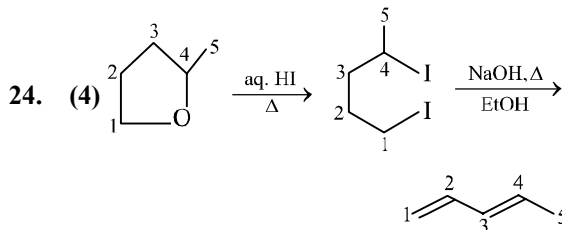
22. (1) The corresponding phenoxide ion will be less basic, hence equilibrium will not lie favourably toward carboxylation.



23. (3)

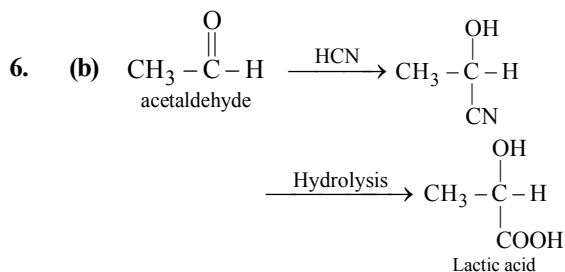
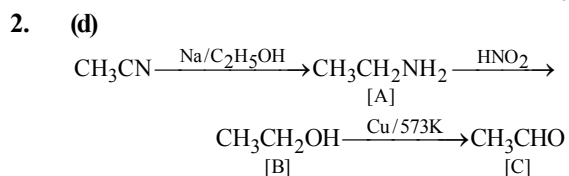
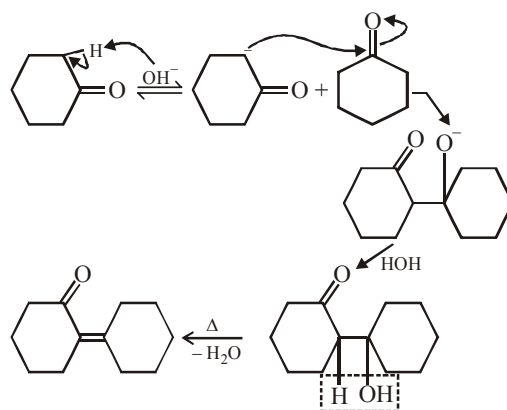
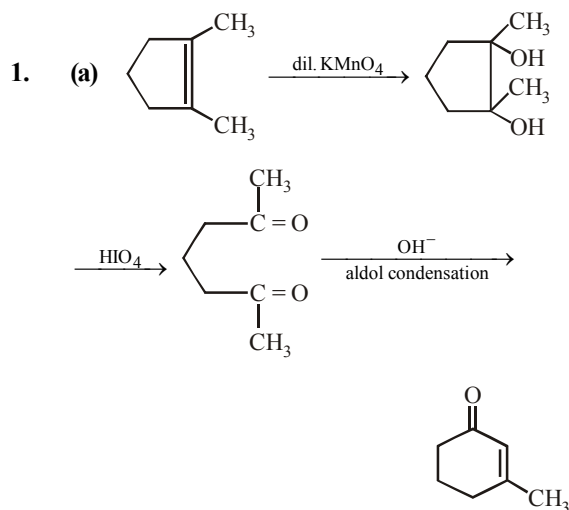


2 chiral carbons and one plane of symmetry
 $\Rightarrow$ 3 stereoisomers.



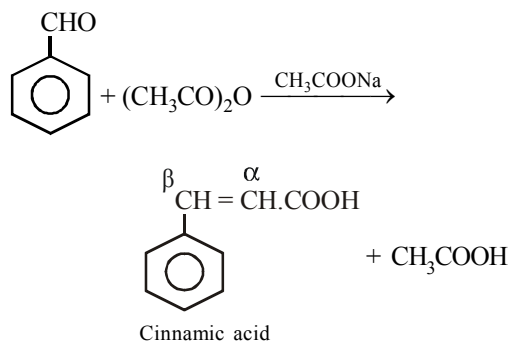
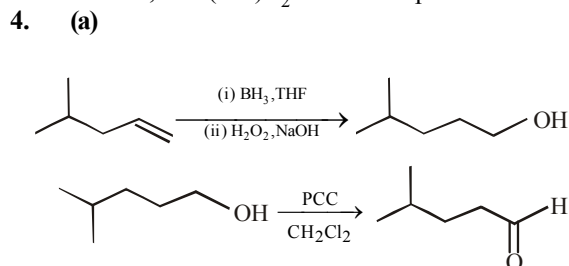
25. (3) 3 acidic hydrogens will produce, 3 moles of methane gas.

Aldehydes, Ketones and Carboxylic Acids



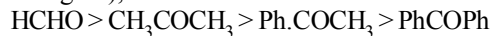
3. (a) ArCOR can be prepared by the combination of ArH + R'COCl and not by ArCOCl + RMgX because here the ArCOR formed will further react with RMgX to form 3° alcohol, ArC(OH)R₂ as the final products.

7. (b) Benzaldehyde forms cinnamic acid as follows.



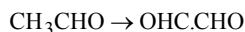
5. (c) In the presence of base, cyclohexanone show aldol condensation.

8. (a) Addition of HCN to carbonyl compounds is nucleophilic addition reaction. The order of reactivity of carbonyl compounds is Aldehydes (smaller to higher) Ketones (smaller to higher), Then

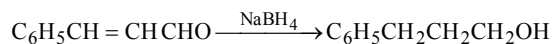


The lower reactivity of Ketones is due to presence of two alkyl group which shows +I effect. The reactivity of Ketones decreases as the size of alkyl group increases.

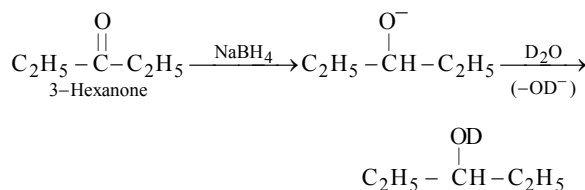
9. (c) Selenium dioxide oxidises aldehydes and ketones at α -position to carbonyl group.



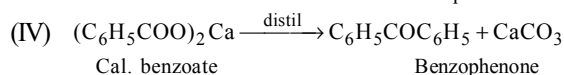
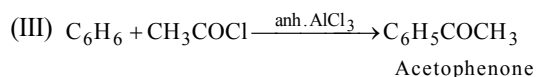
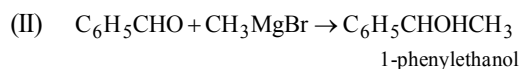
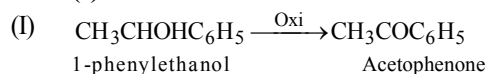
10. (a) Normally NaBH_4 as well as LiAlH_4 reduce only $-\text{CHO}$ group without effecting carbon-carbon double bond, however when it is present in conjugation with benzene ring and aldehydic group it is also reduced along with the reduction of $-\text{CHO}$ group.



11. (c) Remember that during reduction with NaBH_4 , a hydride ion is transferred by NaBH_4 to carbonyl carbon and a proton (or deuteron) is transferred from the solvent (in the second step) to carbonyl oxygen.



12. (c)



13. (a) (C) contains the most reactive methylene group followed by (A) then (B).

14. (d) The intermediate is carbocation which is destabilised by $\text{C}=\text{O}$ group (present on α -carbon to the $-\text{OH}$ group) in the first three cases. In (d), α -hydrogen is more acidic which

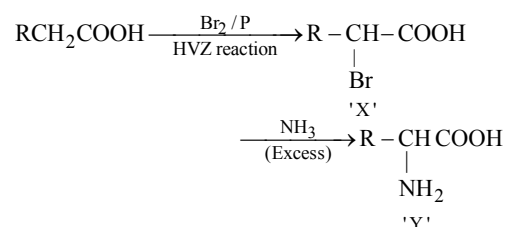
can be removed as water. Moreover, the positive charge on the intermediate carbocation is relatively away from the $\text{C}=\text{O}$ group.

15. (a) Enolic form predominates in compounds containing two carbonyl groups separated by a $-\text{CH}_2$ group. This is due to following two factors.

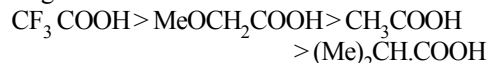
(i) Presence of conjugation which increases stability.

(ii) Formation of intramolecular hydrogen bond between enolic hydroxyl group and second carbonyl group which leads to stabilisation of the molecule. Hence the correct answer is $\text{III} > \text{II} > \text{I}$.

16. (b)

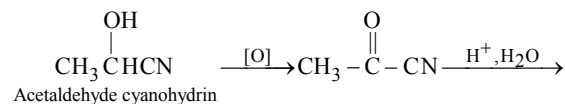


17. (a) The correct order of increasing acid strength

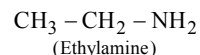
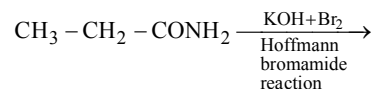
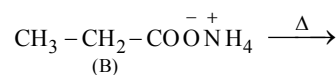
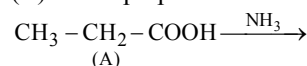


Electron withdrawing groups increase the acid strength and electron donating groups decrease the acid strength.

18. (a)



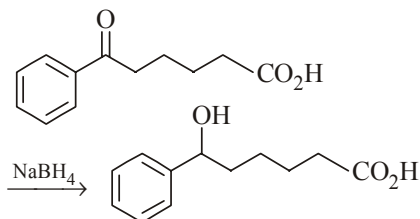
19. (d) Since, C when heated with Br_2 in presence of KOH produces ethylamine, hence it must be propanamide and hence the organic compound (A) will be propanoic acid. The reactions follows.



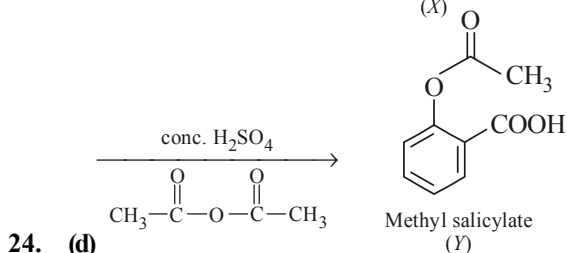
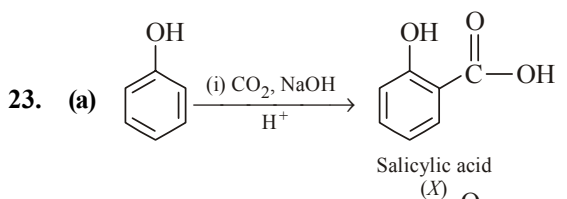
20. (d) Hydrolysis is a nucleophilic substitution. Hence more is the electron deficiency of carbonyl carbon, more will be the ease of hydrolysis; $-\text{NO}_2$ group is strongly electron-withdrawing due to $-\text{M}$ and $-\text{I}$ effects, hence it will produce electron deficiency at carbonyl carbon to the maximum extent.

21. (d)

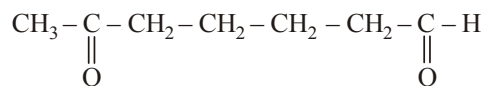
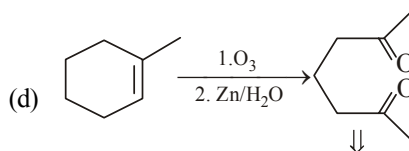
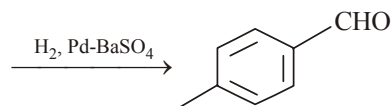
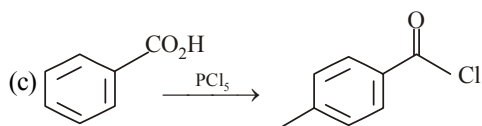
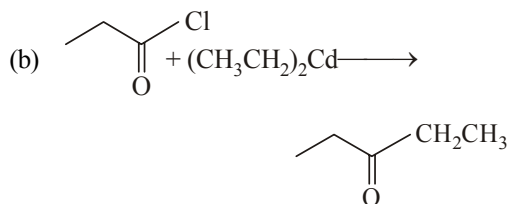
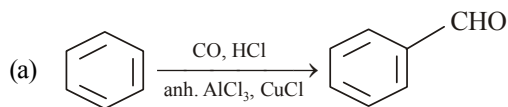
22. (a)



NaBH_4 is a selective reducing agent which reduces carboxyl group of aldehyde and ketone to alcohol, but does not reduce COOH group and double bond.

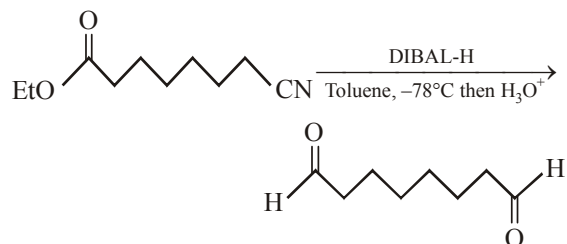


24. (d)



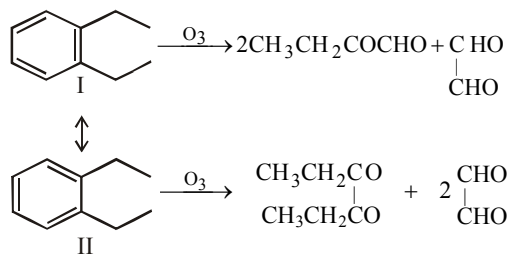
Amongst the above formed product only $\text{CH}_3 - \text{CO} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$ will give reddish brown precipitate in Fehling's solution.

25. (a) DIBAL-H is a reducing agent it reduces RCN to RCHO and esters i.e. RCOOR' to RCHO . Therefore,



26. (3)

1, 2-Diethylbenzene is a resonance hybrid of the two structures which on ozonolysis give different products.

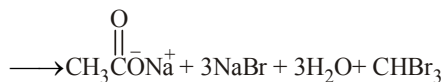


27. (4) $2\text{CH}_3\text{CHO} \longrightarrow$ 1st Product,
 $2\text{CH}_3\text{CH}_2\text{CHO} \longrightarrow$ 2nd Product



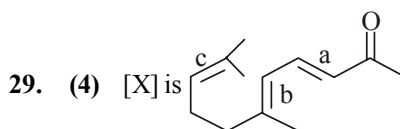
28. (11) $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\text{C}}} - \text{CH}_3 + 3\text{Br}_2 + 4\text{NaOH}$

Acetone



3 moles of bromine produces 1 mole of CHBr_3 .

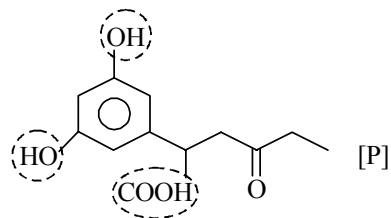
$\therefore$ 9 mole of bromine produces $\frac{1}{3} \times 9 = 3$ mole of CHBr_3 .



Out of a, b, c only a and b can show geometrical isomerism

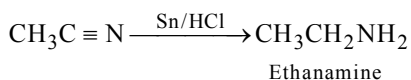
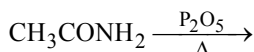
$$\text{GI} = 2^n = 2^2 = 4$$

30. (3) Compound P contains three acidic groups

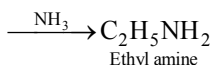
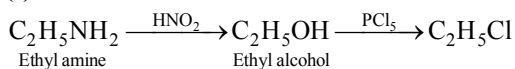


Amines

1. (c) Completing the given reaction we get



2. (b)



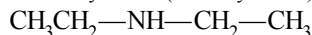
3. (d)
- $\text{CH}_3 - \text{C} \begin{smallmatrix} \text{NH} \\ \text{NH}_2 \end{smallmatrix}$
- is most basic

$(\text{CH}_3)_2\text{NH}$ is less basic than $\text{CH}_3\text{CH}_2\text{NH}_2$ due to steric hindrance because of two bulky $-\text{CH}_3$ groups on nitrogen atom.

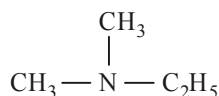
4. (a) Structures of given amines are following



n-Butylamine (Primary amine)



Diethylamine (Secondary amine)

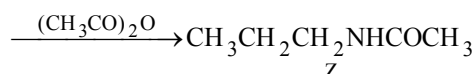
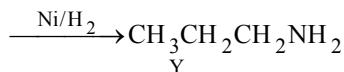
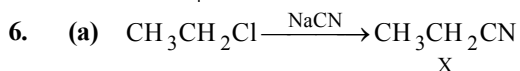


Ethyldimethylamine (Tertiary amine)

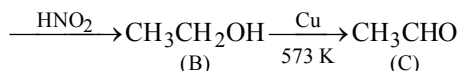
Amines have ability to form intermolecular H-bonding and the extent of H-bonding depends upon number of H-atom present of N-atom. Primary amines have two hydrogen atoms available for hydrogen bond formation, while 2° amines have only one hydrogen atom for hydrogen bond formation. 3° amines do not have intermolecular association due to absence of hydrogen atom.

Thus, order of boiling point of isomeric amines is as follows: $3^\circ < 2^\circ < 1^\circ$.

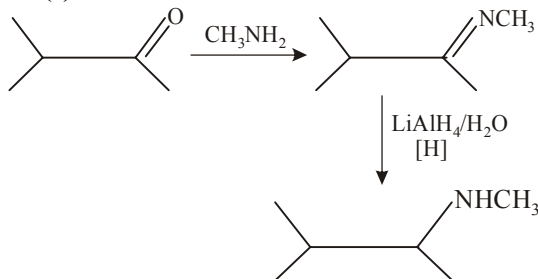
5. (c)
- CH_3NC
- (methyl isocyanide) on reduction with
- LiAlH_4
- gives secondary amine



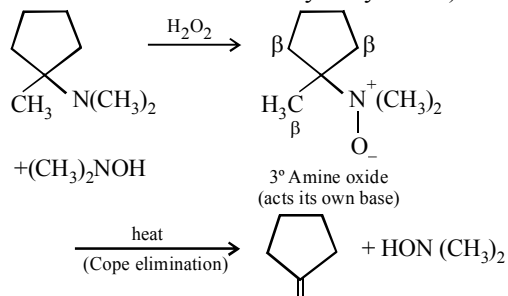
7. (d)
- $\text{CH}_3\text{CN} \xrightarrow{\text{Na/C}_2\text{H}_5\text{OH}} \text{CH}_3\text{CH}_2\text{NH}_2$
- (A)



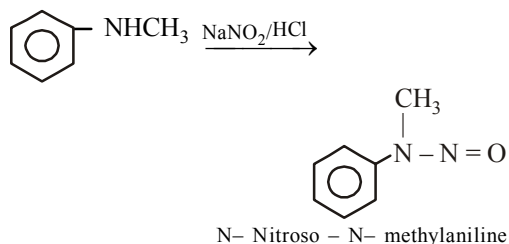
8. (b)



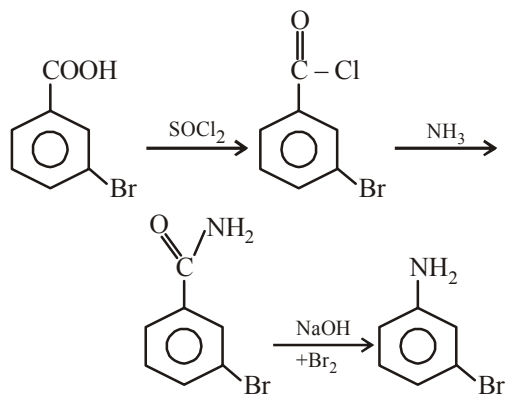
9. (d) The reaction involves Cope elimination (heating of a 3° amine oxide to form an alkene with the elimination of a 2° hydroxylamine).



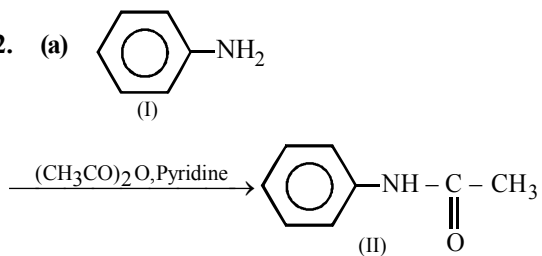
10. (d) Secondary amine with ($\text{NaNO}_2 + \text{HCl}$) gives a nitroso product



11. (c)



12. (a)



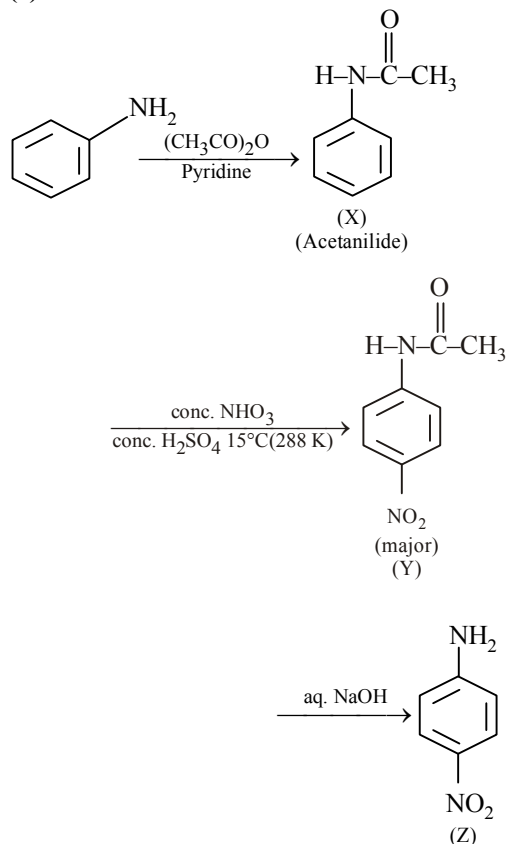
Product (II) is CC(=O)Nc1ccccc1 and

(III) is CCNc1ccccc1

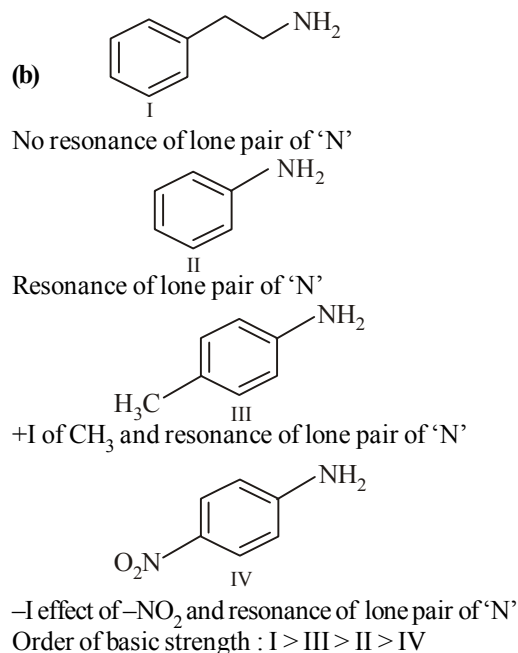
$\therefore \text{III} > \text{I} > \text{II}$, As +I effect increases the basic strength and -R, -I effect shown by $-\text{COCH}_3$ reduces the basic strength.

13. (c) $\text{NaNO}_2, \text{HCl} \Rightarrow \text{NO}^+$ (as E^+).

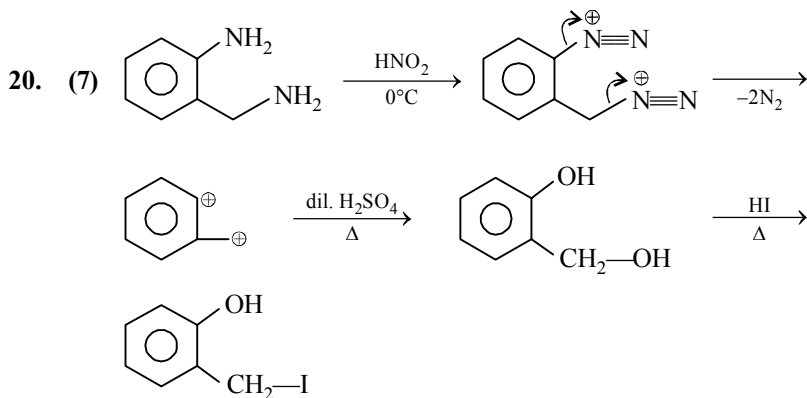
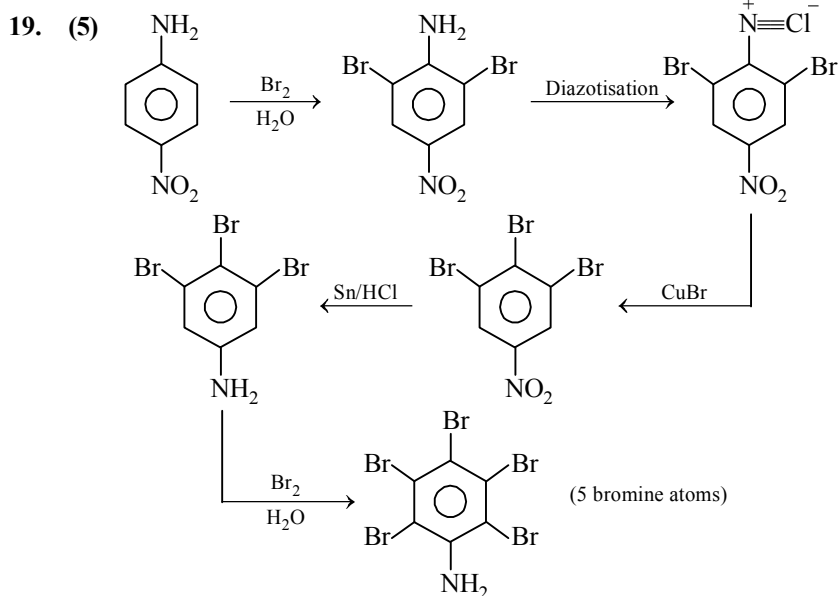
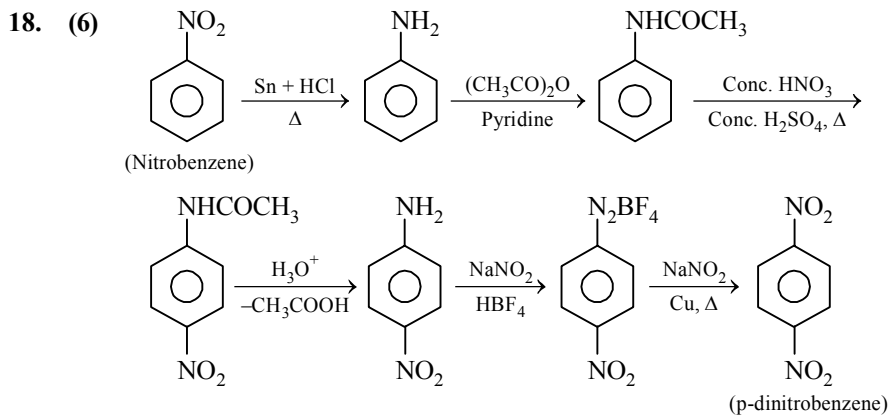
14. (b)



15. (b)



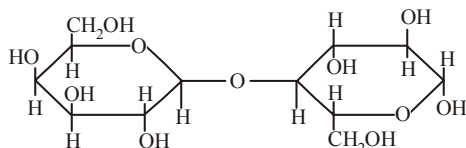
16. (2) Cope elimination is used for removing a nitrogen present outside the ring, while Hofmann elimination is used for removing nitrogen present inside as well as outside the ring.
17. (3) Diazomethane is used for methylating acidic groups; compound IV has enolic —OH group, hence it can also be methylated by CH_2N_2 .



Biomolecules

1. (d) Reaction of D-(+)-glucose with methanolic — HCl leads to formation of methyl glucoside (C_1 —OH group is methylated) which, being acetal, is not hydrolysable by base, so it will not respond Tollen's reagent.

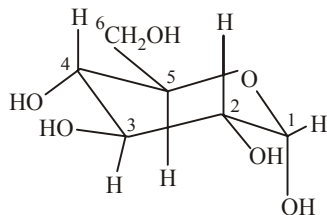
2. (c)



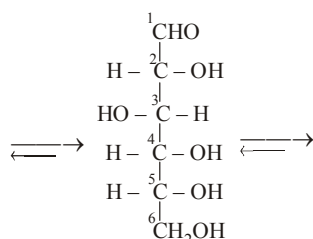
(Lactose)

All reducing sugar shows mutarotation.

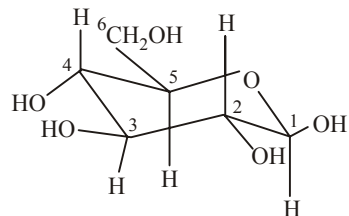
3. (b) Cyclization of the open chain structure of D-(+)-glucose has created a new stereocenter at C_1 which explains the existence of two cyclic forms of D-(+)-glucose, namely α - and β -. These two cyclic forms are diastereomers, such diastereomers which differ only in the configuration of chiral carbon developed on hemiacetal formation (it is C_1 in glucose and C_2 in fructose) are called anomers and the hemiacetal carbon (C_1 or C_2) is called the anomeric carbon.



α -D-(+)-Glucopyranose
m.p. 146°C ; $[\alpha]_D = +112.2^\circ$
(36% at equilibrium)

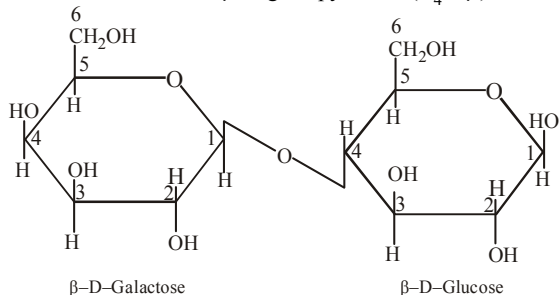


Open chain form of D-(+)-Glucose
(negligible % at equilibrium)



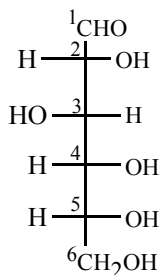
β -D-(+)-Glucopyranose
m.p. 150°C ; $[\alpha]_D = +18.7^\circ$
(67% at equilibrium)

4. (b) Lactose is obtained by the condensation of one molecule of β -D-galactopyranose (C_1 — β) unit and one molecule of β -D-glucopyranose (C_4 — β) unit.

 β -D-Galactose β -D-Glucose**Lactose**

5. (c) Fischer gave the prefix "D" to compounds whose bottom chiral has its OH to the right. So natural glucose is called D-glucose or dextrose.

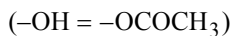
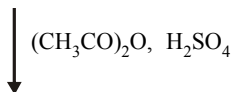
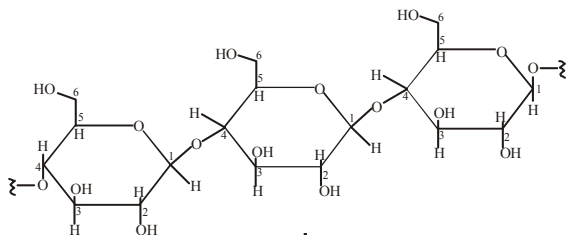
Structure of D-Glucose :



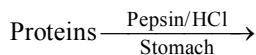
D-Glucose

6. (c) Glycogen is called animal starch and is found in all animal cells. It constitutes the reserve food material.
7. (a)
8. (a) Cellulose is a polysaccharide composed of only D-glucose units. Every adjacent glucose units are joined by β -glycosidic linkage between C_1 of one glucose and C_4 of the next.

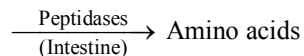
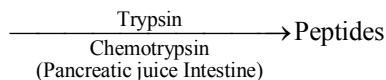
Thus in every glucose units only three -OH groups are free to form triacetate.



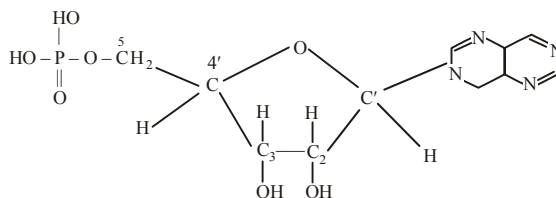
9. (a) Since the compound forms a yellow crystalline solid, i.e. osazone with phenylhydrazine, it may be an aldohexose or a ketohexose. Further, since on reduction, compound forms a mixture of sorbitol and mannitol, it must be a ketohexose, i.e. fructose. Recall that glucose on reduction gives only one alcohol glucitol (Sorbitol)
10. (a) Statement-2 is the correct explanation of statement-1.
11. (c) When the proteins are subjected to the action of heat, mineral acids or alkali, the water soluble form of globular protein changes to water insoluble fibrous protein. This is called denaturation of proteins. During denaturation secondary and tertiary structures of protein destroyed but primary structures remains intact.
12. (b) Pepsin and Trypsin are two enzymes involved in the process (hydrolysis of proteins)



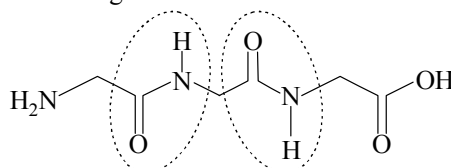
Proteases and Peptones



13. (b) In DNA and RNA heterocyclic base and phosphate ester are at C_1' and C_5' respectively of the sugar molecule.



14. (d) Ascorbic acid - Scurvy
Retinol - Night blindness
Riboflavin - Cracked lips
Thiamine - Beri-Beri
15. (c)
16. (c) Leucine, Lysine and Histidine are essential amino acids.
17. (b) The $\left(\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}- \end{array} \right)$ group represents a peptide linkage.



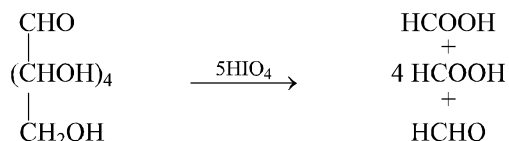
Hence, 2 peptide linkages.

18. (a) Number of H-bonds in each A-T pair = 2
Number of H-bonds in G-C pair = 3
The given double stranded DNA has 5 A-T and 3 G-C pairs.
 $\therefore$ Total number of H-bonds in 5 A-T pair = $5 \times 2 = 10$
Total number of H-bonds in 3 G-C pair = $3 \times 3 = 9$
Average energy of 10 H-bonds in A-T pair = $10x$
Average energy of 9 H-bonds in G-C pair = $9y$.
Thus, the approximate energy required to split the double stranded DNA in two single strands = $10x + 9y$.

19. (d) It is given that C-terminal amino acid is alanine and N-terminal amino acid is chiral, thus glycine can't be N-terminal amino acid as it is an achiral amino acid.
Thus it can have 4 possible sequences of the tetrapeptide as follows:
(i) Ala-Gly-Ser-val
(ii) Ala-Gly-Val-Ser
(iii) Ala-Ser-Gly-Val
(iv) Ala-Val-Gly-Ser
20. (c) In sucrose the two monosaccharides are held together by glycosidic linkage and not by peptide linkage.
21. (3) In strongly alkaline solution of an amino acid, all of its —COOH groups are converted into —COO^- . Thus a strongly alkaline solution of a monoaminodicarboxylic acid will have one —NH_2 and two —COO^- groups, all of which are basic in nature. Further remember that a —NH_2 is more basic than a —COO^- group.
22. (1) Two molecules of an α -amino acid will form only one dipeptide, recall that four different dipeptides are formed when two α -amino acids are different.
23. (36.6)
Let x be the % of α -form in the equilibrium

$$-21x - 133(100 - x) = -92 \times 100$$

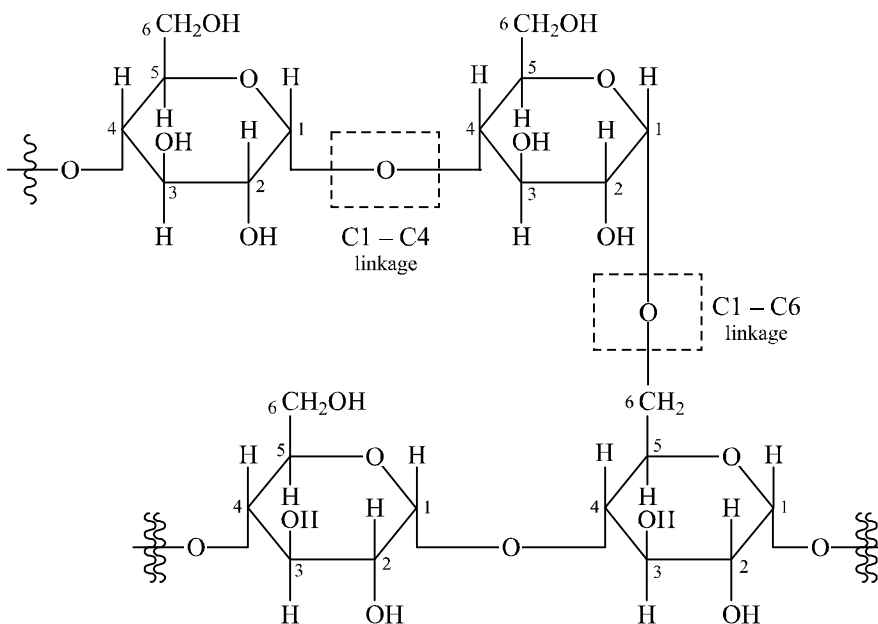
$$x = 36.6\%$$
24. (80) Each mole of glucose will consume 5 moles of HIO_4 .



Glucose has 4 chiral centres No. of optical isomers = $2^4 = 16$.

Total moles of HIO_4 consumed = $16 \times 5 = 80$.

25. (12)



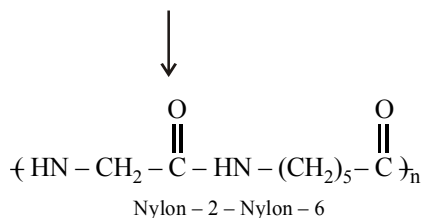
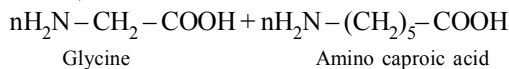
Amylopectin

The straight chain linkage is between C1 – C4 and branching linkage is between C1 – C6.

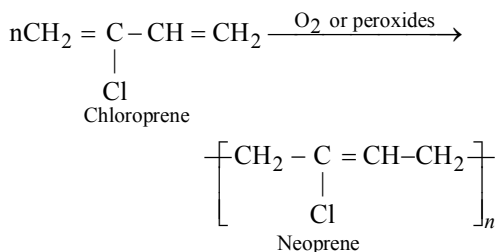
Hence, $a + b + c + d = 1 + 4 + 1 + 6 = 12$

Polymers

- (a) Molecules of a polymer, being large in size, scatter light
- (d)
- (c) Polythene is a linear polymer
- (d) Glyptal is an example of a step growth polymer
- (d) The given compound is a copolymer of hexamethylene diamine and adipic acid. It is actually Nylon-66.
- (b) Biodegradable polymer is Nylon-2-Nylon-6 which is copolymer of glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$) and amino caproic acid ($\text{H}_2\text{N}-(\text{CH}_2)_5-\text{COOH}$).

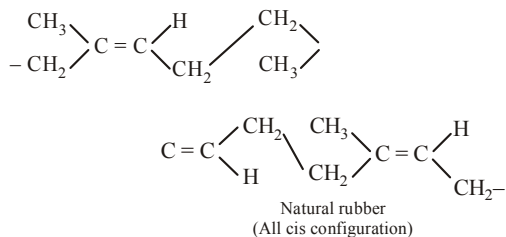


- (a) PVC is an example of thermoplastic polymer.
- (a) The synthetic polymer neoprene (polychloroprene) is prepared by polymerisation of chloroprene.



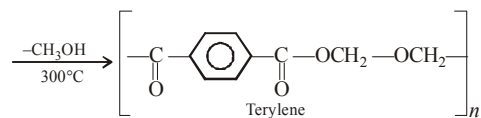
This is inferior to natural rubber but is more resistant to aerial oxidation and to other solvents.

- (b) Since proteins, cellulose and RNA control various activities of plants and animals, they are called biopolymers.
- (a)
- (b)

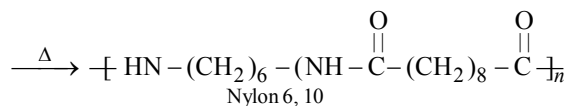
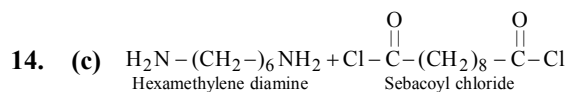


All statements except (b) are correct

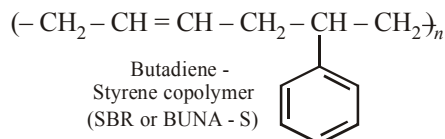
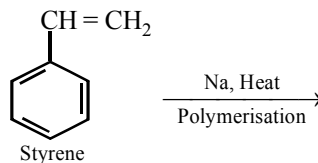
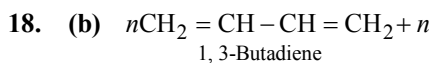
- (a) Terylene is prepared by condensing dimethylterephthalate and ethylene glycol in presence of a weak base, (calcium acetate). Methanol is eliminated during condensation.



- (b) Nylon-66 is an example of first synthetic fibres produced from the simple molecules. It is prepared by condensation polymerisation of adipic acid and hexamethylene diamine.



15. (d) Acrylonitrile is the monomer used in the preparation of orlon.
16. (c) All other three polymers are obtained by addition polymerisation
17. (b) Vinyl alcohol, $\text{CH}_2=\text{CHOH}$, monomer of polyvinyl alcohol exists mainly as CH_3CHO ; hence polyvinyl alcohol is best prepared by the alkaline hydrolysis of polyvinyl acetate which in turn is prepared by the polymerisation of vinyl acetate



19. (b)
20. (d) Vinylidene chloride does not form isotactic, syndiotactic and atactic-polymers. Vinylidene chloride does not contain chiral carbon atoms.

CHAPTER

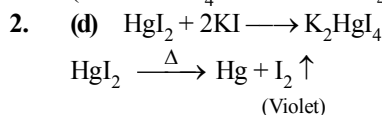
30

Chemistry in Everyday Life

1. (b)
2. (a) Amoxycillin is semisynthetic modification of penicillin
3. (d) 4. (c) 5. (c)
6. (d) Vitamin E is an antioxidant present in edible oils.
7. (a) Substances used for the treatment of malaria are antimalarials, e.g. quinine, chloroquine.
8. (a)
9. (b) Paul Ehrlich, the father of chemotherapy defined it to injure or destroy infection caused by microorganism by the use of drugs without causing any injury to the host.
10. (a)
11. (b) The term “antihistamine” refers only to H_1 antagonists, which is also known as H_1 -receptor antagonists and H_1 -antihistamine.
12. (c) We know that N-acetyl-para-aminophenol (or paracetamol) is an antipyretic which can also be used as an analgesic to relieve pains.
13. (b) Dilute solutions of boric acid and hydrogen peroxide are weak antiseptics.
14. (b) Structurally, biodegradable detergents should contain branched alkyl chain.
15. (c) The correct matching is as follows :

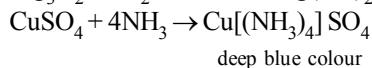
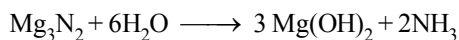
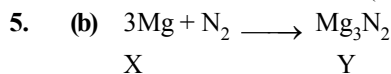
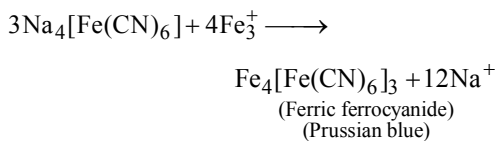
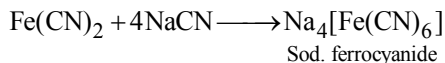
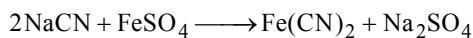
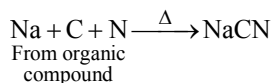
	Column I	Column II
(A)	Sodium perborate	Milk bleaching agent
(B)	Chlorine	Disinfectant
(C)	Bithional	Antiseptic
(D)	Potassium stearate	Soap

1. (d) $\text{Meq of FeSO}_4 = 10 \times \frac{1}{10} = 1$
 (Meq of $\text{FeSO}_4 = \text{Meq of KMnO}_4$)
 Meq. of KMnO_4 needed = 1
 $\therefore$ Meq of 10 mL of 0.02 M = $10 \times 0.1 = 1$
 (1 MK $\text{MnO}_4 = 5e^-$ of KMnO_4 in acid medium)



3. (c)

4. (b) The chemistry of the Lassaigne's test of nitrogen is



6. (d) Mixture of salt, Na_2CO_3 , water when combinedly heated in test tube gives soda extract.
 7. (c) Microcosmic salt $[\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}]$ is also used for bead test.

8. (a) IVth group needs higher S^{2-} ion concentration. In presence of HCl , the dissociation of H_2S decreases hence produces less amount of sulphide ions due to common ion effect, thus HCl decreases the solubility of H_2S which is sufficient to precipitate IInd group radical.

9. (c) Carius method of estimation of halogen: In this method a known weight of organic compound is heated with fuming nitric acid in presence of AgNO_3 . Halogen forms silver halide. Carbon and hydrogen of the compound are oxidised to CO_2 and H_2O .

% of Halogen (X)

$$= \frac{\text{At. wt. of X} \times \text{wt. of AgX} \times 100}{\text{Mol. wt. of AgX} \times \text{wt. of organic compound}}$$

Let the atomic weight of the halogen (X) = x
 the atomic weight of $\text{AgX} = 108 + x$

Substituting the value in the above equation we get,

$$46.78 = \left(\frac{x}{108 + x} \right) \times \frac{2.21 \times 100}{2}$$

$$\frac{46.78 \times 2}{2.21 \times 100} = \frac{x}{108 + x}$$

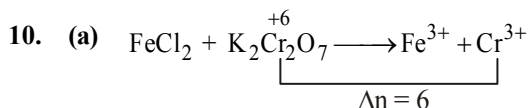
$$0.42335 = \frac{x}{108 + x}$$

$$0.42335(108 + x) = x$$

$$45.7218 + 0.42335x = x$$

$$x = 79.3 \approx 80$$

$\therefore$ Halogen present in organic compound is Bromine.

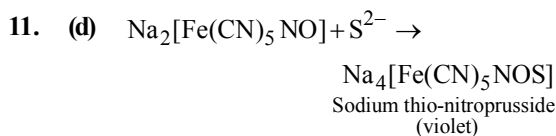


$$\begin{aligned} \text{No. of eq. of FeCl}_2 \\ &= \text{No. of eq. of K}_2\text{Cr}_2\text{O}_7 \\ \text{N}_1\text{V}_1 &= \text{N}_2\text{V}_2 \\ \text{N}_1 \times 10 &= \text{M} \times \text{n} \times 25 \quad [\because \text{N} = \text{M} \times \text{n}] \end{aligned}$$

$$N_1 \times 10 = \frac{n}{V(\text{in L})} \times n_f \times 25$$

$$N_1 \times 10 = \frac{\frac{1.225}{294}}{250 \times 10^{-3}} \times 25$$

$$\begin{aligned} N_1 &= \frac{1.225 \times 25}{294 \times 250 \times 10 \times 10^{-3}} \\ &= \frac{1.225}{10 \times 10 \times 10^{-3} \times 294} \\ N_1 &= \frac{12.25}{294} = 0.25 \end{aligned}$$



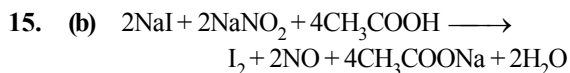
12. (a) % Of sulphur

$$= \frac{32}{233} \times \frac{\text{Mass of BaSO}_4 \text{ formed}}{\text{Mass of substance taken}} \times 100$$

$$\% \text{ of S} = \frac{32}{233} \times \frac{0.699}{0.480} \times 100 = 20 \%$$

13. (d) Yellow colour is due to the formation of $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$

14. (d) Carboxylic acids dissolve in sodium bicarbonate, while phenol does not.



16. (b) K_2SO_4 raises b.pt. and CuSO_4 acts as catalyst.

17. (c) $(\text{C}_2\text{H}_5)_3\text{N}^+\text{HCl}^-$ is an ionic compound, so it will form precipitate of AgCl on adding AgNO_3 . In 2, 4, 6-trinitro-chlorobenzene, $-\text{Cl}$ is activated due to the presence of three $-\text{NO}_2$ groups in *o*- and *p*-positions, so it will be very reactive leading to formation of AgCl on adding HNO_3 and AgNO_3 .

18. (a) H_2SO_4 is dibasic.

$$0.1 \text{ MH}_2\text{SO}_4 = 0.2 \text{ NH}_2\text{SO}_4$$

$$M_{eq} \text{ of } H_2SO_4 \text{ taken} = 100 \times 0.2 = 20$$

$$M_{\text{eq}} \text{ of } \text{H}_2\text{SO}_4 \text{ neutralised by NaOH} = 20 \times 0.5 = 10$$

M_{eq} of H_2SO_4 neutralised by $NH_3 = 12 - 10 = 2$
% of

$$N_2 = \frac{1.4 \times M_{eq} \text{ of acid neutralised by } NH_3}{\text{wt. of organic compound}}$$

$$= \frac{1.4 \times 10}{0.3} = 46.6$$

$$\% \text{ of nitrogen in urea} = \frac{14 \times 2 \times 100}{60} = 46.6$$

19. (b) Percentage of nitrogen by Kjeldahl's method

$$= \frac{14}{1000} \times \frac{\text{Normality of acid} \times \text{vol. of acid used}}{\text{wt. of organic compound}}$$

Mass of organic compound = 0.5 g

Unused acid required = 80 mL of 0.5 M NaOH

$$= 80 \text{ mL of } 0.5 \text{ N NaOH}$$
 $\equiv 40 \text{ mL of } 0.5 \text{ M or } 1 \text{ N H}_2\text{SO}_4$

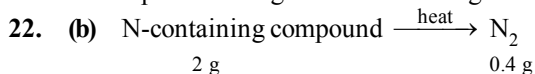
Acid used for absorption of ammonia

$$= (50 - 40) \text{ mL of } 1 \text{ N H}_2\text{SO}_4$$
$$= 10 \text{ mL of } 1 \text{ N H}_2\text{SO}_4$$

$$\% \text{ of nitrogen} = \frac{1.4 \times 1 \times 10}{0.5} = 28\%$$

20. (b) Glycerol is purified by vaccum distillation.

21. (d) Since diazo compounds may lose nitrogen in the form of nitrogen gas, they sometimes do not respond Lassaigue's test for nitrogen.

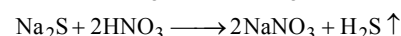
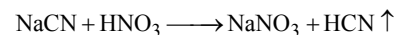


0.4 g of N_2 is obtained from 2 g of compound

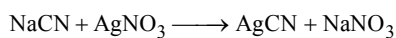
28 g of N_2 is obtained from $\frac{2}{0.4} \times 28 = 140$

Hence molecular weight of the compound is 140 which coincides with $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$ ($72 + 5 + 28 + 35.5 = 140.5$).

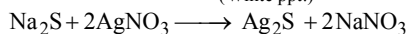
23. (b) If the organic compound also contains nitrogen or sulphur, the Lassaigne's extract on boiling with dil. HNO_3 decomposes sodium cyanide or sodium sulphide formed during fusion.



If cyanide and sulphide ions are not decomposed, they will react with silver nitrate and hence will interfere with the test.



Silver
cyanide
(White ppt.)

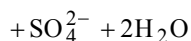


Silver
sulphide
(Black ppt.)

24. (c) IV group reagent



25. (c) $\text{CuSO}_4 + 4\text{NH}_4\text{OH} \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$
Intense blue colour



26. (50) Eq of KMnO_4 used = $\frac{50 \times 1}{1000 \times 10} = 0.005$

$\therefore$ Eq of FAS reacted = 0.005

$\therefore$ weight of FAS needed = $0.005 \times 392 = 1.96 \text{ g}$

$$\% \text{ purity} = \frac{1.96}{3.92} \times 100 = 50\%$$

Thus percentage purity of FAS is 50%

27. (20) % of N in an organic compound

$$= \frac{\text{Mass of nitrogen}}{\text{Mass of organic compounds}} \times 100 \quad \dots (i)$$

Now, 22400 mL of N_2 at STP weigh = 28 g

$\therefore$ 224 mL of N_2 at STP weigh

$$= \frac{28}{22400} \times 224$$

On substituting the value in equation (i) we get,

$$\% \text{N} = \frac{28 \times 224}{22400 \times 1.4} \times 100 = 20\%$$

28. (56) Percentage of N in a compound

$$= \frac{1.4 \times \text{Normality of acid} \times \text{Volume of acid used}}{\text{Mass of the substance taken}}$$

Given, 0.5 M H_2SO_4 is used.

$$\text{Normality} = \text{Molarity} \times n$$

$$\text{where } n = \frac{\text{Mol.mass}}{\text{Eq.mass}} = \frac{98}{49} = 2$$

$$\therefore \text{Normality} = 0.5 \times 2 = 1 \text{ N } \text{H}_2\text{SO}_4$$

Volume of acid used to neutralise $\text{NH}_3 = 10 \text{ cm}^3$

Mass of organic compound taken = 0.25 g

$$\therefore \% \text{N} = \frac{1.4 \times 1 \times 10}{0.25} = 56.$$

29. (16.46) Given wt of compound taken (w) = 0.35 g

Volume of nitrogen collected (V) = 55 ml

Room temperature (t K) = 300 K

Atmospheric pressure (P) = 715 mm

Aq. tension (p) = 15 mm

Calculation -

Volume of N_2 at NTP

$$= \frac{(P - p) \times V}{t} \times \frac{273}{760} \text{ ml}$$

$$= \frac{(715 - 15) \times 55}{300} \times \frac{273}{760} = 46.098 \text{ ml}$$

% of nitrogen

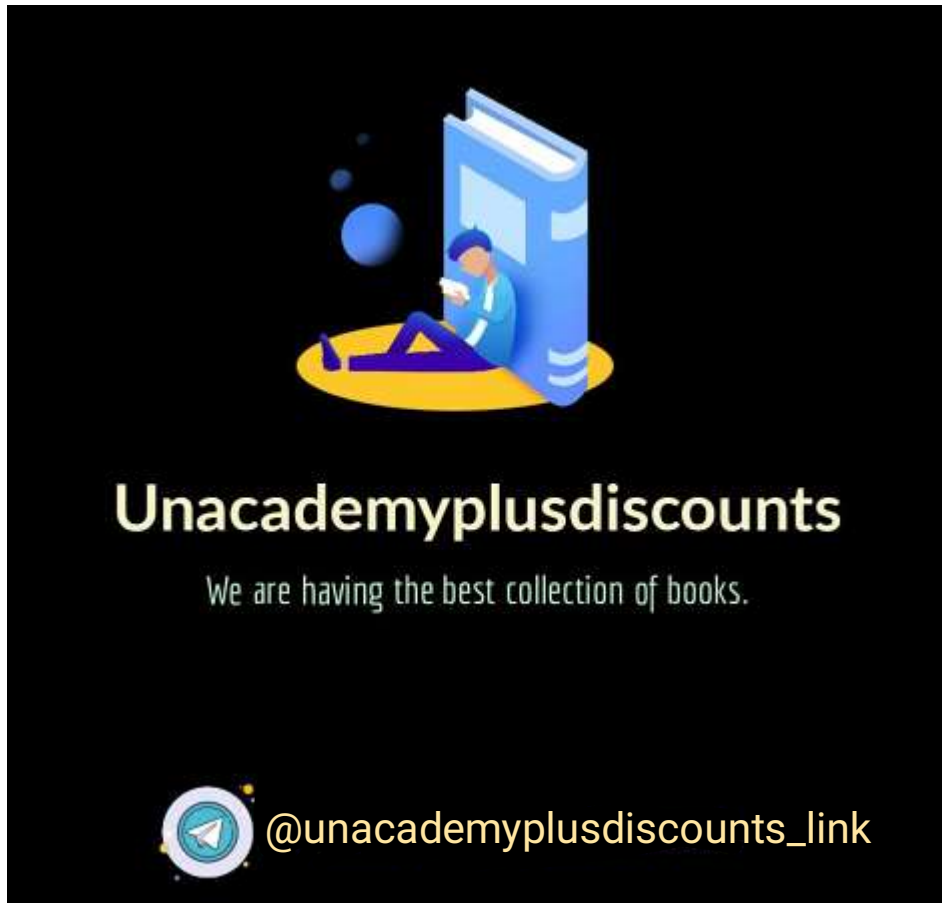
$$= \frac{28 \times \text{vol. of } \text{N}_2 \text{ at NTP} \times 100}{22400 \times \text{wt of organic compound}}$$

$$= \frac{28 \times 46.098 \times 100}{22400 \times 0.35} = 16.46\%$$

30. (13.33) Percentage of

$$P = \frac{62}{222} \times \frac{\text{wt. of } \text{Mg}_2\text{P}_2\text{O}_7}{\text{wt. of compound}} \times 100$$

$$= \frac{62}{222} \times \frac{1.332}{2.79} \times 100 = 13.33\%$$



https://telegram.me/Unacademyplusdiscounts_link

https://telegram.me/Unacademyplusdiscounts_link

https://telegram.me/Unacademyplusdiscounts_link

Join Us Now For all study Materials for free